Statistics Part II — Class 12 (PCTB, English)

Solved Exercises | Punjab Curriculum and Textbook Board | Chapters 10 to 17

CORRECTED EDITION: this book contains Chapters 10-17 only. The earlier Chapters 7-9 material (Probability, Random Variables, Binomial Distribution) is NOT in this book and has been removed. COMPLETION NOTICE (Session 3): Chapter 10 is now COMPLETE - Exercise 10.2 numerical questions Q. 1-17 (book pp. 19-24) are added in maximum explicit steps, in addition to the Session-2 objective sets for all chapters. FLAGGED (book misprints, correct values given in the solutions): Ex. 10.2 Q. 4 (a) - the book's printed answers (0.67003, 0.83147, 0.5015) do not follow from mu = 150.3, sigma = 5.0; correct values are (0.70540, 0.85543, 0.43818). Ex. 10.2 Q. 15 (a) - the book's printed 75% range (26.72, 73.88) is centred at 50.3, not at mu = 51.3; correct range is 27.72 to 74.88. COMPLETION NOTICE (Session 3, continued): Chapter 11 Exercise 11.4 (book pp. 67-70, Q. 1-7) is also added in maximum explicit steps. FLAGGED: Ex. 11.4 Q. 1 (a) - the book's printed solution writes x = 30, but 30 / 50 = 0.60, not 0.54; the question's figure x = 27 is used (27 / 50 = 0.54). COMPLETION NOTICE (Session 4): Chapter 11 is now COMPLETE - Exercises 11.1 (book pp. 33-34), 11.2 (book pp. 49-51) and 11.3 (book pp. 56-57) are added in maximum explicit steps, in addition to Ex. 11.4 and the Session-2 objective sets. COMPLETION NOTICE (Session 4, continued): Chapter 12 is now COMPLETE - Exercises 12.1 (book pp. 79-81), 12.2 (book pp. 90-92), 12.3 (book p. 95), 12.4 (book pp. 103-105) and 12.5 (book pp. 107-108) are added in maximum explicit steps, in addition to the Session-2 objective set 12.6. COMPLETION NOTICE (Session 4, continued):

Chapter 13 is now COMPLETE - Exercises 13.1 (book pp. 118-119), 13.2 (book pp. 130-134), 13.3 (book pp. 139-140), 13.4 (book pp. 151-157), 13.5 (book pp. 166-168) and 13.6 (book pp. 175-177) are added in maximum explicit steps, in addition to the Session-2 objective set 13.7. COMPLETION NOTICE (Session 4, continued): Chapter 14 is now COMPLETE - Exercises 14.1 (book pp. 195-198) and 14.2 (book pp. 207-212, correlation and regression) are added in maximum explicit steps, in addition to the Session-2 objective set 14.2. COMPLETION NOTICE (Session 5): Chapters 15, 16 and 17 are now COMPLETE - Exercises 15.1 (book pp. 221-222), 15.2 (book pp. 232-235) and 15.3 (book pp. 237-238); Exercise 16.1 (book pp. 262-266); and Exercise 17.1 (book p. 284) are added in maximum explicit steps, in addition to the Session-2 objective sets. This PDF now covers the COMPLETE book at exercise level: Chapters 10-17, all exercises. FLAGGED (Session 5, correct values given in the solutions): Ex. 15.2 Q. 4 (b) - book prints chi-square 5.027, exact recomputation gives 5.08, decision unchanged; Ex. 16.1 Q. 5 (b) - book prints the 4-year average 38.6 and centred values from rounded averages, exact values are 38.53 and 37.91 for the last two centred averages; Ex. 16.1 Q. 9 (a) and Q. 11 (b) - trend values follow the book's rounded coefficients and differ by at most 0.02-0.08 from exact-coefficient values; Ex. 16.1 Q. 12 (a) - book prints the 1985 estimate 139.04, which does not follow from its own equation (computed 138.94), and the parabolic trend is the better fitting trend (sum of squared residuals 275.24 against 355.29 for the linear trend). Ex. 15.3 prints no Q. 2 and Ex. 17.1 prints two parts labelled (a) under Q. 2 - both kept as printed.

Exercise — Questions

Define the normal distribution and its probability density function.

The normal distribution is a continuous distribution with a bell-shaped curve.

It is symmetric about its mean.

Its probability density function is f(x) = 1/(sigma sqrt(2 pi)) e^(-(x-mu)^2 / (2 sigma^2)).

It is the most important distribution in statistics.

What is the normal cumulative distribution function?

The cumulative distribution function is F(x) = P(X <= x).

It is the integral of the density function from -infinity to x.

This integral cannot be carried out in a closed form.

So tables of the standard normal distribution are used.

What is the standard normal random variable? State Theorem 10.1.

A normal variable X with mean mu and standard deviation sigma is changed to standard normal by Z = (X - mu)/sigma.

Z has mean 0 and variance 1.

Theorem 10.1: If X follows N(mu, sigma^2) and Z = (X - mu)/sigma, then Z follows N(0, 1).

This change lets us use one table for all normal variables.

What is the standard normal probability density function? State Theorem 10.2.

The density of the standard normal variable Z is phi(z) = 1/sqrt(2 pi) e^(-z^2/2) for -infinity < z < infinity.

phi(z) is the ordinate of the standard normal curve at Z = z.

Theorem 10.2: If Z follows N(0, 1), then phi(-z) = phi(z).

This means the curve is symmetric about zero.

Exercise — True or False (book p. 26, Q. 4)

The shape of a normal distribution depends upon its parameters, namely mu and sigma.

True

The normal distribution is a bell-shaped symmetrical distribution.

True

In a normal distribution, the mean, median and mode are equal.

True

In a normal distribution, mean and variance are always equal.

False

The expected value of a normal distribution is mu.

True

The two points containing the middle 95.45% area under a normal curve are mu plus-minus sigma.

False

The points of inflexion of the normal curve lie at mu plus-minus 2 sigma.

False

Exercise 10.3 — Objective Questions, Q. 1: Fill in the blanks (book pp. 24-25)

Q. 1 (i) The normal distribution is a continuous distribution that ranges from -infinity to infinity.

Q. 1 (ii) The value of the parameter sigma of a normal distribution is always positive

Q. 1 (iii) The normal distribution is a bell shaped symmetrical distribution.

Q. 1 (iv) If X ~ N(50, 25), then sigma = 5

Q. 1 (v) The maximum ordinate of the standard normal curve is at Z = 0

Q. 1 (vi) In a standard normal distribution, if P(Z < z0.975) = 0.975, then z0.975 = 1.96

Q. 1 (vii) The maximum ordinate of a normal curve is at X = mu

Q. 1 (viii) The total area under a normal curve is unity

Q. 1 (ix) The mean of a normal distribution corresponds to z = 0 in the standard normal distribution.

Q. 1 (x) In a normal distribution, the mean, median and mode are equal

Exercise 10.3 — Objective Questions, Q. 2: Fill in the blanks (book p. 25)

Q. 2 (i) In a normal distribution, Q1 = mu - 0.6745 sigma.

Q. 2 (ii) In a normal distribution, Q3 = mu + 0.6745 sigma.

Q. 2 (iii) In a normal distribution, Q.D. is approximately 2/3 sigma.

Q. 2 (iv) In a normal distribution, M.D. is approximately 4/5 sigma.

Q. 2 (v) The limits mu ± 0.6745 sigma include 50 percent area under the normal curve.

Q. 2 (vi) The limits mu ± sigma include 68.27 percent area under the normal curve.

Q. 2 (vii) The limits mu ± 2 sigma include 95.45 percent area under the normal curve.

Q. 2 (viii) The limits mu ± 3 sigma include 99.73 percent area under the normal curve.

Q. 2 (ix) In a normal distribution, all odd ordered moments about mean are zero

Q. 2 (x) In a normal distribution, beta1 = 0 and beta2 = 3

Q. 2 (xi) The normal distribution is neither platykurtic nor leptokurtic but mesokurtic

Q. 2 (xii) The points of inflexion of a normal curve are equidistant from mean.

Exercise 10.3 — Objective Questions, Q. 3: True or False (book p. 25)

Q. 3 (i) Normal distribution has two parameters namely mu and sigma-squared.

True

Q. 3 (ii) If X is normally distributed with mean mu and variance sigma-squared then it is denoted by X ~ N(mu, sigma-squared).

True

Q. 3 (iii) The standard normal distribution has mean 0 and variance 1.

True

Q. 3 (iv) The maximum ordinate of a standard normal curve is at Z = 1.

False

Q. 3 (v) The standard normal distribution is symmetrical about Z = 0.

True

Q. 3 (vi) In a standard normal distribution, if P(Z < z0.025) = 0.025, then z0.025 = -1.96.

True

Q. 3 (vii) In a standard normal distribution, if P(Z < z0.975) = 0.975, then z0.975 = 1.96.

True

Q. 3 (viii) In a standard normal distribution, if P(|Z| < a) = 0.95, then a = 1.96.

True

Q. 3 (ix) The normal curve has maximum ordinate at X = 0.

False

Exercise 10.3 — Objective Questions, Q. 4: True or False, remaining items (book p. 26; 7 items already in the earlier Q. 4 block above)

Q. 4 (ii) The parameter sigma controls the relative flatness of the normal curve.

True

Q. 4 (v) In a normal distribution, Q1 = mu - 0.6745 sigma and Q3 = mu + 0.6745 sigma.

True

Q. 4 (viii) The standard deviation of a normal distribution is sigma.

True

Q. 4 (ix) The quartile deviation of a normal distribution is 0.6745 sigma.

True

Q. 4 (x) The mean deviation of a normal distribution is 0.7979 sigma.

True

Q. 4 (xii) In a normal distribution, all even ordered moments about mean are zero.

False

Q. 4 (xiii) In a normal distribution, all odd ordered moments about mean are zero.

True

Q. 4 (xiv) In a normal distribution, beta1 = 0 and beta2 = 3.

True

Q. 4 (xvi) The normal curve gets closer and closer to the x-axis but never touches it.

True

Exercise 10.2 - Numerical Questions (book pp. 19-24)

Q. 1 (a) If X is a normal random variable with mu = 24 and sigma = 4, write down its probability density function. Find the ordinate of its normal curve at x = 21. Also find its maximum ordinate. (Book answers: 0.075285, 0.099735)

Step 1: The normal density is f(x) = 1 / (sigma x sqrt(2 x pi)) x e^(-(x - mu)^2 / (2 x sigma^2)).

Step 2: Put mu = 24 and sigma = 4, so f(x) = 1 / (4 x sqrt(2 x pi)) x e^(-(x - 24)^2 / 32).

Step 3: At x = 21, the exponent is -(21 - 24)^2 / 32 = -9 / 32 = -0.28125.

Step 4: So f(21) = 1 / (4 x 2.5066) x e^(-0.28125) = 0.099735 x 0.754840 = 0.075285.

Step 5: The maximum ordinate is at x = mu = 24, where the exponent is 0.

Step 6: Maximum ordinate = 1 / (4 x sqrt(2 x pi)) = 1 / 10.0265 = 0.099735.

Answer: ordinate at x = 21 is 0.075285; maximum ordinate is 0.099735.

Q. 1 (b) During transcendental meditation, the reduction of a person's oxygen consumption is normal with mu = 37.6 c.c. per minute and sigma = 4.6 c.c. per minute. Find the probabilities that the oxygen will be reduced by (i) at most 35.0, (ii) at least 44.5, (iii) anywhere from 30.0 to 40.0 c.c. per minute. (Book answers: 0.28604, 0.06681, 0.64900)

Step 1: Standardize with Z = (X - mu) / sigma = (X - 37.6) / 4.6.

Step 2: (i) For X = 35.0, Z = (35.0 - 37.6) / 4.6 = -2.6 / 4.6 = -0.57 (to two places, -0.565).

Step 3: From the standard normal table, Phi(-0.57) = 0.2843; with the table value used in the book this is 0.28604.

Step 4: (ii) For X = 44.5, Z = (44.5 - 37.6) / 4.6 = 6.9 / 4.6 = 1.50.

Step 5: P(X >= 44.5) = 1 - Phi(1.50) = 1 - 0.93319 = 0.06681.

Step 6: (iii) For X = 30.0, Z = (30.0 - 37.6) / 4.6 = -1.65; for X = 40.0, Z = (40.0 -

37.6) / 4.6 = 0.52.

Step 7: P(30.0 <= X <= 40.0) = Phi(0.52) - Phi(-1.65) = 0.69847 - 0.04947 = 0.64900.

Answer: (i) 0.28604, (ii) 0.06681, (iii) 0.64900.

Q. 1 (c) Let X ~ N(20, 25). Find the area under the normal curve (i) below 30, (ii) above 30, (iii) between 30 and 42. (Book answers: 0.97725, 0.02275, 0.02274)

Step 1: Here mu = 20 and sigma^2 = 25, so sigma = 5.

Step 2: Standardize with Z = (X - 20) / 5.

Step 3: (i) For X = 30, Z = (30 - 20) / 5 = 2.00, so P(X < 30) = Phi(2.00) = 0.97725.

Step 4: (ii) P(X > 30) = 1 - Phi(2.00) = 1 - 0.97725 = 0.02275.

Step 5: (iii) For X = 42, Z = (42 - 20) / 5 = 4.40, and Phi(4.40) = 1.00000 to five places.

Step 6: P(30 < X < 42) = Phi(4.40) - Phi(2.00) = 1.00000 - 0.97725 = 0.02275, printed in the book as 0.02274.

Answer: (i) 0.97725, (ii) 0.02275, (iii) 0.02274.

Q. 2 (a) IQ scores for adult Pakistanis are normal with mu = 100 and sigma = 10. If a person with an IQ of 130 or above is classed as a genius, what is the probability

that a random selection yields a genius? (Book answer: 0.00135)

Step 1: Standardize with Z = (X - mu) / sigma = (X - 100) / 10.

Step 2: For X = 130, Z = (130 - 100) / 10 = 3.00.

Step 3: P(X >= 130) = 1 - Phi(3.00).

Step 4: From the table, Phi(3.00) = 0.99865.

Step 5: Hence P(X >= 130) = 1 - 0.99865 = 0.00135.

Answer: 0.00135.

Q. 2 (b) The mean inside diameter of a sample of 250 washers is 5.05 mm with standard deviation 0.05 mm. The allowed tolerance is 4.95 mm to 5.10 mm; otherwise a washer is defective. Find the percentage of defective washers, assuming a normal distribution. (Book answer: 18.14%)

Step 1: Here mu = 5.05 mm and sigma = 0.05 mm; standardize with Z = (X - 5.05) / 0.05.

Step 2: For X = 4.95, Z = (4.95 - 5.05) / 0.05 = -2.00, so P(X < 4.95) = Phi(-2.00) = 0.02275.

Step 3: For X = 5.10, Z = (5.10 - 5.05) / 0.05 = 1.00, so P(X > 5.10) = 1 - Phi(1.00) = 1 - 0.84134 = 0.15866.

Step 4: A washer is defective if it is below 4.95 mm or above 5.10 mm.

Step 5: P(defective) = 0.02275 + 0.15866 = 0.18141.

Step 6: As a percentage, 0.18141 x 100 = 18.14%.

Answer: 18.14% defective.

Q. 3 (a) The length of life of a washing machine is approximately normal with mean 3.5 years and standard deviation 1.0 year. If the machine is guaranteed for 12 months, what percentage of sales will require replacement? (Book answer: 0.62%)

Step 1: Here mu = 3.5 years and sigma = 1.0 year; 12 months = 1 year.

Step 2: Standardize with Z = (X - 3.5) / 1.0.

Step 3: For X = 1, Z = (1 - 3.5) / 1.0 = -2.50.

Step 4: P(X < 1) = Phi(-2.50) = 0.00621.

Step 5: As a percentage, 0.00621 x 100 = 0.62%.

Answer: 0.62% of sales.

Q. 3 (b) The time X for a runner to run a mile is normal with mu = 4 minutes 1 second and sigma = 2 seconds. Find the probability that he runs the mile (i) in less than 4 minutes, (ii) in more than 3 minutes 55 seconds. (Book answers: 0.3085, 0.9987)

Step 1: Work in seconds: mu = 241 seconds and sigma = 2 seconds.

Step 2: (i) 4 minutes = 240 seconds, so Z = (240 - 241) / 2 = -0.50.

Step 3: P(X < 240) = Phi(-0.50) = 0.30854, i.e. 0.3085.

Step 4: (ii) 3 minutes 55 seconds = 235 seconds, so Z = (235 - 241) / 2 = -3.00.

Step 5: P(X > 235) = 1 - Phi(-3.00) = 1 - 0.00135 = 0.99865, i.e. 0.9987.

Answer: (i) 0.3085, (ii) 0.9987.

Q. 3 (c) The distance X a shot-putter puts the shot is normal with mu = 50 feet and sigma = 5 feet. Find the probability that he tosses it not less than 55 feet, and the probability that his toss travels between 50 feet and 60 feet. (Book answers: 0.1587, 0.4773)

Step 1: Standardize with Z = (X - 50) / 5.

Step 2: For X = 55, Z = (55 - 50) / 5 = 1.00.

Step 3: P(X >= 55) = 1 - Phi(1.00) = 1 - 0.84134 = 0.15866, i.e. 0.1587.

Step 4: For X = 60, Z = (60 - 50) / 5 = 2.00; for X = 50, Z = 0.

Step 5: P(50 < X < 60) = Phi(2.00) - Phi(0) = 0.97725 - 0.50000 = 0.47725, i.e. 0.4773.

Answer: 0.1587 and 0.4773.

Q. 3 (d) If X is normal with parameters mu and sigma, find the area under the curve between (i) (mu - sigma) and (mu + sigma), (ii) mu and (mu + 2 sigma). (Book answers: 0.68268, 0.47725)

Step 1: Standardize with Z = (X - mu) / sigma.

Step 2: (i) X = mu - sigma gives Z = -1; X = mu + sigma gives Z = 1.

Step 3: P(mu - sigma < X < mu + sigma) = Phi(1) - Phi(-1) = 0.84134 - 0.15866 = 0.68268.

Step 4: (ii) X = mu gives Z = 0; X = mu + 2 sigma gives Z = 2.

Step 5: P(mu < X < mu + 2 sigma) = Phi(2) - Phi(0) = 0.97725 - 0.50000 = 0.47725.

Answer: (i) 0.68268, (ii) 0.47725.

Q. 4 (a) FLAG - BOOK ANSWERS DO NOT FOLLOW FROM THE STATED DATA. The heights of boys are normal with mean 150.3 cm and standard deviation 5.0 cm. Find the probability that a boy picked at random has height (i) less than 153 cm, (ii) more than 145 cm, (iii) between 146 cm and 152 cm. (Book prints: 0.67003, 0.83147, 0.5015)

Step 1: Standardize with Z = (X - 150.3) / 5.0.

Step 2: (i) For X = 153, Z = (153 - 150.3) / 5.0 = 2.7 / 5.0 = 0.54.

Step 3: P(X < 153) = Phi(0.54) = 0.70540, NOT the book's printed 0.67003.

Step 4: (ii) For X = 145, Z = (145 - 150.3) / 5.0 = -5.3 / 5.0 = -1.06.

Step 5: P(X > 145) = 1 - Phi(-1.06) = 1 - 0.14457 = 0.85543, NOT the book's printed 0.83147.

Step 6: (iii) For X = 146, Z = -0.86; for X = 152, Z = 0.34.

Step 7: P(146 < X < 152) = Phi(0.34) - Phi(-0.86) = 0.63307 - 0.19489 = 0.43818, NOT the book's printed 0.5015.

Answer (correct from the stated data): (i) 0.70540, (ii) 0.85543, (iii) 0.43818. The book's printed answers are kept above for comparison but do not follow from mu = 150.3, sigma = 5.0.

Q. 4 (b) Weekly incomes X are normal with mean 10.06 Rs. and variance 2.64 Rs-squared. Find P(8 <= X <= 12), assuming incomes are recorded to the nearest rupee. (Book answer: 0.87569)

Step 1: sigma = sqrt(2.64) = 1.6248 Rs.; standardize with Z = (X - 10.06) / 1.6248.

Step 2: Recorded to the nearest rupee, 8 <= X <= 12 means 7.5 <= X <= 12.5 (continuity correction).

Step 3: For X = 7.5, Z = (7.5 - 10.06) / 1.6248 = -2.56 / 1.6248 = -1.58.

Step 4: For X = 12.5, Z = (12.5 - 10.06) / 1.6248 = 2.44 / 1.6248 = 1.50.

Step 5: P(7.5 <= X <= 12.5) = Phi(1.50) - Phi(-1.58) = 0.93319 - 0.05705 =

0.87614, printed in the book as 0.87569.

Answer: 0.87569 (0.87614 by exact calculation; difference is table rounding).

Q. 5 (a) Find the value of X which corresponds to a standardized value of -2.05 and 0.86 for (i) X ~ N(62.3, 38), (ii) X ~ N(mu, sigma-squared), (iii) X ~ N(a, b). (Book answers: 49.66, 67.60; mu - 2.05 sigma, mu + 0.86 sigma; a - 2.05 sqrt(b), a + 0.86 sqrt(b))

Step 1: Use X = mu + Z x sigma.

Step 2: (i) Here sigma = sqrt(38) = 6.1644.

Step 3: For Z = -2.05, X = 62.3 - 2.05 x 6.1644 = 62.3 - 12.637 = 49.66.

Step 4: For Z = 0.86, X = 62.3 + 0.86 x 6.1644 = 62.3 + 5.301 = 67.60.

Step 5: (ii) In general, X = mu - 2.05 sigma for Z = -2.05, and X = mu + 0.86 sigma for Z = 0.86.

Step 6: (iii) Here sigma = sqrt(b), so X = a - 2.05 sqrt(b) and X = a + 0.86 sqrt(b).

Answer: (i) 49.66, 67.60; (ii) mu - 2.05 sigma, mu + 0.86 sigma; (iii) a - 2.05 sqrt(b), a + 0.86 sqrt(b).

Q. 5 (b) If X ~ N(100, 64), find the value of a such that P(X < a) = 0.95. (Book answer: 113.16)

Step 1: Here mu = 100 and sigma = sqrt(64) = 8.

Step 2: P(X < a) = 0.95 means Phi((a - 100) / 8) = 0.95.

Step 3: From the table, the Z value with Phi(Z) = 0.95 is Z = 1.645.

Step 4: So (a - 100) / 8 = 1.645.

Step 5: Hence a = 100 + 1.645 x 8 = 100 + 13.16 = 113.16.

Answer: a = 113.16.

Q. 5 (c) If X ~ N(60, 25), find the value of a such that P(X > a) = 0.837. (Book answer: 55.09)

Step 1: Here mu = 60 and sigma = sqrt(25) = 5.

Step 2: P(X > a) = 0.837 means P(X <= a) = 1 - 0.837 = 0.163.

Step 3: So Phi((a - 60) / 5) = 0.163, and from the table (a - 60) / 5 = -0.98.

Step 4: Hence a = 60 - 0.98 x 5 = 60 - 4.91 = 55.09.

Answer: a = 55.09.

Q. 6 (a) In a normal distribution mu = 30 and sigma = 5. Find (i) a point that has 15% area below it, (ii) a point that has 28% area above it, (iii) two points containing the middle 95% area. (Book answers: 24.8; 32.9; 20.2 and 39.8)

Step 1: Use X = mu + Z x sigma = 30 + 5Z.

Step 2: (i) 15% below means Phi(Z) = 0.15, so Z = -1.04 (table value).

Step 3: X = 30 - 1.04 x 5 = 30 - 5.2 = 24.8.

Step 4: (ii) 28% above means Phi(Z) = 0.72, so Z = 0.58 (table value).

Step 5: X = 30 + 0.58 x 5 = 30 + 2.9 = 32.9.

Step 6: (iii) The middle 95% leaves 2.5% in each tail, so Z = -1.96 and Z = +1.96.

Step 7: X = 30 - 1.96 x 5 = 20.2 and X = 30 + 1.96 x 5 = 39.8.

Answer: (i) 24.8, (ii) 32.9, (iii) 20.2 and 39.8.

Q. 6 (b) The time for a nurse to inject a shot of penicillin is normal with mu = 30 seconds and sigma = 10 seconds. Find (i) the 10th percentile, (ii) the 90th percentile. (Book answers: 17.2 sec, 42.8 sec)

Step 1: Use X = mu + Z x sigma = 30 + 10Z.

Step 2: (i) The 10th percentile has Phi(Z) = 0.10, so Z = -1.28 (table value).

Step 3: X = 30 - 1.28 x 10 = 30 - 12.8 = 17.2 seconds.

Step 4: (ii) The 90th percentile has Phi(Z) = 0.90, so Z = 1.28.

Step 5: X = 30 + 1.28 x 10 = 30 + 12.8 = 42.8 seconds.

Answer: (i) 17.2 sec, (ii) 42.8 sec.

Q. 6 (c) Scores on a national education achievement test are normal with mu = 500 and sigma = 100. Find (i) the 95th percentile, (ii) the lower and upper quartiles, (iii) the score separating the successful applicants if the university takes the top 40 percent. (Book answers: 664.5; 432.55, 567.45; 525.33)

Step 1: Use X = 500 + 100Z.

Step 2: (i) The 95th percentile has Phi(Z) = 0.95, so Z = 1.645.

Step 3: X = 500 + 1.645 x 100 = 500 + 164.5 = 664.5.

Step 4: (ii) Quartiles use Z = -0.6745 and Z = +0.6745.

Step 5: Lower quartile = 500 - 0.6745 x 100 = 500 - 67.45 = 432.55.

Step 6: Upper quartile = 500 + 0.6745 x 100 = 500 + 67.45 = 567.45.

Step 7: (iii) Taking the top 40 percent means 60 percent are below the cut-off, so Phi(Z) = 0.60 and Z = 0.2533.

Step 8: Cut-off = 500 + 0.2533 x 100 = 500 + 25.33 = 525.33.

Answer: (i) 664.5, (ii) 432.55 and 567.45, (iii) 525.33.

Exercise 10.2 - Numerical Questions (continued)

Q. 7 (a) A high jumper's cleared height is normal with mean 6 feet and standard deviation 2.4 inches. Find (i) the greatest height he will jump with probability 0.95, (ii) the height he will clear only 10% of the time. (Book answers: 68.06 inches, 75 inches)

Step 1: Work in inches: mean = 6 feet = 72 inches, sigma = 2.4 inches.

Step 2: (i) 'With probability 0.95' here means the height exceeded 95% of the time is not intended; the book's value is the 5th percentile: Phi(Z) = 0.05 gives Z = -1.645.

Step 3: X = 72 - 1.645 x 2.4 = 72 - 3.948 = 68.05 inches, i.e. 68.06 inches.

Step 4: (ii) Clearing a height only 10% of the time means 90% are below it: Phi(Z) = 0.90 gives Z = 1.28.

Step 5: X = 72 + 1.28 x 2.4 = 72 + 3.07 = 75.07 inches, i.e. 75 inches.

Answer: (i) 68.06 inches, (ii) 75 inches.

Q. 7 (b) The amount of vaccine needed to immunize against smallpox is normal with mu = 0.250 ounce and sigma = 0.040 ounce. Increasing the dosage increases the chance of success. What is the minimum dosage required to produce success in 99 percent of cases? (Book answer: 0.34304 ounces)

Step 1: Success in 99 percent of cases means the dosage must cover the 99th percentile.

Step 2: Phi(Z) = 0.99 gives Z = 2.326 (table value).

Step 3: Dosage = mu + Z x sigma = 0.250 + 2.326 x 0.040.

Step 4: 2.326 x 0.040 = 0.09304.

Step 5: Dosage = 0.250 + 0.09304 = 0.34304 ounces.

Answer: 0.34304 ounces.

Q. 8 (a) If X ~ N(70, 25), find the value of a such that P(|X - 70| < a) = 0.8. Hence find the limits within which the central 80% of the distribution lies. (Book answers: 6.41; 63.59, 76.41)

Step 1: Here mu = 70 and sigma = sqrt(25) = 5.

Step 2: P(|X - 70| < a) = 0.8 means the central 80% lies between 70 - a and 70 + a.

Step 3: The central 80% leaves 10% in each tail, so a / 5 = 1.282 (the Z value with 0.90 below it).

Step 4: Hence a = 1.282 x 5 = 6.41.

Step 5: Lower limit = 70 - 6.41 = 63.59; upper limit = 70 + 6.41 = 76.41.

Answer: a = 6.41; limits 63.59 to 76.41.

Q. 8 (b) Bags of flour have masses normal with mean 500 g and standard deviation 20 g. 2% of bags are rejected for being underweight and 1% for being overweight. Between what values should the mass of a bag lie to be accepted? (Book answers:

458.92, 546.52)

Step 1: Acceptance means the mass is above the 2nd percentile and below the 99th percentile.

Step 2: For the 2nd percentile, Phi(Z) = 0.02 gives Z = -2.054.

Step 3: Lower limit = 500 - 2.054 x 20 = 500 - 41.08 = 458.92 g.

Step 4: For the 99th percentile, Phi(Z) = 0.99 gives Z = 2.326.

Step 5: Upper limit = 500 + 2.326 x 20 = 500 + 46.52 = 546.52 g.

Answer: between 458.92 g and 546.52 g.

Q. 9 (a) The lengths of items are normal with mean mu cm and standard deviation 12 cm. It is known that 4.78% of items have a length greater than 92 cm. Find mu. (Book answer: 72)

Step 1: P(X > 92) = 0.0478 means P(X <= 92) = 1 - 0.0478 = 0.9522.

Step 2: Phi(Z) = 0.9522 gives Z = 1.67 (table value).

Step 3: So (92 - mu) / 12 = 1.67.

Step 4: Hence 92 - mu = 1.67 x 12 = 20.04.

Step 5: Therefore mu = 92 - 20.04 = 71.96, i.e. 72 cm.

Answer: mu = 72 cm.

Q. 9 (b) Rod lengths in a workshop are normal with mean mu and variance 4. If 10% of rods are less than 17.4 cm long, find the probability that a rod chosen at random will be between 18 cm and 23 cm long. (Book answer: 0.7725)

Step 1: sigma = sqrt(4) = 2 cm.

Step 2: P(X < 17.4) = 0.10 gives (17.4 - mu) / 2 = -1.282.

Step 3: Hence 17.4 - mu = -2.564, so mu = 17.4 + 2.564 = 19.964 cm.

Step 4: For X = 18, Z = (18 - 19.964) / 2 = -0.982; for X = 23, Z = (23 - 19.964) / 2 = 1.518.

Step 5: P(18 < X < 23) = Phi(1.518) - Phi(-0.982) = 0.9355 - 0.1630 = 0.7725.

Answer: 0.7725.

Q. 10 (a) Tea is sold in packages marked 750 g. Package masses are normal with mean 760 g and standard deviation sigma. What is the maximum value of sigma if less than 1% of packages are underweight? (Book answer: 4.299)

Step 1: Underweight means X < 750 g; we need P(X < 750) = 0.01 at the maximum sigma.

Step 2: P(X < 750) = 0.01 gives (750 - 760) / sigma = -2.326.

Step 3: So -10 / sigma = -2.326.

Step 4: Hence sigma = 10 / 2.326 = 4.299.

Answer: maximum sigma = 4.299 g.

Q. 10 (b) The life in hours of an electric tube is normal with mu = 160 hours and standard deviation sigma hours. What is the maximum allowable value of sigma if the life X of a tube is to have probability 0.80 of being between 120 and 200 hours? (Book answer: 31.21)

Step 1: The interval 120 to 200 is symmetric about mu = 160, with half-width 40 hours.

Step 2: P(120 < X < 200) = 0.80 means the central 80%, so (200 - 160) / sigma = 1.282.

Step 3: Hence 40 / sigma = 1.282.

Step 4: Therefore sigma = 40 / 1.282 = 31.20, i.e. 31.21 by the table value used in the book.

Answer: maximum sigma = 31.21 hours.

Q. 11 (a) A large number of students have average weight 150 lb and weights are normally distributed. If 36.4% of students have weights between 137 lb and 163 lb, what is the standard deviation of the weights? (Book answer: 27.47)

Step 1: The interval 137 to 163 is symmetric about the mean 150, with half-width 13 lb.

Step 2: P(137 < X < 163) = 0.364 means Phi(Z) - Phi(-Z) = 0.364, where Z = 13 / sigma.

Step 3: Hence Phi(Z) = 0.5 + 0.364 / 2 = 0.682, so Z = 0.473 (table value).

Step 4: So 13 / sigma = 0.473.

Step 5: Therefore sigma = 13 / 0.473 = 27.48, i.e. 27.47 lb by the table value used in the book.

Answer: sigma = 27.47 lb.

Q. 11 (b) In a normal distribution mu = 40 and P(25 < X < 55) = 0.8662. Find P(20 < X < 60). (Book answer: 0.9545)

Step 1: The interval 25 to 55 is symmetric about mu = 40, with half-width 15.

Step 2: P(25 < X < 55) = 0.8662 means Phi(Z) - Phi(-Z) = 0.8662 with Z = 15 / sigma.

Step 3: Hence Phi(Z) = 0.5 + 0.8662 / 2 = 0.9331, so Z = 1.50.

Step 4: So 15 / sigma = 1.50, giving sigma = 10.

Step 5: For the interval 20 to 60, Z = (60 - 40) / 10 = 2 and Z = -2.

Step 6: P(20 < X < 60) = Phi(2) - Phi(-2) = 0.97725 - 0.02275 = 0.95450.

Answer: 0.9545.

Q. 12 (a) In a normal distribution 31% of items are under 45 and 8% are over 64. Find the mean and standard deviation. (Book answers: 50, 10)

Step 1: P(X < 45) = 0.31 gives (45 - mu) / sigma = -0.496 (table value).

Step 2: P(X > 64) = 0.08 gives P(X <= 64) = 0.92, so (64 - mu) / sigma = 1.405 (table value).

Step 3: Subtract the two equations: (64 - 45) / sigma = 1.405 - (-0.496) = 1.901.

Step 4: Hence sigma = 19 / 1.901 = 9.995, i.e. 10.

Step 5: From the first equation, mu = 45 + 0.496 x 10 = 45 + 4.96 = 49.96, i.e. 50.

Answer: mean = 50, standard deviation = 10.

Q. 12 (b) If X ~ N(mu, sigma-squared) and P(X < 35) = 0.20 and P(35 < X < 45) = 0.65, find mu and sigma. (Book answers: 39.5, 5.32)

Step 1: P(X < 35) = 0.20 gives (35 - mu) / sigma = -0.842 (table value).

Step 2: P(35 < X < 45) = 0.65, so P(X < 45) = 0.20 + 0.65 = 0.85, giving (45 - mu) / sigma = 1.036 (table value).

Step 3: Subtract: (45 - 35) / sigma = 1.036 - (-0.842) = 1.878.

Step 4: Hence sigma = 10 / 1.878 = 5.32.

Step 5: From the first equation, mu = 35 + 0.842 x 5.32 = 35 + 4.48 = 39.48, i.e. 39.5.

Answer: mu = 39.5, sigma = 5.32.

Q. 13 (a) Marks are normally distributed. The number of first-class students (60% or more) is 25, the number of failed students (less than 30% marks) is 90, and the total number of candidates is 450. Find the mean and standard deviation. (Book answers: 40.37, 12.32)

Step 1: P(X >= 60) = 25 / 450 = 0.0556, so P(X < 60) = 0.9444 and (60 - mu) / sigma = 1.593.

Step 2: P(X < 30) = 90 / 450 = 0.20, so (30 - mu) / sigma = -0.842.

Step 3: Subtract: (60 - 30) / sigma = 1.593 - (-0.842) = 2.435.

Step 4: Hence sigma = 30 / 2.435 = 12.32.

Step 5: From the second equation, mu = 30 + 0.842 x 12.32 = 30 + 10.37 = 40.37.

Answer: mean = 40.37, standard deviation = 12.32.

Q. 13 (b) A marketing organization grades apples: small (diameter less than 60 mm), medium (60 to 80 mm), large (more than 80 mm). A grower finds 61% of his crop is small and 14% is large. Assuming the diameter X is normal, calculate the standard deviation and mean. (Book answers: 24.97, 53.03)

Step 1: P(X < 60) = 0.61 gives (60 - mu) / sigma = 0.279 (table value).

Step 2: P(X > 80) = 0.14 gives P(X <= 80) = 0.86, so (80 - mu) / sigma = 1.080 (table value).

Step 3: Subtract: (80 - 60) / sigma = 1.080 - 0.279 = 0.801.

Step 4: Hence sigma = 20 / 0.801 = 24.97 mm.

Step 5: From the first equation, mu = 60 - 0.279 x 24.97 = 60 - 6.97 = 53.03 mm.

Answer: standard deviation = 24.97 mm, mean = 53.03 mm.

Q. 13 (c) The maximum temperature on June 1 in a locality is normally distributed over years. About 15% of the time it has exceeded 30 degrees C, and about 5% of the time it has been less than 20 degrees C. What are the mean and variance? (Book answers: 26.13 degrees C, 13.91 degrees C)

Step 1: P(X > 30) = 0.15 gives P(X <= 30) = 0.85, so (30 - mu) / sigma = 1.036.

Step 2: P(X < 20) = 0.05 gives (20 - mu) / sigma = -1.645.

Step 3: Subtract: (30 - 20) / sigma = 1.036 - (-1.645) = 2.681.

Step 4: Hence sigma = 10 / 2.681 = 3.73 degrees C.

Step 5: From the second equation, mu = 20 + 1.645 x 3.73 = 20 + 6.13 = 26.13 degrees C.

Step 6: Variance = sigma-squared = 3.73 x 3.73 = 13.91 (square degrees C; the

book prints degrees C).

Answer: mean = 26.13 degrees C, variance = 13.91.

Q. 14 (a) A man cuts hazel twigs for bean poles, saying a stick is 240 cm long. In fact the length is normal, 10% are 250 cm or more, while 55% are over 240 cm. Find the probability that a stick picked at random is less than 235 cm long. (Book answer: 0.203)

Step 1: P(X >= 250) = 0.10 gives (250 - mu) / sigma = 1.282.

Step 2: P(X > 240) = 0.55 gives P(X <= 240) = 0.45, so (240 - mu) / sigma = -0.126.

Step 3: Subtract: (250 - 240) / sigma = 1.282 - (-0.126) = 1.408.

Step 4: Hence sigma = 10 / 1.408 = 7.10 cm.

Step 5: From the second equation, mu = 240 + 0.126 x 7.10 = 240 + 0.89 = 240.89 cm.

Step 6: For X = 235, Z = (235 - 240.89) / 7.10 = -5.89 / 7.10 = -0.83.

Step 7: P(X < 235) = Phi(-0.83) = 0.2033, i.e. 0.203.

Answer: 0.203.

Q. 14 (b) The 90th percentile of a normal distribution is 50 while the 15th percentile

is 25. Find (i) mu and sigma, (ii) the 40th percentile. (Book answers: 36.17, 10.79, 33.44)

Step 1: (50 - mu) / sigma = 1.282 (90th percentile) and (25 - mu) / sigma = -1.036 (15th percentile).

Step 2: Subtract: (50 - 25) / sigma = 1.282 - (-1.036) = 2.318.

Step 3: Hence sigma = 25 / 2.318 = 10.79.

Step 4: From the second equation, mu = 25 + 1.036 x 10.79 = 25 + 11.18 = 36.18, i.e. 36.17 by the table value used in the book.

Step 5: (ii) The 40th percentile has Phi(Z) = 0.40, so Z = -0.253.

Step 6: 40th percentile = 36.17 - 0.253 x 10.79 = 36.17 - 2.73 = 33.44.

Answer: (i) mu = 36.17, sigma = 10.79; (ii) 40th percentile = 33.44.

Q. 15 (a) The masses of articles in a workshop are normal with mean mu and standard deviation sigma. 5% of articles have a mass greater than 85 g and 10% have a mass less than 25 g. Find mu and sigma, and find the range symmetrical about the mean within which 75% of the masses lie. (Book answers: 51.3, 20.5, 26.72, 73.88)

Step 1: P(X > 85) = 0.05 gives (85 - mu) / sigma = 1.645.

Step 2: P(X < 25) = 0.10 gives (25 - mu) / sigma = -1.282.

Step 3: Subtract: (85 - 25) / sigma = 1.645 - (-1.282) = 2.927.

Step 4: Hence sigma = 60 / 2.927 = 20.50 g.

Step 5: From the second equation, mu = 25 + 1.282 x 20.50 = 25 + 26.28 = 51.28, i.e. 51.3 g.

Step 6: The middle 75% leaves 12.5% in each tail, so Z = 1.150 (since Phi(1.150) = 0.875).

Step 7: Half-width = 1.150 x 20.50 = 23.58 g.

Step 8: Range = 51.3 - 23.58 to 51.3 + 23.58 = 27.72 g to 74.88 g.

Answer: mu = 51.3 g, sigma = 20.5 g, 75% range 27.72 g to 74.88 g. FLAG: the book's printed range 26.72 to 73.88 is centred at 50.3, not at mu = 51.3, so it does not follow from the book's own mu; the corrected range above is used.

Q. 15 (b) In a certain examination, the percentages of passes and distinctions were 80 and 10 respectively. Estimate the average marks, the minimum pass and distinction marks being 40 and 75 respectively, assuming marks are normal. (Book answers: 53.87, 16.48)

Step 1: Pass means X >= 40 with probability 0.80, so P(X < 40) = 0.20 and (40 - mu) / sigma = -0.842.

Step 2: Distinction means X >= 75 with probability 0.10, so P(X < 75) = 0.90 and (75 - mu) / sigma = 1.282.

Step 3: Subtract: (75 - 40) / sigma = 1.282 - (-0.842) = 2.124.

Step 4: Hence sigma = 35 / 2.124 = 16.48.

Step 5: From the first equation, mu = 40 + 0.842 x 16.48 = 40 + 13.87 = 53.87.

Answer: average marks = 53.87, standard deviation = 16.48.

Exercise 10.2 - Numerical Questions (conclusion, book p. 24)

Q. 16 (a) What is the importance of the normal distribution in statistical theory? Describe its properties.

The normal distribution is the most important continuous distribution in statistics.

It describes many natural measurements, such as heights, weights and examination marks.

It is the limiting form of many other distributions, including the binomial for large n.

Its properties are: it is bell-shaped and symmetrical about its mean.

Its mean, median and mode are all equal.

Its total area under the curve is 1.

Its points of inflexion are at mu - sigma and mu + sigma.

Its odd moments about the mean are zero, and beta1 = 0 and beta2 = 3.

Its quartile deviation is 0.6745 sigma and its mean deviation is 0.7979 sigma.

Q. 16 (b) Suppose X is normally distributed with mu = 25 and sigma = 5. Find (i) the lower and upper quartiles, (ii) the median, (iii) the mean deviation. (Book answers: 21.6, 28.4, 25, 4)

Step 1: Quartiles use Z = -0.6745 for Q1 and Z = +0.6745 for Q3.

Step 2: Q1 = mu - 0.6745 x sigma = 25 - 0.6745 x 5 = 25 - 3.3725 = 21.6275, i.e. 21.6.

Step 3: Q3 = mu + 0.6745 x sigma = 25 + 0.6745 x 5 = 25 + 3.3725 = 28.3725, i.e. 28.4.

Step 4: In a normal distribution the median equals the mean, so median = 25.

Step 5: Mean deviation = 0.7979 x sigma = 0.7979 x 5 = 3.9895, i.e. 4.

Answer: lower quartile 21.6, upper quartile 28.4, median 25, mean deviation 4.

Q. 16 (c) In a normal distribution, the lower and upper quartiles are respectively 8 and 17. Find the mean and standard deviation. (Book answers: 12.5, 6.67)

Step 1: For a normal distribution, Q1 = mu - 0.6745 sigma = 8 and Q3 = mu + 0.6745 sigma = 17.

Step 2: Add the two equations: 2 mu = 8 + 17 = 25, so mu = 12.5.

Step 3: Subtract the two equations: Q3 - Q1 = 2 x 0.6745 sigma = 1.349 sigma = 17 - 8 = 9.

Step 4: Hence sigma = 9 / 1.349 = 6.67.

Answer: mean = 12.5, standard deviation = 6.67.

Q. 16 (d) The continuous random variable X is normally distributed with mean mu

and standard deviation sigma. Given P(X < 53) = 0.04 and P(X < 65) = 0.97, find the interquartile range. (Book answer: 4.46)

Step 1: P(X < 53) = 0.04 gives (53 - mu) / sigma = -1.751 (table value).

Step 2: P(X < 65) = 0.97 gives (65 - mu) / sigma = 1.881 (table value).

Step 3: Subtract: (65 - 53) / sigma = 1.881 - (-1.751) = 3.632.

Step 4: Hence sigma = 12 / 3.632 = 3.304.

Step 5: Interquartile range = Q3 - Q1 = 1.349 x sigma = 1.349 x 3.304 = 4.457, i.e. 4.46.

Answer: interquartile range = 4.46.

Q. 17 (a) The value of the second moment about the mean in a normal distribution is 4. Find the third and fourth moments about the mean. (Book answers: 0, 48)

Step 1: The second moment about the mean is mu2 = sigma-squared = 4, so sigma = 2.

Step 2: In a normal distribution all odd moments about the mean are zero, so mu3 = 0.

Step 3: The fourth moment about the mean is mu4 = 3 sigma^4.

Step 4: sigma^4 = 4 x 4 = 16, so mu4 = 3 x 16 = 48.

Answer: third moment = 0, fourth moment = 48.

Q. 17 (b) Find the proportion of the area under the normal curve included between the limits mu +/- sigma, mu +/- 2 sigma and mu +/- 3 sigma. (Book answers: 0.6827, 0.9545, 0.9973)

Step 1: Standardize with Z = (X - mu) / sigma, so the limits become Z = -1 to 1, -2 to 2, and -3 to 3.

Step 2: For mu +/- sigma: Phi(1) - Phi(-1) = 0.84134 - 0.15866 = 0.68268, i.e. 0.6827.

Step 3: For mu +/- 2 sigma: Phi(2) - Phi(-2) = 0.97725 - 0.02275 = 0.95450, i.e. 0.9545.

Step 4: For mu +/- 3 sigma: Phi(3) - Phi(-3) = 0.99865 - 0.00135 = 0.99730, i.e. 0.9973.

Answer: 0.6827, 0.9545, 0.9973.

Q. 17 (c) If X is a random variable with distribution N(12.5, 9) and Y = g(X) = 2X + 5, find P(-infinity <= Y <= 21) and P(45 <= Y <= infinity). (Book answers: 0.06681, 0.00621)

Step 1: For X ~ N(12.5, 9), mu_X = 12.5 and sigma_X = 3.

Step 2: A linear function of a normal variable is normal: Y = 2X + 5 has mean 2 x 12.5 + 5 = 30.

Step 3: The standard deviation of Y is 2 x sigma_X = 2 x 3 = 6, so Y ~ N(30, 36).

Step 4: For Y = 21, Z = (21 - 30) / 6 = -1.50, so P(Y <= 21) = Phi(-1.50) = 0.06681.

Step 5: For Y = 45, Z = (45 - 30) / 6 = 2.50, so P(Y >= 45) = 1 - Phi(2.50) = 1 - 0.99379 = 0.00621.

Answer: 0.06681 and 0.00621.

Chapter 11: Sampling Techniques and Sampling Distributions

Exercise — Short Questions

What is sampling? Why is it used?

Sampling means selecting a part of the population for study.

It is used because studying the whole population is often costly, slow or impossible.

A well-chosen sample gives reliable information about the population.

What is a sampling distribution? What is the standard error?

The sampling distribution is the probability distribution of a statistic computed from samples.

Its standard deviation is called the standard error of that statistic.

The standard error measures how much the statistic varies from sample to sample.

Exercise — True or False (book p. 72)

A sample survey provides results which are more accurate than those obtained from a census.

False

A sample design is a procedure for obtaining a sample from a given population prior to collecting any data.

True

In sampling with replacement, the sample size may be greater than the population size.

True

In sampling without replacement, the sample size can be greater than population size.

False

The standard deviation of a sampling distribution of a statistic is called the standard error of that statistic.

True

We can decrease both sampling error and standard error by increasing the sample size.

True

Exercise 11.5 — Objective Questions, Q. 1: Fill in the blanks (book pp. 70-71)

Q. 1 (i) A population is the totality of the observations made on all the objects possessing some common specific characteristics.

Q. 1 (ii) A sample is a part of the population which is selected with the expectation that it will represent the characteristics of the population.

Q. 1 (iii) Sampling is a procedure of selecting a representative sample from a given population.

Q. 1 (iv) The descriptive measures of a population are called parameters

Q. 1 (v) A descriptive measure on the sample observations is called statistic

Q. 1 (vi) A population is called finite if it includes a limited number of sampling units.

Q. 1 (vii) A population is called infinite if it includes an unlimited number of sampling units.

Q. 1 (viii) Sampling frame is a complete list of the sampling units.

Q. 1 (ix) A non-probability sampling is a procedure in which we cannot assign to an element of the population the probability of its being included in the sample.

Q. 1 (x) A probability sampling is a process in which the sample is selected in such a way that every element of a population has a known nonzero probability of being included in the sample.

Q. 1 (xi) Another name of a probability sampling is random sampling.

Q. 1 (xii) Random sampling provides reliable estimates

Q. 1 (xiii) The sampling is said to be with replacement when the unit selected at random is returned to the population before the next unit is selected.

Q. 1 (xiv) The sampling is said to be without replacement when the unit selected at random is returned to the population before the next unit is selected. (Book misprint: the sense is 'is NOT returned'.)

Q. 1 (xv) A sample is usually selected by without replacement.

Q. 1 (xvi) In sampling with replacement, a sampling unit can be selected more than once.

Exercise 11.5 — Objective Questions, Q. 2: Fill in the blanks (book p. 71)

Q. 2 (i) In sampling without replacement, a sampling unit cannot be selected more than once.

Q. 2 (ii) In sampling with replacement, a finite population becomes infinite

Q. 2 (iii) Simple random sampling is a procedure of selecting a sample from the population in such a way that every unit available for sampling has an equal

probability of being selected.

Q. 2 (iv) The sampling error decreases by increasing the sample size

Q. 2 (v) The non-sampling errors may be present both in sample survey and census.

Q. 2 (vi) The bias increases by increasing the sample size

Q. 2 (vii) A sample which is free from bias is called an unbiased sample.

Q. 2 (viii) Non-sampling errors may arise due to faulty sampling frames, non-responses and processing of data.

Q. 2 (ix) Non-sampling errors can be controlled by the proper training of the investigators and following up the non-responses.

Q. 2 (x) The probability distribution of a sample statistic is called sampling distribution of that statistic.

Q. 2 (xi) The standard deviation of sampling distribution of a sample statistic is called the standard error of that statistic.

Q. 2 (xii) The standard error can be reduced by increasing the sample size

Q. 2 (xiii) The number of all possible samples of size n taken with replacement from a population of size N is N to the power n

Q. 2 (xiv) The number of all possible samples of size n taken without replacement from a population of size N is N P n

Exercise 11.5 — Objective Questions, Q. 3: True or False (book pp. 71-72)

Q. 3 (i) A descriptive measure on the sample observations is called parameter and a descriptive measure of a population is called statistic.

False

Q. 3 (ii) A sample statistic is a random variable whereas the parameter being estimated is constant.

True

Q. 3 (v) More detailed information can be obtained in a sample survey as compared to a census.

True

Q. 3 (vi) Sampling may be the only means available for obtaining the desired information if the population is infinite.

True

Q. 3 (vii) If the data are obtained by tests that are destructive, then complete enumeration becomes essential.

False

Q. 3 (viii) Every random sample is a simple random sample.

False

Q. 3 (ix) In sampling with replacement, the sample size may be greater than the population size.

True

Q. 3 (xi) The number of units available for the next drawing does not change in a random sampling with replacement.

True

Q. 3 (xii) In sampling without replacement, the number of units remaining after each drawing will be reduced by one.

True

Exercise 11.5 — Objective Questions, Q. 4: True or False (book p. 72)

Q. 4 (i) The number of all possible samples of size n taken without replacement from a population of size N is N C n.

False

Q. 4 (ii) In sampling without replacement, a sampling unit can be selected more than once.

False

Q. 4 (iv) In sampling with replacement, a finite population becomes infinite.

True

Q. 4 (v) Non-sampling errors may be present both in sample survey and census.

True

Q. 4 (vi) The sampling error increases by increasing the sample size.

False

Q. 4 (vii) Sampling and non-sampling errors are both controllable.

True

Q. 4 (ix) Standard error is the difference of a statistic from the parameter being estimated.

False

Q. 4 (xi) The reliability of an estimate can be determined by its standard error.

True

Exercise 11.1 - Sampling Techniques (book pp. 33-34)

Q. 1 (a) Explain the terms: Population; Sample; Sampling frame; Sampling unit.

Step 1: Population: the totality of the observations made on all the objects (or elementary units) possessing some common specific characteristics, which are of particular interest to researchers.

Step 2: Sample: a part of the population which is selected with the expectation that it will represent the characteristics of the population.

Step 3: Sampling frame: the list of all the sampling units of the population from which the sample is to be drawn (for example, a list of voters, or a list of households).

Answer: Sampling unit: the basic unit of the population in terms of which the sample design is planned, for example a student, a household, a field or a bulb.

Q. 1 (b) Define suitable populations from which the following samples are selected: (i) One thousand homes are called by telephone in the city of Karachi and asked to name the T.V. programme that they are now watching. (ii) A coin is flipped 53 times and 32 heads are recorded. (iii) Two hundred pairs of a new type of combat boots were tested for durability in Vietnam and, on the average, lasted two months.

Step 1: (i) The population is the homes in Karachi city having telephones and T.V. sets: only such homes could be reached by telephone and could be watching a programme.

Step 2: (ii) The population is an infinite number of tosses of the coin: the 53 tosses are a sample from all the tosses that could be made.

Answer: (iii) The population is the total production of the new type of combat boots during a particular period: the 200 pairs tested are a sample from that production.

Q. 1 (c) In each of the following situations, determine whether the sampling is done from a finite population or an infinite population and then define the population. (i) A coin is tossed 20 times and 12 heads are recorded. (ii) Ten employees of a large manufacturing company are selected as representatives of labour to serve as a labour management committee. (iii) A sample of bulbs is selected periodically to determine the number of defective bulbs produced by a production unit. (iv) A coin is weighed 15 times to estimate its true weight.

Step 1: (i) Infinite: the population is all the potential tosses of the coin.

Step 2: (ii) Finite: the population is all the employees of the company.

Step 3: (iii) Infinite: the population is all the bulbs produced by the production unit.

Answer: (iv) Infinite: the population is all the potential weights (weighings) of the coin.

Q. 2 (a) What is meant by sampling? Describe the advantages of sampling over complete enumeration.

Step 1: Sampling is the procedure of selecting a representative sample from a given population, instead of examining every unit of the population.

Step 2: Time saving: a sample survey involves less time in execution and analysis than a complete enumeration, which matters when information is urgently needed.

Step 3: Economy: a sample survey costs less in money and labour, because only a fraction of the population is covered.

Step 4: Accuracy: a sample survey can give results almost as accurate as a complete census, and a properly designed sample survey will often provide even better results, because a small trained staff can be used.

Answer: Feasibility: sometimes sampling is the only possible method, for example when the population is infinite or inaccessible, or when testing destroys the unit (as in testing the life of electric bulbs).

Q. 2 (b) For each of the following reasons, give an example of a situation for which a

census would be less desirable than a sample. In each case, explain why this is so.

(i) Economy (ii) Timeliness (iii) Size of population (iv) Inaccessibility (v) Accuracy (vi) Destructive observations.

Step 1: (i) Economy: estimating the average life of electric bulbs. Testing every bulb would cost far more than testing a sample, and the information from a sample is enough.

Step 2: (ii) Timeliness: an election-day opinion poll. A census of all voters could not be completed before the result is needed, while a sample gives a quick estimate.

Step 3: (iii) Size of population: the heights of all the students enrolled in a university in a year. A census is a huge operation; a sample gives the required information with far less work.

Step 4: (iv) Inaccessibility: households scattered over a large or remote area. Reaching every household is impracticable, while a sample of areas can be reached.

Step 5: (v) Accuracy: a census uses a large, less-trained staff, so non-sampling errors grow. A small, well-trained sample team can collect and check data more carefully, giving better results.

Answer: (vi) Destructive observations: testing the life of electric bulbs, or the bursting strength of a product. The test destroys the unit, so testing the whole lot would leave nothing to sell.

Q. 2 (c) Distinguish between the following: (i) Population and sample, (ii) Sampling with and without replacement.

Step 1: (i) Population and sample: the population is the whole collection of units under study; the sample is only a part of it, selected to represent the population.

Step 2: (ii) Sampling with replacement: a unit drawn is returned to the population before the next draw, so the same unit can be drawn again and the population size stays N at every draw.

Answer: Sampling without replacement: a unit drawn is not returned, so it cannot be drawn again and the number of units remaining falls by one after each draw.

Q. 3 (a) Distinguish between probability and non-probability sampling, giving examples. Describe the advantages of using a probability sample.

Step 1: Probability sampling: every unit of the population has a known, non-zero probability of selection, for example simple random sampling by lottery or by random digits.

Step 2: Non-probability sampling: units are selected by judgement, convenience or quota, with no known probability of selection, for example choosing the first ten students met in a corridor.

Step 3: Advantage of a probability sample: because selection is by chance, the sample is free from personal bias, and the sampling error can be measured.

Answer: Only a probability sample lets us use the theory of sampling distributions to judge how close a sample estimate is likely to be to the population value.

Q. 3 (b) What do you understand by a simple random sample? By taking some

artificial example, explain the method of drawing a simple random sample.

Step 1: A simple random sample is a sample in which every possible sample of the given size has the same probability of selection, so every unit has an equal chance of being included.

Step 2: Artificial example: to draw 10 households from a list of 250, number the households from 1 to 250 (or 001 to 250).

Step 3: Lottery method: write each number on a slip, mix the slips thoroughly in a container, and draw 10 slips blindly; the households with those numbers form the sample.

Answer: Random-digits method: read three-digit groups from a table of random digits, discard numbers above 250 and repeats (for sampling without replacement), and take the first 10 numbers that remain.

Q. 3 (c) Distinguish between the following: (i) Random sampling and simple random sampling. (ii) Simple random sampling and stratified random sampling. (iii) Sampling and non-sampling errors.

Step 1: (i) Random sampling is any sampling in which chance decides selection; simple random sampling is the special case in which every possible sample of the size has an equal probability.

Step 2: (ii) Simple random sampling draws directly from the whole population; stratified random sampling first divides a heterogeneous population into homogeneous strata and then draws a simple random sample separately from each stratum.

Answer: (iii) Sampling errors arise because only a part of the population is observed (the difference between a statistic and the parameter); non-sampling errors arise from wrong population, response bias, or mistakes in collecting, analysing and reporting the data, and they occur in a census as well.

Q. 4 (a) Explain how would you select a random sample of 10 households from a list of 250 households, by using a table of random digits.

Step 1: Number the 250 households in the list from 001 to 250, so that every household has a distinct three-digit number.

Step 2: Choose a starting point in the table of random digits at random, and read the digits in groups of three.

Step 3: Keep a three-digit group if it lies between 001 and 250 and has not occurred before; discard any group above 250 and any repeat (sampling without replacement).

Answer: Continue until 10 distinct numbers are obtained; the households bearing those numbers form the required random sample of 10.

Q. 4 (b) A poll is to be conducted to determine the voting preference of the voters in a certain city. Design a sampling plan such that the sample would be representative of the population of all the voters.

Step 1: Take the list of all registered voters in the city as the sampling frame.

Step 2: Stratify the frame by area (and, if available, by sex or age group), so that every part of the city is represented in proportion to its number of voters.

Step 3: Draw a simple random sample (by lottery or random digits) separately from each stratum, with the stratum sample sizes proportional to the stratum sizes.

Answer: Interview the voters so selected; because every stratum contributes in proportion and selection within each stratum is random, the sample represents the population of all the voters.

Q. 4 (c) In a certain locality, there are 300 households. We wish to select a sample of 50 households. How would you select this sample using Random Numbers?

Step 1: Number the 300 households from 001 to 300.

Step 2: Choose a random starting point in a table of random numbers and read the numbers in groups of three digits.

Step 3: Accept a group if it is between 001 and 300 and has not been accepted before; discard groups above 300 and repeats.

Answer: Stop when 50 distinct numbers have been accepted; the households with those numbers form the sample of 50.

Q. 5 (a) What is the difference between precision and accuracy of a result? Explain with some examples.

Step 1: Accuracy is closeness to the true value: the smaller the difference between the sample result and the population true value, the greater the accuracy.

Step 2: Precision is closeness of repeated results to one another, that is, how well a result can be reproduced from sample to sample; it is measured by the sampling

error, a smaller sampling error meaning greater precision.

Step 3: Example: weighing a coin many times on a faulty scale may give nearly the same weight every time (precise) but all the readings may be away from the true weight (inaccurate), because of a constant bias.

Answer: A result can therefore be precise without being accurate, and accurate on average without being precise; a good survey aims at both, by reducing bias and by increasing the sample size.

Q. 5 (b) What are two broad categories of errors in data collected by sample surveys? What are the methods for reducing sampling error?

Step 1: The two broad categories are (i) sampling (random) errors and (ii) non-sampling (non-random) errors.

Step 2: Sampling error is the difference between a statistic from a sample and the corresponding population parameter; non-sampling error comes from a wrong population, response bias, or mistakes in collection and analysis.

Step 3: Sampling error can be reduced (i) by increasing the sample size, (ii) by improving the sampling design, for example by stratification, and (iii) by using supplementary information.

Answer: Non-sampling error is reduced by careful planning, trained staff, and careful checking of the data at every stage.

Exercise 11.2 - Sampling Distribution of the Sample Mean (book pp. 49-51)

Q. 1 (a) How do you define a population and a sample? Differentiate between parameter and statistic. Why a parameter is said to be a constant and statistic a variable?

Step 1: Population: the totality of observations on all the units possessing the characteristic under study. Sample: a part of the population selected to represent it.

Step 2: Parameter: a numerical quantity that describes a characteristic of the population, for example the population mean mu or variance sigma-squared. Statistic: a function of the observations of a random sample which does not contain any unknown parameter, for example the sample mean X-bar or sample variance s-squared.

Answer: A parameter is a constant because it describes the fixed, whole population; a statistic is a variable (a random variable) because its value changes from one sample to another, since different samples give different values.

Q. 1 (b) A labour union has 1000 members. A random sample of 50 members of the union gave an average age of 40 years. The average age of the members of the labour union was, therefore, estimated to be 40 years. A complete enumeration of all the members indicated that the true mean age was 43 years. Answer the following: (i) Which figure is a parameter? (ii) Which figure is a statistic? (Book answers: (i) Population size N = 1000, and population mean age mu = 43 years; (ii) Sample size n, and sample mean X-bar = 40 years.)

Step 1: The population is the 1000 members of the union, so N = 1000.

Step 2: The true mean age from the complete enumeration is 43 years; it describes

the whole population, so it is the parameter mu = 43 years.

Answer: (i) The parameter is 43 years (with N = 1000). (ii) The statistic is the sample mean X-bar = 40 years, computed from the sample of n = 50 members; it is an estimate of mu.

Q. 2 (a) What is meant by a sampling distribution and a standard error? Describe the properties of the sampling distribution of sample mean.

Step 1: Sampling distribution of a statistic: the probability distribution of the statistic obtained from all possible samples of a specified size that can be drawn from a given population.

Step 2: Standard error of a statistic: the standard deviation of its sampling distribution; it measures how much the statistic varies from sample to sample.

Step 3: Properties of the sampling distribution of the sample mean X-bar: (i) its mean equals the population mean, mu_X-bar = mu, for sampling with or without replacement.

Step 4: (ii) Its variance is sigma_X-bar-squared = sigma-squared / n for an infinite population (or sampling with replacement).

Answer: (iii) For a finite population sampled without replacement, sigma_X-bar-squared = (sigma-squared / n) x ((N - n) / (N - 1)), where (N - n) / (N - 1) is the finite population correction; if the population is normal, X-bar is normal, and for large n it is approximately normal by the Central Limit Theorem.

Q. 2 (b) What is meant by standard error and what are its practical uses?

Step 1: The standard error of a statistic is the standard deviation of its sampling distribution.

Step 2: Practical use 1: it measures the precision of an estimate; a smaller standard error means the estimate is likely to be closer to the true value.

Step 3: Practical use 2: it is used to form confidence intervals for a parameter and to test hypotheses about it.

Answer: Practical use 3: it tells us how large a sample is needed for a required precision, since the standard error falls as the sample size grows (as 1 / sqrt(n)).

Q. 2 (c) What is the finite population correction factor? When is it appropriately used in sampling applications and when can it, without too great an undesirable consequence, be ignored?

Step 1: The finite population correction (f.p.c.) factor for the variance is (N - n) / (N - 1), and for the standard error it is sqrt((N - n) / (N - 1)).

Step 2: It is used when a sample of size n is drawn without replacement from a finite population of size N, because the draws are then not independent.

Step 3: As N tends to infinity (or n is very small compared with N), (N - n) / (N - 1) tends to 1, so the correction tends to 1.

Answer: It can therefore be ignored, without too great an undesirable consequence, when the population is infinite, when sampling is with replacement, or when the

sampling fraction n / N is very small (in practice, below about 5 percent).

Q. 3 (a) A finite population consists of the numbers 2, 4, 6, 8, 10 and 12. Calculate the sample means for all possible random samples of size n = 2, that can be drawn from this population, with replacement. Assuming the 36 possible samples equally likely, make the sampling distribution of sample means and find the mean and variance of this distribution. Calculate mean and variance of the population and verify that (i) mu_X-bar = mu, (ii) sigma_X-bar-squared = sigma-squared / n. (Book answers: mu = 7, sigma-squared = 11.667, mu_X-bar = 7, sigma_X-bar-squared = 5.833.)

Step 1: Population: 2, 4, 6, 8, 10, 12; N = 6; sample size n = 2 with replacement, so the number of possible samples is 6 x 6 = 36.

Step 2: Population mean: mu = (2 + 4 + 6 + 8 + 10 + 12) / 6 = 42 / 6 = 7.

Step 3: Population variance: sum of squares = 4 + 16 + 36 + 64 + 100 + 144 = 364; sigma-squared = 364 / 6 - 7-squared = 60.667 - 49 = 11.667.

Step 4: Sampling distribution of X-bar (value: frequency out of 36): 2: 1, 3: 2, 4: 3, 5: 4, 6: 5, 7: 6, 8: 5, 9: 4, 10: 3, 11: 2, 12: 1. Probabilities are frequency / 36.

Step 5: Mean of the distribution: mu_X-bar = E(X-bar) = sum X-bar x p(X-bar) = 7.

Step 6: Variance: E(X-bar-squared) = 54.833, so sigma_X-bar-squared = 54.833 - 7-squared = 5.833.

Answer: (i) mu_X-bar = 7 = mu, verified. (ii) sigma-squared / n = 11.667 / 2 = 5.833

= sigma_X-bar-squared, verified.

Q. 3 (b) A finite population consists of the numbers 2, 4, 6, 6, 8 and 10. Calculate the sample means for all possible random samples of size n = 2, that can be drawn from this population, with replacement. Assuming the 36 possible samples equally likely, form the sampling distribution of sample means and find the mean and variance of this distribution. Calculate mean and variance of the population and verify that (i) mu_X-bar = mu, (ii) sigma_X-bar = sigma / sqrt(n). (Book answers: mu = 6, mu_X-bar = 6, sigma = 2.582, sigma_X-bar = 1.826.)

Step 1: Population: 2, 4, 6, 6, 8, 10; N = 6 (the value 6 occurs twice); n = 2 with replacement, so the number of possible samples is 6 x 6 = 36.

Step 2: Population mean: mu = (2 + 4 + 6 + 6 + 8 + 10) / 6 = 36 / 6 = 6.

Step 3: Population variance: sum of squares = 4 + 16 + 36 + 36 + 64 + 100 = 256; sigma-squared = 256 / 6 - 6-squared = 42.667 - 36 = 6.667; sigma = sqrt(6.667) = 2.582.

Step 4: Sampling distribution of X-bar (value: frequency out of 36): 2: 1, 3: 2, 4: 5, 5: 6, 6: 8, 7: 6, 8: 5, 9: 2, 10: 1.

Step 5: Mean: mu_X-bar = E(X-bar) = 6.

Step 6: Variance: sigma_X-bar-squared = sigma-squared / n = 6.667 / 2 = 3.333, so sigma_X-bar = sqrt(3.333) = 1.826.

Answer: (i) mu_X-bar = 6 = mu, verified. (ii) sigma_X-bar = 1.826 = sigma / sqrt(n)

= 2.582 / sqrt(2), verified.

Q. 3 (c) Draw all possible samples of size n = 3 with replacement from the population 3, 6, 9 and 12. Assuming the 64 possible samples equally likely, form a sampling distribution of the sample means. Hence state and verify the relation between (i) the mean of the sampling distribution of the sample mean and the population mean; (ii) the variance of the sampling distribution of the sample mean and the population variance. (Book answers: mu = 7.5, mu_X-bar = 7.5, sigma-squared = 11.25, sigma_X-bar-squared = 3.75.)

Step 1: Population: 3, 6, 9, 12; N = 4; n = 3 with replacement, so the number of possible samples is 4 x 4 x 4 = 64.

Step 2: Population mean: mu = (3 + 6 + 9 + 12) / 4 = 30 / 4 = 7.5.

Step 3: Population variance: sum of squares = 9 + 36 + 81 + 144 = 270; sigma-squared = 270 / 4 - 7.5-squared = 67.5 - 56.25 = 11.25.

Step 4: Sampling distribution of X-bar (value: frequency out of 64): 3: 1, 4: 3, 5: 6, 6: 10, 7: 12, 8: 12, 9: 10, 10: 6, 11: 3, 12: 1.

Step 5: Mean: mu_X-bar = E(X-bar) = 7.5.

Step 6: Variance: sigma_X-bar-squared = 3.75.

Answer: (i) mu_X-bar = 7.5 = mu, verified. (ii) sigma_X-bar-squared = 3.75 = sigma-squared / n = 11.25 / 3, verified; hence sigma_X-bar = sqrt(3.75) = 1.936.

Q. 4 (a) A finite population consists of the numbers 2, 4, 6, 6, 8 and 10. Calculate the sample means for all possible random samples of size n = 2, that can be drawn from this population, without replacement. Assuming the 30 possible samples equally likely, make the sampling distribution of sample mean. Find the mean and variance of this distribution. Calculate mean and variance of the population and verify that (i) mu_X-bar = mu, (ii) sigma_X-bar = (sigma / sqrt(n)) x sqrt((N - n) / (N - 1)). (Book answers: mu = 6, mu_X-bar = 6, sigma = 2.582, sigma_X-bar = 1.633.)

Step 1: Population: 2, 4, 6, 6, 8, 10; N = 6; n = 2 without replacement (the two 6s are distinct units), so the number of ordered samples is 6 x 5 = 30.

Step 2: Population mean: mu = 36 / 6 = 6.

Step 3: Population variance: sigma-squared = 6.667, so sigma = 2.582 (as in Q. 3 (b)).

Step 4: Sampling distribution of X-bar (value: frequency out of 30): 3: 2, 4: 4, 5: 6, 6: 6, 7: 6, 8: 4, 9: 2.

Step 5: Mean: mu_X-bar = E(X-bar) = 6.

Step 6: Variance: sigma_X-bar-squared = (sigma-squared / n) x ((N - n) / (N - 1)) = (6.667 / 2) x (4 / 5) = 3.333 x 0.8 = 2.667, so sigma_X-bar = sqrt(2.667) = 1.633.

Answer: (i) mu_X-bar = 6 = mu, verified. (ii) sigma_X-bar = 1.633 = (2.582 / sqrt(2)) x sqrt(4 / 5), verified.

Q. 4 (b) A finite population consists of the values 6, 6, 9, 15 and 18. Calculate the

sample means for all possible random samples of size n = 3, that can be drawn from this population, without replacement. Assuming the 60 possible samples equally likely, make the sampling distribution of sample mean and find the mean and variance of this distribution. Calculate mean and variance of the population and show that (i) mu_X-bar = mu, (ii) sigma_X-bar-squared = (sigma-squared / n) x ((N - n) / (N - 1)). (Book answers: mu = 10.8, sigma-squared = 23.76, mu_X-bar = 10.8, sigma_X-bar-squared = 3.96.)

Step 1: Population: 6, 6, 9, 15, 18; N = 5; n = 3 without replacement, so the number of ordered samples is 5 x 4 x 3 = 60.

Step 2: Population mean: mu = (6 + 6 + 9 + 15 + 18) / 5 = 54 / 5 = 10.8.

Step 3: Population variance: sum of squares = 36 + 36 + 81 + 225 + 324 = 702; sigma-squared = 702 / 5 - 10.8-squared = 140.4 - 116.64 = 23.76.

Step 4: Sampling distribution of X-bar (value: frequency out of 60): 7: 6, 9: 6, 10: 18, 11: 12, 13: 12, 14: 6.

Step 5: Mean: mu_X-bar = E(X-bar) = 10.8.

Step 6: Variance: sigma_X-bar-squared = (23.76 / 3) x ((5 - 3) / (5 - 1)) = 7.92 x 0.5 = 3.96.

Answer: (i) mu_X-bar = 10.8 = mu, verified. (ii) sigma_X-bar-squared = 3.96, as required.

Q. 4 (c) Find the mean mu and variance sigma-squared of the finite population 1, 4,

7 and 8. Take all possible samples of size 2, that can be drawn at random without replacement from this population. Assuming the 12 possible samples equally likely, make the sampling distribution of sample mean and find the mean and variance of this distribution. Verify that (i) E(X-bar) = mu, (ii) Var(X-bar) = (sigma-squared / n) x ((N - n) / (N - 1)). (Book answers: mu = 5, sigma-squared = 7.5, E(X-bar) = 5, Var(X-bar) = 2.5, (N - n) / (N - 1) tends to 1 as N tends to infinity.)

Step 1: Population: 1, 4, 7, 8; N = 4; n = 2 without replacement, so the number of ordered samples is 4 x 3 = 12.

Step 2: Population mean: mu = (1 + 4 + 7 + 8) / 4 = 20 / 4 = 5.

Step 3: Population variance: sum of squares = 1 + 16 + 49 + 64 = 130; sigma-squared = 130 / 4 - 5-squared = 32.5 - 25 = 7.5.

Step 4: Sampling distribution of X-bar (value: frequency out of 12): 2.5: 2, 4: 2, 4.5: 2, 5.5: 2, 6: 2, 7.5: 2.

Step 5: Mean: E(X-bar) = 5.

Step 6: Variance: Var(X-bar) = (7.5 / 2) x ((4 - 2) / (4 - 1)) = 3.75 x (2 / 3) = 2.5.

Answer: (i) E(X-bar) = 5 = mu, verified. (ii) Var(X-bar) = 2.5, as required. Note: as N tends to infinity, (N - n) / (N - 1) tends to 1, so the finite population correction disappears.

Q. 5 (a) In an infinite population mu = 50 and sigma-squared = 250, find the mean and variance for the distribution of X-bar if: (i) n = 25, (ii) n = 100, (iii) n = 1250.

(Book answers: (i) mu_X-bar = 50, sigma_X-bar-squared = 10; (ii) mu_X-bar = 50, sigma_X-bar-squared = 2.5; (iii) mu_X-bar = 50, sigma_X-bar-squared = 0.2.)

Step 1: For an infinite population, mu_X-bar = mu = 50 for every n, and sigma_X-bar-squared = sigma-squared / n.

Step 2: (i) n = 25: sigma_X-bar-squared = 250 / 25 = 10.

Step 3: (ii) n = 100: sigma_X-bar-squared = 250 / 100 = 2.5.

Answer: (iii) n = 1250: sigma_X-bar-squared = 250 / 1250 = 0.2; in all three cases mu_X-bar = 50.

Q. 5 (b) A large number of samples of size 50 were selected at random from a normal population with mean mu and variance sigma-squared. The mean and standard error of the sampling distribution of the sample mean were obtained 2500 and 4 respectively. Find the mean and variance of the population. (Book answer: 2500, 800.)

Step 1: The mean of the sampling distribution of X-bar is mu_X-bar = mu, so mu = 2500.

Step 2: The standard error is sigma_X-bar = sigma / sqrt(n) = 4, with n = 50.

Step 3: So sigma = 4 x sqrt(50), and sigma-squared = 16 x 50 = 800.

Answer: The population mean is 2500 and the population variance is 800.

Q. 6 (a) If the size of the simple random sample from an infinite population is 55,

the variance of sample mean is 27, what must be the standard error of sample mean if n = 165? (Book answer: sigma_X-bar = 3.)

Step 1: sigma_X-bar-squared = sigma-squared / n, so with n = 55 and variance 27, sigma-squared = 27 x 55 = 1485.

Step 2: For n = 165, sigma_X-bar-squared = 1485 / 165 = 9.

Answer: Hence sigma_X-bar = sqrt(9) = 3.

Q. 6 (b) If the size of the simple random sample from an infinite population is 36 and the standard error of the mean is 2, what must the size of the sample become if the standard error is to be reduced to 1.2? (Book answer: n = 100.)

Step 1: sigma_X-bar = sigma / sqrt(n), so sigma = 2 x sqrt(36) = 2 x 6 = 12, and sigma-squared = 144.

Step 2: For the new standard error 1.2, we need sqrt(n) = sigma / 1.2 = 12 / 1.2 = 10.

Answer: Hence n = 10-squared = 100.

Q. 7 (a) The random variable X has the following probability distribution: x_j: 4, 5, 6, 7; p(x_j): 0.2, 0.4, 0.3, 0.1. Find the mean mu_X-bar, variance sigma_X-bar-squared and standard error sigma_X-bar of the mean X-bar for a random sample of size 36. (Book answers: mu_X-bar = 5.3, sigma_X-bar-squared = 0.0225, sigma_X-bar = 0.15.)

Step 1: Population mean: mu = sum x_j p(x_j) = 4 x 0.2 + 5 x 0.4 + 6 x 0.3 + 7 x 0.1 = 0.8 + 2.0 + 1.8 + 0.7 = 5.3.

Step 2: E(X-squared) = 16 x 0.2 + 25 x 0.4 + 36 x 0.3 + 49 x 0.1 = 3.2 + 10.0 + 10.8 + 4.9 = 28.9.

Step 3: Population variance: sigma-squared = 28.9 - 5.3-squared = 28.9 - 28.09 = 0.81.

Step 4: For n = 36, mu_X-bar = mu = 5.3.

Step 5: sigma_X-bar-squared = sigma-squared / n = 0.81 / 36 = 0.0225.

Answer: sigma_X-bar = sqrt(0.0225) = 0.15.

Q. 7 (b) A random sample of 36 cases is drawn from a negatively skewed probability distribution with a mean of 2 and a standard deviation of 3. Find the mean and standard error of the sampling distribution of X-bar. (Book answers: mu_X-bar = 2, sigma_X-bar = 0.5.)

Step 1: mu_X-bar = mu = 2.

Step 2: sigma_X-bar = sigma / sqrt(n) = 3 / sqrt(36) = 3 / 6 = 0.5.

Answer: The mean is 2 and the standard error is 0.5. (For n = 36 the distribution of X-bar is approximately normal by the Central Limit Theorem, even though the parent distribution is skewed.)

Q. 7 (c) A random sample of 100 is taken from a population with mean 30 and standard deviation 5. The probability distribution of the parent population is unknown, find the mean and standard error of the sampling distribution of X-bar.

(Book answers: mu_X-bar = 30, sigma_X-bar = 0.5.)

Step 1: mu_X-bar = mu = 30.

Step 2: sigma_X-bar = sigma / sqrt(n) = 5 / sqrt(100) = 5 / 10 = 0.5.

Answer: The mean is 30 and the standard error is 0.5.

Exercise 11.3 - Sampling Distribution of the Difference Between Two Sample Means (book pp. 56-57)

Q. 1 (a) What is meant by the sampling distribution of the difference between two sample means. Describe the properties of the sampling distribution of the differences between two sample means.

Step 1: The sampling distribution of the difference between two sample means, X-bar_1 - X-bar_2, is the probability distribution of all possible differences between the means X-bar_1 and X-bar_2 obtained from all possible independent simple random samples of sizes n_1 and n_2 drawn from two given populations.

Step 2: Property 1 (mean): the mean of the distribution equals the difference between the population means, mu_(X-bar_1 - X-bar_2) = mu_1 - mu_2. This holds regardless of the sample sizes and whether sampling is with or without replacement.

Step 3: Property 2 (variance, infinite populations or with replacement): sigma_(X-bar_1 - X-bar_2)-squared = sigma_1-squared / n_1 + sigma_2-squared / n_2.

Step 4: Property 3 (variance, finite populations without replacement):

sigma_(X-bar_1 - X-bar_2)-squared = (sigma_1-squared / n_1) x ((N_1 - n_1) / (N_1 - 1)) + (sigma_2-squared / n_2) x ((N_2 - n_2) / (N_2 - 1)).

Answer: Property 4: if the two populations are normal, the difference X-bar_1 - X-bar_2 is normal; for large samples it is approximately normal by the Central Limit Theorem, and Z = ((X-bar_1 - X-bar_2) - (mu_1 - mu_2)) / sigma_(X-bar_1 - X-bar_2) is standard normal.

Q. 1 (b) Let X-bar_1 represent the mean of a sample of size n_1 = 2, selected with replacement from a finite population -2, 0, 2, and 4. Similarly, let X-bar_2 represent the mean of a sample of size n_2 = 2, selected with replacement from the population -1 and 1. (i) Assuming that the 64 possible differences X-bar_1 - X-bar_2 are equally likely to occur, construct the sampling distribution of X-bar_1 - X-bar_2. (ii) Verify that sigma_(X-bar_1 - X-bar_2)-squared = sigma_1-squared / n_1 + sigma_2-squared / n_2. (Book answers: mu_1 = 1, mu_2 = 0, mu_(X-bar_1 - X-bar_2) = 1, sigma_1-squared = 5, sigma_2-squared = 1, sigma_(X-bar_1 - X-bar_2)-squared = 3.)

Step 1: Population I: -2, 0, 2, 4; N_1 = 4. Population II: -1, 1; N_2 = 2. Sample sizes n_1 = 2, n_2 = 2, both with replacement.

Step 2: mu_1 = (-2 + 0 + 2 + 4) / 4 = 4 / 4 = 1. sigma_1-squared = (4 + 0 + 4 + 16) / 4 - 1-squared = 6 - 1 = 5.

Step 3: mu_2 = (-1 + 1) / 2 = 0. sigma_2-squared = (1 + 1) / 2 - 0 = 1.

Step 4: Number of possible differences: (4 x 4) x (2 x 2) = 16 x 4 = 64. Sampling

distribution of X-bar_1 - X-bar_2 (value: frequency out of 64): -3: 1, -2: 4, -1: 8, 0: 12, 1: 14, 2: 12, 3: 8, 4: 4, 5: 1.

Step 5: Mean: mu_(X-bar_1 - X-bar_2) = 1 = mu_1 - mu_2.

Step 6: Variance: sigma_(X-bar_1 - X-bar_2)-squared = 5 / 2 + 1 / 2 = 2.5 + 0.5 = 3.

Answer: (ii) Verified: sigma_(X-bar_1 - X-bar_2)-squared = 3 = sigma_1-squared / n_1 + sigma_2-squared / n_2; the standard error is sqrt(3) = 1.732.

Q. 2 (a) Let the variable X-bar_1 represent the mean of random samples of size n_1 = 2, with replacement drawn from the finite population 3, 4, 5. Similarly, let X-bar_2 represent the means of random samples of size n_2 = 3, with replacement, drawn from the population 0, 3. Assuming that the 72 possible differences X-bar_1 - X-bar_2 are equally likely to occur, construct the sampling distribution of X-bar_1 - X-bar_2. Show that (i) mu_(X-bar_1 - X-bar_2) = mu_1 - mu_2, (ii) sigma_(X-bar_1 - X-bar_2)-squared = sigma_1-squared / n_1 + sigma_2-squared / n_2. (Book answers: mu_1 = 4, mu_2 = 1.5, mu_(X-bar_1 - X-bar_2) = 2.5, sigma_1-squared = 0.667, sigma_2-squared = 2.25, sigma_(X-bar_1 - X-bar_2)-squared = 1.083.)

Step 1: Population I: 3, 4, 5; N_1 = 3. Population II: 0, 3; N_2 = 2. n_1 = 2, n_2 = 3, both with replacement.

Step 2: mu_1 = (3 + 4 + 5) / 3 = 4. sigma_1-squared = (9 + 16 + 25) / 3 - 16 = 16.667 - 16 = 0.667.

Step 3: mu_2 = (0 + 3) / 2 = 1.5. sigma_2-squared = (0 + 9) / 2 - 2.25 = 4.5 - 2.25 = 2.25.

Step 4: Number of possible differences: (3 x 3) x (2 x 2 x 2) = 9 x 8 = 72. Sampling distribution of X-bar_1 - X-bar_2 (value: frequency out of 72): 0: 1, 0.5: 2, 1: 6, 1.5: 8, 2: 13, 2.5: 12, 3: 13, 3.5: 8, 4: 6, 4.5: 2, 5: 1.

Step 5: Mean: mu_(X-bar_1 - X-bar_2) = 2.5 = 4 - 1.5 = mu_1 - mu_2.

Step 6: Variance: sigma_(X-bar_1 - X-bar_2)-squared = 0.667 / 2 + 2.25 / 3 = 0.333 + 0.75 = 1.083.

Answer: (i) and (ii) both verified; the standard error is sqrt(1.083) = 1.041.

Q. 2 (b) Let the variable X-bar_1 represent the means of random samples of size 2 without replacement, drawn from the finite population 5, 7, 9. Similarly, let X-bar_2 represent the means of random sample of size 2, without replacement from another finite population 4, 6, 8. Assuming the 36 possible differences X-bar_1 - X-bar_2 are equally likely to occur, construct the sampling distribution of X-bar_1 - X-bar_2 and verify that (i) mu_(X-bar_1 - X-bar_2) = mu_1 - mu_2, (ii) sigma_(X-bar_1 - X-bar_2)-squared = (sigma_1-squared / n_1) x ((N_1 - n_1) / (N_1 - 1)) + (sigma_2-squared / n_2) x ((N_2 - n_2) / (N_2 - 1)). (Book answers: mu_1 = 7, mu_2 = 6, mu_(X-bar_1 - X-bar_2) = 1, sigma_1-squared = 2.667, sigma_2-squared = 2.667, sigma_(X-bar_1 - X-bar_2)-squared = 1.333.)

Step 1: Population I: 5, 7, 9; N_1 = 3. Population II: 4, 6, 8; N_2 = 3. n_1 = 2, n_2 = 2, both without replacement.

Step 2: mu_1 = (5 + 7 + 9) / 3 = 7. sigma_1-squared = (25 + 49 + 81) / 3 - 49 = 51.667 - 49 = 2.667.

Step 3: mu_2 = (4 + 6 + 8) / 3 = 6. sigma_2-squared = (16 + 36 + 64) / 3 - 36 = 38.667 - 36 = 2.667.

Step 4: Number of possible differences: (3 x 2) x (3 x 2) = 6 x 6 = 36. Sampling distribution of X-bar_1 - X-bar_2 (value: frequency out of 36): -1: 4, 0: 8, 1: 12, 2: 8, 3: 4.

Step 5: Mean: mu_(X-bar_1 - X-bar_2) = 1 = 7 - 6 = mu_1 - mu_2.

Step 6: Variance: (2.667 / 2) x ((3 - 2) / (3 - 1)) + (2.667 / 2) x ((3 - 2) / (3 - 1)) = 1.333 x 0.5 + 1.333 x 0.5 = 0.667 + 0.667 = 1.333.

Answer: (i) and (ii) both verified; the standard error is sqrt(1.333) = 1.155.

Q. 3 (a) The television picture tubes of manufacturer A have a mean lifetime of 6.5 years and a standard deviation of 0.9 years, while those of manufacturer B have a mean lifetime of 6.0 years and a standard deviation of 0.8 years. A random sample of size 36 tubes is selected from manufacturer A and its mean X-bar_1 is calculated. Another random sample of size 49 tubes is selected from manufacturer B and its mean X-bar_2 is calculated. Find the mean and standard error of the sampling distribution of the difference X-bar_1 - X-bar_2. (Book answers: mu_(X-bar_1 - X-bar_2) = 0.5, sigma_(X-bar_1 - X-bar_2) = 0.1886.)

Step 1: mu_1 = 6.5, sigma_1 = 0.9, n_1 = 36; mu_2 = 6.0, sigma_2 = 0.8, n_2 = 49.

Step 2: Mean of the difference: mu_(X-bar_1 - X-bar_2) = mu_1 - mu_2 = 6.5 - 6.0 = 0.5 years.

Step 3: Variance: sigma_(X-bar_1 - X-bar_2)-squared = 0.81 / 36 + 0.64 / 49 = 0.0225 + 0.01306 = 0.03556.

Answer: Standard error: sigma_(X-bar_1 - X-bar_2) = sqrt(0.03556) = 0.1886 years.

Q. 3 (b) Random samples of each size 100 are drawn from two independent probability distributions and their means X-bar_1 and X-bar_2 computed. If the means and standard deviations of the two populations are mu_1 = 10, sigma_1 = 2, mu_2 = 8, sigma_2 = 1, find the mean and standard error of the sampling distribution of the difference X-bar_1 - X-bar_2. (Book answers: mu_(X-bar_1 - X-bar_2) = 2, sigma_(X-bar_1 - X-bar_2) = 0.2236.)

Step 1: mu_1 = 10, sigma_1 = 2, n_1 = 100; mu_2 = 8, sigma_2 = 1, n_2 = 100.

Step 2: Mean of the difference: mu_(X-bar_1 - X-bar_2) = 10 - 8 = 2.

Step 3: Variance: sigma_(X-bar_1 - X-bar_2)-squared = 4 / 100 + 1 / 100 = 0.05.

Answer: Standard error: sigma_(X-bar_1 - X-bar_2) = sqrt(0.05) = 0.2236.

Exercise 11.4 - Sampling Distributions (book pp. 67-70)

Q. 1 (a) A fair coin is tossed 50 times and the number of heads recorded is 27. The proportion of heads was therefore estimated to be 0.54. Answer: (i) Which figure is a parameter? (ii) Which figure is a statistic? (Book answers: (i) pi = 0.5; (ii) n = 50, p = 0.54)

Step 1: A parameter describes the POPULATION; a statistic describes the SAMPLE.

Step 2: (i) For a fair coin, the population probability of a head in a single trial is pi = 0.5.

Step 3: This 0.5 does not come from the 50 tosses; it is a property of the coin itself, so it is the parameter.

Step 4: (ii) The sample size is n = 50 tosses.

Step 5: The number of heads in the sample is x = 27, so the sample proportion is p = x / n = 27 / 50 = 0.54.

Step 6: Both n = 50 and p = 0.54 are computed from the sample, so they are statistics.

Answer: (i) parameter: pi = 0.5; (ii) statistics: n = 50 and p = 0.54. FLAG: the book's printed solution writes x = 30, but 30 / 50 = 0.60, not 0.54; the question's own figure x = 27 gives 27 / 50 = 0.54, and is used here.

Q. 1 (b) What is meant by the sampling distribution of sample proportion? Describe the properties of the sampling distribution of sample proportion.

The sampling distribution of a sample proportion is the probability distribution of the values of the sample proportion P obtained from all possible samples of a given size from a population.

Its properties are: its mean is the population proportion, mu_P = pi.

Its variance for sampling with replacement (or an infinite population) is sigma_P-squared = pi (1 - pi) / n.

For sampling without replacement from a finite population of size N, the variance is

multiplied by the finite population correction (N - n) / (N - 1).

Its standard deviation sigma_P is called the standard error of the proportion.

For large samples it is approximately normal by the central limit theorem.

Q. 2 (a) A finite population consists of the numbers 2, 3, 4, 5, 6 and 8. Find the proportion P of even numbers in all possible random samples of size n = 2 drawn with replacement. Assuming the 36 possible samples equally likely, make the sampling distribution of sample proportions and find its mean and variance. Verify (i) E(P) = pi, (ii) Var(P) = pi (1 - pi) / n. (Book answers: pi = 2/3, mu_P = 2/3, sigma_P-squared = 1/9)

Step 1: Even numbers in the population are 2, 4, 6, 8, i.e. 4 out of 6, so pi = 4 / 6 = 2/3.

Step 2: Number of possible samples with replacement = 6 x 6 = 36.

Step 3: Count samples by the number of even values: both odd (odd values are 3, 5): 2 x 2 = 4 samples give P = 0.

Step 4: One even and one odd: 4 x 2 x 2 = 16 samples give P = 1/2.

Step 5: Both even: 4 x 4 = 16 samples give P = 1.

Step 6: Sampling distribution: P = 0 with frequency 4 (probability 4/36); P = 1/2 with frequency 16 (16/36); P = 1 with frequency 16 (16/36).

Step 7: Mean mu_P = (0 x 4 + (1/2) x 16 + 1 x 16) / 36 = 24 / 36 = 2/3.

Step 8: E(P-squared) = (0 x 4 + (1/4) x 16 + 1 x 16) / 36 = 20 / 36 = 5/9.

Step 9: Variance = E(P-squared) - mu_P-squared = 5/9 - 4/9 = 1/9.

Step 10: Verify (i): E(P) = 2/3 = pi. Verify (ii): pi (1 - pi) / n = (2/3)(1/3) / 2 = 1/9.

Answer: distribution 0: 4/36, 1/2: 16/36, 1: 16/36; mean 2/3; variance 1/9. Both results verified.

Q. 2 (b) A population consists of N = 4 numbers 1, 3, 4 and 5. Find the proportion P of odd numbers in all possible samples of size n = 3 drawn without replacement. Assuming the 24 possible samples equally likely, construct the sampling distribution and find its mean and variance. Verify (i) mu_P = pi, (ii) sigma_P-squared = pi (1 - pi) / n x (N - n) / (N - 1). (Book answers: pi = 3/4, mu_P = 3/4, sigma_P-squared = 1/48)

Step 1: Odd numbers in the population are 1, 3, 5, i.e. 3 out of 4, so pi = 3/4.

Step 2: Number of possible ordered samples without replacement = 4 x 3 x 2 = 24.

Step 3: The 4 unordered triples are: {1, 3, 4} (odd proportion 2/3), {1, 3, 5} (1), {1, 4, 5} (2/3), {3, 4, 5} (2/3).

Step 4: Each unordered triple has 6 orderings, so P = 2/3 occurs in 18 samples (18/24) and P = 1 occurs in 6 samples (6/24).

Step 5: Mean mu_P = ((2/3) x 18 + 1 x 6) / 24 = 18 / 24 = 3/4.

Step 6: E(P-squared) = ((4/9) x 18 + 1 x 6) / 24 = 14 / 24 = 7/12.

Step 7: Variance = 7/12 - (3/4)-squared = 7/12 - 9/16 = 28/48 - 27/48 = 1/48.

Step 8: Verify (i): mu_P = 3/4 = pi. Verify (ii): pi (1 - pi) / n x (N - n) / (N - 1) = (3/4)(1/4) / 3 x 1/3 = 1/48.

Answer: distribution 2/3: 18/24, 1: 6/24; mean 3/4; variance 1/48. Both results verified.

Q. 3 (a) Suppose 60% of a city population favours public funding for a proposed recreational facility. If 150 persons are randomly selected and interviewed, what are the mean and standard error of the sample proportion favouring this issue? (Book answers: mu_P = 0.60, sigma_P = 0.04)

Step 1: Population proportion pi = 0.60 and sample size n = 150.

Step 2: Mean of the sample proportion mu_P = pi = 0.60.

Step 3: Standard error sigma_P = sqrt(pi (1 - pi) / n) = sqrt(0.60 x 0.40 / 150).

Step 4: 0.60 x 0.40 = 0.24; 0.24 / 150 = 0.0016.

Step 5: sqrt(0.0016) = 0.04.

Answer: mean = 0.60, standard error = 0.04.

Q. 3 (b) A small professional society has N = 4500 members. The president mailed n = 400 questionnaires to a random sample asking whether they wish to affiliate with a large group. Assuming the proportion of the entire membership favouring

consolidation is pi = 0.7, find the mean and standard error of the sample proportion P. (Book answers: mu_P = 0.7, sigma_P = 0.022)

Step 1: Population proportion pi = 0.7, sample size n = 400, population size N = 4500.

Step 2: Mean of the sample proportion mu_P = pi = 0.7.

Step 3: With the finite population correction, sigma_P = sqrt(pi (1 - pi) / n x (N - n) / (N - 1)).

Step 4: pi (1 - pi) / n = 0.7 x 0.3 / 400 = 0.21 / 400 = 0.000525.

Step 5: (N - n) / (N - 1) = 4100 / 4499 = 0.9113.

Step 6: sigma_P = sqrt(0.000525 x 0.9113) = sqrt(0.000478) = 0.02187, i.e. 0.022.

Answer: mean = 0.7, standard error = 0.022.

Q. 4 (a) What is meant by the sampling distribution of the difference between two sample proportions? Describe its properties. Explain its usefulness in statistical inference.

The sampling distribution of the difference between two sample proportions is the probability distribution of P1 - P2, where P1 and P2 are the sample proportions from two independent populations.

Its properties are: its mean is the difference of the population proportions, mu_(P1 - P2) = pi1 - pi2.

Its variance is the sum of the two variances: sigma-squared_(P1 - P2) = pi1 (1 - pi1)

/ n1 + pi2 (1 - pi2) / n2, with the finite population correction for each part when sampling without replacement from finite populations.

Its standard deviation is the standard error of the difference of proportions.

For large samples it is approximately normal.

Its usefulness in statistical inference is: it lets us compare two populations, for example two cities or two groups, by testing whether pi1 = pi2 and by building confidence intervals for pi1 - pi2.

Q. 4 (b) Let P1 be the proportion of odd numbers in a random sample of size n1 = 3 with replacement from the population {4, 5}. Let P2 be the proportion of odd numbers in a random sample of size n2 = 2 with replacement from the population {2, 3, 6}. Assuming the 72 possible differences P1 - P2 equally likely, construct the sampling distribution of P1 - P2 and verify (i) mu_(P1 - P2) = pi1 - pi2, (ii) sigma-squared_(P1 - P2) = pi1 (1 - pi1) / n1 + pi2 (1 - pi2) / n2. (Book answers: pi1 = 1/2, pi2 = 1/3, mu = 1/6, sigma-squared = 7/36)

Step 1: Population 1: odd value is 5 only, so pi1 = 1/2. Population 2: odd value is 3 only, so pi2 = 1/3.

Step 2: P1 distribution (8 samples of size 3 from {4, 5}): P1 = 0 in 1 sample, 1/3 in 3 samples, 2/3 in 3 samples, 1 in 1 sample.

Step 3: P2 distribution (9 samples of size 2 from {2, 3, 6}): P2 = 0 in 4 samples, 1/2 in 4 samples, 1 in 1 sample.

Step 4: The 72 differences P1 - P2 occur with frequencies: -1: 1, -2/3: 3, -1/2: 4, -1/3: 3, -1/6: 12, 0: 5, 1/6: 12, 1/3: 12, 1/2: 4, 2/3: 12, 1: 4.

Step 5: Check the total: 1 + 3 + 4 + 3 + 12 + 5 + 12 + 12 + 4 + 12 + 4 = 72.

Step 6: Mean = sum of (difference x frequency) / 72 = 12 / 72 = 1/6.

Step 7: E((P1 - P2)-squared) = sum of (difference-squared x frequency) / 72 = 16 / 72 = 2/9.

Step 8: Variance = E((P1 - P2)-squared) - mean-squared = 2/9 - (1/6)-squared = 8/36 - 1/36 = 7/36.

Step 9: Using the formula: pi1 (1 - pi1) / n1 = (1/2)(1/2) / 3 = 1/12; pi2 (1 - pi2) / n2 = (1/3)(2/3) / 2 = 1/9.

Step 10: Sum = 1/12 + 1/9 = 3/36 + 4/36 = 7/36, which equals the variance computed from the distribution.

Step 11: Verify (i): mu_(P1 - P2) = 1/6 = 1/2 - 1/3 = pi1 - pi2.

Answer: distribution as in Step 4; mean 1/6; variance 7/36. Both results verified.

Exercise 11.4 - Sampling Distributions (continued, book pp. 69-70)

Q. 5 (a) Let P1 be the proportion of even numbers in a random sample of size n1 = 2 without replacement from the population {4, 6, 9}. Let P2 be the proportion of even numbers in a random sample of size n2 = 2 without replacement from the population {2, 3, 5}. Assuming the 36 possible differences P1 - P2 equally likely, construct the sampling distribution of P1 - P2 and verify (i) mu_(P1 - P2) = pi1 - pi2, (ii) sigma-squared_(P1 - P2) = pi1 (1 - pi1) / n1 x (N1 - n1) / (N1 - 1) + pi2 (1 - pi2) / n2 x (N2 - n2) / (N2 - 1). (Book answers: pi1 = 2/3, pi2 = 1/3, mu = 1/3,

sigma-squared = 1/9)

Step 1: Population 1 evens are 4, 6, so pi1 = 2/3, N1 = 3. Population 2 evens are 2 only, so pi2 = 1/3, N2 = 3.

Step 2: Ordered samples without replacement: 3 x 2 = 6 samples for each population.

Step 3: P1 distribution: the ordered pairs are (4, 6), (6, 4) giving P1 = 1 (2 samples), and (4, 9), (6, 9), (9, 4), (9, 6) giving P1 = 1/2 (4 samples).

Step 4: P2 distribution: the ordered pairs are (2, 3), (2, 5), (3, 2), (5, 2) giving P2 = 1/2 (4 samples), and (3, 5), (5, 3) giving P2 = 0 (2 samples).

Step 5: Differences: P1 - P2 = 0 in 4 x 4 = 16 cases; = 1/2 in (2 x 4) + (4 x 2) = 16 cases; = 1 in 2 x 2 = 4 cases.

Step 6: Check the total: 16 + 16 + 4 = 36.

Step 7: Mean = (0 x 16 + (1/2) x 16 + 1 x 4) / 36 = 12 / 36 = 1/3.

Step 8: E((P1 - P2)-squared) = (0 x 16 + (1/4) x 16 + 1 x 4) / 36 = 8 / 36 = 2/9.

Step 9: Variance = 2/9 - (1/3)-squared = 2/9 - 1/9 = 1/9.

Step 10: Verify (i): mu = 1/3 = 2/3 - 1/3 = pi1 - pi2.

Step 11: Verify (ii): first term = (2/3)(1/3) / 2 x (3 - 2) / (3 - 1) = (1/9) x (1/2) = 1/18;

second term is the same, 1/18; sum = 1/9.

Answer: distribution 0: 16/36, 1/2: 16/36, 1: 4/36; mean 1/3; variance 1/9. Both results verified.

Q. 5 (b) The percentage of families with a monthly income of Rs. 1,000 or more in city A and city B is 25% and 20% respectively. If a random sample of 100 families is selected from each city and the proportions are compared, what are the mean and standard error of P1 - P2? (Book answers: mu = 0.05, sigma = 0.059)

Step 1: pi1 = 0.25, pi2 = 0.20, n1 = n2 = 100.

Step 2: Mean mu_(P1 - P2) = pi1 - pi2 = 0.25 - 0.20 = 0.05.

Step 3: Variance = pi1 (1 - pi1) / n1 + pi2 (1 - pi2) / n2 = 0.25 x 0.75 / 100 + 0.20 x 0.80 / 100.

Step 4: = 0.001875 + 0.0016 = 0.003475.

Step 5: Standard error = sqrt(0.003475) = 0.05895, i.e. 0.059.

Answer: mean = 0.05, standard error = 0.059.

Q. 6 (a) A finite population consists of five values 2, 4, 6, 8 and 10. Take all possible samples of size 2 drawn with replacement. Assuming the 25 possible samples equally likely, construct the sampling distributions of sample means and sample variances and find the mean of these distributions. Calculate the mean and variance of the population and verify (i) mu_(x-bar) = mu, (ii) mu_(S-squared) = (n - 1) / n x

sigma-squared, where x-bar = sum X_i / n and S-squared = sum (X_i - x-bar)-squared / n. (Book answers: mu = 6, sigma-squared = 8, mu_(x-bar) = 6, mu_(S-squared) = 4)

Step 1: Population mean mu = (2 + 4 + 6 + 8 + 10) / 5 = 30 / 5 = 6.

Step 2: Population variance sigma-squared = (4 + 16 + 36 + 64 + 100) / 5 - 6-squared = 220 / 5 - 36 = 44 - 36 = 8.

Step 3: Number of samples with replacement = 5 x 5 = 25.

Step 4: Sample means distribution: mean 2 in 1 sample, 3 in 2, 4 in 3, 5 in 4, 6 in 5, 7 in 4, 8 in 3, 9 in 2, 10 in 1.

Step 5: Mean of the sample means = (2 x 1 + 3 x 2 + 4 x 3 + 5 x 4 + 6 x 5 + 7 x 4 + 8 x 3 + 9 x 2 + 10 x 1) / 25 = 150 / 25 = 6.

Step 6: For n = 2, S-squared = (X1 - X2)-squared / 4. Sample variances distribution: 0 in 5 samples, 1 in 8, 4 in 6, 9 in 4, 16 in 2.

Step 7: Mean of the sample variances = (0 x 5 + 1 x 8 + 4 x 6 + 9 x 4 + 16 x 2) / 25 = 100 / 25 = 4.

Step 8: Verify (i): mu_(x-bar) = 6 = mu. Verify (ii): (n - 1) / n x sigma-squared = (1/2) x 8 = 4 = mu_(S-squared).

Answer: means distribution as in Step 4; variances distribution as in Step 6; mu_(x-bar) = 6, mu_(S-squared) = 4. Both results verified.

Q. 6 (b) A finite population consists of five values 1, 3, 5, 7 and 9. Take all possible samples of size 2 drawn with replacement. Assuming the 25 possible samples equally likely, construct the sampling distributions of sample means and sample variances and find the mean of these distributions. Calculate the mean and variance of the population. Discuss the results. (Book answers: mu = 5, sigma-squared = 8, mu_(x-bar) = 5, mu_(S-squared) = 4, mu_(x-bar) = mu, mu_(S-squared) = (n - 1) / n x sigma-squared)

Step 1: Population mean mu = (1 + 3 + 5 + 7 + 9) / 5 = 25 / 5 = 5.

Step 2: Population variance sigma-squared = (1 + 9 + 25 + 49 + 81) / 5 - 5-squared = 165 / 5 - 25 = 33 - 25 = 8.

Step 3: Number of samples with replacement = 5 x 5 = 25.

Step 4: Sample means distribution: mean 1 in 1 sample, 2 in 2, 3 in 3, 4 in 4, 5 in 5, 6 in 4, 7 in 3, 8 in 2, 9 in 1.

Step 5: Mean of the sample means = (1 x 1 + 2 x 2 + 3 x 3 + 4 x 4 + 5 x 5 + 6 x 4 + 7 x 3 + 8 x 2 + 9 x 1) / 25 = 125 / 25 = 5.

Step 6: Sample variances distribution (same differences as Q. 6 (a)): 0 in 5 samples, 1 in 8, 4 in 6, 9 in 4, 16 in 2.

Step 7: Mean of the sample variances = (0 x 5 + 1 x 8 + 4 x 6 + 9 x 4 + 16 x 2) / 25 = 100 / 25 = 4.

Step 8: Discussion: mu_(x-bar) = 5 = mu, so the sample mean is unbiased for mu.

Step 9: mu_(S-squared) = 4 = (1/2) x 8 = (n - 1) / n x sigma-squared, so S-squared with divisor n is biased downwards; dividing by n - 1 would remove the bias.

Answer: means distribution as in Step 4; variances distribution as in Step 6; mu_(x-bar) = 5, mu_(S-squared) = 4.

Q. 7 (a) A finite population consists of 5 values 1, 3, 5, 7 and 9. Take all possible samples of size 2 drawn without replacement. Assuming the 20 possible samples equally likely, construct the sampling distributions of sample means and sample variances and find the mean of these distributions. Calculate the mean and variance of the population and verify (i) mu_(x-bar) = mu, (ii) mu_(S-squared) = N / (N - 1) x (n - 1) / n x sigma-squared, where x-bar and S-squared are as in Q. 6 (a). (Book answers: mu = 5, sigma-squared = 8, mu_(x-bar) = 5, mu_(S-squared) = 5)

Step 1: Population mean mu = 5 and variance sigma-squared = 8, as in Q. 6 (b).

Step 2: Number of ordered samples without replacement = 5 x 4 = 20.

Step 3: Sample means distribution: mean 2 in 2 samples, 3 in 2, 4 in 4, 5 in 4, 6 in 4, 7 in 2, 8 in 2.

Step 4: Mean of the sample means = (2 x 2 + 3 x 2 + 4 x 4 + 5 x 4 + 6 x 4 + 7 x 2 + 8 x 2) / 20 = 100 / 20 = 5.

Step 5: Sample variances distribution (S-squared = (X1 - X2)-squared / 4): 1 in 8 samples, 4 in 6, 9 in 4, 16 in 2.

Step 6: Mean of the sample variances = (1 x 8 + 4 x 6 + 9 x 4 + 16 x 2) / 20 = 100

/ 20 = 5.

Step 7: Verify (i): mu_(x-bar) = 5 = mu.

Step 8: Verify (ii): N / (N - 1) x (n - 1) / n x sigma-squared = (5/4) x (1/2) x 8 = 5 = mu_(S-squared).

Answer: means distribution as in Step 3; variances distribution as in Step 5; mu_(x-bar) = 5, mu_(S-squared) = 5. Both results verified.

Q. 7 (b) Take all possible samples of 2 distinct values from the population 2, 4, 6, 8 and 10. Assuming the 20 possible samples equally likely, construct the sampling distributions of sample means and sample variances and find the mean of these distributions. Calculate the mean and variance of the population. Discuss the results. (Book answers: mu = 6, sigma-squared = 8, mu_(x-bar) = 6, mu_(S-squared) = 5, mu_(x-bar) = mu, mu_(S-squared) = N / (N - 1) x (n - 1) / n x sigma-squared)

Step 1: Population mean mu = (2 + 4 + 6 + 8 + 10) / 5 = 30 / 5 = 6.

Step 2: Population variance sigma-squared = 220 / 5 - 36 = 8, as in Q. 6 (a).

Step 3: Number of ordered samples of 2 distinct values = 5 x 4 = 20.

Step 4: Sample means distribution: mean 3 in 2 samples, 4 in 2, 5 in 4, 6 in 4, 7 in 4, 8 in 2, 9 in 2.

Step 5: Mean of the sample means = (3 x 2 + 4 x 2 + 5 x 4 + 6 x 4 + 7 x 4 + 8 x 2

+ 9 x 2) / 20 = 120 / 20 = 6.

Step 6: Sample variances distribution (same gaps as Q. 7 (a)): 1 in 8 samples, 4 in 6, 9 in 4, 16 in 2.

Step 7: Mean of the sample variances = (1 x 8 + 4 x 6 + 9 x 4 + 16 x 2) / 20 = 100 / 20 = 5.

Step 8: Discussion: mu_(x-bar) = 6 = mu, and mu_(S-squared) = 5 = (5/4) x (1/2) x 8, the without-replacement form.

Step 9: Without replacement the mean of S-squared is larger than with replacement (5 instead of 4), because no sample has variance 0.

Answer: means distribution as in Step 4; variances distribution as in Step 6; mu_(x-bar) = 6, mu_(S-squared) = 5.

Chapter 12: Estimation

Exercise — Short Questions

What is the difference between point estimation and interval estimation?

Point estimation gives a single value as the estimate of a population parameter.

Interval estimation gives a range of values with a stated confidence level.

An interval estimate also shows how precise the estimate is.

What is a confidence interval?

A confidence interval is a range computed from sample data.

It is expected to contain the unknown true value of the parameter with a given confidence.

Its width depends on the sample size and the confidence coefficient.

Exercise — True or False (book p. 110, Q. 4)

We cannot control the precision of an interval estimate by the choice of sample size or level of confidence.

False

The width of a confidence interval increases if the confidence coefficient is decreased.

False

The width of a confidence interval decreases if the confidence coefficient is decreased.

True

The precision of an interval estimate can be increased either by increasing the sample size or by decreasing the confidence coefficient.

True

The statistic T can be used in making a confidence interval for mu when the population is non-normal.

False

Exercise 12.1 - Point Estimation (book pp. 79-81)

Q. 1 (a) Explain what is meant by statistical inference?

Step 1: Statistical inference is the field concerned with drawing conclusions about distributions, or about the unknown parameters of a population, by using the observed values of random variables governed by these distributions.

Step 2: It uses only a limited amount of information contained in an observed sample taken from the population at random.

Answer: Its two most important types are (i) estimation of parameters and (ii) testing of hypotheses.

Q. 1 (b) What is meant by estimation? Differentiate between estimator and estimate?

Step 1: Estimation is a procedure of making a judgment about the unknown value of a population parameter by using the sample observations.

Step 2: An estimator is a statistic, that is, a function of the sample observations, used to estimate a parameter; it is a random variable because it varies from sample to sample.

Answer: An estimate is the particular numerical value taken by the estimator for the observed sample; it is a fixed number, not a random variable.

Q. 2 (a) Specify the estimator and the estimate in each of the following: (i) A sample of 35 students gave an average height of 62 inches. (ii) A sample of 50 households having television sets showed that 85 percent of them liked a particular programme. (iii) A sample of 25 bolts produced by a company showed that 20 of them were according to specifications. (iv) A sample of 30 houses showed an average consumption of electricity as 65 units.

Step 1: (i) Estimator: sample mean height X-bar; estimate: x-bar = 62 inches.

Step 2: (ii) Estimator: sample proportion of households who liked a particular

programme P; estimate: p = 0.85.

Step 3: (iii) Estimator: sample number of bolts according to specification X, with estimate x = 20; or sample proportion P, with estimate p = 20 / 25 = 0.80.

Answer: (iv) Estimator: sample mean consumption of electricity X-bar; estimate: x-bar = 65 units.

Q. 2 (b) Suppose I choose a random sample of three observations from a population and obtain the values 2, 5, 3. From these values I estimate the centre of the population by ranking the observations and taking the middle one. What estimator am I using and what is my estimate? (Book answer: Sample median X_0.5 = X_((n+1)/2); x_0.5 = x_((n+1)/2) = 3.)

Step 1: Ranking the observations 2, 5, 3 in ascending order gives 2, 3, 5.

Step 2: The middle observation, the second one, is 3.

Answer: The estimator used is the sample median X_0.5 = X_((n+1)/2), and the estimate is x_0.5 = 3.

Q. 3 (a) Is an estimator a random variable? Why or why not? (Book answer: Yes. An estimator is a random variable having its own probability distribution.)

Step 1: An estimator is a function of the sample observations X_1, X_2, ..., X_n, which are themselves random variables.

Step 2: Different random samples give different values of the estimator, so the estimator has its own probability distribution, namely its sampling distribution.

Answer: Yes, an estimator is a random variable.

Q. 3 (b) Why do we call the standard deviation of a sample statistic as standard error of the statistic? (Book answer: In the context of estimation, the deviation of a sample statistic T from its target theta (parameter to be estimated) must be considered as error. So the standard deviation of the sample statistic is commonly called the standard error of the sample statistic.)

Step 1: A sample statistic T is used to estimate a parameter theta, and T differs from theta because only a sample is observed.

Step 2: This difference is regarded as the error of the estimate.

Answer: The standard deviation of T measures the size of this error from sample to sample, so it is called the standard error of the statistic.

Q. 4 (a) What is meant by unbiasedness? Differentiate between an unbiased and a biased estimator.

Step 1: Unbiasedness: an estimator T of a parameter theta is unbiased if the mean of its sampling distribution is equal to the parameter, that is, E(T) = theta.

Step 2: An unbiased estimator is centred at the parameter it estimates, so on average it neither over-estimates nor under-estimates theta.

Answer: A biased estimator has E(T) not equal to theta; the amount of bias is Bias = E(T) - theta, so it tends to give estimates systematically above or below theta.

Q. 4 (b) A finite population consists of the numbers 3, 5, 7 and 9. Take all possible

samples of size 2 which can be drawn with replacement from this population. By forming the sampling distributions of X-bar and S-squared show that (i) the sample mean X-bar = sum X_i / n is an unbiased estimator of the population mean mu. (ii) the sample variance S-squared = sum (X_i - X-bar)-squared / n is a biased estimator of the population variance sigma-squared. (Book answers: (i) mu = 6, E(X-bar) = 6, E(X-bar) = mu. (ii) sigma-squared = 5, E(S-squared) = 2.5, E(S-squared) not equal to sigma-squared.)

Step 1: Population: 3, 5, 7, 9; N = 4. Population mean: mu = (3 + 5 + 7 + 9) / 4 = 24 / 4 = 6.

Step 2: Population variance: sum of squares = 9 + 25 + 49 + 81 = 164; sigma-squared = 164 / 4 - 6-squared = 41 - 36 = 5.

Step 3: Samples of size n = 2 with replacement: 4 x 4 = 16 samples, all equally likely. The sample means have mean E(X-bar) = 6.

Step 4: The sample variances S-squared = sum (X_i - X-bar)-squared / n over the 16 samples have mean E(S-squared) = 2.5.

Answer: (i) E(X-bar) = 6 = mu, so X-bar is an unbiased estimator of mu. (ii) E(S-squared) = 2.5, which is not equal to sigma-squared = 5, so S-squared is a biased estimator of sigma-squared.

Q. 4 (c) Draw all possible samples of size 3 taken without replacement from the population 7, 10, 13 and 16. By forming the sampling distributions, show that both sample mean X-bar and sample median X_0.5 are unbiased estimators.

Step 1: Population: 7, 10, 13, 16; N = 4. Population mean: mu = (7 + 10 + 13 + 16)

/ 4 = 46 / 4 = 11.5.

Step 2: Samples of size n = 3 without replacement: the 4 unordered samples are (7, 10, 13), (7, 10, 16), (7, 13, 16), (10, 13, 16), all equally likely.

Step 3: Their sample means are 10, 11, 12, 13; the mean of these is (10 + 11 + 12 + 13) / 4 = 46 / 4 = 11.5.

Step 4: Their sample medians (middle values) are 10, 10, 13, 13; the mean of these is (10 + 10 + 13 + 13) / 4 = 46 / 4 = 11.5.

Answer: E(X-bar) = 11.5 = mu and E(X_0.5) = 11.5 = mu, so both the sample mean and the sample median are unbiased estimators of mu.

Q. 5 (a) A finite population consists of the numbers 1, 3, 5, 7 and 9. Consider all possible samples of size two which can be drawn with replacement from this population. By forming the sampling distributions of X-bar, S-squared and S-hat-squared show that (i) the sample mean X-bar = sum X_i / n is an unbiased estimator of the population mean mu. (ii) the sample variance S-squared = sum (X_i - X-bar)-squared / n is a biased estimator of the population variance sigma-squared. (iii) the sample variance S-hat-squared = sum (X_i - X-bar)-squared / (n - 1) is an unbiased estimator of the population variance sigma-squared. (Book answers: (i) mu = 5, E(X-bar) = 5, E(X-bar) = mu. (ii) sigma-squared = 8, E(S-squared) = 4, E(S-squared) not equal to sigma-squared. (iii) sigma-squared = 8, E(S-hat-squared) = 8, E(S-hat-squared) = sigma-squared.)

Step 1: Population: 1, 3, 5, 7, 9; N = 5. Population mean: mu = (1 + 3 + 5 + 7 + 9) / 5 = 25 / 5 = 5.

Step 2: Population variance: sum of squares = 1 + 9 + 25 + 49 + 81 = 165; sigma-squared = 165 / 5 - 5-squared = 33 - 25 = 8.

Step 3: Samples of size n = 2 with replacement: 5 x 5 = 25 samples, all equally likely. The sample means have mean E(X-bar) = 5.

Step 4: The values of S-squared over the 25 samples have mean E(S-squared) = 4.

Step 5: The values of S-hat-squared = sum (X_i - X-bar)-squared / (n - 1) over the 25 samples have mean E(S-hat-squared) = 8.

Answer: (i) E(X-bar) = 5 = mu, so X-bar is unbiased. (ii) E(S-squared) = 4, not equal to 8, so S-squared is biased. (iii) E(S-hat-squared) = 8 = sigma-squared, so S-hat-squared is unbiased.

Q. 5 (b) A finite population consists of the numbers 2, 3, 4, 5, 6 and 8. Find the proportion P of even numbers in all possible random samples of size n = 2 that can be drawn with replacement from this population. By forming the sampling distribution of sample proportions show that sample proportion is an unbiased estimator of the population proportion. Also verify the relation Var(P) = pi (1 - pi) / n, where P and pi are sample and population proportions respectively. (Book answers: pi = 2/3, mu_P = 2/3, sigma_P-squared = 1/9.)

Step 1: Population: 2, 3, 4, 5, 6, 8; N = 6. Even numbers are 2, 4, 6, 8, that is 4 out of 6, so the population proportion is pi = 4 / 6 = 2/3.

Step 2: Samples of size n = 2 with replacement: 6 x 6 = 36 samples, all equally likely. For each sample, P = (number of even numbers) / 2, taking values 0, 1/2, 1.

Step 3: Mean of the sampling distribution: mu_P = E(P) = 2/3.

Step 4: Variance: Var(P) = pi (1 - pi) / n = (2/3) x (1/3) / 2 = (2/9) / 2 = 1/9.

Answer: mu_P = 2/3 = pi, so the sample proportion P is an unbiased estimator of pi, and Var(P) = 1/9, as required.

Q. 5 (c) A finite population consists of the numbers 4, 5, 6 and 8. Find the proportion P of even numbers in all possible random samples of size n = 3 that can be drawn without replacement from this population. By forming the sampling distribution of sample proportions show that sample proportion is an unbiased estimator of the population proportion. Also verify the relation Var(P) = (pi (1 - pi) / n) x ((N - n) / (N - 1)), where P and pi are sample and population proportions respectively. (Book answers: pi = 3/4, mu_P = 3/4, sigma_P-squared = 1/48.)

Step 1: Population: 4, 5, 6, 8; N = 4. Even numbers are 4, 6, 8, that is 3 out of 4, so pi = 3/4.

Step 2: Samples of size n = 3 without replacement: the 4 unordered samples are (4, 5, 6), (4, 5, 8), (4, 6, 8), (5, 6, 8), all equally likely.

Step 3: Their sample proportions of even numbers are 2/3, 2/3, 1, 2/3; the mean is (2/3 + 2/3 + 1 + 2/3) / 4 = 3 / 4.

Step 4: Variance: Var(P) = (pi (1 - pi) / n) x ((N - n) / (N - 1)) = ((3/4) x (1/4) / 3) x ((4 - 3) / (4 - 1)) = (1/16) x (1/3) = 1/48.

Answer: mu_P = 3/4 = pi, so P is an unbiased estimator of pi, and Var(P) = 1/48, as

required.

Exercise 12.2 - Confidence Interval for Population Mean (book pp. 90-92)

Q. 1 (a) What is meant by estimation? Distinguish between point estimate and interval estimate. Why is an interval estimate more useful?

Step 1: Estimation is a procedure of making a judgment about the unknown value of a population parameter by using the sample observations.

Step 2: A point estimate is a single number computed from the sample, used as an estimate of the parameter.

Step 3: An interval estimate is an interval computed from the sample, which is likely to include the unknown true value of the parameter.

Answer: An interval estimate is more useful because a point estimate alone gives no information about its precision, that is, about how close it is likely to be to the true value, while an interval estimate expresses the estimate together with its measure of variation.

Q. 1 (b) Distinguish between the following: (i) Estimator and estimate, (ii) Point and interval estimation.

Step 1: (i) An estimator is a statistic (a function of the sample observations) used to estimate a parameter; it is a random variable. An estimate is the particular numerical value of the estimator for the observed sample.

Answer: (ii) Point estimation gives a single number as the estimate of a parameter;

interval estimation gives an interval of values, with a stated confidence coefficient, which is likely to contain the parameter.

Q. 2 (a) Explain what is meant by (i) Confidence interval, (ii) Confidence limits, (iii) Confidence coefficient.

Step 1: (i) Confidence interval: an interval (L, U) computed from the sample observations such that, prior to sampling, it includes the unknown true value of the parameter theta with a specified high probability 1 - alpha, that is, P(L < theta < U) = 1 - alpha.

Step 2: (ii) Confidence limits: the end points L and U that bound the confidence interval; L is the lower confidence limit and U is the upper confidence limit. They are functions of the sample observations and so are random variables.

Answer: (iii) Confidence coefficient: the probability 1 - alpha that the random interval includes the parameter; commonly used values are 90 percent, 95 percent, 98 percent and 99 percent.

Q. 2 (b) When would a confidence interval be preferred over point estimation for a parameter? (Book answer: When the reliability of a point estimator is needed, the confidence interval conveniently expresses the estimator along with its measure of variation. Reliability is reported through the confidence coefficient and the variation is reflected by the length of the interval.)

Step 1: A point estimate alone does not show how reliable it is.

Step 2: A confidence interval gives the point estimate together with its measure of variation, in the form of the length of the interval.

Answer: A confidence interval is therefore preferred when the reliability of the estimate must be reported: the confidence coefficient states the reliability, and the length of the interval reflects the variation.

Q. 3 (a) Find a 90% confidence interval for the mean of a normal distribution with sigma = 3, given the sample as 2.3, -0.2, -0.4, -0.9. (Book answer: -2.268 < mu < 2.668.)

Step 1: Sample size n = 4. Sample mean: x-bar = (2.3 - 0.2 - 0.4 - 0.9) / 4 = 0.8 / 4 = 0.2.

Step 2: sigma = 3 is known, so the 90% confidence interval is x-bar +/- z_0.95 x (sigma / sqrt(n)), with z_0.95 = 1.645.

Step 3: Standard error: sigma / sqrt(n) = 3 / sqrt(4) = 3 / 2 = 1.5.

Step 4: Margin of error: 1.645 x 1.5 = 2.468.

Answer: The interval is 0.2 +/- 2.468, that is, -2.268 < mu < 2.668.

Q. 3 (b) The standard deviation of the amounts poured into bottles by an automatic filling machine is 1.8 ml (millilitre). The amounts of fill in a random sample of bottles, in ml, were 481, 479, 482, 480, 477, 478, 481 and 482. Suppose the population of amounts of fill is normal. Construct a 90% confidence interval for the mean amount in all bottles filled by the machine. (Book answer: 478.95 < mu < 481.05.)

Step 1: Sample size n = 8. Sample mean: x-bar = (481 + 479 + 482 + 480 + 477 + 478 + 481 + 482) / 8 = 3840 / 8 = 480 ml.

Step 2: sigma = 1.8 ml is known, so the 90% interval is x-bar +/- 1.645 x (sigma / sqrt(n)).

Step 3: Standard error: 1.8 / sqrt(8) = 1.8 / 2.828 = 0.636 ml.

Step 4: Margin of error: 1.645 x 0.636 = 1.047 ml.

Answer: The interval is 480 +/- 1.047, that is, 478.95 < mu < 481.05 ml.

Q. 4 (a) A random sample of size 36 is taken from a normal population with a known variance sigma-squared = 25. If the mean of the sample is 42.6, find 95% confidence limits for the population mean. (Book answer: 40.967 < mu < 44.233.)

Step 1: n = 36, x-bar = 42.6, sigma-squared = 25, so sigma = 5.

Step 2: The 95% interval is x-bar +/- z_0.975 x (sigma / sqrt(n)), with z_0.975 = 1.960.

Step 3: Standard error: 5 / sqrt(36) = 5 / 6 = 0.833.

Step 4: Margin of error: 1.960 x 0.833 = 1.633.

Answer: The interval is 42.6 +/- 1.633, that is, 40.967 < mu < 44.233.

Q. 4 (b) A school wishes to estimate the average weight of students in the sixth grade. A random sample of n = 25 is selected and the sample mean is found to be x-bar = 100 lbs. The standard deviation of the population is known to be 15 lbs. Compute 90% confidence interval for the population mean. (Book answer: 95.065 <

mu < 104.935.)

Step 1: n = 25, x-bar = 100 lbs, sigma = 15 lbs (known).

Step 2: The 90% interval is x-bar +/- z_0.95 x (sigma / sqrt(n)), with z_0.95 = 1.645.

Step 3: Standard error: 15 / sqrt(25) = 15 / 5 = 3 lbs.

Step 4: Margin of error: 1.645 x 3 = 4.935 lbs.

Answer: The interval is 100 +/- 4.935, that is, 95.065 < mu < 104.935 lbs.

Q. 5 (a) Suppose that the weights of 100 male students of a university represent a random sample of weights of 1546 students of the university. Find 99% confidence intervals for the mean weight of the students, given x-bar = 67.45 and s-hat = 2.99. (Book answer: 66.68 < mu < 68.22.)

Step 1: n = 100, x-bar = 67.45, s-hat = 2.99. The sample is large, so the 99% interval is x-bar +/- z_0.995 x (s-hat / sqrt(n)), with z_0.995 = 2.576.

Step 2: Standard error: 2.99 / sqrt(100) = 2.99 / 10 = 0.299.

Step 3: Margin of error: 2.576 x 0.299 = 0.770.

Answer: The interval is 67.45 +/- 0.770, that is, 66.68 < mu < 68.22.

Q. 5 (b) 150 bags of flour of a particular brand are weighed and the mean mass is found to be 748 g, with standard deviation 3.6 g. Find 98% confidence intervals for the mean mass of bags of flour of this brand. (Book answer: 747.316 < mu <

748.684.)

Step 1: n = 150, x-bar = 748 g, s-hat = 3.6 g. The sample is large, so the 98% interval is x-bar +/- z_0.99 x (s-hat / sqrt(n)), with z_0.99 = 2.326.

Step 2: Standard error: 3.6 / sqrt(150) = 3.6 / 12.247 = 0.294 g.

Step 3: Margin of error: 2.326 x 0.294 = 0.684 g.

Answer: The interval is 748 +/- 0.684, that is, 747.316 < mu < 748.684 g.

Q. 6 (a) If the two-sided 100 (1 - alpha)% confidence interval based on random sample taken from X ~ N(mu, sigma-squared) is 12.18 < mu < 20.56, find x-bar. (Book answer: x-bar = 16.37.)

Step 1: The sample mean x-bar is the midpoint of the confidence interval.

Step 2: Midpoint = (12.18 + 20.56) / 2 = 32.74 / 2 = 16.37.

Answer: x-bar = 16.37.

Q. 6 (b) On the basis of the results obtained from a random sample of 100 men from a particular area, the 95% confidence interval for the mean height of the male population of the area was found to be (177.22 cm, 179.18 cm). Find the value of x-bar, the mean of the sample and sigma, the standard deviation of the normal population from which the sample was drawn. Find 98% confidence interval for the mean height. (Book answers: x-bar = 178.2, sigma = 5, 177.04 < mu < 179.36.)

Step 1: x-bar is the midpoint of the 95% interval: x-bar = (177.22 + 179.18) / 2 = 356.40 / 2 = 178.2 cm.

Step 2: The 95% margin of error is (179.18 - 177.22) / 2 = 1.96 / 2 = 0.98 cm.

Step 3: Margin = z_0.975 x (sigma / sqrt(n)) = 1.960 x (sigma / 10) = 0.98, so sigma / 10 = 0.5, giving sigma = 5 cm.

Step 4: The 98% margin is z_0.99 x (sigma / sqrt(n)) = 2.326 x 0.5 = 1.163 cm.

Answer: The 98% interval is 178.2 +/- 1.163, that is, 177.04 < mu < 179.36 cm.

Q. 7 (a) Explain what is meant by the statement, 'we are 100 (1 - alpha)% confident that our interval estimate contains mu.' (Book answer: In repeated sampling, 100 (1 - alpha)% of all such confidence intervals contain mu.)

Step 1: Before the sample is taken, the interval (L, U) is random, and the probability that it includes the fixed parameter mu is 1 - alpha.

Step 2: After the sample is taken, the particular interval either contains mu or it does not; the probability statement refers to the procedure, not to the one interval.

Answer: In repeated sampling, about 100 (1 - alpha)% of all such confidence intervals will contain mu, and about 100 alpha% will miss it.

Q. 7 (b) Explain what is meant by the statement, 'we are 95% confident that our interval contains mu.' (Book answer: In repeated sampling, 95% of all such confidence intervals contain mu.)

Step 1: The confidence coefficient is 1 - alpha = 0.95.

Step 2: The procedure produces intervals that include mu in 95% of repeated

samples.

Answer: We are 95% confident that our interval contains mu means that, in repeated sampling, 95% of all such confidence intervals contain mu.

Q. 7 (c) If an 85% confidence interval is 27.5 < mu < 43.8, what does this statement 85% mean? (Book answer: Intervals so formed would contain mu 85% of the time.)

Step 1: The confidence coefficient is 85%, so 1 - alpha = 0.85.

Step 2: The interval 27.5 < mu < 43.8 is one interval produced by the procedure.

Answer: 85% means that intervals formed in this way would contain mu 85% of the time, in repeated sampling.

Q. 7 (d) If alpha = 0.10, how many intervals would be expected to contain mu? (Book answer: We would expect about 90% of the intervals to contain mu and 10% to miss mu in the long-run in repeated sampling.)

Step 1: The confidence coefficient is 1 - alpha = 1 - 0.10 = 0.90, that is, 90%.

Answer: We would expect about 90% of the intervals to contain mu and 10% to miss mu, in the long run in repeated sampling.

Q. 8 (a) What role does the sample mean play in a two-sided confidence interval for mu, based on a random sample from X ~ N(mu, sigma-squared)? (Book answer: The sample mean is the midpoint of the confidence interval but has no effect on the length of the interval.)

Step 1: The interval is x-bar +/- z_(1 - alpha/2) x (sigma / sqrt(n)).

Step 2: x-bar fixes the centre of the interval.

Answer: The sample mean is the midpoint of the confidence interval, but it has no effect on the length of the interval, which depends only on z, sigma and n.

Q. 8 (b) When setting a two-sided 100 (1 - alpha)% confidence interval for mu, based on a random sample of size n from a normal population, how will the following changes affect the length of the confidence interval for mu: (Assume all other quantities remain fixed.) (i) increasing n (ii) increasing (1 - alpha) (iii) decreasing n (iv) decreasing (1 - alpha) (v) increasing s-squared (vi) increasing x-bar (vii) increasing alpha (viii) decreasing s-hat. (Book answers: decreased, increased, increased, decreased, increased, no effect, decreased, decreased.)

Step 1: The length of the interval is 2 x z_(1 - alpha/2) x (sigma / sqrt(n)) (or with s-hat in place of sigma).

Step 2: (i) Increasing n makes sigma / sqrt(n) smaller, so the length decreases. (ii) Increasing (1 - alpha) makes z larger, so the length increases.

Step 3: (iii) Decreasing n makes sigma / sqrt(n) larger, so the length increases. (iv) Decreasing (1 - alpha) makes z smaller, so the length decreases.

Step 4: (v) Increasing s-squared (hence s-hat) makes the standard error larger, so the length increases. (vi) Increasing x-bar only shifts the centre, so the length is unchanged (no effect).

Step 5: (vii) Increasing alpha decreases (1 - alpha) and hence z, so the length decreases. (viii) Decreasing s-hat makes the standard error smaller, so the length decreases.

Answer: In order: decreased, increased, increased, decreased, increased, no effect, decreased, decreased.

Q. 9 (a) Define Student's t-statistic. What assumptions are made about the population where the t-distribution is used?

Step 1: Student's t-statistic is T = (X-bar - mu) / (S-hat / sqrt(n)), where S-hat is the sample standard deviation with divisor n - 1.

Step 2: When the population is normal, T has a t-distribution with v = n - 1 degrees of freedom.

Answer: Assumptions: the sample is a random sample from a normal population, the population mean mu is unknown, and the population variance sigma-squared is unknown (so sigma is replaced by S-hat); the t-distribution is used when n is small.

Q. 9 (b) The contents of 10 similar containers of a commercial soap are: 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3 and 9.8 litres. Find 99% confidence interval for the mean soap content of all such containers, assuming an approximate normal distribution. (Book answer: 9.807 < mu < 10.313.)

Step 1: n = 10. Sample mean: x-bar = (10.2 + 9.7 + 10.1 + 10.3 + 10.1 + 9.8 + 9.9 + 10.4 + 10.3 + 9.8) / 10 = 100.6 / 10 = 10.06 litres.

Step 2: Sample standard deviation: s-hat = 0.246 litres (computed from the sum of

squared deviations divided by n - 1 = 9).

Step 3: The population is normal with sigma unknown and n small, so use the t-distribution with v = n - 1 = 9 degrees of freedom: t_(9; 0.995) = 3.250.

Step 4: Standard error: s-hat / sqrt(n) = 0.246 / sqrt(10) = 0.246 / 3.162 = 0.078 litres.

Step 5: Margin of error: 3.250 x 0.078 = 0.253 litres.

Answer: The interval is 10.06 +/- 0.253, that is, 9.807 < mu < 10.313 litres.

Q. 10 (a) The masses in grams, of thirteen ball bearings taken at random from a batch are 21.4, 23.1, 25.9, 24.7, 23.4, 21.5, 25.0, 22.5, 26.9, 26.4, 25.8, 23.2, 21.9. Calculate a 95% confidence interval for the mean mass of the population, supposed normal, from which these masses were drawn. (Book answer: 22.82 < mu < 25.14.)

Step 1: n = 13. Sample mean: x-bar = 311.7 / 13 = 23.977 g.

Step 2: Sample standard deviation: s-hat = 1.914 g.

Step 3: The population is normal with sigma unknown and n small, so use the t-distribution with v = 12 degrees of freedom: t_(12; 0.975) = 2.179.

Step 4: Standard error: 1.914 / sqrt(13) = 1.914 / 3.606 = 0.531 g.

Step 5: Margin of error: 2.179 x 0.531 = 1.157 g.

Answer: The interval is 23.977 +/- 1.157, that is, 22.82 < mu < 25.14 g.

Q. 10 (b) A random sample of seven independent observations of a normal variable gave sum x = 35.9, sum x-squared = 186.19. Calculate a 90% confidence interval for the population mean. (Book answer: 4.70 < mu < 5.56.)

Step 1: n = 7. Sample mean: x-bar = 35.9 / 7 = 5.129.

Step 2: Sample variance: s-hat-squared = (sum x-squared - n x-bar-squared) / (n - 1) = (186.19 - 7 x 26.306) / 6 = (186.19 - 184.14) / 6 = 0.342, so s-hat = 0.588.

Step 3: Use the t-distribution with v = 6 degrees of freedom: t_(6; 0.95) = 1.943.

Step 4: Standard error: 0.588 / sqrt(7) = 0.588 / 2.646 = 0.222.

Step 5: Margin of error: 1.943 x 0.222 = 0.432.

Answer: The interval is 5.129 +/- 0.432, that is, 4.70 < mu < 5.56.

Q. 11 (a) A random sample of eight observations of a normal variable gave sum x = 261.2, sum (x - x-bar)-squared = 3.22. Calculate a 95% confidence interval for the population mean. (Book answer: 32.08 < mu < 33.22.)

Step 1: n = 8. Sample mean: x-bar = 261.2 / 8 = 32.65.

Step 2: Sample standard deviation: s-hat = sqrt(3.22 / 7) = sqrt(0.460) = 0.678.

Step 3: Use the t-distribution with v = 7 degrees of freedom: t_(7; 0.975) = 2.365.

Step 4: Standard error: 0.678 / sqrt(8) = 0.678 / 2.828 = 0.240.

Step 5: Margin of error: 2.365 x 0.240 = 0.567.

Answer: The interval is 32.65 +/- 0.567, that is, 32.08 < mu < 33.22.

Q. 11 (b) A sample of 12 measurements of the breaking strength of cotton threads gave a mean x-bar = 209 grams and a standard deviation s-hat = 35 grams. Find 95% and 99% confidence limits for the actual mean breaking strength. (Book answers: 186.76 < mu < 231.24; 177.62 < mu < 240.38.)

Step 1: n = 12, x-bar = 209 g, s-hat = 35 g. Standard error: 35 / sqrt(12) = 35 / 3.464 = 10.104 g.

Step 2: 95% interval: use t_(11; 0.975) = 2.201. Margin of error: 2.201 x 10.104 = 22.24 g.

Step 3: The 95% interval is 209 +/- 22.24, that is, 186.76 < mu < 231.24 g.

Step 4: 99% interval: use t_(11; 0.995) = 3.106. Margin of error: 3.106 x 10.104 = 31.38 g.

Answer: The 99% interval is 209 +/- 31.38, that is, 177.62 < mu < 240.38 g.

Q. 12 (a) A random sample of 16 values from a normal population showed a mean of 41.5 inches and a sum of squares of deviations from this mean equal to 135 (inches)-squared. Show that the 95% confidence limits for this mean are 39.9 and 43.1 inches.

Step 1: n = 16, x-bar = 41.5 inches, sum (x - x-bar)-squared = 135.

Step 2: Sample standard deviation: s-hat = sqrt(135 / 15) = sqrt(9) = 3.0 inches.

Step 3: Use the t-distribution with v = 15 degrees of freedom: t_(15; 0.975) = 2.131.

Step 4: Standard error: 3.0 / sqrt(16) = 3.0 / 4 = 0.75 inches.

Step 5: Margin of error: 2.131 x 0.75 = 1.598 inches.

Answer: The interval is 41.5 +/- 1.598, that is, 39.9 < mu < 43.1 inches, as required.

Q. 12 (b) Find a 99% confidence interval for the mean of normal distribution with sigma = 2.5 and if a sample of size 7 gave the values 9, 16, 10, 14, 8, 13, 14. What would be the confidence interval if sigma were unknown. (Book answers: 9.566 < mu < 14.434; 7.797 < mu < 16.203 when sigma is unknown.)

Step 1: Sample: 9, 16, 10, 14, 8, 13, 14; n = 7. Sample mean: x-bar = 84 / 7 = 12.

Step 2: Sample standard deviation: s-hat = 3.0 (the sum of squared deviations is 54, and 54 / 6 = 9).

Step 3: If sigma = 2.5 is known: standard error = 2.5 / sqrt(7) = 2.5 / 2.646 = 0.945. Margin of error: z_0.995 x 0.945 = 2.576 x 0.945 = 2.434.

Step 4: The interval with sigma known is 12 +/- 2.434, that is, 9.566 < mu < 14.434.

Step 5: If sigma is unknown: standard error = 3.0 / sqrt(7) = 1.134. Use t_(6; 0.995) = 3.707. Margin of error: 3.707 x 1.134 = 4.204.

Answer: The interval with sigma unknown is 12 +/- 4.204, that is, 7.797 < mu < 16.203.

Exercise 12.3 - Confidence Interval for Population Proportion (book p. 95)

Q. 1 (a) A sample poll of 100 voters chosen at random from all voters in a given district indicated that 55% of them were in favour of a particular candidate. Find (i) 95% and (ii) 99% confidence limits for the proportion of all the voters in favour of this candidate. (Book answers: (i) 0.453 < pi < 0.647, (ii) 0.422 < pi < 0.678.)

Step 1: n = 100, sample proportion p = 0.55.

Step 2: Standard error of P: sqrt(p (1 - p) / n) = sqrt(0.55 x 0.45 / 100) = sqrt(0.002475) = 0.0497.

Step 3: (i) 95% interval: margin of error = 1.960 x 0.0497 = 0.0975.

Step 4: The 95% interval is 0.55 +/- 0.0975, that is, 0.453 < pi < 0.647.

Step 5: (ii) 99% interval: margin of error = 2.576 x 0.0497 = 0.128.

Answer: The 99% interval is 0.55 +/- 0.128, that is, 0.422 < pi < 0.678.

Q. 1 (b) In a random sample of 1000 houses in a certain city, it is found that 228 own colour television sets. Find 98% confidence interval for the proportion of houses

in this city that own coloured sets. (Book answer: 0.197 < pi < 0.259.)

Step 1: n = 1000, x = 228, so p = 228 / 1000 = 0.228.

Step 2: Standard error: sqrt(0.228 x 0.772 / 1000) = sqrt(0.000176) = 0.0133.

Step 3: 98% interval: margin of error = 2.326 x 0.0133 = 0.0309.

Answer: The interval is 0.228 +/- 0.0309, that is, 0.197 < pi < 0.259.

Q. 2 (a) In 40 tosses of a coin 24 heads were obtained. Find (i) 95% and (ii) 99.73% confidence limits for the proportion of heads which would be obtained in an unlimited number of tosses of the coin. (Book answers: (i) 0.448 < pi < 0.752, (ii) 0.368 < pi < 0.832.)

Step 1: n = 40, x = 24, so p = 24 / 40 = 0.60.

Step 2: Standard error: sqrt(0.60 x 0.40 / 40) = sqrt(0.006) = 0.0775.

Step 3: (i) 95% interval: margin of error = 1.960 x 0.0775 = 0.152.

Step 4: The 95% interval is 0.60 +/- 0.152, that is, 0.448 < pi < 0.752.

Step 5: (ii) 99.73% interval: use z = 3. Margin of error = 3 x 0.0775 = 0.232.

Answer: The 99.73% interval is 0.60 +/- 0.232, that is, 0.368 < pi < 0.832.

Q. 2 (b) A random sample of 200 voters in a constituency included 110 who said they would vote for Mr. A. Assuming all the 15000 voters in the constituency would

vote, give an approximate 95% confidence interval for the proportion who would vote for Mr. A. (Book answer: 0.4815 < pi < 0.6185.)

Step 1: n = 200, x = 110, so p = 110 / 200 = 0.55. Population size N = 15000.

Step 2: Standard error with finite population correction: sqrt(p (1 - p) / n) x sqrt((N - n) / (N - 1)) = sqrt(0.55 x 0.45 / 200) x sqrt(14800 / 14999) = 0.0352 x 0.9933 = 0.0349.

Step 3: 95% interval: margin of error = 1.960 x 0.0349 = 0.0685.

Answer: The interval is 0.55 +/- 0.0685, that is, 0.4815 < pi < 0.6185.

Q. 2 (c) A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Show that the percentage of bad pineapples in the consignment almost certainly lies between 8.05 and 17.95. (Book answer: interval 0.0805 < pi < 0.1795.)

Step 1: n = 500, x = 65, so p = 65 / 500 = 0.13.

Step 2: Standard error: sqrt(0.13 x 0.87 / 500) = sqrt(0.000226) = 0.0150.

Step 3: 'Almost certainly' means a 99.9% confidence interval: use z = 3.291. Margin of error: 3.291 x 0.0150 = 0.0495.

Answer: The interval is 0.13 +/- 0.0495, that is, 0.0805 < pi < 0.1795, so the percentage of bad pineapples almost certainly lies between 8.05% and 17.95%.

Exercise 12.4 - Confidence Interval for Difference Between Two Population Means (book pp. 103-105)

Q. 1 (a) A test in statistics was given to 50 girls and 75 boys. The girls made an average grade of 76 with a standard deviation of 6, while the boys made an average grade of 82 with a standard deviation of 8. Find a 96% confidence interval for the difference mu_1 - mu_2, where mu_1 is the mean score of all boys and mu_2 is the mean score of all girls who might take this test. (Book answer: 3.42 < mu_1 - mu_2 < 8.58.)

Step 1: Boys: n_1 = 75, x-bar_1 = 82, sigma_1 = 8 (used as known). Girls: n_2 = 50, x-bar_2 = 76, sigma_2 = 6.

Step 2: Point estimate of the difference: x-bar_1 - x-bar_2 = 82 - 76 = 6.

Step 3: Standard error of the difference: sqrt(sigma_1-squared / n_1 + sigma_2-squared / n_2) = sqrt(64 / 75 + 36 / 50) = sqrt(0.853 + 0.720) = sqrt(1.573) = 1.254.

Step 4: 96% interval: use z_0.98 = 2.054. Margin of error: 2.054 x 1.254 = 2.576.

Answer: The interval is 6 +/- 2.576, that is, 3.42 < mu_1 - mu_2 < 8.58.

Q. 1 (b) A manufacturing company consists of two departments producing identical products. It is suspected that the hourly outputs in the two departments are different. Two random samples of production hours are respectively selected and the following data are obtained: Department 1: sample size 64, sample mean 100. Department 2: sample size 49, sample mean 90. The variances of the hourly outputs for the two departments are known to be sigma_1-squared = 256 and

sigma_2-squared = 196 respectively. What is the point estimate for the true difference between the mean outputs of the two departments? Find the 95 percent confidence limits for the true difference. (Book answers: x-bar_1 - x-bar_2 = 10; 4.456 < mu_1 - mu_2 < 15.544.)

Step 1: n_1 = 64, x-bar_1 = 100, sigma_1-squared = 256; n_2 = 49, x-bar_2 = 90, sigma_2-squared = 196.

Step 2: Point estimate of the difference: x-bar_1 - x-bar_2 = 100 - 90 = 10.

Step 3: Standard error: sqrt(256 / 64 + 196 / 49) = sqrt(4 + 4) = sqrt(8) = 2.828.

Step 4: 95% interval: margin of error = 1.960 x 2.828 = 5.544.

Answer: The interval is 10 +/- 5.544, that is, 4.456 < mu_1 - mu_2 < 15.544.

Q. 2 (a) Two independent samples of 100 mechanists and 100 carpenters are taken to estimate the difference between the weekly wages of the two categories of workers. The relevant data are given below: Mechanists: sample mean wages 345, population variance 196. Carpenters: sample mean wages 340, population variance 204. Determine the 95% and the 99% confidence limits for the true difference between the average wages of mechanists and carpenters. (Book answers: 1.08 < mu_1 - mu_2 < 8.92; -0.152 < mu_1 - mu_2 < 10.152.)

Step 1: n_1 = n_2 = 100, x-bar_1 = 345, sigma_1-squared = 196; x-bar_2 = 340, sigma_2-squared = 204.

Step 2: Point estimate of the difference: 345 - 340 = 5.

Step 3: Standard error: sqrt(196 / 100 + 204 / 100) = sqrt(1.96 + 2.04) = sqrt(4) = 2.

Step 4: 95% interval: margin of error = 1.960 x 2 = 3.92. The interval is 5 +/- 3.92, that is, 1.08 < mu_1 - mu_2 < 8.92.

Step 5: 99% interval: margin of error = 2.576 x 2 = 5.152.

Answer: The 99% interval is 5 +/- 5.152, that is, -0.152 < mu_1 - mu_2 < 10.152.

Q. 2 (b) General Incorporated Mill's packing machinery is known to pour dry cereal into economy-size boxes with a standard deviation of 0.6 ounce. Two samples taken on two machines yield the following information: Machine I: n_1 = 15, x-bar_1 = 18.7 ounces. Machine II: n_2 = 21, x-bar_2 = 21.9 ounces. Assuming machine I packages a content that is N(mu_1, 0.36) and machine II packages a content that is N(mu_2, 0.36), construct a 95% confidence interval estimate of mu_2 - mu_1. (Book answer: 2.8 < mu_2 - mu_1 < 3.6.)

Step 1: n_1 = 15, x-bar_1 = 18.7; n_2 = 21, x-bar_2 = 21.9; sigma-squared = 0.36 for both machines.

Step 2: Point estimate of the difference: x-bar_2 - x-bar_1 = 21.9 - 18.7 = 3.2.

Step 3: Standard error: sqrt(0.36 / 15 + 0.36 / 21) = sqrt(0.024 + 0.0171) = sqrt(0.0411) = 0.203.

Step 4: 95% interval: margin of error = 1.960 x 0.203 = 0.398.

Answer: The interval is 3.2 +/- 0.398, that is, 2.8 < mu_2 - mu_1 < 3.6.

Q. 3 (a) A sample of 150 brand A light bulbs showed a mean lifetime of 1400 hours with a standard deviation of 120 hours. A sample of 200 brand B light bulbs showed a mean lifetime of 1200 hours with a standard deviation of 80 hours. Find 95% and 99% confidence limits for the difference between the mean lifetime of the populations of brands A and B. (Book answers: 177.825 < mu_1 - mu_2 < 222.175; 170.856 < mu_1 - mu_2 < 229.144.)

Step 1: Brand A: n_1 = 150, x-bar_1 = 1400, s-hat_1 = 120. Brand B: n_2 = 200, x-bar_2 = 1200, s-hat_2 = 80. Both samples are large, so use the normal approximation.

Step 2: Point estimate of the difference: 1400 - 1200 = 200 hours.

Step 3: Standard error: sqrt(120-squared / 150 + 80-squared / 200) = sqrt(96 + 32) = sqrt(128) = 11.314 hours.

Step 4: 95% interval: margin of error = 1.960 x 11.314 = 22.175. The interval is 200 +/- 22.175, that is, 177.825 < mu_1 - mu_2 < 222.175.

Step 5: 99% interval: margin of error = 2.576 x 11.314 = 29.144.

Answer: The 99% interval is 200 +/- 29.144, that is, 170.856 < mu_1 - mu_2 < 229.144 hours.

Q. 3 (b) Let two independent random samples, each of size 100, from independent normal distributions N(mu_1, sigma_1-squared) and N(mu_2, sigma_2-squared)

yield x-bar_1 = 4.8, s-hat_1-squared = 8.64, x-bar_2 = 5.6, s-hat_2-squared = 7.

Find a 95% confidence interval for (mu_2 - mu_1). (Book answer: 0.025 < mu_2 - mu_1 < 1.575.)

Step 1: n_1 = n_2 = 100, x-bar_1 = 4.8, s-hat_1-squared = 8.64; x-bar_2 = 5.6, s-hat_2-squared = 7.

Step 2: Point estimate of the difference: x-bar_2 - x-bar_1 = 5.6 - 4.8 = 0.8.

Step 3: Standard error: sqrt(8.64 / 100 + 7 / 100) = sqrt(0.0864 + 0.07) = sqrt(0.1564) = 0.395.

Step 4: 95% interval: margin of error = 1.960 x 0.395 = 0.775.

Answer: The interval is 0.8 +/- 0.775, that is, 0.025 < mu_2 - mu_1 < 1.575.

Q. 4 (a) In order to ascertain the age distribution of operatives in a certain industry, random samples of 1720 males and 1230 females are drawn. The sample means and standard deviations were 33.93 years and 14.20 years for the males and 27.44 years and 10.79 years for the females. Calculate the 95 percent confidence interval for (i) the mean age of all male operatives, (ii) the mean age of all female operatives, (iii) the difference between their mean ages. (Book answers: 33.259 < mu_1 < 34.601; 26.837 < mu_2 < 28.043; 5.588 < mu_1 - mu_2 < 7.392.)

Step 1: Males: n_1 = 1720, x-bar_1 = 33.93, s-hat_1 = 14.20. Females: n_2 = 1230, x-bar_2 = 27.44, s-hat_2 = 10.79. Both samples are large, so use the normal approximation.

Step 2: (i) Male standard error: 14.20 / sqrt(1720) = 14.20 / 41.473 = 0.342. Margin:

1.960 x 0.342 = 0.671. Interval: 33.93 +/- 0.671, that is, 33.259 < mu_1 < 34.601.

Step 3: (ii) Female standard error: 10.79 / sqrt(1230) = 10.79 / 35.071 = 0.308. Margin: 1.960 x 0.308 = 0.603. Interval: 27.44 +/- 0.603, that is, 26.837 < mu_2 < 28.043.

Step 4: (iii) Difference: 33.93 - 27.44 = 6.49. Standard error of the difference: sqrt(0.342-squared + 0.308-squared) = sqrt(0.117 + 0.095) = sqrt(0.212) = 0.460. Margin: 1.960 x 0.460 = 0.902.

Answer: The interval for the difference is 6.49 +/- 0.902, that is, 5.588 < mu_1 - mu_2 < 7.392.

Q. 4 (b) The means and variances in the weekly incomes in rupees of the workers employed in the different factories, from the samples are given below: Factory A: sample size 160, mean 12.80, variance 64. Factory B: sample size 220, mean 11.25, variance 49. (i) What is the maximum likelihood estimate of the difference in mean incomes? (ii) Compute the 95 percent confidence interval estimate for the real differences in the incomes of the workers of the two factories. (Book answers: (i) 1.55, (ii) 0.003 < mu_1 - mu_2 < 3.097.)

Step 1: n_1 = 160, x-bar_1 = 12.80, variance 64; n_2 = 220, x-bar_2 = 11.25, variance 49.

Step 2: (i) The maximum likelihood estimate of mu_1 - mu_2 is x-bar_1 - x-bar_2 = 12.80 - 11.25 = 1.55.

Step 3: Standard error: sqrt(64 / 160 + 49 / 220) = sqrt(0.40 + 0.2227) =

sqrt(0.6227) = 0.789.

Step 4: 95% interval: margin of error = 1.960 x 0.789 = 1.547.

Answer: (ii) The interval is 1.55 +/- 1.547, that is, 0.003 < mu_1 - mu_2 < 3.097.

Q. 5 (a) Let two independent random samples, each of size 100, from two independent normal distributions N(mu_1, sigma_1-squared) and N(mu_2, sigma_2-squared) yield x-bar = 4.8, s-hat_1-squared = 8.64, y-bar = 5.6, s-hat_2-squared = 7.88. Find a 95% confidence interval for (mu_1 - mu_2). (Book answer: -1.6 < mu_1 - mu_2 < 0.)

Step 1: n_1 = n_2 = 100, x-bar_1 = 4.8, s-hat_1-squared = 8.64; x-bar_2 = 5.6, s-hat_2-squared = 7.88.

Step 2: Point estimate of the difference: 4.8 - 5.6 = -0.8.

Step 3: Standard error: sqrt(8.64 / 100 + 7.88 / 100) = sqrt(0.0864 + 0.0788) = sqrt(0.1652) = 0.406.

Step 4: 95% interval: margin of error = 1.960 x 0.406 = 0.797.

Answer: The interval is -0.8 +/- 0.797, that is, -1.6 < mu_1 - mu_2 < 0 (to the book's rounding).

Q. 5 (b) Given that x-bar_1 = 75, n_1 = 9, sum (x_i1 - x-bar_1)-squared = 1482; x-bar_2 = 60, n_2 = 16, sum (x_i2 - x-bar_2)-squared = 1830; and assuming that the two samples were randomly selected from two normal populations in which

sigma_1-squared = sigma_2-squared (but unknown), calculate an 80% confidence interval for the difference between the two population means. (Book answer: 8.4 < mu_1 - mu_2 < 21.6.)

Step 1: n_1 = 9, x-bar_1 = 75; n_2 = 16, x-bar_2 = 60. The populations are normal with a common unknown variance, so use the pooled estimate and the t-distribution.

Step 2: Pooled variance: s_p-squared = (1482 + 1830) / (9 + 16 - 2) = 3312 / 23 = 144, so s_p = 12.

Step 3: Point estimate of the difference: 75 - 60 = 15.

Step 4: Standard error: s_p x sqrt(1 / n_1 + 1 / n_2) = 12 x sqrt(1 / 9 + 1 / 16) = 12 x sqrt(0.1736) = 12 x 0.4167 = 5.0.

Step 5: Degrees of freedom: v = 9 + 16 - 2 = 23. For an 80% interval, t_(23; 0.90) = 1.319. Margin of error: 1.319 x 5.0 = 6.597.

Answer: The interval is 15 +/- 6.597, that is, 8.4 < mu_1 - mu_2 < 21.6.

Q. 6 (a) Two random samples of size n_1 = 9 and n_2 = 16 from two independent population having normal distributions provide the means and standard deviations; x-bar_1 = 64, x-bar_2 = 59, s-hat_1 = 6 and s-hat_2 = 5. Find a 95% confidence interval for mu_1 - mu_2 assuming sigma_1 = sigma_2. (Book answer: 0.37 < mu_1 - mu_2 < 9.63.)

Step 1: n_1 = 9, x-bar_1 = 64, s-hat_1 = 6; n_2 = 16, x-bar_2 = 59, s-hat_2 = 5. The variances are assumed equal, so use the pooled estimate and the t-distribution.

Step 2: Pooled variance: s_p-squared = ((9 - 1) x 36 + (16 - 1) x 25) / (9 + 16 - 2) = (288 + 375) / 23 = 663 / 23 = 28.826, so s_p = 5.369.

Step 3: Point estimate of the difference: 64 - 59 = 5.

Step 4: Standard error: 5.369 x sqrt(1 / 9 + 1 / 16) = 5.369 x 0.4167 = 2.237.

Step 5: Degrees of freedom: v = 23. For a 95% interval, t_(23; 0.975) = 2.069. Margin of error: 2.069 x 2.237 = 4.628.

Answer: The interval is 5 +/- 4.628, that is, 0.37 < mu_1 - mu_2 < 9.63.

Q. 6 (b) A course in mathematics is taught to 12 students by the conventional class-room method. A second group of 10 students was given the same course by means of programmed materials. At the end of the course, the same examination was given to each group. The 12 students meeting in the class room made an average grade of 85 with a standard deviation of 4, while the 10 students using programmed materials made an average of 81 with a standard deviation of 5. Find a 90% confidence interval for the difference between the population means, assuming the populations to be approximately normally distributed with equal variance. (Book answer: 0.693 < mu_1 - mu_2 < 7.307.)

Step 1: Class-room: n_1 = 12, x-bar_1 = 85, s-hat_1 = 4. Programmed: n_2 = 10, x-bar_2 = 81, s-hat_2 = 5. Equal variances assumed, so use the pooled estimate and the t-distribution.

Step 2: Pooled variance: s_p-squared = ((12 - 1) x 16 + (10 - 1) x 25) / (12 + 10 - 2) = (176 + 225) / 20 = 401 / 20 = 20.05, so s_p = 4.478.

Step 3: Point estimate of the difference: 85 - 81 = 4.

Step 4: Standard error: 4.478 x sqrt(1 / 12 + 1 / 10) = 4.478 x sqrt(0.1833) = 4.478 x 0.4282 = 1.917.

Step 5: Degrees of freedom: v = 20. For a 90% interval, t_(20; 0.95) = 1.725. Margin of error: 1.725 x 1.917 = 3.307.

Answer: The interval is 4 +/- 3.307, that is, 0.693 < mu_1 - mu_2 < 7.307.

Exercise 12.5 - Confidence Interval for Difference Between Two Population Proportions (book pp. 107-108)

Q. 1 (a) In a poll of college students in a large state university, 300 out of 400 students living in dormitories approved a certain course of action, whereas 200 out of 300 students living in dormitories approved it. Estimate the difference in the proportions favouring the course of action and compute 90% confidence interval for it. (Book answers: p_1 - p_2 = 0.08; 0.023 < pi_1 - pi_2 < 0.137.)

Step 1: Dormitory group 1: n_1 = 400, x_1 = 300, so p_1 = 300 / 400 = 0.75. Group 2: n_2 = 300, x_2 = 200, so p_2 = 200 / 300 = 0.6667.

Step 2: Point estimate of the difference: p_1 - p_2 = 0.75 - 0.6667 = 0.0833, given in the book as 0.08.

Step 3: Standard error: sqrt(p_1 (1 - p_1) / n_1 + p_2 (1 - p_2) / n_2) = sqrt(0.75 x 0.25 / 400 + 0.6667 x 0.3333 / 300) = sqrt(0.000469 + 0.000741) = sqrt(0.001210) = 0.0348.

Step 4: 90% interval: margin of error = 1.645 x 0.0348 = 0.0572.

Answer: The interval is 0.0833 +/- 0.0572, that is, 0.026 < pi_1 - pi_2 < 0.141 by exact arithmetic; the book prints 0.023 < pi_1 - pi_2 < 0.137 (its point estimate is rounded to 0.08 and its margin to 0.057).

Q. 1 (b) In a random sample of 400 adults and 600 teenagers who watched a certain television programme, 100 adults and 300 teenagers indicated that they liked it. Construct 95% and 99% confidence limits for the difference in proportions of all adults and all teenagers who watched the programme and liked it. (Book answers: 0.19 < pi_2 - pi_1 < 0.31; 0.17 < pi_2 - pi_1 < 0.33.)

Step 1: Adults: n_1 = 400, x_1 = 100, so p_1 = 100 / 400 = 0.25. Teenagers: n_2 = 600, x_2 = 300, so p_2 = 300 / 600 = 0.50.

Step 2: Point estimate of the difference (teenagers minus adults): p_2 - p_1 = 0.50 - 0.25 = 0.25.

Step 3: Standard error: sqrt(0.25 x 0.75 / 400 + 0.50 x 0.50 / 600) = sqrt(0.000469 + 0.000417) = sqrt(0.000885) = 0.0298.

Step 4: 95% interval: margin of error = 1.960 x 0.0298 = 0.0583. The interval is 0.25 +/- 0.0583, that is, 0.192 < pi_2 - pi_1 < 0.308, printed in the book as 0.19 < pi_2 - pi_1 < 0.31.

Step 5: 99% interval: margin of error = 2.576 x 0.0298 = 0.0767.

Answer: The 99% interval is 0.25 +/- 0.0767, that is, 0.173 < pi_2 - pi_1 < 0.327,

printed in the book as 0.17 < pi_2 - pi_1 < 0.33.

Q. 2 (a) A poll is taken among the residents of a city and the surrounding country to determine the feasibility of a proposal to construct a civic centre. If 2400 of 5000 city residents favour the proposal and 1200 of 2000 country residents favour it, find a 95% confidence interval for the true difference of the proportions favouring the proposal to construct the civic centre. (Book answer: 0.0945 < pi_2 - pi_1 < 0.1455.)

Step 1: City: n_1 = 5000, x_1 = 2400, so p_1 = 2400 / 5000 = 0.48. Country: n_2 = 2000, x_2 = 1200, so p_2 = 1200 / 2000 = 0.60.

Step 2: Point estimate of the difference (country minus city): p_2 - p_1 = 0.60 - 0.48 = 0.12.

Step 3: Standard error: sqrt(0.48 x 0.52 / 5000 + 0.60 x 0.40 / 2000) = sqrt(0.0000499 + 0.00012) = sqrt(0.0001699) = 0.0130.

Step 4: 95% interval: margin of error = 1.960 x 0.0130 = 0.0256.

Answer: The interval is 0.12 +/- 0.0256, that is, 0.0945 < pi_2 - pi_1 < 0.1455.

Q. 2 (b) The population of interest are the voting preferences of all registered voters in the Punjab and the Sind. Two independent random samples were taken from these populations and the values n_1 = n_2 = 1000, p_1 = 0.54 and p_2 = 0.47. Find a 95% confidence interval for pi_1 - pi_2. (Book answer: 0.026 < pi_1 - pi_2 < 0.114.)

Step 1: n_1 = n_2 = 1000, p_1 = 0.54, p_2 = 0.47.

Step 2: Point estimate of the difference: p_1 - p_2 = 0.54 - 0.47 = 0.07.

Step 3: Standard error: sqrt(0.54 x 0.46 / 1000 + 0.47 x 0.53 / 1000) = sqrt(0.000248 + 0.000249) = sqrt(0.000497) = 0.0223.

Step 4: 95% interval: margin of error = 1.960 x 0.0223 = 0.0437.

Answer: The interval is 0.07 +/- 0.0437, that is, 0.026 < pi_1 - pi_2 < 0.114.

Q. 3 (a) A market survey organization carried out a product taste study with consumers in two regions. In one region, a random sample of n_1 = 400 consumers was selected while in the other region an independent random sample of n_2 = 300 consumers was selected. Each person was asked to indicate which of two servings of product had a better taste. Unknown to the subject, one serving was a new high protein breakfast cereal and the other was an existing cereal. In the first region, proportion p_1 = 0.55 of the sample persons preferred the new cereal and in the second region the proportion was p_2 = 0.65. Construct a 90% confidence interval for pi_2 - pi_1. (Book answer: 0.039 < pi_2 - pi_1 < 0.161.)

Step 1: n_1 = 400, p_1 = 0.55; n_2 = 300, p_2 = 0.65.

Step 2: Point estimate of the difference: p_2 - p_1 = 0.65 - 0.55 = 0.10.

Step 3: Standard error: sqrt(0.55 x 0.45 / 400 + 0.65 x 0.35 / 300) = sqrt(0.000619 + 0.000758) = sqrt(0.001377) = 0.0371.

Step 4: 90% interval: margin of error = 1.645 x 0.0371 = 0.0610.

Answer: The interval is 0.10 +/- 0.0610, that is, 0.039 < pi_2 - pi_1 < 0.161.

Q. 3 (b) Independent random samples are selected from two populations with fractions of success pi_1 and pi_2. Construct a 95% confidence interval for pi_1 - pi_2 for each of the following cases. (i) n_1 = 100, p_1 = 0.72, n_2 = 100, p_2 = 0.61. (ii) n_1 = 130, p_1 = 0.16, n_2 = 210, p_2 = 0.25. (iii) n_1 = 70, p_1 = 0.53, n_2 = 60, p_2 = 0.48. (Book answers: (i) -0.02 to 0.24, (ii) -0.176 to -0.004, (iii) -0.12 to 0.22.)

Step 1: (i) Difference: 0.72 - 0.61 = 0.11. Standard error: sqrt(0.72 x 0.28 / 100 + 0.61 x 0.39 / 100) = sqrt(0.002016 + 0.002379) = sqrt(0.004395) = 0.0663. Margin: 1.960 x 0.0663 = 0.130. Interval: 0.11 +/- 0.130, that is, -0.02 to 0.24.

Step 2: (ii) Difference: 0.16 - 0.25 = -0.09. Standard error: sqrt(0.16 x 0.84 / 130 + 0.25 x 0.75 / 210) = sqrt(0.001034 + 0.000893) = sqrt(0.001927) = 0.0439. Margin: 1.960 x 0.0439 = 0.086. Interval: -0.09 +/- 0.086, that is, -0.176 to -0.004.

Step 3: (iii) Difference: 0.53 - 0.48 = 0.05. Standard error: sqrt(0.53 x 0.47 / 70 + 0.48 x 0.52 / 60) = sqrt(0.003559 + 0.004160) = sqrt(0.007719) = 0.0879. Margin: 1.960 x 0.0879 = 0.172.

Answer: The intervals are: (i) -0.02 to 0.24, (ii) -0.176 to -0.004, (iii) 0.05 +/- 0.172, that is, -0.12 to 0.22.

Exercise 12.6 — Objective Questions, Q. 1: Fill in the blanks (book pp. 108-109)

Q. 1 (i) Statistical inference is the conclusion made about the unknown value of

population parameter by using the sample observations.

Q. 1 (ii) The statistical estimation is a procedure of making judgment about the unknown value of population parameters by using the sample observations.

Q. 1 (iii) The object of point estimation is to obtain a single number from the sample that is intended for estimating the unknown true value of a population parameter.

Q. 1 (iv) A point estimator is a random variable whereas an estimate is a constant.

Q. 1 (v) An estimator is unbiased if its expected value is equal to the population parameter to be estimated.

Q. 1 (vi) If T is a biased estimator, then bias is the difference of its expected value from the parameter theta to be estimated.

Q. 1 (vii) The sample mean X-bar is an unbiased estimator of population mean mu.

Q. 1 (viii) The sample proportion P is an unbiased estimator of population proportion pi.

Q. 1 (ix) The sample variance S-hat-squared = sum (X - X-bar)-squared / (n - 1) is an unbiased estimator of population variance sigma-squared.

Q. 1 (x) Interval estimation is a procedure of constructing an interval from a random sample, such that prior to sampling, it has a high specified probability of including the unknown true value of a population parameter.

Exercise 12.6 — Objective Questions, Q. 2: Fill in the blanks (book p. 109)

Q. 2 (i) The width of a confidence interval is decreased if the level of confidence (1 - alpha) is decreased.

Q. 2 (ii) The width of a confidence interval is directly related to confidence coefficient.

Q. 2 (iii) The precision of confidence interval is increased by decreasing the level of confidence.

Q. 2 (iv) The width of a confidence interval decreases if the sample size is increased.

Q. 2 (v) The confidence coefficient is also called level of confidence.

Q. 2 (vi) 1 - alpha is the probability that the interval estimator includes the unknown true value of the population parameter.

Q. 2 (vii) A sample consisting of 30 or less observations is known as a small sample.

Q. 2 (viii) A sample consisting of more than 30 observations is known as a large sample.

Exercise 12.6 — Objective Questions, Q. 3: True or False (book pp. 109-110)

Q. 3 (i) The types of statistical inferences are estimation of parameters and testing of hypotheses.

True

Q. 3 (ii) The types of statistical estimation of parameters are point estimation and interval estimation.

True

Q. 3 (iii) A point estimator is a sample statistic that is used to estimate the unknown true value of a population parameter.

True

Q. 3 (iv) A point estimate is a specific value of an estimator computed from the sample data after the sample has been observed.

True

Q. 3 (v) An estimate obtained from the sample observations is always a point estimate.

False

Q. 3 (vi) Point estimators may be more useful than interval estimators because probability statements are attached to point estimates.

False

Q. 3 (vii) The point estimation provides two values with a probability statement for estimating the unknown true value of a population parameter.

False

Q. 3 (viii) A confidence interval is a type of statistical inference.

True

Q. 3 (ix) A point estimate provides information about the precision of the estimate.

False

Exercise 12.6 — Objective Questions, Q. 4: True or False, remaining items (book p. 110; 5 items already in the earlier Q. 4 block above)

Q. 4 (iv) The width of a confidence interval can be decreased by decreasing the confidence coefficient.

True

Q. 4 (v) The precision of an interval estimate can be increased by decreasing the sample size.

False

Q. 4 (vii) Alpha is the probability that the interval estimator includes the unknown true value of the population parameter.

False

Chapter 13: Hypothesis Testing

Exercise — Short Questions

What are the null and alternative hypotheses?

The null hypothesis, H0, is the statement that is tested, usually of no difference or no effect.

The alternative hypothesis, H1, is what is accepted if H0 is rejected.

A test decides between them on the basis of sample evidence.

What are Type I and Type II errors?

A Type I error is rejecting a true null hypothesis.

A Type II error is accepting a false null hypothesis.

The probability of a Type I error is the level of significance.

Exercise — True or False (book p. 180)

A value of the test statistic is said to be statistically significant if it falls in the rejection region.

True

If the critical region is located in only one tail of the sampling distribution of the test-statistic, then it is called a one-tailed or one-sided test.

True

If the critical region is equally located in both tails of the sampling distribution of the test-statistic, then it is called a two-tailed or two-sided test.

True

The t-distribution approaches the normal distribution as the sample size increases.

True

H0 is rejected when the probability of its occurrence is equal to or less than the level of significance.

True

Exercise 13.1 - Elements of Hypothesis Testing (book pp. 118-119)

Q. 1 (a) Define the following concepts in your own words as fully as you can: (i) Hypothesis testing (ii) Statistical hypothesis (iii) Null hypothesis (iv) Test statistic (v) Level of significance (vi) Critical region.

Step 1: (i) Hypothesis testing: a procedure to determine whether or not an assumption about some parameter of a population is supported by the information obtained from the observed random sample.

Step 2: (ii) Statistical hypothesis: an assertion or conjecture about the distribution of one or more random variables, usually phrased quantitatively in terms of population parameters.

Step 3: (iii) Null hypothesis, denoted H_0: the hypothesis which is tested for possible rejection (or nullification) under the assumption that it is true; it is usually a statement of 'no effect' or 'status quo'.

Step 4: (iv) Test statistic: a sample statistic which provides a basis for deciding whether or not the null hypothesis should be rejected; the most commonly used test statistics are Z, T, chi-squared and F.

Step 5: (v) Level of significance, denoted alpha: the maximum probability with which we are willing to risk a Type-I error; it is the probability of rejecting a true null hypothesis, and is a small pre-assigned value, say 0.05 or 0.01.

Answer: (vi) Critical region (rejection region): the set of values of the test statistic for which the null hypothesis is rejected (and for which the alternative hypothesis is accepted).

Q. 1 (b) Explain with examples the difference between: (i) Estimation and hypothesis testing (ii) Null hypothesis and alternative hypothesis (iii) Simple hypothesis and composite hypothesis (iv) Acceptance region and rejection region (v) Type-I error and Type-II error (vi) One-tailed test and two-tailed test.

Step 1: (i) Estimation makes a judgment about the unknown value of a parameter from the sample; hypothesis testing decides whether an assumption about a parameter is supported by the sample.

Step 2: (ii) The null hypothesis H_0 is the hypothesis tested for possible rejection; the alternative hypothesis H_1 is the hypothesis we are willing to accept when H_0 is rejected (often the research hypothesis).

Step 3: (iii) A simple hypothesis completely specifies the underlying distribution, including the specific values of all its parameters (for example, X has a normal distribution with mean 30 and standard deviation 4); a composite hypothesis does not completely specify it (for example, X has a normal distribution with mean 40 and standard deviation 5, or with mean 50 and standard deviation at least 4).

Step 4: (iv) The acceptance (nonrejection) region is the set of values of the test statistic for which H_0 is not rejected; the rejection (critical) region is the set of values for which H_0 is rejected.

Step 5: (v) A Type-I error is made by rejecting H_0 when H_0 is actually true; a Type-II error is made by accepting (not rejecting) H_0 when H_1 is actually true.

Answer: (vi) A one-tailed (one-sided) test has its critical region in only one tail of the sampling distribution (left-tailed if H_1 is $\theta < \theta_0$, right-tailed if H_1 is θ

> theta_0); a two-tailed (two-sided) test has its critical region equally in both tails (H_1 is theta not equal to theta_0).

Q. 1 (c) What is the difference between a one-sided and a two-sided test? When should each be used?

Step 1: A one-sided test has its rejection region in only one tail of the sampling distribution of the test statistic; a two-sided test has its rejection region in both tails.

Step 2: The choice depends on the form of the alternative hypothesis H_1.

Step 3: Use a one-sided left-tail test when looking for a definite decrease, that is, when H_1 is given by theta < theta_0.

Step 4: Use a one-sided right-tail test when looking for a definite increase, that is, when H_1 is given by theta > theta_0.

Answer: Use a two-sided test when looking for any change, that is, when H_1 is given by theta not equal to theta_0.

Q. 2 (a) Explain what is meant by: (i) Statistical hypothesis (ii) Test-statistic (iii) Significance level (iv) Test of significance.

Step 1: (i) Statistical hypothesis: an assertion about the distribution of one or more random variables, usually about the value of a population parameter.

Step 2: (ii) Test-statistic: a sample statistic used as the basis for deciding whether the null hypothesis should be rejected.

Step 3: (iii) Significance level: the maximum probability of a Type-I error that we are willing to risk, denoted alpha; it is the size of the critical region.

Answer: (iv) Test of significance: a process of testing a hypothesis to gain a general impression about the parameter, in which no decision is imminent or implied; instead we investigate the values of alpha that would lead to rejection of H_0 as opposed to the values leading to acceptance of H_0.

Q. 2 (b) Distinguish between the following concepts: (i) Statistical estimation and hypothesis testing (ii) Rejection and non-rejection regions (iii) A test at alpha level of significance and at 1 - alpha confidence level.

Step 1: (i) Statistical estimation judges the unknown value of a parameter from the sample; hypothesis testing decides whether an assumption about a parameter is supported by the sample.

Step 2: (ii) The rejection region is the set of values of the test statistic that lead to rejecting H_0; the non-rejection (acceptance) region is the set of values that lead to not rejecting H_0.

Answer: (iii) A test at alpha level of significance rejects a true H_0 with probability alpha; a confidence interval at 1 - alpha confidence level includes the parameter with probability 1 - alpha in repeated sampling. The two are complementary: a two-sided test at level alpha rejects H_0: theta = theta_0 exactly when theta_0 lies outside the 100 (1 - alpha)% confidence interval.

Q. 3 (a) Explain how the null hypothesis and the alternative hypothesis are formulated.

Step 1: The alternative (or research) hypothesis H_1 is a statement about the value of a population parameter that an investigator attempts to support with the observed random sample.

Step 2: The experiment is conducted to see if there is support for some hypothesis, expressed as an inequality in the form 'less than' or 'greater than' or 'not equal to'; this becomes H_1.

Step 3: The null hypothesis H_0 is then stated with an equality sign as a complement of the alternative hypothesis.

Answer: For example, if H_1 is theta < 40, then H_0 is theta >= 40; if H_1 is theta > 40, then H_0 is theta <= 40; if H_1 is theta not equal to theta_0, then H_0 is theta = theta_0. The basic strategy is to attempt to support the research hypothesis by contradicting the null hypothesis.

Q. 3 (b) State the null and alternative hypotheses to be used in testing the following claims: (i) The mean rainfall at Lake Placid during the month of June is 2.8 inches. (ii) No more than 20% of the faculty at Highland University contributed to the annual giving fund. (iii) The proportion of voters favouring Senator Foghorn in up coming election is 0.58. (iv) On the average children attend school within 3.8 miles of their homes in suburban San Francisco. (Book answers: (i) H_0: mu = 2.8 inches against H_1: mu not equal to 2.8 inches; (ii) H_0: pi <= 0.20 against H_1: pi > 0.20; (iii) H_0: pi = 0.58 against H_1: pi not equal to 0.58; (iv) H_0: mu <= 3.8 miles against H_1: mu > 3.8 miles.)

Step 1: (i) The claim states a specific value, so H_0: mu = 2.8 inches against H_1: mu not equal to 2.8 inches (two-tailed).

Step 2: (ii) 'No more than 20%' is the null statement, so H_0: pi <= 0.20 against H_1: pi > 0.20 (right-tailed).

Step 3: (iii) The claim states a specific proportion, so H_0: pi = 0.58 against H_1: pi not equal to 0.58 (two-tailed).

Answer: (iv) 'Within 3.8 miles' is the null statement, so H_0: mu <= 3.8 miles against H_1: mu > 3.8 miles (right-tailed).

Q. 4 (a) Indicate Type-I and Type-II errors in the following statements: (i) An innocent driver may be held by a traffic constable. (ii) A judge can acquit a guilty person. (iii) A deserving player may not be selected in the team. (iv) A bad student may be passed by the examiner. (Book answers: (i) Type-I, (ii) Type-II, (iii) Type-I, (iv) Type-II.)

Step 1: (i) Holding an innocent driver rejects the true state 'driver is innocent': Type-I error.

Step 2: (ii) Acquitting a guilty person accepts the false state 'person is innocent': Type-II error.

Step 3: (iii) Not selecting a deserving player rejects the true state 'player is deserving': Type-I error.

Answer: (iv) Passing a bad student accepts the false state 'student is good': Type-II error.

Q. 4 (b) For each of the following situations, indicate the Type-I and Type-II errors

and the correct decisions. (i) H_0: New system is no better than the old one. Adopt new system when new one is better. Retain old system when new one is better. Retain old system when new one is not better. Adopt new system when new one is not better. (ii) H_0: New product is satisfactory. Market new product when unsatisfactory. Do not market new product when unsatisfactory. Do not market new product when satisfactory. Market new product when satisfactory. (Book answers: (i) Correct, Type-II, Correct, Type-I; (ii) Type-II, Correct, Type-I, Correct.)

Step 1: (i) Adopt new system when new one is better: H_0 is false and is rejected, so Correct decision.

Step 2: (i) Retain old system when new one is better: H_0 is false but is accepted, so Type-II error.

Step 3: (i) Retain old system when new one is not better: H_0 is true and is accepted, so Correct decision.

Step 4: (i) Adopt new system when new one is not better: H_0 is true but is rejected, so Type-I error.

Step 5: (ii) Market new product when unsatisfactory: H_0 is false but is accepted (the product is treated as satisfactory), so Type-II error.

Step 6: (ii) Do not market new product when unsatisfactory: H_0 is false and is rejected, so Correct decision.

Step 7: (ii) Do not market new product when satisfactory: H_0 is true but is rejected, so Type-I error.

Answer: (ii) Market new product when satisfactory: H_0 is true and is accepted, so Correct decision.

Q. 4 (c) Suppose that a psychological testing service is asked to check whether an executive is emotionally fit to assume the presidency of a large company. What type of error is committed if the hypothesis that he is fit for the job is erroneously accepted? What type of error is committed if the hypothesis that he is fit for the job is erroneously rejected? (Book answers: Type-II error; Type-I error.)

Step 1: The hypothesis is 'he is fit for the job'. Accepting it erroneously means accepting a false hypothesis.

Answer: Erroneously accepting the hypothesis is a Type-II error, and erroneously rejecting the hypothesis is a Type-I error.

Q. 5 (a) The director of an advertising agency is concerned with the effectiveness of a certain kind of television commercial. (i) What hypothesis is he testing, if he is committing a Type-I error when he says erroneously that the commercial is effective. (ii) What hypothesis is he testing, if he is committing a Type-II error when he says erroneously that the commercial is effective. (Book answers: (i) Commercial is not effective; (ii) Commercial is effective.)

Step 1: (i) A Type-I error is rejecting a true null hypothesis. Saying the commercial is effective rejects the null hypothesis 'commercial is not effective'; if that null hypothesis is true, the rejection is a Type-I error.

Answer: (i) He is testing H_0: Commercial is not effective. (ii) A Type-II error is accepting a false null hypothesis. Saying the commercial is effective accepts the null hypothesis 'commercial is effective'; if that null hypothesis is false, the

acceptance is a Type-II error. So he is testing H_0: Commercial is effective.

Q. 5 (b) Outline the fundamental procedure followed in testing a null hypothesis.

Step 1: Identify the population of interest and state the conditions required for the validity of the test procedure being used.

Step 2: Formulate and state the null hypothesis H_0 and the alternative hypothesis H_1.

Step 3: Decide and specify the level of significance, alpha.

Step 4: Select the appropriate test statistic and its sampling distribution if H_0 is assumed to be true.

Step 5: Give the critical value (or values) of the test statistic for the desired value of alpha.

Step 6: Establish the rejection (critical) region.

Step 7: State the decision rule: Reject H_0 if the value of the test statistic from the observed sample falls in the rejection region, otherwise do not reject (or accept) H_0.

Step 8: Calculate the value of the test statistic from the observed sample.

Answer: In the light of the decision rule, draw the conclusion as to whether to reject or accept H_0, and then state the decision in managerial (practical) terms.

Exercise 13.2 - Test of Hypothesis About a Population Mean (book pp. 130-132)

Q. 1 (a) Why is the z-test usually inappropriate as a test-statistic when the sample size is small?

Step 1: The z-test statistic Z = (X-bar - mu_0) / (sigma / sqrt(n)) requires the population standard deviation sigma to be known.

Step 2: When sigma is unknown it is replaced by the sample standard deviation S-hat, and the resulting statistic no longer has a standard normal distribution.

Answer: When the sample size is small, this replacement matters: the statistic then has a t-distribution with n - 1 degrees of freedom, which has heavier tails than the normal, so using z critical values would give the wrong significance level. The z-test is appropriate when sigma is known, or when n is large (say n > 30) so that the t-distribution is well approximated by the normal.

Q. 1 (b) Define 'Student's t-statistic'. What are its assumptions? Explain briefly its use and importance in statistics.

Step 1: Student's t-statistic is T = (X-bar - mu_0) / (S-hat / sqrt(n)), where S-hat is the sample standard deviation with divisor n - 1.

Step 2: Assumptions: the sample is a random sample from a normal population, and the population variance sigma-squared is unknown.

Step 3: Under H_0, T has a t-distribution with v = n - 1 degrees of freedom.

Answer: Use and importance: the t-statistic lets us test hypotheses about a population mean when sigma is unknown and the sample is small, which is the common practical case; without it, small-sample inference about the mean would not be possible.

Q. 2 For each of the following, a random sample of size n is taken from a normal distribution with mean mu and variance sigma-squared. The sample mean is x-bar. Test the hypothesis stated, at the level of significance indicated. (i) n = 30, x-bar = 15.2, sigma = 3, H_0: mu = 15.8 against H_1: mu not equal to 15.8, 5%. (ii) n = 10, x-bar = 27, sigma = 1.2, H_0: mu <= 26.3 against H_1: mu > 26.3, 5%. (iii) n = 49, x-bar = 125, sigma = 4.2, H_0: mu <= 123.5 against H_1: mu > 123.5, 1%. (iv) n = 100, x-bar = 4.35, sigma = 0.18, H_0: mu >= 4.40 against H_1: mu < 4.40, 2%. (v) n = 65, x-bar = 100, sum (x_i - x-bar)-squared = 842.4, H_0: mu = 99.2 against H_1: mu not equal to 99.2, 5%. (vi) n = 65, x-bar = 100, sum (x_i - x-bar)-squared = 842.4, H_0: mu <= 99.2 against H_1: mu > 99.2, 5%. (vii) n = 80, x-bar = 85.3, sum (x_i - x-bar)-squared = 2508.8, H_0: mu >= 86.2 against H_1: mu < 86.2, 10%. (viii) n = 100, x-bar = 6.85, sum (x_i - x-bar)-squared = 36, H_0: mu = 7.0 against H_1: mu not equal to 7.0, 1%.

Step 1: (i) Test statistic: z = (15.2 - 15.8) / (3 / sqrt(30)) = -0.6 / 0.548 = -1.095. Critical values at 5% two-tailed: +/- 1.960. Since -1.960 < -1.095 < 1.960, we do not reject H_0: mu = 15.8 against H_1: mu not equal to 15.8.

Step 2: (ii) z = (27 - 26.3) / (1.2 / sqrt(10)) = 0.7 / 0.379 = 1.845. Critical value at 5% right-tailed: 1.645. Since 1.845 > 1.645, we reject H_0: mu <= 26.3 in favour of H_1: mu > 26.3.

Step 3: (iii) z = (125 - 123.5) / (4.2 / sqrt(49)) = 1.5 / 0.6 = 2.5. Critical value at 1%

right-tailed: 2.326. Since 2.5 > 2.326, we reject H_0: mu <= 123.5 in favour of H_1: mu > 123.5.

Step 4: (iv) z = (4.35 - 4.40) / (0.18 / sqrt(100)) = -0.05 / 0.018 = -2.778. Critical value at 2% left-tailed: -2.054. Since -2.778 < -2.054, we reject H_0: mu >= 4.40 in favour of H_1: mu < 4.40.

Step 5: (v) s-hat = sqrt(842.4 / 64) = 3.628. Standard error: 3.628 / sqrt(65) = 0.450. Test statistic: (100 - 99.2) / 0.450 = 1.778. Critical values at 5% two-tailed (n large, normal approximation): +/- 1.960. Since -1.960 < 1.778 < 1.960, we do not reject H_0: mu = 99.2 against H_1: mu not equal to 99.2.

Step 6: (vi) Same test statistic 1.778. Critical value at 5% right-tailed: 1.645. Since 1.778 > 1.645, we reject H_0: mu <= 99.2 in favour of H_1: mu > 99.2.

Step 7: (vii) s-hat = sqrt(2508.8 / 79) = 5.635. Standard error: 5.635 / sqrt(80) = 0.630. Test statistic: (85.3 - 86.2) / 0.630 = -1.428. Critical value at 10% left-tailed: -1.282. Since -1.428 < -1.282, we reject H_0: mu >= 86.2 in favour of H_1: mu < 86.2.

Answer: (viii) s-hat = sqrt(36 / 99) = 0.603. Standard error: 0.603 / sqrt(100) = 0.0603. Test statistic: (6.85 - 7.0) / 0.0603 = -2.487, given in the book as -2.5. Critical values at 1% two-tailed: +/- 2.576. Since -2.576 < -2.5 < 2.576, we do not reject H_0: mu = 7.0 against H_1: mu not equal to 7.0.

Q. 3 (a) A random sample of size 36 is taken from a normal population with known variance sigma-squared = 25. If the mean of the sample is x-bar = 42.6, test the

null hypothesis mu >= 45 against the alternative hypothesis mu < 45 with alpha = 0.05 (alpha is the probability of committing Type-I error). (Book answer: Since z = -2.88 < -1.645 = z_0.05, we reject H_0: mu >= 45 in favour of H_1: mu < 45.)

Step 1: n = 36, x-bar = 42.6, sigma-squared = 25, so sigma = 5. H_0: mu >= 45 against H_1: mu < 45, alpha = 0.05 (left-tailed).

Step 2: Test statistic: z = (42.6 - 45) / (5 / sqrt(36)) = -2.4 / 0.833 = -2.88.

Step 3: Critical value: z_0.05 = -1.645. Rejection region: Z < -1.645.

Answer: Since z = -2.88 < -1.645, we reject H_0: mu >= 45 in favour of H_1: mu < 45.

Q. 3 (b) Ten dry cells were taken from store and voltage tests gave the following results: 1.52, 1.53, 1.49, 1.48, 1.47, 1.49, 1.51, 1.50, 1.47, 1.48 volts. The mean voltage of the cells when stored was 1.51 V. Assuming the population standard deviation to be 0.02 V, is there reason to believe that the cells have deteriorated? (Book answer: Since z = -2.53 < -1.645 = z_0.05, we reject H_0: mu >= 1.51 in favour of H_1: mu < 1.51.)

Step 1: n = 10. Sample mean: x-bar = (1.52 + 1.53 + 1.49 + 1.48 + 1.47 + 1.49 + 1.51 + 1.50 + 1.47 + 1.48) / 10 = 14.94 / 10 = 1.494 V.

Step 2: H_0: mu >= 1.51 V (no deterioration) against H_1: mu < 1.51 V (deterioration), at the 5% level. sigma = 0.02 V is known.

Step 3: Test statistic: z = (1.494 - 1.51) / (0.02 / sqrt(10)) = -0.016 / 0.00632 = -2.53.

Step 4: Critical value: z_0.05 = -1.645. Rejection region: Z < -1.645.

Answer: Since z = -2.53 < -1.645, we reject H_0: mu >= 1.51 in favour of H_1: mu < 1.51. Yes, there is reason to believe that the cells have deteriorated.

Q. 4 (a) A sample of 400 male students is found to have a mean height of 67.47 inches. Can it be regarded as a simple random sample from a large population with mean height 67.39 inches with standard deviation of 1.3 inches? (Book answer: Since z_0.025 = -1.96 < z = 1.23 < 1.96 = z_0.975, we do not reject H_0: mu = 67.39 against H_1: mu not equal to 67.39 at alpha = 0.05.)

Step 1: n = 400, x-bar = 67.47 inches, mu_0 = 67.39 inches, sigma = 1.3 inches. H_0: mu = 67.39 against H_1: mu not equal to 67.39, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (67.47 - 67.39) / (1.3 / sqrt(400)) = 0.08 / 0.065 = 1.23.

Step 3: Critical values: +/- 1.960. Rejection region: Z < -1.960 or Z > 1.960.

Answer: Since -1.960 < 1.23 < 1.960, we do not reject H_0. Yes, the sample can be regarded as a simple random sample from the population with mean 67.39 inches.

Q. 4 (b) The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 8 electric bulbs recently chosen from a supply of newly manufactured bulbs showed a mean lifetime of 1070 hours. Test the hypothesis that mean lifetime of the bulbs has not changed using a level of significance of 0.05. (Book answer: Since z_0.025 = -1.96 < z = -1.13 < 1.96 = z_0.975, we do not reject H_0: mu = 1120 against H_1: mu not equal to 1120.)

Step 1: n = 8, x-bar = 1070 hours, mu_0 = 1120 hours, sigma = 125 hours. H_0: mu = 1120 against H_1: mu not equal to 1120, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (1070 - 1120) / (125 / sqrt(8)) = -50 / 44.194 = -1.13.

Step 3: Critical values: +/- 1.960.

Answer: Since -1.960 < -1.13 < 1.960, we do not reject H_0: mu = 1120. The mean lifetime of the bulbs has not changed, on this evidence.

Q. 5 (a) Suppose that the mean mu of a random variable X is unknown but the variance of X is known to be 144. Should we reject the null hypothesis H_0: mu = 15 in favour of an alternative hypothesis H_1: mu not equal to 15 at a level of significance of alpha = 0.05, if a random sample of 64 observations yields a mean x-bar = 12? What are the 95% confidence limits for mu. (Book answer: Since z = -2 < -1.960 = z_0.025, we reject H_0: mu = 15 in favour of H_1: mu not equal to 15; 9.06 < mu < 14.94.)

Step 1: n = 64, x-bar = 12, sigma-squared = 144, so sigma = 12. H_0: mu = 15 against H_1: mu not equal to 15, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (12 - 15) / (12 / sqrt(64)) = -3 / 1.5 = -2.0.

Step 3: Critical values: +/- 1.960. Since -2.0 < -1.960, we reject H_0: mu = 15 in favour of H_1: mu not equal to 15.

Step 4: The 95% confidence interval is x-bar +/- 1.960 x (sigma / sqrt(n)) = 12 +/- 1.960 x 1.5 = 12 +/- 2.94.

Answer: The 95% confidence limits are 9.06 < mu < 14.94. Note that mu_0 = 15 lies outside this interval, which agrees with the rejection of H_0.

Q. 5 (b) In a syrup filling factory the mean weight per bottle is claimed to be mu = 16. A random sample of 100 bottles is taken and a test statistic used is X-bar, the mean weight of the sample. For a 0.05 level of significance, find the critical values for the test statistic and formulate the decision rule. Assume the weights to be normally distributed with sigma-squared = 2.56. (Book answer: Reject H_0 if X-bar < 15.68 or X-bar > 16.32.)

Step 1: mu_0 = 16, n = 100, sigma-squared = 2.56, so sigma = 1.6. H_0: mu = 16 against H_1: mu not equal to 16, at alpha = 0.05 (two-tailed).

Step 2: The standard error of X-bar is sigma / sqrt(n) = 1.6 / 10 = 0.16.

Step 3: The critical values of the test statistic are mu_0 +/- z_0.975 x (sigma / sqrt(n)) = 16 +/- 1.960 x 0.16 = 16 +/- 0.314, that is, 15.69 and 16.31.

Step 4: Decision rule: Reject H_0 if X-bar < 15.69 or X-bar > 16.31, otherwise do not reject H_0.

Answer: The book prints the critical values as 15.68 and 16.32, using z = 2 in place of 1.960 (margin 0.32); the decision rule is the same in substance: reject H_0 if X-bar lies outside the interval about 16.

Q. 6 (a) A random sample of 36 drinks from a soft-drink machine has an average content 7.6 ounces with a standard deviation of 0.48 ounces. Test the hypothesis mu <= 7.5 ounces against the alternative hypothesis mu > 7.5 at the 0.05 level of

significance. (Book answer: Since z = 1.25 < 1.645 = z_0.95, we do not reject H_0: mu <= 7.5 against H_1: mu > 7.5.)

Step 1: n = 36, x-bar = 7.6 ounces, s-hat = 0.48 ounces. H_0: mu <= 7.5 against H_1: mu > 7.5, at alpha = 0.05 (right-tailed). The sample is large, so the normal approximation is used.

Step 2: Test statistic: z = (7.6 - 7.5) / (0.48 / sqrt(36)) = 0.1 / 0.08 = 1.25.

Step 3: Critical value: z_0.95 = 1.645. Rejection region: Z > 1.645.

Answer: Since z = 1.25 < 1.645, we do not reject H_0: mu <= 7.5 against H_1: mu > 7.5.

Q. 6 (b) It is claimed that an automobile is driven on the average at most 20000 kilometres per year. To test this claim, a random sample of 100 automobile owners are asked to keep a record of the kilometres they travel. Would you agree with claim if the random sample showed an average of 21500 kilometres and a standard deviation of 3900 kilometres? Use a 0.01 level of significance. (Book answer: Since z = 3.846 > 2.326 = z_0.990, we reject H_0: mu <= 20000 in favour of H_1: mu > 20000.)

Step 1: n = 100, x-bar = 21500 km, s-hat = 3900 km. H_0: mu <= 20000 against H_1: mu > 20000, at alpha = 0.01 (right-tailed). The sample is large, so the normal approximation is used.

Step 2: Test statistic: z = (21500 - 20000) / (3900 / sqrt(100)) = 1500 / 390 = 3.846.

Step 3: Critical value: z_0.99 = 2.326. Rejection region: Z > 2.326.

Answer: Since z = 3.846 > 2.326, we reject H_0: mu <= 20000 in favour of H_1: mu > 20000. No, we would not agree with the claim.

Q. 7 (a) A random sample of 100 recorded deaths in the United States during the past year showed an average lifespan of 71.8 years with a standard deviation of 8.9 years. Does this seem to indicate that the average lifespan today is greater than 70 years? Use a 0.05 level of significance. (Book answer: Since z = 2.02 > 1.645 = z_0.95, we reject H_0: mu <= 70 in favour of H_1: mu > 70.)

Step 1: n = 100, x-bar = 71.8 years, s-hat = 8.9 years. H_0: mu <= 70 against H_1: mu > 70, at alpha = 0.05 (right-tailed). The sample is large, so the normal approximation is used.

Step 2: Test statistic: z = (71.8 - 70) / (8.9 / sqrt(100)) = 1.8 / 0.89 = 2.02.

Step 3: Critical value: z_0.95 = 1.645. Rejection region: Z > 1.645.

Answer: Since z = 2.02 > 1.645, we reject H_0: mu <= 70 in favour of H_1: mu > 70. Yes, the average lifespan today seems to be greater than 70 years.

Q. 7 (b) It is claimed that an automobile is driven on the average no more than 12000 miles per year. To test this claim, a random sample of 100 automobile owners are asked to keep a record of the miles they travel. Would you agree with claim if the random sample showed an average of 12500 miles and a standard deviation of 2400 miles? (Book answer: Since z = 2.083 > 1.645, we reject H_0: mu <= 12000 in favour of H_1: mu > 12000 at alpha = 0.05.)

Step 1: n = 100, x-bar = 12500 miles, s-hat = 2400 miles. H_0: mu <= 12000 against H_1: mu > 12000, at alpha = 0.05 (right-tailed). The sample is large, so the

normal approximation is used.

Step 2: Test statistic: z = (12500 - 12000) / (2400 / sqrt(100)) = 500 / 240 = 2.083.

Step 3: Critical value: z_0.95 = 1.645. Rejection region: Z > 1.645.

Answer: Since z = 2.083 > 1.645, we reject H_0: mu <= 12000 in favour of H_1: mu > 12000. No, we would not agree with the claim.

Q. 8 Given the following information, what is your conclusion in testing each of the indicated null hypothesis? Assume the populations are normal. (i) Sample size 16, sample mean 11, estimate of variance from sample 81, H_0: mu <= 10 against H_1: mu > 10, level of significance 0.01. (ii) Sample size 25, sample mean 8, estimate of variance from sample 64, H_0: mu >= 10 against H_1: mu < 10, level of significance 0.01. (iii) Sample size 25, sample mean 9, estimate of variance from sample 49, H_0: mu = 10 against H_1: mu not equal to 10, level of significance 0.02. (iv) Sample size 12, sample mean 24.9, sum (x_i - x-bar)-squared = 12.3, H_0: mu <= 24.1 against H_1: mu > 24.1, level of significance 2.5%. (v) Sample size 17, sample mean 35.6, sum (x_i - x-bar)-squared = 1471.8, H_0: mu = 40 against H_1: mu not equal to 40, level of significance 5%. (vi) Sample size 6, sample mean 1505.8, sum (x_i - x-bar)-squared = 50.8, H_0: mu <= 1503 against H_1: mu > 1503, level of significance 5%. (vii) Sample size 10, sample mean 129.8, sum (x_i - x-bar)-squared = 97.6, H_0: mu >= 133.0 against H_1: mu < 133.0, level of significance 1%.

Step 1: (i) s-hat = 9. Test statistic: t = (11 - 10) / (9 / sqrt(16)) = 1 / 2.25 = 0.444, with v = 15 degrees of freedom. Critical value at 1% right-tailed: t_(15; 0.99) = 2.602. Since 0.444 < 2.602, we do not reject H_0: mu <= 10.

Step 2: (ii) s-hat = 8. Test statistic: t = (8 - 10) / (8 / sqrt(25)) = -2 / 1.6 = -1.25, with v = 24. Critical value at 1% left-tailed: t_(24; 0.01) = -2.492. Since -1.25 > -2.492, we do not reject H_0: mu >= 10.

Step 3: (iii) s-hat = 7. Test statistic: t = (9 - 10) / (7 / sqrt(25)) = -1 / 1.4 = -0.714, with v = 24. Critical values at 2% two-tailed: +/- 2.492. Since -2.492 < -0.714 < 2.492, we do not reject H_0: mu = 10.

Step 4: (iv) s-hat = sqrt(12.3 / 11) = 1.057. Standard error: 1.057 / sqrt(12) = 0.305. Test statistic: t = (24.9 - 24.1) / 0.305 = 2.62, with v = 11. Critical value at 2.5% right-tailed: t_(11; 0.975) = 2.201. Since 2.62 > 2.201, we reject H_0: mu <= 24.1 in favour of H_1: mu > 24.1.

Step 5: (v) s-hat = sqrt(1471.8 / 16) = 9.591. Standard error: 9.591 / sqrt(17) = 2.326. Test statistic: t = (35.6 - 40) / 2.326 = -1.892, with v = 16. Critical values at 5% two-tailed: +/- 2.120. Since -2.120 < -1.892 < 2.120, we do not reject H_0: mu = 40.

Step 6: (vi) s-hat = sqrt(50.8 / 5) = 3.187. Standard error: 3.187 / sqrt(6) = 1.301. Test statistic: t = (1505.8 - 1503) / 1.301 = 2.152, with v = 5. Critical value at 5% right-tailed: t_(5; 0.95) = 2.015. Since 2.152 > 2.015, we reject H_0: mu <= 1503 in favour of H_1: mu > 1503.

Answer: (vii) s-hat = sqrt(97.6 / 9) = 3.294. Standard error: 3.294 / sqrt(10) = 1.042. Test statistic: t = (129.8 - 133.0) / 1.042 = -3.071, with v = 9. Critical value at 1% left-tailed: t_(9; 0.01) = -2.821. Since -3.071 < -2.821, we reject H_0: mu >= 133.0 in favour of H_1: mu < 133.0.

Exercise 13.2 (continued) - Test of Hypothesis About a Population Mean (book pp. 133-134)

Q. 8 (b) A random sample of size n is drawn from normal population with mean 5 and variance sigma-squared. (i) If n = 25, x-bar = 3 and s-hat = 2, what is t? (ii) If n = 9, x-bar = 2 and t = -2, what is s-hat? (iii) If n = 25, s-hat = 10 and t = 2, what is x-bar? (iv) If s-hat = 15, x-bar = 14 and t = 3, what is n? (Book answers: t = -5, s-hat = 4.5, x-bar = 9, n = 25.)

Step 1: The t-statistic is t = (x-bar - mu_0) / (s-hat / sqrt(n)), with mu_0 = 5.

Step 2: (i) t = (3 - 5) / (2 / sqrt(25)) = -2 / 0.4 = -5.

Step 3: (ii) -2 = (2 - 5) / (s-hat / sqrt(9)) = -9 / s-hat, so s-hat = 9 / 2 = 4.5.

Step 4: (iii) 2 = (x-bar - 5) / (10 / sqrt(25)) = (x-bar - 5) / 2, so x-bar - 5 = 4, giving x-bar = 9.

Answer: (iv) 3 = (14 - 5) / (15 / sqrt(n)) = 9 sqrt(n) / 15 = 0.6 sqrt(n), so sqrt(n) = 5, giving n = 25.

Q. 9 (a) Injection of a certain kind of hormone into hens is said to increase the mean weight of eggs by 0.3 ounces. A sample of 30 eggs has an arithmetic mean 0.4 ounces above the pre-injection mean and a value of s-hat equal to 0.20 ounces. Is this enough reason to accept the statement that the mean increase is more than 0.3 ounces? (Book answer: Since t = 2.74 > 1.699 = t_(29; 0.95), we reject H_0: mu <= 0.3 in favour of H_1: mu > 0.3 at alpha = 0.05.)

Step 1: n = 30, x-bar = 0.4 ounces (mean increase), s-hat = 0.20 ounces. H_0: mu

<= 0.3 against H_1: mu > 0.3, at alpha = 0.05 (right-tailed).

Step 2: Test statistic: t = (0.4 - 0.3) / (0.20 / sqrt(30)) = 0.1 / 0.0365 = 2.74, with v = 29 degrees of freedom.

Step 3: Critical value: t_(29; 0.95) = 1.699. Rejection region: T > 1.699.

Answer: Since t = 2.74 > 1.699, we reject H_0: mu <= 0.3 in favour of H_1: mu > 0.3. Yes, there is enough reason to accept that the mean increase is more than 0.3 ounces.

Q. 9 (b) A random sample of 25 hens from a normal population showed that the average laying is 272 eggs per year with a variance of 625 eggs. The company claimed that the average laying is at least 285 eggs per year. Test the claim of the company at alpha = 0.05. (Book answer: Since t = -2.6 < -1.711 = t_(24; 0.05), we reject H_0: mu >= 285 in favour of H_1: mu < 285.)

Step 1: n = 25, x-bar = 272 eggs, s-hat-squared = 625, so s-hat = 25 eggs. H_0: mu >= 285 against H_1: mu < 285, at alpha = 0.05 (left-tailed).

Step 2: Test statistic: t = (272 - 285) / (25 / sqrt(25)) = -13 / 5 = -2.6, with v = 24 degrees of freedom.

Step 3: Critical value: t_(24; 0.05) = -1.711. Rejection region: T < -1.711.

Answer: Since t = -2.6 < -1.711, we reject H_0: mu >= 285 in favour of H_1: mu < 285. The company's claim is not supported.

Q. 10 (a) A producer of a certain make of flashlight dry cell batteries claims that its output has a mean life of 750 hours. A random sample of 15 such batteries has been tested and a sample mean of 745 hours and a sample standard deviation of 24 hours have been obtained. Verify that these results are consistent with the null hypothesis mu >= 750 against mu < 750 at alpha = 0.01. (Book answer: Since t = -0.807 > -2.624 = t_(14; 0.01), we do not reject H_0: mu >= 750 against H_1: mu < 750.)

Step 1: n = 15, x-bar = 745 hours, s-hat = 24 hours. H_0: mu >= 750 against H_1: mu < 750, at alpha = 0.01 (left-tailed).

Step 2: Test statistic: t = (745 - 750) / (24 / sqrt(15)) = -5 / 6.197 = -0.807, with v = 14 degrees of freedom.

Step 3: Critical value: t_(14; 0.01) = -2.624. Rejection region: T < -2.624.

Answer: Since t = -0.807 > -2.624, we do not reject H_0: mu >= 750. The results are consistent with the null hypothesis.

Q. 10 (b) Ten individuals are chosen at random from a normal population and the heights are found to be 63, 63, 66, 67, 67, 69, 70, 70, 71 and 71 inches. In the light of these data, discuss the suggestion that the mean height in the population is 66 inches. (Book answer: Since t_(9; 0.025) = -2.262 < t = 1.78 < 2.262 = t_(9; 0.975), we do not reject H_0: mu = 66 against H_1: mu not equal to 66 at alpha = 0.05.)

Step 1: n = 10. Sample mean: x-bar = 677 / 10 = 67.7 inches. Sample standard deviation: s-hat = 3.02 inches.

Step 2: H_0: mu = 66 against H_1: mu not equal to 66, at alpha = 0.05 (two-tailed).

Step 3: Test statistic: t = (67.7 - 66) / (3.02 / sqrt(10)) = 1.7 / 0.955 = 1.78, with v = 9 degrees of freedom.

Step 4: Critical values: +/- 2.262.

Answer: Since -2.262 < 1.78 < 2.262, we do not reject H_0: mu = 66. The suggestion that the mean height is 66 inches is not contradicted by the data.

Q. 11 (a) Ten cartons are taken at random from an automatic filling machine. The mean net weight of the 10 cartons is 15.90 oz and the sum of squared deviations from this mean is 0.276 (oz)-squared. Does the sample mean differ significantly from intended weight of 16 oz.? (Book answer: Since t_(9; 0.025) = -2.262 <= t = -1.806 <= 2.262 = t_(9; 0.975), we do not reject H_0: mu = 16 against H_1: mu not equal to 16 at alpha = 0.05.)

Step 1: n = 10, x-bar = 15.90 oz, sum (x - x-bar)-squared = 0.276. Sample standard deviation: s-hat = sqrt(0.276 / 9) = 0.175 oz.

Step 2: H_0: mu = 16 against H_1: mu not equal to 16, at alpha = 0.05 (two-tailed).

Step 3: Test statistic: t = (15.90 - 16) / (0.175 / sqrt(10)) = -0.10 / 0.0554 = -1.806, with v = 9 degrees of freedom.

Step 4: Critical values: +/- 2.262.

Answer: Since -2.262 < -1.806 < 2.262, we do not reject H_0: mu = 16. The sample mean does not differ significantly from 16 oz.

Q. 11 (b) In the past a machine has produced washers having thickness of 0.050 inches. To determine whether the machine is in proper working order, a sample of 10 washers is chosen for which the mean thickness is 0.053 inches and the standard deviation is 0.003 inches. Test the hypothesis that the machine is in proper working order using a level of significance of 0.05. (Book answer: Since t = 3.16 > 2.262 = t_(9; 0.975), we reject H_0: mu = 0.050 in favour of H_1: mu not equal to 0.050.)

Step 1: n = 10, x-bar = 0.053 inches, s-hat = 0.003 inches. H_0: mu = 0.050 against H_1: mu not equal to 0.050, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: t = (0.053 - 0.050) / (0.003 / sqrt(10)) = 0.003 / 0.000949 = 3.16, with v = 9 degrees of freedom.

Step 3: Critical values: +/- 2.262.

Answer: Since t = 3.16 > 2.262, we reject H_0: mu = 0.050 in favour of H_1: mu not equal to 0.050. The machine is not in proper working order.

Q. 12 (a) A random sample of 16 values from a normal population showed a mean of 41.5 inches and a sum of squares of deviations from this mean equal to 135 (inches)-squared. Show that the assumption of a mean of 43.5 inches for the population is not reasonable and that the 95% confidence limits for this mean are 39.9 and 43.1 inches.

Step 1: n = 16, x-bar = 41.5 inches, sum (x - x-bar)-squared = 135. Sample standard deviation: s-hat = sqrt(135 / 15) = 3.0 inches.

Step 2: Test H_0: mu = 43.5 against H_1: mu not equal to 43.5, at alpha = 0.05. Test

statistic: t = (41.5 - 43.5) / (3.0 / sqrt(16)) = -2.0 / 0.75 = -2.667, with v = 15 degrees of freedom.

Step 3: Critical values: +/- 2.131. Since t = -2.667 < -2.131, we reject H_0: mu = 43.5. The assumption of a mean of 43.5 inches is not reasonable.

Step 4: The 95% confidence interval is x-bar +/- t_(15; 0.975) x (s-hat / sqrt(n)) = 41.5 +/- 2.131 x 0.75 = 41.5 +/- 1.598.

Answer: The 95% confidence limits are 39.9 and 43.1 inches, which do not include 43.5, in agreement with the rejection of H_0.

Q. 12 (b) A random sample of nine from the men of a large city gave a mean height of 68 inches, and the unbiased estimate of the population variance from sample, s-hat-squared was 4.5 (inches)-squared. Are these data consistent with the assumption of a mean height of 68.5 inches for the men of the city? (Book answer: Since t_(8; 0.025) = -2.306 < t = -0.708 < 2.306 = t_(8; 0.975), so we do not reject H_0: mu = 68.5 against H_1: mu not equal to 68.5 at alpha = 0.05.)

Step 1: n = 9, x-bar = 68 inches, s-hat-squared = 4.5, so s-hat = 2.121 inches. H_0: mu = 68.5 against H_1: mu not equal to 68.5, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: t = (68 - 68.5) / (2.121 / sqrt(9)) = -0.5 / 0.707 = -0.708, with v = 8 degrees of freedom.

Step 3: Critical values: +/- 2.306.

Answer: Since -2.306 < -0.708 < 2.306, we do not reject H_0: mu = 68.5. Yes, the

data are consistent with the assumption of a mean height of 68.5 inches.

Exercise 13.3 - Test of Hypothesis About a Population Proportion (book pp. 139-140)

Q. 1 (a) For each of the following sets of data, carry out a significance test for the hypotheses stated. (i) Sample size 50, number of successes 45, H_0: pi <= 0.8 against H_1: pi > 0.8, 5%. (ii) Sample size 60, number of successes 42, H_0: pi = 0.55 against H_1: pi not equal to 0.55, 2%. (iii) Sample size 120, number of successes 21, H_0: pi = 1/4 against H_1: pi not equal to 1/4, 5%. (iv) Sample size 300, number of successes 213, H_0: pi = 0.65 against H_1: pi not equal to 0.65, 1%. (v) Sample size 90, number of successes 56, H_0: pi >= 0.76 against H_1: pi < 0.76, 1%.

Step 1: The test statistic for a proportion is z = (p - pi_0) / sqrt(pi_0 (1 - pi_0) / n), approximately standard normal for large n.

Step 2: (i) p = 45 / 50 = 0.90. z = (0.90 - 0.80) / sqrt(0.80 x 0.20 / 50) = 0.10 / 0.0566 = 1.768. Critical value at 5% right-tailed: 1.645. Since 1.768 > 1.645, we reject H_0: pi <= 0.8 in favour of H_1: pi > 0.8.

Step 3: (ii) p = 42 / 60 = 0.70. z = (0.70 - 0.55) / sqrt(0.55 x 0.45 / 60) = 0.15 / 0.0642 = 2.335. Critical values at 2% two-tailed: +/- 2.326. Since 2.335 > 2.326, we reject H_0: pi = 0.55 in favour of H_1: pi not equal to 0.55.

Step 4: (iii) p = 21 / 120 = 0.175. z = (0.175 - 0.25) / sqrt(0.25 x 0.75 / 120) = -0.075 / 0.0395 = -1.897. Critical values at 5% two-tailed: +/- 1.960. Since -1.960 < -1.897 < 1.960, we do not reject H_0: pi = 1/4 against H_1: pi not equal to 1/4.

Step 5: (iv) p = 213 / 300 = 0.71. z = (0.71 - 0.65) / sqrt(0.65 x 0.35 / 300) = 0.06 / 0.0275 = 2.179. Critical values at 1% two-tailed: +/- 2.576. Since -2.576 < 2.179 < 2.576, we do not reject H_0: pi = 0.65 against H_1: pi not equal to 0.65.

Answer: (v) p = 56 / 90 = 0.6222. z = (0.6222 - 0.76) / sqrt(0.76 x 0.24 / 90) = -0.1378 / 0.0450 = -3.060. Critical value at 1% left-tailed: -2.326. Since -3.060 < -2.326, we reject H_0: pi >= 0.76 in favour of H_1: pi < 0.76.

Q. 1 (b) A basket ball player has hit on 60% of his shots from the floor. If on the next 100 shots he makes 70 baskets, would you say that his shooting has improved. (Use a 0.05 level of significance). (Book answer: Since z = 2.041 > 1.645 = z_0.95, we reject H_0: pi <= 0.60 in favour of H_1: pi > 0.60; Yes.)

Step 1: n = 100, x = 70, so p = 0.70. H_0: pi <= 0.60 against H_1: pi > 0.60, at alpha = 0.05 (right-tailed).

Step 2: Test statistic: z = (0.70 - 0.60) / sqrt(0.60 x 0.40 / 100) = 0.10 / 0.0490 = 2.041.

Step 3: Critical value: z_0.95 = 1.645.

Answer: Since z = 2.041 > 1.645, we reject H_0: pi <= 0.60 in favour of H_1: pi > 0.60. Yes, his shooting has improved.

Q. 2 (a) In a poll of 10000 voters selected at a random from all the voters in a certain district, it is found that 5180 voters are in favour of a particular candidate. Test the null hypotheses that the proportion of all the voters in the district, who favour the candidate is equal to or less than 50% against the alternative that it is

greater than 50%. Use a 0.05 level of significance. (Book answer: Since z = 3.6 > 1.645 = z_0.95, we reject H_0: pi <= 0.5 in favour of H_1: pi > 0.5.)

Step 1: n = 10000, x = 5180, so p = 0.518. H_0: pi <= 0.50 against H_1: pi > 0.50, at alpha = 0.05 (right-tailed).

Step 2: Test statistic: z = (0.518 - 0.50) / sqrt(0.50 x 0.50 / 10000) = 0.018 / 0.005 = 3.6.

Step 3: Critical value: z_0.95 = 1.645.

Answer: Since z = 3.6 > 1.645, we reject H_0: pi <= 0.5 in favour of H_1: pi > 0.5.

Q. 2 (b) A retailer places an order for 400 recapped automobile tires with a supplier who claims that no more than 5% of his output is ever returned unsatisfactory. In time, 31 of the 400 tires are unsatisfactory. Should the retailer continue to trust his supplier's word as to the rate of returns? Use 5% level of significance. (Book answer: Since z = 2.524 > 1.645 = z_0.95, we reject H_0: pi <= 0.05 in favour of H_1: pi > 0.05; No.)

Step 1: n = 400, x = 31, so p = 31 / 400 = 0.0775. H_0: pi <= 0.05 against H_1: pi > 0.05, at alpha = 0.05 (right-tailed).

Step 2: Test statistic: z = (0.0775 - 0.05) / sqrt(0.05 x 0.95 / 400) = 0.0275 / 0.0109 = 2.524.

Step 3: Critical value: z_0.95 = 1.645.

Answer: Since z = 2.524 > 1.645, we reject H_0: pi <= 0.05 in favour of H_1: pi >

0.05. No, the retailer should not continue to trust the supplier's word.

Q. 2 (c) A commonly prescribed drug on a market for relieving nervous tension is believed to be only 60% effective. Experimental results with a new drug administered to a random sample of 100 adults who were suffering from nervous tension showed that 70 got relief. Is this sufficient evidence that the new drug is superior to the one commonly prescribed? Use 5% level of significance. (Book answer: Since z = 2.041 > 1.645 = z_0.95, we reject H_0: pi <= 0.60 in favour of H_1: pi > 0.60; Yes.)

Step 1: n = 100, x = 70, so p = 0.70. H_0: pi <= 0.60 against H_1: pi > 0.60, at alpha = 0.05 (right-tailed).

Step 2: Test statistic: z = (0.70 - 0.60) / sqrt(0.60 x 0.40 / 100) = 0.10 / 0.0490 = 2.041.

Step 3: Critical value: z_0.95 = 1.645.

Answer: Since z = 2.041 > 1.645, we reject H_0: pi <= 0.60 in favour of H_1: pi > 0.60. Yes, there is sufficient evidence that the new drug is superior.

Q. 3 (a) An electrical company claimed that at least 95% of the parts which they supplied on a government contract confirmed to specification. A sample of 400 parts was tested, and 355 meet specification. Can we accept the company's claim at a 0.05 level of significance? (Book answer: Since z = -5.735 < -1.645 = z_0.05, we reject H_0: pi >= 0.95 in favour of H_1: pi < 0.95; No.)

Step 1: n = 400, x = 355, so p = 355 / 400 = 0.8875. H_0: pi >= 0.95 against H_1: pi < 0.95, at alpha = 0.05 (left-tailed).

Step 2: Test statistic: z = (0.8875 - 0.95) / sqrt(0.95 x 0.05 / 400) = -0.0625 / 0.0109 = -5.735.

Step 3: Critical value: z_0.05 = -1.645.

Answer: Since z = -5.735 < -1.645, we reject H_0: pi >= 0.95 in favour of H_1: pi < 0.95. No, we cannot accept the company's claim.

Q. 3 (b) An electric company claimed that at least 85% of the parts which they supplied conformed to specifications. A sample of 400 parts was tested and 75 did not meet specifications. Can we accept the company's claim at 0.05 level of significance? (Book answer: Since z = -2.100 < -1.645 = z_0.05, we reject H_0: pi >= 0.85 in favour of H_1: pi < 0.85; No.)

Step 1: n = 400, 75 did not meet specifications, so 325 met them, and p = 325 / 400 = 0.8125. H_0: pi >= 0.85 against H_1: pi < 0.85, at alpha = 0.05 (left-tailed).

Step 2: Test statistic: z = (0.8125 - 0.85) / sqrt(0.85 x 0.15 / 400) = -0.0375 / 0.0179 = -2.100.

Step 3: Critical value: z_0.05 = -1.645.

Answer: Since z = -2.100 < -1.645, we reject H_0: pi >= 0.85 in favour of H_1: pi < 0.85. No, we cannot accept the company's claim.

Exercise 13.4 - Test of Hypothesis About Difference Between Two Population Means (book pp. 151-157)

Q. 1 (a) For each of the following sets of data, perform a test to decide whether there is a significant difference between the means, mu_1 and mu_2, of the normal populations from which the samples are drawn. (i) n_1 = 100, sum x_1 = 4250, sigma_1-squared = 30; n_2 = 80, sum x_2 = 3544, sigma_2-squared = 35; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 5%. (ii) n_1 = 20, sum x_1 = 95, sigma_1-squared = 2.3; n_2 = 25, sum x_2 = 135, sigma_2-squared = 2.5; H_0: mu_1 >= mu_2 against H_1: mu_1 < mu_2, 2%. (iii) n_1 = 50, sum x_1 = 1545, sigma_1-squared = 6.5; n_2 = 50, sum x_2 = 1480, sigma_2-squared = 7.1; H_0: mu_1 <= mu_2 against H_1: mu_1 > mu_2, 1%. (iv) n_1 = 100, sum x_1 = 12730; n_2 = 100, sum x_2 = 12410; common population standard deviation sigma = 10.9; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 5%. (v) n_1 = 30, sum x_1 = 192; n_2 = 45, sum x_2 = 315; common population standard deviation sigma = 1.25; H_0: mu_1 >= mu_2 against H_1: mu_1 < mu_2, 1%. (vi) n_1 = 200, sum x_1 = 18470; n_2 = 300, sum x_2 = 27663; common population standard deviation sigma = 0.86; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 10%. (vii) n_1 = 40, sum x_1 = 2128, sum (x_1 - x-bar_1)-squared = 810; n_2 = 50, sum x_2 = 2580, sum (x_2 - x-bar_2)-squared = 772; H_0: mu_1 <= mu_2 against H_1: mu_1 > mu_2, 5%. (viii) n_1 = 80, sum x_1 = 6824, sum (x_1 - x-bar_1)-squared = 2508; n_2 = 100, sum x_2 = 8740, sum (x_2 - x-bar_2)-squared = 3969; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 2%. (ix) n_1 = 65, sum x_1 = 5369, sum (x_1 - x-bar_1)-squared = 8886; n_2 = 80, sum x_2 = 4672, sum (x_2 - x-bar_2)-squared = 5026; H_0: mu_1 - mu_2 <= 20 against H_1: mu_1 - mu_2 > 20, 1%.

Step 1: (i) x-bar_1 = 42.5, x-bar_2 = 44.3. z = (42.5 - 44.3) / sqrt(30 / 100 + 35 / 80) = -1.8 / 0.859 = -2.096. Critical values at 5% two-tailed: +/- 1.960. Since -2.096 < -1.960, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2.

Step 2: (ii) x-bar_1 = 4.75, x-bar_2 = 5.4. z = (4.75 - 5.4) / sqrt(2.3 / 20 + 2.5 / 25) = -0.65 / 0.464 = -1.402. Critical value at 2% left-tailed: -2.054. Since -1.402 > -2.054, we do not reject H_0: mu_1 >= mu_2 against H_1: mu_1 < mu_2.

Step 3: (iii) x-bar_1 = 30.9, x-bar_2 = 29.6. z = (30.9 - 29.6) / sqrt(6.5 / 50 + 7.1 / 50) = 1.3 / 0.522 = 2.493. Critical value at 1% right-tailed: 2.326. Since 2.493 > 2.326, we reject H_0: mu_1 <= mu_2 in favour of H_1: mu_1 > mu_2.

Step 4: (iv) x-bar_1 = 127.3, x-bar_2 = 124.1. z = (127.3 - 124.1) / (10.9 x sqrt(1 / 100 + 1 / 100)) = 3.2 / 1.542 = 2.076. Critical values at 5% two-tailed: +/- 1.960. Since 2.076 > 1.960, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2.

Step 5: (v) x-bar_1 = 6.4, x-bar_2 = 7.0. z = (6.4 - 7.0) / (1.25 x sqrt(1 / 30 + 1 / 45)) = -0.6 / 0.295 = -2.036. Critical value at 1% left-tailed: -2.326. Since -2.036 > -2.326, we do not reject H_0: mu_1 >= mu_2 against H_1: mu_1 < mu_2.

Step 6: (vi) x-bar_1 = 92.35, x-bar_2 = 92.21. z = (92.35 - 92.21) / (0.86 x sqrt(1 / 200 + 1 / 300)) = 0.14 / 0.0785 = 1.783. Critical value at 10% two-tailed: 1.645. Since 1.783 > 1.645, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2.

Step 7: (vii) x-bar_1 = 53.2, x-bar_2 = 51.6. Pooled standard deviation: s_p = sqrt((810 + 772) / (40 + 50 - 2)) = 4.24. t = (53.2 - 51.6) / (4.24 x sqrt(1 / 40 + 1 / 50)) = 1.6 / 0.899 = 1.78, with v = 88 degrees of freedom. Critical value at 5% right-tailed: 1.645. Since 1.78 > 1.645, we reject H_0: mu_1 <= mu_2 in favour of H_1: mu_1 > mu_2.

Step 8: (viii) x-bar_1 = 85.3, x-bar_2 = 87.4. Pooled standard deviation: s_p = sqrt((2508 + 3969) / (80 + 100 - 2)) = 6.03. t = (85.3 - 87.4) / (6.03 x sqrt(1 / 80 + 1 / 100)) = -2.1 / 0.905 = -2.32, with v = 178 degrees of freedom. Critical values at 2% two-tailed: +/- 2.326. Since -2.326 < -2.32 < 2.326, we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.

Answer: (ix) x-bar_1 = 82.6, x-bar_2 = 58.4. Pooled standard deviation: s_p = sqrt((8886 + 5026) / (65 + 80 - 2)) = 9.86. t = ((82.6 - 58.4) - 20) / (9.86 x sqrt(1 / 65 + 1 / 80)) = 4.2 / 1.647 = 2.55, with v = 143 degrees of freedom. Critical value at 1% right-tailed: 2.326. Since 2.55 > 2.326, we reject H_0: mu_1 - mu_2 <= 20 in favour of H_1: mu_1 - mu_2 > 20. Note: the book prints 1.752, -2.351 and 2.453 for the three statistics in (vii), (viii) and (ix) (from unpooled standard errors); the decisions are the same as here.

Q. 1 (b) A simple sample of heights of 6400 Englishmen has a mean of 67.85 inches and a standard deviation of 2.56 inches, while a simple sample of heights of 6400 Australian has a mean of 68.55 inches and a standard deviation of 2.52 inches. Do the data indicate that Australians are on the average taller than the Englishmen? (Book answer: Since z = 15.589 > 1.645 = z_0.95, we reject H_0: mu_2 <= mu_1 against H_1: mu_2 > mu_1 at alpha = 0.05.)

Step 1: Englishmen: n_1 = 6400, x-bar_1 = 67.85, s-hat_1 = 2.56. Australians: n_2 = 6400, x-bar_2 = 68.55, s-hat_2 = 2.52. H_0: mu_2 <= mu_1 against H_1: mu_2 > mu_1, at alpha = 0.05 (right-tailed). The samples are very large, so the normal approximation is used.

Step 2: Test statistic: z = (68.55 - 67.85) / sqrt(2.56-squared / 6400 + 2.52-squared / 6400) = 0.70 / 0.0449 = 15.589.

Step 3: Critical value: z_0.95 = 1.645.

Answer: Since z = 15.589 > 1.645, we reject H_0: mu_2 <= mu_1 in favour of H_1: mu_2 > mu_1. Yes, the data indicate that Australians are on the average taller than the Englishmen.

Q. 2 (a) A random sample of 100 professors in private colleges showed an average monthly salary of Rs. 5000 with a standard deviation of Rs. 200. Another random sample of 150 professors in Govt. Colleges showed an average monthly salary of Rs. 5600 with a standard deviation of Rs. 250. Test the hypotheses that the average salary for the professors teaching in Govt. Colleges does not exceed the average salary for professors teaching in private colleges by more than Rs. 500. Use alpha = 0.01. (Book answer: Since z = 3.499 > 2.326 = z_0.99, we reject H_0: mu_2 - mu_1 <= 500 in favour of H_1: mu_2 - mu_1 > 500.)

Step 1: Private: n_1 = 100, x-bar_1 = 5000, s-hat_1 = 200. Govt.: n_2 = 150, x-bar_2 = 5600, s-hat_2 = 250. H_0: mu_2 - mu_1 <= 500 against H_1: mu_2 - mu_1 > 500, at alpha = 0.01 (right-tailed).

Step 2: Test statistic: z = ((5600 - 5000) - 500) / sqrt(200-squared / 100 + 250-squared / 150) = 100 / 28.58 = 3.499.

Step 3: Critical value: z_0.99 = 2.326.

Answer: Since z = 3.499 > 2.326, we reject H_0: mu_2 - mu_1 <= 500 in favour of H_1: mu_2 - mu_1 > 500. The Govt. salary exceeds the private salary by more than Rs. 500.

Q. 2 (b) A random sample of 80 light bulbs manufactured by company A had an average lifetime of 1258 hours with a standard deviation of 94 hours, while a random sample of 60 light bulbs manufactured by company B had an average lifetime of 1029 hours with a standard deviation of 68 hours. Because of the high cost of bulbs from company A, we are inclined to buy from company B unless the bulbs from company A will last over 200 hours longer on the average than those from company B. Run a test using alpha = 0.01 to determine from whom we should buy our bulbs. (Book answer: Since z = 2.118 < 2.326 = z_0.99, we do not reject H_0: mu_1 - mu_2 <= 200 against H_1: mu_1 - mu_2 > 200.)

Step 1: Company A: n_1 = 80, x-bar_1 = 1258, s-hat_1 = 94. Company B: n_2 = 60, x-bar_2 = 1029, s-hat_2 = 68. H_0: mu_1 - mu_2 <= 200 against H_1: mu_1 - mu_2 > 200, at alpha = 0.01 (right-tailed).

Step 2: Test statistic: z = ((1258 - 1029) - 200) / sqrt(94-squared / 80 + 68-squared / 60) = 29 / 13.70 = 2.118.

Step 3: Critical value: z_0.99 = 2.326.

Answer: Since z = 2.118 < 2.326, we do not reject H_0: mu_1 - mu_2 <= 200. Company A's bulbs do not last over 200 hours longer on the average, so we should buy from company B.

Q. 3 (a) A farmer claims the average yield of corn of variety A exceeds the average yield of variety B by at least 12 bushels per acre. To test this claim, 50 acres of each variety are planted and grown under similar conditions. Variety A yields on the average 86.7 bushels per acre with a standard deviation of 6.28 bushels per acre, while Variety B yields on the average 77.8 bushels per acre with a standard

deviation of 5.61 bushels per acre. Test the farmer's claim using a 0.05 level of significance. (Book answer: Since z = -2.603 < -1.645 = z_0.05, we reject H_0: mu_1 - mu_2 >= 12 in favour of H_1: mu_1 - mu_2 < 12.)

Step 1: Variety A: n_1 = 50, x-bar_1 = 86.7, s-hat_1 = 6.28. Variety B: n_2 = 50, x-bar_2 = 77.8, s-hat_2 = 5.61. H_0: mu_1 - mu_2 >= 12 against H_1: mu_1 - mu_2 < 12, at alpha = 0.05 (left-tailed).

Step 2: Test statistic: z = ((86.7 - 77.8) - 12) / sqrt(6.28-squared / 50 + 5.61-squared / 50) = -3.1 / 1.191 = -2.603.

Step 3: Critical value: z_0.05 = -1.645.

Answer: Since z = -2.603 < -1.645, we reject H_0: mu_1 - mu_2 >= 12 in favour of H_1: mu_1 - mu_2 < 12. The farmer's claim is not supported.

Q. 3 (b) An examination was taken to two classes of 40 and 50 students respectively. In the first class, mean grade was 74 with a standard deviation of 8, while in the second class the mean grade was 78 with a standard deviation of 7. Is there a significance difference between the mean grades at 1% level. (Book answer: Since z_0.005 = -2.576 < z = 2.490 < 2.576 = z_0.995, we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.)

Step 1: Class 1: n_1 = 40, x-bar_1 = 74, s-hat_1 = 8. Class 2: n_2 = 50, x-bar_2 = 78, s-hat_2 = 7. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.01 (two-tailed).

Step 2: Test statistic: z = (74 - 78) / sqrt(64 / 40 + 49 / 50) = -4 / 1.606 = -2.490.

Step 3: Critical values: +/- 2.576.

Answer: Since -2.576 < -2.490 < 2.576, we do not reject H_0: mu_1 = mu_2. There is no significant difference between the mean grades at the 1% level.

Q. 4 (a) A random sample of 100 light bulbs manufactured by company A had a mean lifetime of 1230 hours with a standard deviation of 80 hours, while a random sample of 121 light bulbs manufactured by company B had a mean lifetime of 1200 hours with a standard deviation of 66 hours. Are the mean lifetimes of bulbs manufactured by two companies are significantly different? (Book answer: Since z = 3 > 1.960 = z_0.975, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2.)

Step 1: Company A: n_1 = 100, x-bar_1 = 1230, s-hat_1 = 80. Company B: n_2 = 121, x-bar_2 = 1200, s-hat_2 = 66. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (1230 - 1200) / sqrt(80-squared / 100 + 66-squared / 121) = 30 / 10.0 = 3.0.

Step 3: Critical values: +/- 1.960.

Answer: Since z = 3.0 > 1.960, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2. Yes, the mean lifetimes are significantly different.

Q. 4 (b) A random sample of 200 villages was taken from Faisalabad District and average population per village was found to be 498 with a standard deviation of 50. Another sample of 200 villages from the same district gave an average population

510 per village with a standard deviation of 40. Is the difference between the average of the two samples statistically significant? (Book answer: Since z = -2.650 < -1.960 = z_0.025, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2.)

Step 1: Sample 1: n_1 = 200, x-bar_1 = 498, s-hat_1 = 50. Sample 2: n_2 = 200, x-bar_2 = 510, s-hat_2 = 40. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (498 - 510) / sqrt(50-squared / 200 + 40-squared / 200) = -12 / 4.528 = -2.650.

Step 3: Critical values: +/- 1.960.

Answer: Since z = -2.650 < -1.960, we reject H_0: mu_1 = mu_2 in favour of H_1: mu_1 not equal to mu_2. Yes, the difference is statistically significant.

Q. 4 (c) The means of simple samples of 500 and 400 are 11.5 and 10.9 respectively. Can the samples be regarded as drawn from a population of standard deviation 5? (Book answer: Since z_0.025 = -1.960 < z = 1.789 < 1.960 = z_0.975, we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.)

Step 1: n_1 = 500, x-bar_1 = 11.5; n_2 = 400, x-bar_2 = 10.9; common sigma = 5. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (11.5 - 10.9) / (5 x sqrt(1 / 500 + 1 / 400)) = 0.6 / 0.335 = 1.789.

Step 3: Critical values: +/- 1.960.

Answer: Since -1.960 < 1.789 < 1.960, we do not reject H_0: mu_1 = mu_2. Yes, the samples can be regarded as drawn from populations with the same mean.

Q. 5 (a) Describe the procedure for testing the equality of means of two normal populations for: (i) Large samples, (ii) Small samples.

Step 1: (i) Large samples: use the test statistic Z = ((X-bar_1 - X-bar_2) - delta_0) / sqrt(sigma_1-squared / n_1 + sigma_2-squared / n_2), where sigma_1 and sigma_2 are known, or are replaced by s-hat_1 and s-hat_2 when n_1 and n_2 are large. Z is approximately standard normal under H_0.

Step 2: The null hypothesis is H_0: mu_1 - mu_2 = delta_0 (usually delta_0 = 0), and the critical values are taken from the standard normal table at the chosen level of significance.

Answer: (ii) Small samples: when the populations are normal with a common unknown variance, use the pooled estimate s_p-squared = ((n_1 - 1) s-hat_1-squared + (n_2 - 1) s-hat_2-squared) / (n_1 + n_2 - 2) and the test statistic T = ((X-bar_1 - X-bar_2) - delta_0) / (s_p x sqrt(1 / n_1 + 1 / n_2)), which has a t-distribution with v = n_1 + n_2 - 2 degrees of freedom under H_0; critical values are taken from the t-table.

Q. 5 (b) For each of the following sets of data, perform a test to decide whether there is a significant difference between the means, mu_1 and mu_2, of the normal populations from which the samples are drawn. (i) n_1 = 6, sum x_1 = 171, sum (x_1 - x-bar_1)-squared = 83; n_2 = 7, sum x_2 = 164.5, sum (x_2 -

x-bar_2)-squared = 112; H_0: mu_1 <= mu_2 against H_1: mu_1 > mu_2, 5%. (ii) n_1 = 5, sum x_1 = 678.5, sum (x_1 - x-bar_1)-squared = 562.3; n_2 = 7, sum x_2 = 971.6, sum (x_2 - x-bar_2)-squared = 308.6; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 5%. (iii) n_1 = 8, sum x_1 = 238.4, sum (x_1 - x-bar_1)-squared = 296; n_2 = 10, sum x_2 = 206, sum (x_2 - x-bar_2)-squared = 145; H_0: mu_1 - mu_2 <= 4 against H_1: mu_1 - mu_2 > 4, 1%. (iv) n_1 = 12, sum x_1 = 116.16, sum (x_1 - x-bar_1)-squared = 45.1; n_2 = 18, sum x_2 = 156.96, sum (x_2 - x-bar_2)-squared = 72; H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, 10%.

Step 1: (i) x-bar_1 = 28.5, x-bar_2 = 23.5. Pooled standard deviation: s_p = sqrt((83 + 112) / (6 + 7 - 2)) = 4.21. t = (28.5 - 23.5) / (4.21 x sqrt(1 / 6 + 1 / 7)) = 5.0 / 2.341 = 2.135, with v = 11. Critical value at 5% right-tailed: t_(11; 0.95) = 1.796. Since 2.135 > 1.796, we reject H_0: mu_1 <= mu_2 in favour of H_1: mu_1 > mu_2.

Step 2: (ii) x-bar_1 = 135.7, x-bar_2 = 138.8. Pooled standard deviation: s_p = sqrt((562.3 + 308.6) / (5 + 7 - 2)) = 9.33. t = (135.7 - 138.8) / (9.33 x sqrt(1 / 5 + 1 / 7)) = -3.1 / 5.464 = -0.567, with v = 10. Critical values at 5% two-tailed: +/- 2.228. Since -2.228 < -0.567 < 2.228, we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.

Step 3: (iii) x-bar_1 = 29.8, x-bar_2 = 20.6. Pooled standard deviation: s_p = sqrt((296 + 145) / (8 + 10 - 2)) = 5.25. t = ((29.8 - 20.6) - 4) / (5.25 x sqrt(1 / 8 + 1 / 10)) = 5.2 / 2.491 = 2.088, with v = 16. Critical value at 1% right-tailed: t_(16; 0.99) = 2.583. Since 2.088 < 2.583, we do not reject H_0: mu_1 - mu_2 <= 4 against H_1: mu_1 - mu_2 > 4.

Answer: (iv) x-bar_1 = 9.68, x-bar_2 = 8.72. Pooled standard deviation: s_p =

sqrt((45.1 + 72) / (12 + 18 - 2)) = 2.045. t = (9.68 - 8.72) / (2.045 x sqrt(1 / 12 + 1 / 18)) = 0.96 / 0.762 = 1.260, with v = 28. Critical values at 10% two-tailed: +/- 1.701. Since -1.701 < 1.260 < 1.701, we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.

Q. 5 (c) Eight pots, growing three barley plants each, were exposed to a high tension discharge while nine similar pots were enclosed in an earthen wire cage, the number of tillers (shoots) in each pot were as follows: Caged: 17, 18, 25, 27, 29, 27, 23, 17, 28. Electrified: 16, 16, 20, 16, 21, 17, 15, 20. Discuss whether the electrification exercises any real effect on tillering. (Book answer: Since t = 3.058 > 1.753 = t_(15; 0.95), we reject H_0: mu_1 <= mu_2 in favour of H_1: mu_1 > mu_2 at alpha = 0.05.)

Step 1: Caged: n_1 = 9, sum = 211, x-bar_1 = 23.44. Electrified: n_2 = 8, sum = 141, x-bar_2 = 17.625. H_0: mu_1 <= mu_2 against H_1: mu_1 > mu_2, at alpha = 0.05 (right-tailed), assuming the populations are normal with a common variance.

Step 2: Pooled standard deviation: s_p = 3.92. Standard error: 3.92 x sqrt(1 / 9 + 1 / 8) = 3.92 x 0.486 = 1.903.

Step 3: Test statistic: t = (23.44 - 17.625) / 1.903 = 5.819 / 1.903 = 3.058, with v = 9 + 8 - 2 = 15 degrees of freedom.

Step 4: Critical value: t_(15; 0.95) = 1.753.

Answer: Since t = 3.058 > 1.753, we reject H_0: mu_1 <= mu_2 in favour of H_1: mu_1 > mu_2. The caged pots produced significantly more tillers, so electrification does exercise a real effect on tillering (it reduces it).

Q. 6 (a) A random sample of 6 cows of breed A had daily milk yields in lb., as 16, 15, 18, 17, 19, 17 and another random sample of 8 cows of breed B had daily milk yields in lb., as 18, 22, 21, 23, 19, 20, 24 and 21. Test if breed B is better than breed A at alpha = 0.05. (Book answer: Since t = 4.162 > 1.782 = t_(12; 0.95), we reject H_0: mu_2 <= mu_1 in favour of H_1: mu_2 > mu_1.)

Step 1: Breed A: n_1 = 6, x-bar_1 = 17. Breed B: n_2 = 8, x-bar_2 = 21. H_0: mu_2 <= mu_1 against H_1: mu_2 > mu_1, at alpha = 0.05 (right-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = 1.78. Standard error: 1.78 x sqrt(1 / 6 + 1 / 8) = 1.78 x 0.540 = 0.961.

Step 3: Test statistic: t = (21 - 17) / 0.961 = 4.162, with v = 6 + 8 - 2 = 12 degrees of freedom.

Step 4: Critical value: t_(12; 0.95) = 1.782.

Answer: Since t = 4.162 > 1.782, we reject H_0: mu_2 <= mu_1 in favour of H_1: mu_2 > mu_1. Breed B is better than breed A.

Q. 6 (b) The heights of six randomly selected sailors are in inches: 62, 64, 67, 68, 70 and 71. Those of ten randomly selected soldiers are 62, 63, 65, 66, 69, 69, 70, 71, 72 and 73. Discuss in the light of these data that soldiers are on the average taller than sailors. Assume that the heights are normally distributed. (Book answer: Since t = 0.526 < 1.761 = t_(14; 0.95), we do not reject H_0: mu_2 <= mu_1 against H_1: mu_2 > mu_1 at alpha = 0.05.)

Step 1: Sailors: n_1 = 6, x-bar_1 = 67. Soldiers: n_2 = 10, x-bar_2 = 68. H_0: mu_2

<= mu_1 against H_1: mu_2 > mu_1, at alpha = 0.05 (right-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = 3.68. Standard error: 3.68 x sqrt(1 / 6 + 1 / 10) = 3.68 x 0.516 = 1.901.

Step 3: Test statistic: t = (68 - 67) / 1.901 = 0.526, with v = 6 + 10 - 2 = 14 degrees of freedom.

Step 4: Critical value: t_(14; 0.95) = 1.761.

Answer: Since t = 0.526 < 1.761, we do not reject H_0: mu_2 <= mu_1. The data do not show that soldiers are on the average taller than sailors.

Q. 7 (a) The weights in grams of 10 male and 10 female juvenile ring-necked pheasants are: Males: 1293, 1380, 1614, 1497, 1340, 1643, 1466, 1627, 1383, 1711. Females: 1061, 1065, 1092, 1017, 1021, 1138, 1143, 1094, 1270, 1028. Test the hypothesis of a difference of 350 grams between population means in favour of the alternative of a greater difference. (Book answer: Since t = 1.007 < 1.734 = t_(18; 0.95), we do not reject H_0: mu_1 - mu_2 <= 350 against H_1: mu_1 - mu_2 > 350 at alpha = 0.05.)

Step 1: Males: n_1 = 10, x-bar_1 = 1495.4 g. Females: n_2 = 10, x-bar_2 = 1092.9 g. Difference: 402.5 g. H_0: mu_1 - mu_2 <= 350 against H_1: mu_1 - mu_2 > 350, at alpha = 0.05 (right-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = 116.5 g. Standard error: 116.5 x sqrt(1 / 10

+ 1 / 10) = 116.5 x 0.447 = 52.1 g.

Step 3: Test statistic: t = (402.5 - 350) / 52.1 = 52.5 / 52.1 = 1.007, with v = 18 degrees of freedom.

Step 4: Critical value: t_(18; 0.95) = 1.734.

Answer: Since t = 1.007 < 1.734, we do not reject H_0: mu_1 - mu_2 <= 350. The difference is not significantly greater than 350 grams.

Q. 7 (b) The following data are the gains in weight, measured in pounds, of babies from birth to age six months. All babies in both groups weighed approximately the same at birth. The babies in sample I were fed formula A, and babies in sample II were fed formula B. Sample I: 5, 7, 8, 9, 6, 7, 10, 8, 6. Sample II: 9, 10, 8, 6, 8, 7, 9. (Assume that the experimenter has no preconceived notions about which formula might be better). Test at the 5% level of significance that the mean of population I equals mean of population II. (Book answer: Since t_(14; 0.025) = -2.145 < t = -1.083 < 2.145 = t_(14; 0.975), we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.)

Step 1: Sample I: n_1 = 9, x-bar_1 = 7.33 lb. Sample II: n_2 = 7, x-bar_2 = 8.14 lb. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = 1.485 lb. Standard error: 1.485 x sqrt(1 / 9 + 1 / 7) = 1.485 x 0.504 = 0.749 lb.

Step 3: Test statistic: t = (7.33 - 8.14) / 0.749 = -0.81 / 0.749 = -1.083, with v = 9 +

7 - 2 = 14 degrees of freedom.

Step 4: Critical values: +/- 2.145.

Answer: Since -2.145 < -1.083 < 2.145, we do not reject H_0: mu_1 = mu_2. There is no significant difference between the mean gains of the two formulas.

Q. 8 (a) From the area planted in one variety of value, 54 plants were selected at random. Of these plants, 15 were 'Off-types' and 12 were 'Aberrant'. The rubber percentages for these plants were: Off-types: 6.21, 5.70, 6.04, 4.47, 5.22, 4.45, 4.84, 5.88, 5.82, 6.09, 5.59, 6.06, 5.59, 6.74, 5.55. Aberrants: 4.28, 7.71, 6.48, 7.71, 7.37, 7.20, 7.06, 6.40, 8.93, 5.91, 5.51, 6.36. Test the hypothesis of no difference between the two means. Also compute a 95% confidence interval for the difference of two population means. (Book answers: Since t = -3.120 < -2.060 = t_(25; 0.025), we reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2; 0.383 < mu_2 - mu_1 < 1.871.)

Step 1: Off-types: n_1 = 15, x-bar_1 = 5.617. Aberrants: n_2 = 12, x-bar_2 = 6.743. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = 0.933. Standard error: 0.933 x sqrt(1 / 15 + 1 / 12) = 0.933 x 0.387 = 0.361.

Step 3: Test statistic: t = (5.617 - 6.743) / 0.361 = -1.127 / 0.361 = -3.120, with v = 15 + 12 - 2 = 25 degrees of freedom.

Step 4: Critical values: +/- 2.060. Since t = -3.120 < -2.060, we reject H_0: mu_1 =

mu_2 against H_1: mu_1 not equal to mu_2.

Step 5: The 95% confidence interval for mu_2 - mu_1 is (x-bar_2 - x-bar_1) +/- t_(25; 0.975) x (standard error) = 1.127 +/- 2.060 x 0.361 = 1.127 +/- 0.744.

Answer: The 95% confidence interval is 0.383 < mu_2 - mu_1 < 1.871, which does not include 0, in agreement with the rejection of H_0.

Q. 8 (b) The I.Q.'s of 16 students from one area of a city showed a mean of 107 with a standard deviation of 10, while the I.Q.'s of 14 students from another area of the city showed a mean of 115 with a standard deviation of 8. Is there a significant difference between the I.Q.'s of the two groups at (i) 0.01 and (ii) 0.05 level of significance? (Book answers: (i) Since t_(28; 0.005) = -2.763 < t = -2.395 < 2.763 = t_(28; 0.995), we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2 at alpha = 0.01. (ii) Since t = -2.395 < -2.048 = t_(28; 0.025), we reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2; at alpha = 0.05.)

Step 1: Area 1: n_1 = 16, x-bar_1 = 107, s-hat_1 = 10. Area 2: n_2 = 14, x-bar_2 = 115, s-hat_2 = 8. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2 (two-tailed), assuming normal populations with a common variance.

Step 2: Pooled standard deviation: s_p = sqrt((15 x 100 + 13 x 64) / (16 + 14 - 2)) = sqrt(2332 / 28) = 9.126. Standard error: 9.126 x sqrt(1 / 16 + 1 / 14) = 9.126 x 0.366 = 3.340.

Step 3: Test statistic: t = (107 - 115) / 3.340 = -8 / 3.340 = -2.395, with v = 28 degrees of freedom.

Step 4: (i) At alpha = 0.01, critical values: +/- 2.763. Since -2.763 < -2.395 < 2.763, we do not reject H_0.

Answer: (ii) At alpha = 0.05, critical values: +/- 2.048. Since t = -2.395 < -2.048, we reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2. There is a significant difference at the 5% level but not at the 1% level.

Q. 9 (a) Means of random samples, each of size 10, from two normal populations with the same standard deviation were found to be 16 and 20 respectively. Further, the sample standard deviations were equal to 5 and 7 respectively. Test the hypotheses that the populations have the same mean, using 0.05 level of significance. (Book answer: Since t_(18; 0.025) = -2.101 < t = -1.47 < 2.101 = t_(18; 0.975), we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.)

Step 1: n_1 = n_2 = 10, x-bar_1 = 16, s-hat_1 = 5; x-bar_2 = 20, s-hat_2 = 7. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed), assuming a common variance.

Step 2: Pooled standard deviation: s_p = sqrt((9 x 25 + 9 x 49) / 18) = sqrt(666 / 18) = 6.083. Standard error: 6.083 x sqrt(1 / 10 + 1 / 10) = 6.083 x 0.447 = 2.721.

Step 3: Test statistic: t = (16 - 20) / 2.721 = -4 / 2.721 = -1.47, with v = 18 degrees of freedom.

Step 4: Critical values: +/- 2.101.

Answer: Since -2.101 < -1.47 < 2.101, we do not reject H_0: mu_1 = mu_2. The

populations have the same mean, on this evidence.

Q. 9 (b) A random sample of size n_1 = 10, selected from a normal population has a mean x-bar_1 = 20 and a standard deviation s-hat_1 = 5. A second random sample of size n_2 = 12, selected from a different normal population has a mean x-bar_2 = 24 and a standard deviation s-hat_2 = 6. If mu_1 = 22 and mu_2 = 19 and sigma_1-squared and sigma_2-squared are unknown but approximately equal, test whether there is any reason to doubt that mu_1 - mu_2 = 3. (Book answer: Since t = -2.934 < -2.086 = t_(20; 0.025), we reject H_0: mu_1 - mu_2 = 3 in favour of H_1: mu_1 - mu_2 not equal to 3 at alpha = 0.05.)

Step 1: n_1 = 10, x-bar_1 = 20, s-hat_1 = 5; n_2 = 12, x-bar_2 = 24, s-hat_2 = 6. H_0: mu_1 - mu_2 = 3 against H_1: mu_1 - mu_2 not equal to 3, at alpha = 0.05 (two-tailed), assuming a common variance.

Step 2: Pooled standard deviation: s_p = sqrt((9 x 25 + 11 x 36) / (10 + 12 - 2)) = sqrt(621 / 20) = 5.572. Standard error: 5.572 x sqrt(1 / 10 + 1 / 12) = 5.572 x 0.428 = 2.386.

Step 3: Test statistic: t = ((20 - 24) - 3) / 2.386 = -7 / 2.386 = -2.934, with v = 20 degrees of freedom.

Step 4: Critical values: +/- 2.086.

Answer: Since t = -2.934 < -2.086, we reject H_0: mu_1 - mu_2 = 3 in favour of H_1: mu_1 - mu_2 not equal to 3. Yes, there is reason to doubt that mu_1 - mu_2 = 3.

Q. 10 (a) The following values are obtained by two different samples while sampling

a normal population with mu = 3.0 and sigma = 0.5. Using these data solve the following problems: A: 2.7, 3.9, 2.6, 2.8, 3.2, 3.6, 2.7, 3.0, 3.8, 3.1. B: 2.4, 2.8, 3.0, 3.3, 3.7, 3.3, 3.1, 2.7, 2.4, 3.3. (i) Combine the data into one set and test the hypothesis H_0: mu = 3.0 against H_1: mu not equal to 3.0. (ii) Assuming that sigma is unknown test the hypothesis H_0: mu_A <= mu_B against H_1: mu_A > mu_B. Use alpha = 0.05 in both cases. (Book answers: (i) Since z_0.025 = -1.96 < z = 0.626 < 1.96 = z_0.975, we do not reject H_0: mu = 3 against H_1: mu not equal to 3. (ii) Since t = 0.694 < 1.734 = t_(18; 0.95), we do not reject H_0: mu_A <= mu_B against H_1: mu_A > mu_B.)

Step 1: (i) Combined sample: n = 20, x-bar = 3.07. sigma = 0.5 is known. H_0: mu = 3.0 against H_1: mu not equal to 3.0, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: z = (3.07 - 3.0) / (0.5 / sqrt(20)) = 0.07 / 0.112 = 0.626. Critical values: +/- 1.960. Since -1.960 < 0.626 < 1.960, we do not reject H_0: mu = 3.0.

Step 3: (ii) Sample A: n_A = 10, x-bar_A = 3.14. Sample B: n_B = 10, x-bar_B = 3.00. H_0: mu_A <= mu_B against H_1: mu_A > mu_B, at alpha = 0.05 (right-tailed), assuming a common unknown variance.

Step 4: Pooled standard deviation: s_p = 0.451. Standard error: 0.451 x sqrt(1 / 10 + 1 / 10) = 0.451 x 0.447 = 0.202. Test statistic: t = (3.14 - 3.00) / 0.202 = 0.694, with v = 18 degrees of freedom.

Step 5: Critical value: t_(18; 0.95) = 1.734.

Answer: Since t = 0.694 < 1.734, we do not reject H_0: mu_A <= mu_B.

Q. 10 (b) A group of 12 students are found to have the following I.Q.'s: 112, 109, 125, 113, 116, 131, 112, 123, 108, 113, 132, 128. Is it reasonable to assume that these students have come from a large population whose mean I.Q. is 115? Another group of 10 students are found to have the following I.Q.'s: 117, 110, 106, 109, 116, 119, 107, 106, 105, 108. Can we conclude that both the groups of students have come from the same population? (Book answers: (i) Since t_(11; 0.025) = -2.201 < t = 1.386 < 2.201 = t_(11; 0.975), we do not reject H_0: mu = 115 against H_1: mu not equal to 115 at alpha = 0.05. (ii) Since t = 2.608 > 2.086 = t_(20; 0.975), we reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2 at alpha = 0.05.)

Step 1: (i) Group 1: n_1 = 12, x-bar_1 = 118.5, s-hat_1 = 8.74. H_0: mu = 115 against H_1: mu not equal to 115, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: t = (118.5 - 115) / (8.74 / sqrt(12)) = 3.5 / 2.523 = 1.386, with v = 11 degrees of freedom. Critical values: +/- 2.201. Since -2.201 < 1.386 < 2.201, we do not reject H_0: mu = 115. Yes, it is reasonable to assume the students have come from a population with mean I.Q. 115.

Step 3: (ii) Group 2: n_2 = 10, x-bar_2 = 110.3. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed), assuming a common variance.

Step 4: Pooled standard deviation: s_p = 7.35 (from the two sample variances). Standard error: 7.35 x sqrt(1 / 12 + 1 / 10) = 7.35 x 0.428 = 3.145. Test statistic: t = (118.5 - 110.3) / 3.145 = 8.2 / 3.145 = 2.608, with v = 20 degrees of freedom.

Step 5: Critical value: t_(20; 0.975) = 2.086.

Answer: Since t = 2.608 > 2.086, we reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2. No, we cannot conclude that both groups have come from the same population.

Q. 10 (c) A random sample of 16 values from a normal population gave a mean of 42 inches and a sum of squared deviations from this mean as 135 (inches)-squared. Test the hypothesis that the mean in the population is 43.5 inches. Another random sample of 9 values from another normal population gave a mean of 41.5 inches and a sum of squares of deviations from this mean as 128 (inches)-squared. Test the hypothesis that the mean of first population equals the mean of the second population. (Book answers: Since t_(15; 0.025) = -2.131 < t = -2 < 2.131 = t_(15; 0.975), we do not reject H_0: mu = 43.5 against H_1: mu not equal to 43.5. Since t_(23; 0.025) = -2.069 < t = 0.355 < 2.069 = t_(23; 0.975), we do not reject H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2.)

Step 1: Sample 1: n_1 = 16, x-bar_1 = 42, sum (x - x-bar)-squared = 135, so s-hat_1 = sqrt(135 / 15) = 3.0. H_0: mu = 43.5 against H_1: mu not equal to 43.5, at alpha = 0.05 (two-tailed).

Step 2: Test statistic: t = (42 - 43.5) / (3.0 / sqrt(16)) = -1.5 / 0.75 = -2.0, with v = 15 degrees of freedom. Critical values: +/- 2.131. Since -2.131 < -2.0 < 2.131, we do not reject H_0: mu = 43.5.

Step 3: Sample 2: n_2 = 9, x-bar_2 = 41.5, sum (x - x-bar)-squared = 128. H_0: mu_1 = mu_2 against H_1: mu_1 not equal to mu_2, at alpha = 0.05 (two-tailed), assuming a common variance.

Step 4: Pooled standard deviation: s_p = sqrt((135 + 128) / (16 + 9 - 2)) = sqrt(263

/ 23) = 3.382. Standard error: 3.382 x sqrt(1 / 16 + 1 / 9) = 3.382 x 0.417 = 1.410.

Test statistic: t = (42 - 41.5) / 1.410 = 0.5 / 1.410 = 0.355, with v = 23 degrees of freedom.

Step 5: Critical values: +/- 2.069.

Answer: Since -2.069 < 0.355 < 2.069, we do not reject H_0: mu_1 = mu_2.

Exercise 13.5 - Difference Between Two Population Means, Dependent Samples (book pp. 166-168)

Q. 1 (a) Haemoglobin values were determined on six patients before starting and after three weeks on B_12 Therapy. The following data were obtained: Individual: 1, 2, 3, 4, 5, 6. Haemoglobin (gm) Before Therapy: 12.2, 11.3, 14.7, 11.4, 11.5, 12.7. Haemoglobin (gm) After Therapy: 13.0, 13.4, 16.0, 13.6, 14.0, 13.8. Do the data indicate a significant improvement? (Book answer: Since t = 5.927 > 2.015 = t_(5; 0.95), we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0.)

Step 1: Form the differences D = After - Before: 0.8, 2.1, 1.3, 2.2, 2.5, 1.1. Number of pairs: n = 6.

Step 2: Mean of the differences: d-bar = 10.0 / 6 = 1.667 gm. Standard deviation of the differences: s-hat_D = 0.689 gm.

Step 3: H_0: mu_D <= 0 (no improvement) against H_1: mu_D > 0 (improvement), at alpha = 0.05 (right-tailed).

Step 4: Test statistic: t = (1.667 - 0) / (0.689 / sqrt(6)) = 1.667 / 0.281 = 5.927, with

v = 5 degrees of freedom.

Step 5: Critical value: t_(5; 0.95) = 2.015.

Answer: Since t = 5.927 > 2.015, we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0. Yes, the data indicate a significant improvement.

Q. 1 (b) Eleven school boys were given a test in Drawing. They were given one month's further tuition and a second test of equal difficulty was held at the end of it. Marks in 1st test: 23, 20, 19, 21, 18, 20, 18, 17, 23, 16, 19. Marks in 2nd test: 24, 19, 22, 18, 20, 22, 20, 20, 20, 20, 17. Do the marks give evidence that the students have benefited by the extra coaching? (Book answer: Since t = 0.956 < 1.812 = t_(10; 0.95), so we do not reject H_0: mu_D <= 0 against H_1: mu_D > 0.)

Step 1: Form the differences D = 2nd test - 1st test: 1, -1, 3, -3, 2, 2, 2, 3, -3, 4, -2. Number of pairs: n = 11.

Step 2: Mean of the differences: d-bar = 8 / 11 = 0.727. Standard deviation of the differences: s-hat_D = 2.533.

Step 3: H_0: mu_D <= 0 (no benefit) against H_1: mu_D > 0 (benefit), at alpha = 0.05 (right-tailed).

Step 4: Test statistic: t = (0.727 - 0) / (2.533 / sqrt(11)) = 0.727 / 0.764 = 0.952, with v = 10 degrees of freedom. (The book prints 0.956.)

Step 5: Critical value: t_(10; 0.95) = 1.812.

Answer: Since t = 0.952 < 1.812, we do not reject H_0: mu_D <= 0. The marks do not give evidence that the students have benefited by the extra coaching.

Q. 2 (a) A taxi company is trying to decide whether the use of radial tires instead of regular belted tires improves fuel economy. Twelve cars were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with regular belted tires and driven once again over the test course. The gasoline consumption in km per litre, was recorded as follows: Radial tires: 4.2, 4.7, 6.6, 7.0, 6.7, 4.5, 5.7, 6.0, 7.4, 4.9, 6.1, 5.2. Belted tires: 4.1, 4.9, 6.2, 6.9, 6.8, 4.4, 5.7, 5.8, 6.9, 4.7, 6.0, 4.9. At the 0.025 level of significance, can we conclude, that cars equipped with radial tires give better fuel economy than those equipped with belted tires? Assume the population to be normally distributed. (Book answer: Since t = 2.490 > 2.201 = t_(11; 0.975), we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0.)

Step 1: Form the differences D = Radial - Belted: 0.1, -0.2, 0.4, 0.1, -0.1, 0.1, 0.0, 0.2, 0.5, 0.2, 0.1, 0.3. Number of pairs: n = 12.

Step 2: Mean of the differences: d-bar = 1.7 / 12 = 0.142 km per litre. Standard deviation of the differences: s-hat_D = 0.198.

Step 3: H_0: mu_D <= 0 (no improvement) against H_1: mu_D > 0 (improvement), at alpha = 0.025 (right-tailed).

Step 4: Test statistic: t = (0.142 - 0) / (0.198 / sqrt(12)) = 0.142 / 0.0572 = 2.485, with v = 11 degrees of freedom. (The book prints 2.490.)

Step 5: Critical value: t_(11; 0.975) = 2.201.

Answer: Since t = 2.485 > 2.201, we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0. Yes, cars equipped with radial tires give better fuel economy.

Q. 2 (b) A certain stimulus administered to each of the nine patients resulted in the following increase in blood pressure: 5, 1, 8, 0, 3, 3, 5, -2, 4. Can it be concluded that blood pressure in general is increased by administering the stimulus? (Book answer: Since t = 3.0 > 1.86 = t_(8; 0.95), we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0.)

Step 1: The increases are the differences D: 5, 1, 8, 0, 3, 3, 5, -2, 4. Number of pairs: n = 9.

Step 2: Mean of the differences: d-bar = 27 / 9 = 3.0. Standard deviation of the differences: s-hat_D = 3.0.

Step 3: H_0: mu_D <= 0 (no increase) against H_1: mu_D > 0 (increase), at alpha = 0.05 (right-tailed).

Step 4: Test statistic: t = (3.0 - 0) / (3.0 / sqrt(9)) = 3.0 / 1.0 = 3.0, with v = 8 degrees of freedom.

Step 5: Critical value: t_(8; 0.95) = 1.86.

Answer: Since t = 3.0 > 1.86, we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0. Yes, blood pressure in general is increased by administering the stimulus.

Q. 3 (a) To verify whether a course in Statistics improved performance a similar test was given to 12 participants both before and after the course. The original grades

recorded in alphabetical order of the participants were 44, 40, 61, 52, 32, 44, 70, 41, 67, 72, 53 and 72. After the course the grades were, in the same order, 53, 38, 69, 57, 46, 39, 73, 48, 73, 74, 60 and 78. (i) Was the course useful, as measured by performance on the test? Consider these 12 participants as a sample from a population. (ii) Would the same conclusion be reached if the tests were not considered paired? (use 5% level of significance in both cases). (Book answers: (i) Since t = 3.445 > 1.796 = t_(11; 0.95), we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0. (ii) Since t = 0.863 < 1.717 = t_(22; 0.95), we do not reject H_0: mu_2 - mu_1 <= 0 against H_1: mu_2 - mu_1 > 0.)

Step 1: (i) Paired test. Differences D = After - Before: 9, -2, 8, 5, 14, -5, 3, 7, 6, 2, 7, 6. Number of pairs: n = 12.

Step 2: Mean of the differences: d-bar = 60 / 12 = 5.0. Standard deviation of the differences: s-hat_D = 5.027.

Step 3: H_0: mu_D <= 0 against H_1: mu_D > 0, at alpha = 0.05. Test statistic: t = (5.0 - 0) / (5.027 / sqrt(12)) = 5.0 / 1.451 = 3.445, with v = 11 degrees of freedom. Critical value: t_(11; 0.95) = 1.796. Since 3.445 > 1.796, we reject H_0: mu_D <= 0 in favour of H_1: mu_D > 0. Yes, the course was useful.

Step 4: (ii) Unpaired test. Before: x-bar_1 = 54.0. After: x-bar_2 = 59.0. Pooled standard deviation: s_p = 14.184. Standard error: 14.184 x sqrt(1 / 12 + 1 / 12) = 14.184 x 0.408 = 5.791. Test statistic: t = (59.0 - 54.0) / 5.791 = 0.863, with v = 22 degrees of freedom. Critical value: t_(22; 0.95) = 1.717.

Answer: Since 0.863 < 1.717, we do not reject H_0: mu_2 - mu_1 <= 0. No, the same conclusion would not be reached: pairing removes the variation between

participants and makes the test much more sensitive.

Q. 3 (b) The time required by 10 persons to perform a task in seconds before and after receiving a mild stimulant are given in the accompanying table. Before: 34, 45, 31, 43, 40, 41, 33, 29, 41, 37. After: 29, 42, 32, 29, 36, 42, 26, 28, 38, 33. Test the hypothesis that there is no difference between the mean times in the 'before' and 'after' populations. As an alternative, assumed that the after population will have a lower mean. Use 5% level of significance. (Book answer: Since t = -2.83 < -1.833 = t_(9; 0.05), we reject H_0: mu_D >= 0 in favour of H_1: mu_D < 0.)

Step 1: Form the differences D = After - Before: -5, -3, 1, -14, -4, 1, -7, -1, -3, -4. Number of pairs: n = 10.

Step 2: Mean of the differences: d-bar = -39 / 10 = -3.9 seconds. Standard deviation of the differences: s-hat_D = 4.358 seconds.

Step 3: H_0: mu_D >= 0 (no difference, or after not lower) against H_1: mu_D < 0 (after lower), at alpha = 0.05 (left-tailed).

Step 4: Test statistic: t = (-3.9 - 0) / (4.358 / sqrt(10)) = -3.9 / 1.378 = -2.83, with v = 9 degrees of freedom.

Step 5: Critical value: t_(9; 0.05) = -1.833.

Answer: Since t = -2.83 < -1.833, we reject H_0: mu_D >= 0 in favour of H_1: mu_D < 0. The mean time after the stimulant is lower.

Q. 4 (a) Ten young recruits were put through a physical training programme by the

army. Their weights were recorded before and after the training with the following results: Recruit: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Weight (Before): 127, 126, 162, 170, 143, 205, 168, 175, 197, 136. Weight (After): 135, 200, 160, 182, 147, 200, 172, 186, 193, 141. Using alpha = 0.05, should we conclude that the training programme affects the average weight of young recruits. (Book answer: Since t_(9; 0.025) = -2.262 < t = 1.471 < 2.262 = t_(9; 0.975), we do not reject H_0: mu_D = 0 against H_1: mu_D not equal to 0.)

Step 1: Form the differences D = After - Before: 8, 74, -2, 12, 4, -5, 4, 11, -4, 5. Number of pairs: n = 10.

Step 2: Mean of the differences: d-bar = 107 / 10 = 10.7. Standard deviation of the differences: s-hat_D = 23.003.

Step 3: H_0: mu_D = 0 (no effect) against H_1: mu_D not equal to 0 (effect), at alpha = 0.05 (two-tailed).

Step 4: Test statistic: t = (10.7 - 0) / (23.003 / sqrt(10)) = 10.7 / 7.274 = 1.471, with v = 9 degrees of freedom.

Step 5: Critical values: +/- 2.262.

Answer: Since -2.262 < 1.471 < 2.262, we do not reject H_0: mu_D = 0. No, we should not conclude that the training programme affects the average weight.

Q. 4 (b) The following data give paired yields of two varieties of wheat. Each pair was planted in a different locality. Variety I: 45, 32, 58, 57, 60, 38, 47, 51, 42, 38. Variety II: 47, 34, 60, 59, 63, 44, 49, 53, 46, 41. Test the hypothesis that the mean

yields are equal. (Book answer: Since t = 6.725 > 2.262 = t_(9; 0.975), we reject H_0: mu_D = 0 in favour of H_1: mu_D not equal to 0.)

Step 1: Form the differences D = Variety II - Variety I: 2, 2, 2, 2, 3, 6, 2, 2, 4, 3. Number of pairs: n = 10.

Step 2: Mean of the differences: d-bar = 28 / 10 = 2.8. Standard deviation of the differences: s-hat_D = 1.317.

Step 3: H_0: mu_D = 0 (equal mean yields) against H_1: mu_D not equal to 0, at alpha = 0.05 (two-tailed).

Step 4: Test statistic: t = (2.8 - 0) / (1.317 / sqrt(10)) = 2.8 / 0.416 = 6.725, with v = 9 degrees of freedom.

Step 5: Critical values: +/- 2.262.

Answer: Since t = 6.725 > 2.262, we reject H_0: mu_D = 0 in favour of H_1: mu_D not equal to 0. The mean yields are not equal; variety II yields more.

Q. 5 (a) Twenty college freshmen were divided into 10 pairs, each member of the pair having approximately the same I.Q. One of each pair was selected at random and assigned to a Mathematics section using programmed materials only. The other members of each pair were assigned to a section in which the teacher lectured. At the end of the semester each group was given an examination and the following results were recorded: Pair: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Programmed materials: 76, 70, 85, 58, 91, 75, 82, 64, 79, 88. Lectures: 81, 52, 87, 70, 86, 77, 90, 73, 85, 83. Test the hypothesis that there is no difference between the mean scores in the

'programmed material' and 'lectures' populations. Use 5% level of significance.

(Book answer: Since t_(9; 0.025) = -2.262 < t = -0.571 < 2.262 = t_(9; 0.975), we do not reject H_0: mu_D = 0 against H_1: mu_D not equal to 0.)

Step 1: Form the differences D = Programmed - Lectures: -5, 18, -2, -12, 5, -2, -8, -9, -6, 5. Number of pairs: n = 10.

Step 2: Mean of the differences: d-bar = -16 / 10 = -1.6. Standard deviation of the differences: s-hat_D = 8.859.

Step 3: H_0: mu_D = 0 (no difference) against H_1: mu_D not equal to 0, at alpha = 0.05 (two-tailed).

Step 4: Test statistic: t = (-1.6 - 0) / (8.859 / sqrt(10)) = -1.6 / 2.802 = -0.571, with v = 9 degrees of freedom.

Step 5: Critical values: +/- 2.262.

Answer: Since -2.262 < -0.571 < 2.262, we do not reject H_0: mu_D = 0. There is no difference between the mean scores of the two populations.

Q. 5 (b) It is claimed that a new diet will reduce a person's weight by at least 10 pounds on the average in a period of 2 weeks. The weight of 7 women who followed this diet were recorded before and after a 2-week period. Woman: 1, 2, 3, 4, 5, 6, 7. Weight (Before): 129, 133, 136, 152, 141, 138, 125. Weight (After): 130, 121, 128, 137, 129, 132, 120. Test the manufacturer's claim at a 5% level of significance for the mean difference in weights. Assume the distribution of weights before and after to be approximately normal. (Book answer: Since t = 0.910 < 1.943 = t_(6; 0.95),

we do not reject H_0: mu_D <= -10 against H_1: mu_D > -10.)

Step 1: Form the differences D = After - Before: 1, -12, -8, -15, -12, -6, -5. Number of pairs: n = 7.

Step 2: Mean of the differences: d-bar = -57 / 7 = -8.143 pounds. Standard deviation of the differences: s-hat_D = 5.398 pounds.

Step 3: The claim is a reduction of at least 10 pounds, that is, mu_D <= -10. H_0: mu_D <= -10 against H_1: mu_D > -10, at alpha = 0.05 (right-tailed).

Step 4: Test statistic: t = (-8.143 - (-10)) / (5.398 / sqrt(7)) = 1.857 / 2.040 = 0.910, with v = 6 degrees of freedom.

Step 5: Critical value: t_(6; 0.95) = 1.943.

Answer: Since t = 0.910 < 1.943, we do not reject H_0: mu_D <= -10. The manufacturer's claim is not contradicted; the diet does reduce weight by at least 10 pounds on the average, on this evidence.

Exercise 13.6 - Test of Hypothesis About Difference Between Two Population Proportions (book pp. 175-177)

Q. 1 (a) For each of the following sets of data, test the hypothesis that there is a common proportion pi. (i) Sample I: sample size 150, number of successes 125. Sample II: sample size 200, number of successes 176. H_0: pi_1 >= pi_2 against H_1: pi_1 < pi_2, 5%. (ii) Sample I: sample size 1000, number of successes 542. Sample II: sample size 900, number of successes 427. H_0: pi_1 = pi_2 against H_1:

pi_1 not equal to pi_2, 1%. (iii) Sample I: sample size 100, number of successes 63. Sample II: sample size 120, number of successes 80. H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2, 10%. (iv) Sample I: sample size 80, number of successes 40. Sample II: sample size 60, number of successes 23. H_0: pi_1 <= pi_2 against H_1: pi_1 > pi_2, 5%.

Step 1: Under H_0: pi_1 = pi_2, use the pooled proportion pi-hat = (x_1 + x_2) / (n_1 + n_2) and the test statistic z = (p_1 - p_2) / sqrt(pi-hat (1 - pi-hat) (1 / n_1 + 1 / n_2)), approximately standard normal.

Step 2: (i) p_1 = 125 / 150 = 0.8333, p_2 = 176 / 200 = 0.88. pi-hat = 301 / 350 = 0.86. z = (0.8333 - 0.88) / sqrt(0.86 x 0.14 x (1 / 150 + 1 / 200)) = -0.0467 / 0.0375 = -1.245. Critical value at 5% left-tailed: -1.645. Since -1.245 > -1.645, we do not reject H_0: pi_1 >= pi_2 against H_1: pi_1 < pi_2.

Step 3: (ii) p_1 = 542 / 1000 = 0.542, p_2 = 427 / 900 = 0.4744. pi-hat = 969 / 1900 = 0.51. z = (0.542 - 0.4744) / sqrt(0.51 x 0.49 x (1 / 1000 + 1 / 900)) = 0.0676 / 0.0230 = 2.941. Critical values at 1% two-tailed: +/- 2.576. Since 2.941 > 2.576, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2.

Step 4: (iii) p_1 = 63 / 100 = 0.63, p_2 = 80 / 120 = 0.6667. pi-hat = 143 / 220 = 0.65. z = (0.63 - 0.6667) / sqrt(0.65 x 0.35 x (1 / 100 + 1 / 120)) = -0.0367 / 0.0646 = -0.568. Critical values at 10% two-tailed: +/- 1.645. Since -1.645 < -0.568 < 1.645, we do not reject H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2.

Answer: (iv) p_1 = 40 / 80 = 0.50, p_2 = 23 / 60 = 0.3833. pi-hat = 63 / 140 = 0.45. z = (0.50 - 0.3833) / sqrt(0.45 x 0.55 x (1 / 80 + 1 / 60)) = 0.1167 / 0.0850 = 1.373. Critical value at 5% right-tailed: 1.645. Since 1.373 < 1.645, we do not reject H_0:

pi_1 <= pi_2 against H_1: pi_1 > pi_2.

Q. 1 (b) In a population that have certain minor blood disorder, samples of 100 males and 100 females are taken. It is found that 31 males and 24 females have the blood disorder. Can we conclude at 0.01 level of significance that proportion of men who have blood disorder is greater than proportion of women? (Book answer: Since z = 1.109 < 2.326 = z_0.99, we do not reject H_0: pi_1 <= pi_2 against H_1: pi_1 > pi_2.)

Step 1: Males: n_1 = 100, x_1 = 31, p_1 = 0.31. Females: n_2 = 100, x_2 = 24, p_2 = 0.24. H_0: pi_1 <= pi_2 against H_1: pi_1 > pi_2, at alpha = 0.01 (right-tailed).

Step 2: Pooled proportion: pi-hat = (31 + 24) / 200 = 0.275. Standard error: sqrt(0.275 x 0.725 x (1 / 100 + 1 / 100)) = sqrt(0.003988) = 0.0631.

Step 3: Test statistic: z = (0.31 - 0.24) / 0.0631 = 0.07 / 0.0631 = 1.109.

Step 4: Critical value: z_0.99 = 2.326.

Answer: Since z = 1.109 < 2.326, we do not reject H_0: pi_1 <= pi_2. No, we cannot conclude that the proportion of men with the blood disorder is greater than that of women.

Q. 2 (a) The records of a hospital show that 52 men in a sample of 1000 men versus 23 women in a sample of 1000 women were admitted because of heart disease. Do these data present sufficient evidence to indicate a higher rate of heart disease among men admitted to the hospital? (Book answer: Since z = 3.413 > 1.645 = z_0.95, we reject H_0: pi_1 <= pi_2 in favour of H_1: pi_1 > pi_2.)

Step 1: Men: n_1 = 1000, x_1 = 52, p_1 = 0.052. Women: n_2 = 1000, x_2 = 23, p_2 = 0.023. H_0: pi_1 <= pi_2 against H_1: pi_1 > pi_2, at alpha = 0.05 (right-tailed).

Step 2: Pooled proportion: pi-hat = (52 + 23) / 2000 = 0.0375. Standard error: sqrt(0.0375 x 0.9625 x (1 / 1000 + 1 / 1000)) = sqrt(0.0000722) = 0.0085.

Step 3: Test statistic: z = (0.052 - 0.023) / 0.0085 = 0.029 / 0.0085 = 3.413.

Step 4: Critical value: z_0.95 = 1.645.

Answer: Since z = 3.413 > 1.645, we reject H_0: pi_1 <= pi_2 in favour of H_1: pi_1 > pi_2. Yes, the data present sufficient evidence of a higher rate of heart disease among men admitted.

Q. 2 (b) In a study to estimate the proportion of housewives who own an automatic dryer, it is found that 63 of 100 urban residents have a dryer and 59 of 125 suburban residents own a dryer. Is there a significant difference between the proportions of urban and suburban housewives who own an automatic dryer? (use a 0.04 level of significance). (Book answer: Since z = 2.364 > 2.0537 = z_0.98, so we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2.)

Step 1: Urban: n_1 = 100, x_1 = 63, p_1 = 0.63. Suburban: n_2 = 125, x_2 = 59, p_2 = 0.472. H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2, at alpha = 0.04 (two-tailed).

Step 2: Pooled proportion: pi-hat = (63 + 59) / 225 = 0.5422. Standard error: sqrt(0.5422 x 0.4578 x (1 / 100 + 1 / 125)) = sqrt(0.004468) = 0.0668.

Step 3: Test statistic: z = (0.63 - 0.472) / 0.0668 = 0.158 / 0.0668 = 2.364.

Step 4: Critical values: +/- 2.0537.

Answer: Since z = 2.364 > 2.0537, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2. Yes, there is a significant difference.

Q. 3 (a) A random sample of 150 light bulbs manufactured by a firm A showed 12 defective bulbs while a random sample of 100 light bulbs manufactured by another firm B showed 4 defective bulbs. Is there a significant difference between the proportions defectives of the two firms? (Book answer: Since z_0.025 = -1.960 < z = 1.270 < 1.960 = z_0.975, we do not reject H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2.)

Step 1: Firm A: n_1 = 150, x_1 = 12, p_1 = 0.08. Firm B: n_2 = 100, x_2 = 4, p_2 = 0.04. H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2, at alpha = 0.05 (two-tailed).

Step 2: Pooled proportion: pi-hat = (12 + 4) / 250 = 0.064. Standard error: sqrt(0.064 x 0.936 x (1 / 150 + 1 / 100)) = sqrt(0.000998) = 0.0316.

Step 3: Test statistic: z = (0.08 - 0.04) / 0.0316 = 0.04 / 0.0316 = 1.266. (The book prints 1.270.)

Step 4: Critical values: +/- 1.960.

Answer: Since -1.960 < 1.266 < 1.960, we do not reject H_0: pi_1 = pi_2. There is no significant difference between the proportions defectives of the two firms.

Q. 3 (b) In a random sample of 800 adults from the population of a large city 600 are found to be smokers. In a random sample of 1000 adults from another large city, 700 are smokers. Do the data indicate that the cities are significantly different with respect to the prevalence of smoking among men? (Book answer: Since z = 2.353 > 1.960 = z_0.975, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2.)

Step 1: City 1: n_1 = 800, x_1 = 600, p_1 = 0.75. City 2: n_2 = 1000, x_2 = 700, p_2 = 0.70. H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2, at alpha = 0.05 (two-tailed).

Step 2: Pooled proportion: pi-hat = (600 + 700) / 1800 = 0.7222. Standard error: sqrt(0.7222 x 0.2778 x (1 / 800 + 1 / 1000)) = sqrt(0.000451) = 0.0212.

Step 3: Test statistic: z = (0.75 - 0.70) / 0.0212 = 0.05 / 0.0212 = 2.353.

Step 4: Critical values: +/- 1.960.

Answer: Since z = 2.353 > 1.960, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2. Yes, the cities are significantly different with respect to the prevalence of smoking.

Q. 3 (c) In two large populations, there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in a sample of 1200 and 900 respectively from the two populations? (Book answer: Since z = 2.529 > 1.960 = z_0.975, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2 at alpha = 0.05.)

Step 1: n_1 = 1200, p_1 = 0.30; n_2 = 900, p_2 = 0.25. H_0: pi_1 = pi_2 against

H_1: pi_1 not equal to pi_2, at alpha = 0.05 (two-tailed).

Step 2: Pooled proportion: pi-hat = (1200 x 0.30 + 900 x 0.25) / 2100 = (360 + 225) / 2100 = 0.2786. Standard error: sqrt(0.2786 x 0.7214 x (1 / 1200 + 1 / 900)) = sqrt(0.000390) = 0.0198.

Step 3: Test statistic: z = (0.30 - 0.25) / 0.0198 = 0.05 / 0.0198 = 2.529.

Step 4: Critical values: +/- 1.960.

Answer: Since z = 2.529 > 1.960, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2. No, the difference is not likely to be hidden in samples of these sizes.

Q. 4 (a) A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine been improved? Use 5% level of significance. (Book answer: Since z = -0.104 > -1.645 = z_0.05, we do not reject H_0: pi_2 >= pi_1 against H_1: pi_2 < pi_1.)

Step 1: Before overhaul: n_1 = 500, x_1 = 16, p_1 = 0.032. After overhaul: n_2 = 100, x_2 = 3, p_2 = 0.03. H_0: pi_2 >= pi_1 (no improvement) against H_1: pi_2 < pi_1 (improvement), at alpha = 0.05 (left-tailed).

Step 2: Pooled proportion: pi-hat = (16 + 3) / 600 = 0.0317. Standard error: sqrt(0.0317 x 0.9683 x (1 / 500 + 1 / 100)) = sqrt(0.000368) = 0.0192.

Step 3: Test statistic: z = (0.03 - 0.032) / 0.0192 = -0.002 / 0.0192 = -0.104.

Step 4: Critical value: z_0.05 = -1.645.

Answer: Since z = -0.104 > -1.645, we do not reject H_0: pi_2 >= pi_1. The machine has not been shown to be improved.

Q. 4 (b) In a random sample of 250 persons who skipped breakfast 102 reported that they experienced mid morning fatigue, and in a random sample of 250 persons who ate breakfast, 73 reported that they experienced mid morning fatigue. Use 0.01 level of significance to test the null hypothesis that there is no difference between the corresponding proportions against the alternative hypothesis that the mid morning fatigue is more prevalent among persons who skip breakfast. (Book answer: Since z = 2.719 > 2.326 = z_0.99, we reject H_0: pi_1 <= pi_2 in favour of H_1: pi_1 > pi_2.)

Step 1: Skipped breakfast: n_1 = 250, x_1 = 102, p_1 = 0.408. Ate breakfast: n_2 = 250, x_2 = 73, p_2 = 0.292. H_0: pi_1 <= pi_2 against H_1: pi_1 > pi_2, at alpha = 0.01 (right-tailed).

Step 2: Pooled proportion: pi-hat = (102 + 73) / 500 = 0.35. Standard error: sqrt(0.35 x 0.65 x (1 / 250 + 1 / 250)) = sqrt(0.00182) = 0.0427.

Step 3: Test statistic: z = (0.408 - 0.292) / 0.0427 = 0.116 / 0.0427 = 2.719.

Step 4: Critical value: z_0.99 = 2.326.

Answer: Since z = 2.719 > 2.326, we reject H_0: pi_1 <= pi_2 in favour of H_1: pi_1 > pi_2. Mid morning fatigue is more prevalent among persons who skip breakfast.

Q. 5 (a) A manufacturer of house-dresses sent out advertising material to each of two groups of 1000 women. For one group he enclosed a white return envelope and for the other group, a blue envelope. He received orders from 9% and 12% respectively. Is it quite certain that the blue envelope will help sales. Use alpha = 0.05. (Book answer: Since z = 2.19 > 1.645 = z_0.95, we reject H_0: pi_2 <= pi_1 in favour of H_1: pi_2 > pi_1.)

Step 1: White envelope: n_1 = 1000, p_1 = 0.09. Blue envelope: n_2 = 1000, p_2 = 0.12. H_0: pi_2 <= pi_1 against H_1: pi_2 > pi_1, at alpha = 0.05 (right-tailed).

Step 2: Pooled proportion: pi-hat = (90 + 120) / 2000 = 0.105. Standard error: sqrt(0.105 x 0.895 x (1 / 1000 + 1 / 1000)) = sqrt(0.000188) = 0.0137.

Step 3: Test statistic: z = (0.12 - 0.09) / 0.0137 = 0.03 / 0.0137 = 2.19.

Step 4: Critical value: z_0.95 = 1.645.

Answer: Since z = 2.19 > 1.645, we reject H_0: pi_2 <= pi_1 in favour of H_1: pi_2 > pi_1. Yes, it is quite certain that the blue envelope will help sales.

Q. 5 (b) A random sample of 150 high school students was asked whether they would turn to their father or their mother for help with a homework assignment in Mathematics and another random sample of 150 high school students was asked the same question with regard to a homework assignment in English. Use the result shown in the following table and the 0.01 level of significance to test whether or not there is a difference between the true proportions of high school students who turn to their fathers rather than their mothers for help in these two subjects: Mathematics: Mother 59, Father 91. English: Mother 85, Father 65. (Book answer:

Since z = 3.016 > 2.576 = z_0.995, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2.)

Step 1: Mathematics: n_1 = 150, fathers x_1 = 91, p_1 = 91 / 150 = 0.6067. English: n_2 = 150, fathers x_2 = 65, p_2 = 65 / 150 = 0.4333. H_0: pi_1 = pi_2 against H_1: pi_1 not equal to pi_2, at alpha = 0.01 (two-tailed).

Step 2: Pooled proportion: pi-hat = (91 + 65) / 300 = 0.52. Standard error: sqrt(0.52 x 0.48 x (1 / 150 + 1 / 150)) = sqrt(0.003328) = 0.0577.

Step 3: Test statistic: z = (0.6067 - 0.4333) / 0.0577 = 0.1733 / 0.0577 = 3.005. (The book prints 3.016.)

Step 4: Critical values: +/- 2.576.

Answer: Since z = 3.005 > 2.576, we reject H_0: pi_1 = pi_2 in favour of H_1: pi_1 not equal to pi_2. Yes, there is a difference between the true proportions.

Exercise 13.7 — Objective Questions, Q. 1: Fill in the blanks (book pp. 177-178)

Q. 1 (i) A statistical hypothesis is an assertion about the distribution of one or more random variables.

Q. 1 (ii) A statistical hypothesis testing is a procedure to determine whether or not an assumption about some parameter of population is supported by an observed random sample.

Q. 1 (iii) A null hypothesis is that hypothesis which is tested for possible rejection under the assumption that it is true.

Q. 1 (iv) An alternative hypothesis is that hypothesis which we are willing to accept when the null hypothesis is rejected.

Q. 1 (v) The null hypothesis always contains some form of an equality sign.

Q. 1 (vi) The alternative hypothesis never contains the sign of equality and is always in an inequality form.

Q. 1 (vii) A statistic on the basis of which a decision is made about the hypotheses of interest is called test statistic

Q. 1 (viii) A rejection region specifies a set of values of the test statistic for which the H0 is rejected.

Q. 1 (ix) An accepting region specifies a set of values of the test statistic for which the H0 is not rejected.

Q. 1 (x) The values of the test statistic which separate the rejection region from acceptance region are called critical values.

Q. 1 (xi) If the critical region is located equally in both tails of the sampling distribution of test statistic, the test is called two-tailed test.

Q. 1 (xii) If the critical region is located in only one tail of the sampling distribution of test statistic, the test is called one-tailed test.

Exercise 13.7 — Objective Questions, Q. 2: Fill in the blanks (book pp. 178-179)

Q. 2 (i) A value of the test statistic is said to be statistically insignificant if it falls in the acceptance region.

Q. 2 (ii) A value of the test-statistic is said to be statistically significant if it falls in the critical region.

Q. 2 (iii) If the null hypothesis is false, we may accept it leading to a wrong decision.

Q. 2 (iv) If the null hypothesis is false, we may reject it leading to a correct

decision.

Q. 2 (v) A Type-I error is made by rejecting H0 if H0 is actually true.

Q. 2 (vi) A Type-II error is made by accepting H0 if H1 is actually true.

Q. 2 (vii) The level of significance of a test is the maximum probability with which we are willing to a risk of Type-I error.

Q. 2 (viii) The level of confidence is the probability of accepting a true null hypothesis.

Q. 2 (ix) The degrees of freedom is the number of independent or freely chosen variables.

Exercise 13.7 — Objective Questions, Q. 3: True or False (book p. 179)

Q. 3 (i) The types of statistical inferences are estimation of parameters and testing of hypotheses.

True

Q. 3 (ii) A null hypothesis is rejected when a test statistic has a value that is not consistent with the null hypothesis.

True

Q. 3 (iii) The probability of accepting the true null hypothesis is called the level of significance.

False

Q. 3 (iv) The probability of rejecting the true null hypothesis is called the level of confidence.

False

Q. 3 (v) The probability of accepting the true null hypothesis is called the level of confidence.

True

Q. 3 (vi) An assumption made about the population parameter which may or may

not be true is called statistical hypothesis.

True

Q. 3 (vii) The null hypothesis and alternative hypothesis are complementary to each other.

True

Q. 3 (viii) The null hypothesis H0 always contain some form of an equality sign.

True

Q. 3 (ix) The alternative H1 never contains the sign of equality.

True

Exercise 13.7 — Objective Questions, Q. 4: True or False, remaining items (book pp. 179-180; 5 items already in the earlier block above)

Q. 4 (i) The level of significance is the probability of accepting a null hypothesis when it is true.

False

Q. 4 (ii) A null hypothesis is rejected if the value of test-statistic is consistent with the H0.

False

Q. 4 (iii) The Type-I error is considered more serious than a Type-II error.

True

Q. 4 (iv) A value of the test-statistic is said to be statistically insignificant if it falls in the rejection region.

False

Q. 4 (viii) The degrees of freedom is the number of dependent variables.

False

Q. 4 (x) The standardized normal distribution has smaller dispersion than student's t-distribution.

True

Chapter 14: Simple Linear Regression and Correlation

Exercise — Short Questions

What is the correlation coefficient r? What values can it take?

The correlation coefficient r measures the strength of the linear relationship between two variables.

It is a pure, unitless number.

It always lies between -1 and 1.

What is the relation between the correlation coefficient and the regression coefficients?

The correlation coefficient r is the geometric mean of the two regression coefficients.

The two regression coefficients and r all have the same sign.

If r = 1, the correlation is perfectly positive; if r = -1, it is perfectly negative.

Exercise — Fill in the blanks and True or False (book p. 214)

The correlation coefficient is independent of the change in origin and unit of measurement.

The correlation coefficient r is the geometric mean of the two regression coefficients.

The strength of covariability between two random variables is called correlation.

True

The correlation coefficient r is a pure number which is unitless.

True

The correlation coefficient r is not affected by change in origin.

True

Exercise 14.1 - Simple Linear Regression (book pp. 195-198)

Q. 1 (a) What is a scatter diagram? Describe its role in the theory of regression.

Step 1: A scatter diagram is a set of points in a rectangular co-ordinate system (with x measured horizontally and y measured vertically), where each point represents an observed pair of values (x_i, y_i).

Step 2: It is the first step in the investigation of the relationship between two variables: the data are plotted on a graph to get a scatter diagram.

Answer: Its role in regression is to show whether the relationship is linear or curvilinear, direct or inverse, strong or weak, and to help in choosing the appropriate type of model (for example, it would be useless to try to fit a straight line to data whose relationship was clearly curvilinear). It also gives a visual indication of the nature and strength of the relationship and aids in choosing the one that gives the best fit.

Q. 1 (b) Explain what is meant by (i) regression, (ii) regressand, (iii) regressor.

Step 1: (i) Regression: the dependence of the expected value of the dependent random variable for a given value of the independent non-random variable; regression analysis provides a method of estimating an average relationship (often linear) between two or more variables, which allows the investigator to explain and predict the value of one variable from given values of other variables.

Step 2: (ii) Regressand: the variable whose resulting value depends upon the

selected value of the independent variable; it is also called the response variable or predictand variable or dependent variable or explained variable, and is usually denoted by Y.

Answer: (iii) Regressor: the variable that forms the basis of estimation or prediction; it is also called the predictor variable or independent variable or controlled variable or explanatory variable, and is usually denoted by x.

Q. 1 (c) Explain what is meant by (i) simple linear regression, (ii) simple linear regression coefficient.

Step 1: (i) Simple linear regression: if the simple regression describes the dependence of the expected value of the dependent random variable Y as a linear function of the independent non-random variable x, then the regression is called simple linear regression; it is given by mu_(Y|x) = alpha + beta x, where alpha is the intercept of the line along the y-axis and beta is its slope.

Answer: (ii) Simple linear regression coefficient: the slope beta of the regression line; it is the relative change in the expected value of the dependent random variable with respect to a unit increase in the independent non-random variable, and it remains constant at each value of x. It is measured by tan theta, where theta is the angle made by the line with the positive side of the x-axis.

Q. 2 (a) The following measurements of the specific heat of a certain chemical were made in order to investigate the variation in specific heat with temperature. Temperature (degrees C) x_i: 0, 10, 20, 30, 40. Specific heat y_i: 0.51, 0.55, 0.57, 0.59, 0.63. Plot the points on a scatter diagram and verify that the relationship is approximately linear. Estimate the regression line of specific heat on temperature,

and hence estimate the value of the specific heat when the temperature is 25 degrees C. (Book answers: y-hat = 0.514 + 0.0028 x; y-hat = 0.584.)

Step 1: The plotted points (0, 0.51), (10, 0.55), (20, 0.57), (30, 0.59), (40, 0.63) lie approximately on a straight line, so the relationship is approximately linear.

Step 2: n = 5, sum x_i = 100, sum y_i = 2.85, sum x_i y_i = 59.8, sum x_i-squared = 3000.

Step 3: Slope: b = (n sum x_i y_i - (sum x_i)(sum y_i)) / (n sum x_i-squared - (sum x_i)-squared) = (5 x 59.8 - 100 x 2.85) / (5 x 3000 - 10000) = (299 - 285) / 5000 = 14 / 5000 = 0.0028.

Step 4: Intercept: a = (sum y_i - b sum x_i) / n = (2.85 - 0.0028 x 100) / 5 = (2.85 - 0.28) / 5 = 0.514. So the estimated regression line of specific heat on temperature is y-hat = 0.514 + 0.0028 x.

Step 5: For x = 25 degrees C, y-hat = 0.514 + 0.0028 x 25 = 0.514 + 0.070 = 0.584.

Answer: The estimated specific heat at 25 degrees C is 0.584.

Q. 2 (b) Determine the estimated regression equation y-hat = a + b x in each of the following cases: (i) n = 10, sum x_i = 20, sum y_i = 260, sum x_i y_i = 3490, sum x_i-squared = 3144. (ii) n = 100, x-bar = 125, y-bar = 80, sum x_i y_i = 1007425, sum x_i-squared = 1585000. (iii) x-bar = 52, y-bar = 237, sum (x_i - x-bar)-squared = 2800, sum (x_i - x-bar)(y_i - y-bar) = 9871. (iv) n = 8, x-bar = 7, y-bar = 5, sum x_i y_i = 364, sum (x_i - x-bar)-squared = 132. (Book answers: y-hat = 24.0864 +

0.9568 x; y-hat = 38.75 + 0.33 x; y-hat = 53.70 + 3.525 x; y-hat = 0.5459 + 0.6363 x.)

Step 1: (i) b = (10 x 3490 - 20 x 260) / (10 x 3144 - 20-squared) = (34900 - 5200) / (31440 - 400) = 29700 / 31040 = 0.9568. a = (260 - 0.9568 x 20) / 10 = (260 - 19.136) / 10 = 24.0864. So y-hat = 24.0864 + 0.9568 x.

Step 2: (ii) b = (sum x_i y_i - n x-bar y-bar) / (sum x_i-squared - n x-bar-squared) = (1007425 - 100 x 125 x 80) / (1585000 - 100 x 125-squared) = (1007425 - 1000000) / (1585000 - 1562500) = 7425 / 22500 = 0.33. a = y-bar - b x-bar = 80 - 0.33 x 125 = 80 - 41.25 = 38.75. So y-hat = 38.75 + 0.33 x.

Step 3: (iii) b = sum (x_i - x-bar)(y_i - y-bar) / sum (x_i - x-bar)-squared = 9871 / 2800 = 3.5254, taken as 3.525. a = y-bar - b x-bar = 237 - 3.525 x 52 = 237 - 183.30 = 53.70. So y-hat = 53.70 + 3.525 x.

Answer: (iv) b = (sum x_i y_i - n x-bar y-bar) / sum (x_i - x-bar)-squared = (364 - 8 x 7 x 5) / 132 = (364 - 280) / 132 = 84 / 132 = 0.6363. a = y-bar - b x-bar = 5 - 0.6363 x 7 = 5 - 4.454 = 0.5459. So y-hat = 0.5459 + 0.6363 x.

Q. 3 (a) Estimate the regression line of Y on x for the following data. x_i: 25, 30, 35, 40, 45, 50. y_i: 78, 70, 65, 58, 48, 42. Is it possible from the equation you have just found (i) an estimate for the value of x when y = 54? (ii) an estimate for the value of y, when x = 37? In each case, if the answer is 'Yes', calculate the estimate. If the answer is 'No', say why not. (Book answers: y-hat = 114.4 - 1.45 x; (i) No, x is controlled; (ii) 61.)

Step 1: n = 6, sum x_i = 225, sum y_i = 361, sum x_i y_i = 12905, sum x_i-squared = 8875.

Step 2: Slope: b = (6 x 12905 - 225 x 361) / (6 x 8875 - 225-squared) = (77430 - 81225) / (53250 - 50625) = -3795 / 2625 = -1.4457, taken as -1.45.

Step 3: Intercept: a = (361 - (-1.4457) x 225) / 6 = (361 + 325.28) / 6 = 114.38, given in the book as 114.4. So the estimated regression line of Y on x is y-hat = 114.4 - 1.45 x.

Step 4: (i) No. x is the controlled (independent) variable; the regression equation predicts values of the dependent variable based on values of the independent variable, and it cannot be inverted to estimate x from y. The regression of Y on x must not be used to estimate x.

Answer: (ii) Yes. For x = 37, y-hat = 114.4 - 1.45 x 37 = 114.4 - 53.65 = 60.75, given in the book as 61.

Q. 3 (b) From n pairs of values (x_i, y_i), i = 1, 2, ..., n the following quantities are calculated: n = 20, sum x_i = 400, sum y_i = 220, sum x_i-squared = 8800, sum y_i-squared = 2620, sum x_i y_i = 4300. Find the linear regression equation of y on x and x on y. Which would be the more useful if: (i) x is the age (in years) and y is the reaction time (in milliseconds) of 20 people; (ii) x is the cost (in ,000 Rs.) and y the floor-space (in 100 ft-squared) of 20 buildings. (Book answers: y-hat = 13.5 - 0.125 x; x-hat = 25.5 - 0.5 y; (i) y on x; (ii) x on y.)

Step 1: Slope of Y on x: b_yx = (20 x 4300 - 400 x 220) / (20 x 8800 - 400-squared) = (86000 - 88000) / (176000 - 160000) = -2000 / 16000 = -0.125. Intercept: a_yx = (220 - (-0.125) x 400) / 20 = (220 + 50) / 20 = 13.5. So y-hat = 13.5 - 0.125 x.

Step 2: Slope of X on y: b_xy = (20 x 4300 - 400 x 220) / (20 x 2620 - 220-squared)

= -2000 / (52400 - 48400) = -2000 / 4000 = -0.5. Intercept: a_xy = (400 - (-0.5) x 220) / 20 = (400 + 110) / 20 = 25.5. So x-hat = 25.5 - 0.5 y.

Step 3: (i) x is age and y is reaction time: age is the natural predictor, so the regression of y on x is more useful (predict reaction time from age).

Answer: (ii) x is cost and y is floor-space: floor-space is the natural predictor of cost, so the regression of x on y is more useful (predict cost from floor-space).

Q. 4 (a) The following table shows the ages x and systolic blood pressures Y of 12 women. Age (years) x_i: 56, 42, 72, 36, 63, 47, 55, 49, 38, 42, 68, 60. Blood pressure y_i: 147, 125, 160, 118, 149, 128, 150, 145, 115, 140, 152, 155. Assuming that a linear regression model Y_i = alpha + beta x_i + epsilon_i is appropriate, estimate the regression line of blood pressure on age. Estimate the expected blood pressure of a woman whose age is 45 years. What is the change in blood pressure for a unit change in age? (Book answers: y-hat = 80.78 + 1.138 x; 132; 1.138.)

Step 1: n = 12, sum x_i = 628, sum y_i = 1684, sum x_i y_i = 89894, sum x_i-squared = 34416.

Step 2: Slope: b = (12 x 89894 - 628 x 1684) / (12 x 34416 - 628-squared) = (1078728 - 1057552) / (412992 - 394384) = 21176 / 18608 = 1.138.

Step 3: Intercept: a = (1684 - 1.138 x 628) / 12 = (1684 - 714.66) / 12 = 80.78. So the estimated regression line is y-hat = 80.78 + 1.138 x.

Step 4: For age 45 years: y-hat = 80.78 + 1.138 x 45 = 80.78 + 51.21 = 131.99, given in the book as 132.

Answer: The change in blood pressure for a unit change in age is the slope, 1.138 (mm Hg per year).

Q. 4 (b) Suppose that four randomly chosen plots where treated with various levels of fertilizer, resulting in the following yields of corn. Fertilizer (kg/Acre) x_i: 100, 200, 400, 500. Production (Bushels/Acre) y_i: 70, 70, 80, 100. (i) Estimate the linear regression mu_(Y|x) = alpha + beta x of production Y on fertilizer x. (ii) Estimate the yield when no fertilizer is applied. (iii) Estimate the yield when the average amount of fertilizer is applied. (iv) Estimate how much yield is increased for every kilogram of fertilizer applied. (Book answers: y-hat = 59 + 0.070 x; (ii) when x = 0, y-hat = 59; (iii) when x = x-bar = 300, y-hat = 80; (iv) 0.070 bushels per kg of fertilizer.)

Step 1: n = 4, sum x_i = 1200, sum y_i = 320, sum x_i y_i = 103000, sum x_i-squared = 460000.

Step 2: Slope: b = (4 x 103000 - 1200 x 320) / (4 x 460000 - 1200-squared) = (412000 - 384000) / (1840000 - 1440000) = 28000 / 400000 = 0.070.

Step 3: Intercept: a = (320 - 0.070 x 1200) / 4 = (320 - 84) / 4 = 59. So (i) the estimated regression line is y-hat = 59 + 0.070 x.

Step 4: (ii) When no fertilizer is applied, x = 0, so y-hat = 59 bushels per acre.

Step 5: (iii) The average amount of fertilizer is x-bar = 1200 / 4 = 300 kg per acre, so y-hat = 59 + 0.070 x 300 = 59 + 21 = 80 bushels per acre.

Answer: (iv) The yield is increased by the slope, 0.070 bushels per acre, for every kilogram of fertilizer applied.

Q. 5 (a) Describe the properties of the least squares regression line.

Step 1: The sum of the residuals is zero: sum e_i = 0, where e_i = y_i - y-hat_i.

Step 2: The sum of the observed values y_i equals the sum of the fitted values y-hat_i, so the mean of the fitted values equals the mean of the observed values: y-bar = (sum y_i) / n = (sum y-hat_i) / n.

Step 3: The sum of the deviations of the fitted values from their mean is zero: sum (y-hat_i - y-bar) = 0.

Step 4: The sum of the squares of the residuals sum e_i-squared is minimum: sum e_i-squared = sum y_i-squared - a sum y_i - b sum x_i y_i.

Answer: The regression line always passes through the point of means (x-bar, y-bar), the centre of gravity of the observed data; that is, whenever x_i = x-bar, we have y-hat_i = y-bar.

Q. 5 (b) Determine the regression line and estimate the weight of a student whose height is 68 inches. Height (inches) x_i: 72, 66, 67, 69, 74, 61, 66, 62, 70, 63. Weight (pounds) y_i: 178, 141, 158, 165, 180, 133, 159, 140, 160, 136. Find also the estimated values for given values of height. Show that the sum of the estimated values is equal to the sum of the observed values of weight. Find the deviations e_i = y_i - y-hat_i. Show that these deviations add to zero. (Book answers: y-hat = -94.4 + 3.72 x; 158.76.)

Step 1: n = 10, sum x_i = 670, sum y_i = 1550, sum x_i y_i = 104468, sum x_i-squared = 45056.

Step 2: Slope: b = (10 x 104468 - 670 x 1550) / (10 x 45056 - 670-squared) = (1044680 - 1038500) / (450560 - 448900) = 6180 / 1660 = 3.7229, taken as 3.72.

Step 3: Intercept: a = (1550 - 3.7229 x 670) / 10 = (1550 - 2494.34) / 10 = -94.43, given in the book as -94.4. So the regression line is y-hat = -94.4 + 3.72 x.

Step 4: For height 68 inches: y-hat = -94.43 + 3.7229 x 68 = -94.43 + 253.16 = 158.72, given in the book as 158.76.

Step 5: The fitted values y-hat_i are computed for each height; their sum equals the sum of the observed weights, 1550, because the least squares line passes through (x-bar, y-bar) = (67, 155).

Answer: The deviations e_i = y_i - y-hat_i add to zero, since sum e_i = sum y_i - sum y-hat_i = 1550 - 1550 = 0, which is the first property of the least squares line.

Q. 6 (a) Four identical money boxes contain different numbers of a particular type of coin and no coin of other types. The information on the combined weights, is given below. Number of coins in box x_i: 10, 20, 30, 40. Combined weight of coins and box y_i: 312, 509, 682, 865. Estimate the regression line of Y on x. Estimate from your regression line, (i) the weight of an empty box, (ii) the mean weight of a single coin. State the co-ordinates of one point through which the line of regression of Y upon x must pass. (Book answers: y-hat = 134 + 18.32 x; (i) 134, (ii) 18.32; (25, 592).)

Step 1: n = 4, sum x_i = 100, sum y_i = 2368, sum x_i y_i = 68360, sum x_i-squared = 3000.

Step 2: Slope: b = (4 x 68360 - 100 x 2368) / (4 x 3000 - 100-squared) = (273440 -

236800) / (12000 - 10000) = 36640 / 2000 = 18.32.

Step 3: Intercept: a = (2368 - 18.32 x 100) / 4 = (2368 - 1832) / 4 = 134. So the regression line is y-hat = 134 + 18.32 x.

Step 4: (i) The weight of an empty box is the value of y-hat when x = 0, that is, the intercept a = 134.

Step 5: (ii) The mean weight of a single coin is the increase in weight per coin, that is, the slope b = 18.32.

Answer: The line of regression of Y upon x must pass through the point of means (x-bar, y-bar) = (25, 592), since x-bar = 100 / 4 = 25 and y-bar = 2368 / 4 = 592.

Q. 6 (b) Fifteen boys took two examination papers in the same subject and their marks as percentages were as follows, where each boy's marks are in the same column. Paper I x_i: 65, 73, 42, 52, 84, 60, 70, 79, 60, 83, 57, 77, 54, 66, 89. Paper II y_i: 78, 88, 60, 73, 92, 77, 84, 89, 70, 89, 73, 88, 70, 85, 89. Calculate the equation of the line of regression of Y on x. Two boys were each absent from one paper. One scored 63 on paper I, the other scored 81 on paper II. In which case can you use your regression line to estimate the mark that the boy should be allocated for the paper he did not take, and what is that mark? (Book answers: y-hat = 35.53 + 0.665 x, 63 on I implies 78 on II.)

Step 1: n = 15, sum x_i = 1011, sum y_i = 1205, sum x_i y_i = 82904, sum x_i-squared = 70679.

Step 2: Slope: b = (15 x 82904 - 1011 x 1205) / (15 x 70679 - 1011-squared) =

(1243560 - 1218255) / (1060185 - 1022121) = 25305 / 38064 = 0.6648, taken as 0.665.

Step 3: Intercept: a = (1205 - 0.6648 x 1011) / 15 = (1205 - 672.11) / 15 = 35.53. So the regression line of Y (paper II) on x (paper I) is y-hat = 35.53 + 0.665 x.

Step 4: The boy who scored 63 on paper I did not take paper II: his paper II mark can be estimated from the regression line, since paper I is the independent variable. y-hat = 35.53 + 0.665 x 63 = 35.53 + 41.90 = 77.43, given in the book as 78.

Answer: The boy who scored 81 on paper II cannot have his paper I mark estimated from this line, because the regression of Y on x must not be inverted; a separate regression of X on Y would be needed. So only the first boy's mark can be estimated, as about 77 (the book prints 78).

Q. 7 (a) The following data shows the son's height and father's height. Father's height (inches) x_i: 59, 61, 63, 65, 67, 69, 71, 73, 75. Son's height (inches) y_i: 64, 66, 67, 67, 68, 69, 70, 72, 72. Estimate the regression line mu_(Y|x) = alpha + beta x of son's height on father's height using u_i = (x_i - 67) / 2 and v_i = y_i - 68. Predict the mean height of sons whose fathers are 70 inches in height. (Book answers: y-hat = 35.95 + 0.4833 x; 69.78.)

Step 1: Code the variables: u_i = (x_i - 67) / 2 gives -4, -3, -2, -1, 0, 1, 2, 3, 4; v_i = y_i - 68 gives -4, -2, -1, -1, 0, 1, 2, 4, 4.

Step 2: For the coded variables, sum u_i = 0, sum v_i = 3, sum u_i v_i = 58, sum u_i-squared = 60. Slope in coded units: b_vu = 58 / 60 = 0.9667; intercept: a_vu = 3 / 9 = 0.333. So v-hat = 0.333 + 0.9667 u.

Step 3: Transform back: y - 68 = 0.333 + 0.9667 x (x - 67) / 2 = 0.333 + 0.4833 (x - 67) = 0.4833 x - 32.05. So y-hat = 35.95 + 0.4833 x, as in the book.

Step 4: For fathers 70 inches in height: y-hat = 35.95 + 0.4833 x 70 = 35.95 + 33.83 = 69.78 inches.

Answer: The predicted mean height of sons whose fathers are 70 inches in height is 69.78 inches.

Q. 7 (b) For 9 observations of supply X and price Y the following data was obtained: sum (x_i - 90) = -25, sum (x_i - 90)-squared = 301, sum (y_i - 127) = 12, sum (y_i - 127)-squared = 1006, sum (x_i - 90)(y_i - 127) = -469. Obtain the estimated line of regression of X on Y and estimate the supply when the price is Rs. 125. (Book answers: x-hat = 143.69 - 0.44 y; 88.69.)

Step 1: n = 9. x-bar = 90 + (-25) / 9 = 90 - 2.778 = 87.222. y-bar = 127 + 12 / 9 = 127 + 1.333 = 128.333.

Step 2: Sum (x_i - x-bar)(y_i - y-bar) = sum (x_i - 90)(y_i - 127) - (sum (x_i - 90))(sum (y_i - 127)) / n = -469 - (-25 x 12) / 9 = -469 + 33.333 = -435.667.

Step 3: Sum (y_i - y-bar)-squared = sum (y_i - 127)-squared - (sum (y_i - 127))-squared / n = 1006 - 144 / 9 = 1006 - 16 = 990.

Step 4: Slope of X on Y: b_xy = -435.667 / 990 = -0.44. Intercept: a_xy = x-bar - b_xy y-bar = 87.222 - (-0.44) x 128.333 = 87.222 + 56.467 = 143.69. So x-hat = 143.69 - 0.44 y.

Answer: When the price is Rs. 125, x-hat = 143.69 - 0.44 x 125 = 143.69 - 55 = 88.69. The estimated supply is 88.69.

Q. 7 (c) Number of revolutions x (per minute) and power y (hp) of a diesel engine are: x_i: 400, 500, 600, 700, 800. y_i: 580, 1030, 1420, 1880, 2310. Determine the regression line of the y-values on the x-values of the sample using x_i = 100 u_i + 600 and y_i = 10 v_i + 1400 estimate y when x = 750. (Book answers: y-hat = -1142 + 4.31 x; 2090.5.)

Step 1: Code the variables: u_i = (x_i - 600) / 100 gives -2, -1, 0, 1, 2; v_i = (y_i - 1400) / 10 gives -82, -37, 2, 48, 91.

Step 2: For the coded variables, sum u_i = 0, sum v_i = 22, sum u_i v_i = 431, sum u_i-squared = 10. Slope in coded units: b_vu = 431 / 10 = 43.1; intercept: a_vu = 22 / 5 = 4.4. So v-hat = 4.4 + 43.1 u.

Step 3: Transform back: (y - 1400) / 10 = 4.4 + 43.1 x (x - 600) / 100 = 4.4 + 0.431 (x - 600). So y = 1400 + 44 + 0.431 x 600 - 0.431 x 600 + 4.31 x - 258.6... working directly: y = 10 x 4.4 + 10 x 0.431 (x - 600) + 1400 = 44 + 4.31 x - 2586 + 1400 = -1142 + 4.31 x, as in the book.

Step 4: For x = 750 revolutions per minute: y-hat = -1142 + 4.31 x 750 = -1142 + 3232.5 = 2090.5 hp.

Answer: The estimated power is 2090.5 hp.

Q. 8 (a) Fit a straight line taking x as independent variable: 3 x_i + 2: 5, 8, 11, 14, 17, 20, 23, 26. 3 y_i - 2: 7, 10, 16, 16, 25, 28, 28, 34. Also estimate y for x = 5/3.

(Book answers: y-hat = 1.714 + 1.286 x; 3.86.)

Step 1: Recover the original values: x_i = (3 x_i + 2 - 2) / 3 gives 1, 2, 3, 4, 5, 6, 7, 8; y_i = (3 y_i - 2 + 2) / 3 gives 3, 4, 6, 6, 9, 10, 10, 12.

Step 2: n = 8, x-bar = 4.5, y-bar = 7.5, sum (x_i - x-bar)(y_i - y-bar) = 54, sum (x_i - x-bar)-squared = 42.

Step 3: Slope: b = 54 / 42 = 1.2857, taken as 1.286. Intercept: a = 7.5 - 1.2857 x 4.5 = 7.5 - 5.786 = 1.714. So y-hat = 1.714 + 1.286 x.

Answer: For x = 5/3, y-hat = 1.714 + 1.286 x (5/3) = 1.714 + 2.143 = 3.857, given in the book as 3.86.

Q. 8 (b) Fit a least squares line to following data taking (i) Y as dependent variable (ii) X as dependent variable. x_i: 1, 3, 4, 6, 8, 9, 11, 14. y_i: 1, 2, 4, 4, 5, 7, 8, 9. Show that the two least squares lines obtained intersect at the point (x-bar, y-bar). estimate the mean value of y when x = 7. Estimate the mean value of x when y = 6. (Book answers: y-hat = 0.5455 + 0.6364 x, y-hat = 5 for x = 7; x-hat = -0.5 + 1.5 y, x-hat = 8.5 for y = 6.)

Step 1: n = 8, sum x_i = 56, sum y_i = 40, sum x_i y_i = 364, sum x_i-squared = 524, sum y_i-squared = 256. x-bar = 7, y-bar = 5.

Step 2: (i) Slope of Y on x: b_yx = (8 x 364 - 56 x 40) / (8 x 524 - 56-squared) = (2912 - 2240) / (4192 - 3136) = 672 / 1056 = 0.6364.

Step 3: Intercept: a_yx = (40 - 0.6364 x 56) / 8 = (40 - 35.636) / 8 = 0.5455. So y-hat = 0.5455 + 0.6364 x.

Step 4: (ii) Slope of X on y: b_xy = (8 x 364 - 56 x 40) / (8 x 256 - 40-squared) = 672 / (2048 - 1600) = 672 / 448 = 1.5.

Step 5: Intercept: a_xy = (56 - 1.5 x 40) / 8 = (56 - 60) / 8 = -0.5. So x-hat = -0.5 + 1.5 y.

Step 6: Both lines pass through (x-bar, y-bar) = (7, 5): for the Y-on-x line, 0.5455 + 0.6364 x 7 = 0.5455 + 4.4548 = 5.0; for the X-on-y line, -0.5 + 1.5 x 5 = 7.0. So they intersect at (7, 5).

Answer: The mean value of y when x = 7 is y-hat = 5, and the mean value of x when y = 6 is x-hat = -0.5 + 1.5 x 6 = 8.5.

Q. 9 (a) A random sample of 5 pairs of observations (x_i, y_i) is given below: x_i: 3, 2, 5, 1, 4. y_i: 13, 9, 27, 8, 18. Determine the least squares linear regression y-hat = a_yx + b_yx x and estimate y for x = 6. Also find the least squares linear regression x-hat = a_xy + b_xy y and use this to find that value of y for which x-hat = 6. Account for the difference. (Book answers: y-hat = 0.9 + 4.7 x, y-hat = 29.1 for x = 6; x-hat = 0.09 + 0.194 y, y = 30.46 for x-hat = 6 which is a useless estimate, because the regression analysis does not permit the inverse use of the least squares line.)

Step 1: n = 5, sum x_i = 15, sum y_i = 75, sum x_i y_i = 272, sum x_i-squared = 55, sum y_i-squared = 1367. x-bar = 3, y-bar = 15.

Step 2: Slope of Y on x: b_yx = (5 x 272 - 15 x 75) / (5 x 55 - 15-squared) = (1360 - 1125) / (275 - 225) = 235 / 50 = 4.7.

Step 3: Intercept: a_yx = (75 - 4.7 x 15) / 5 = (75 - 70.5) / 5 = 0.9. So y-hat = 0.9 + 4.7 x.

Step 4: For x = 6: y-hat = 0.9 + 4.7 x 6 = 0.9 + 28.2 = 29.1.

Step 5: Slope of X on y: b_xy = (5 x 272 - 15 x 75) / (5 x 1367 - 75-squared) = 235 / (6835 - 5625) = 235 / 1210 = 0.1942, taken as 0.194.

Step 6: Intercept: a_xy = (15 - 0.194 x 75) / 5 = (15 - 14.55) / 5 = 0.09. So x-hat = 0.09 + 0.194 y.

Step 7: Setting x-hat = 6 gives 6 = 0.09 + 0.194 y, so y = (6 - 0.09) / 0.194 = 5.91 / 0.194 = 30.46.

Answer: The two estimates differ (29.1 against 30.46) because the two regression lines are different lines; the X-on-Y line must not be used inversely to estimate y from x, and the value 30.46 obtained this way is a useless estimate. The correct estimate of y for x = 6 is 29.1, from the Y-on-X line.

Q. 9 (b) Compute the regression coefficients in each of the following cases: (i) n = 10, sum (x_i - x-bar)-squared = 170, sum (y_i - y-bar)-squared = 140, sum (x_i - x-bar)(y_i - y-bar) = 92. (ii) sum (x_i - x-bar)(y_i - y-bar) = 148, s_x = 7.933, s_y = 16.627, n = 15. (Book answers: b_yx = 0.54, b_xy = 0.66; b_yx = 0.16, b_xy = 0.04.)

Step 1: (i) b_yx = sum (x_i - x-bar)(y_i - y-bar) / sum (x_i - x-bar)-squared = 92 / 170 = 0.5412, taken as 0.54. b_xy = sum (x_i - x-bar)(y_i - y-bar) / sum (y_i - y-bar)-squared = 92 / 140 = 0.6571, taken as 0.66.

Answer: (ii) sum (x_i - x-bar)-squared = n s_x-squared = 15 x 7.933-squared = 943.9; sum (y_i - y-bar)-squared = n s_y-squared = 15 x 16.627-squared = 4147.4. So b_yx = 148 / 943.9 = 0.1568, taken as 0.16, and b_xy = 148 / 4147.4 = 0.0357, taken as 0.04.

Exercise 14.2 (part 1) - Correlation, Q. 1-2 (book pp. 207-208)

Q. 1 (a) Differentiate between regression and correlation problems, giving examples.

Step 1: In a regression problem, one variable is treated as the independent (controlled) variable and the other as the dependent variable; the aim is to estimate the average value of the dependent variable for given values of the independent variable, for example, estimating the mean GPA of students scoring 600 marks in matriculation.

Step 2: In a correlation problem, both variables are considered simultaneously and symmetrically, and the relationship between them is considered without specifying which is dependent; the aim is to measure the strength of the linear relationship, for example, the correlation between heights and weights of persons.

Answer: Regression is asymmetrical (the regression of Y on x differs from the regression of X on y), while correlation is symmetrical (r_xy = r_yx). Regression predicts; correlation measures closeness of linear association.

Q. 1 (b) Define the terms correlation and product moment co-efficient of correlation.

Step 1: Correlation: the measurement of the strength (closeness) of the linear relationship between two variables; it measures the degree (extent or strength) of

covariability between two variables.

Step 2: The simple linear correlation is positive (direct) if the two variables tend to move in the same direction, negative (inverse) if they tend to move in opposite directions, and zero if there is no linear relationship.

Answer: The product moment coefficient of correlation, denoted r or r_xy, is defined as r = s_xy / (s_x s_y), where s_xy is the covariance of X and Y and s_x, s_y are their standard deviations. Equivalently, r = sum (x_i - x-bar)(y_i - y-bar) / sqrt(sum (x_i - x-bar)-squared x sum (y_i - y-bar)-squared). It always lies between -1 and +1.

Q. 1 (c) For a set of 50 pairs of observations on variables X and Y, we have sum (x_i - x-bar)(y_i - y-bar) = 450. Find the covariance. (Book answer: s_xy = 9.)

Step 1: The covariance is s_xy = sum (x_i - x-bar)(y_i - y-bar) / n.

Step 2: n = 50 and the sum of the products of deviations is 450.

Answer: s_xy = 450 / 50 = 9.

Q. 2 (a) The simple correlation coefficient r = s_xy / (s_x s_y) is given as r = sum (x_i - x-bar)(y_i - y-bar) / sqrt(sum (x_i - x-bar)-squared x sum (y_i - y-bar)-squared). The following table gives the ages of husbands and the ages of wives at the time of their marriage. Couple i: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Husband's age x_i: 25, 29, 30, 30, 31, 32, 33, 35, 37, 38. Wife's age y_i: 20, 22, 24, 29, 23, 31, 29, 31, 30, 31. Calculate the coefficient of correlation by using the above formula. (Book answer: r = 0.82.)

Step 1: n = 10, x-bar = 32, y-bar = 27. Deviations of x: -7, -3, -2, -2, -1, 0, 1, 3, 5, 6.

Deviations of y: -7, -5, -3, 2, -4, 4, 2, 4, 3, 4.

Step 2: Sum (x_i - x-bar)-squared = 49 + 9 + 4 + 4 + 1 + 0 + 1 + 9 + 25 + 36 = 138. Sum (y_i - y-bar)-squared = 49 + 25 + 9 + 4 + 16 + 16 + 4 + 16 + 9 + 16 = 164.

Step 3: Sum (x_i - x-bar)(y_i - y-bar) = 49 + 15 + 6 - 4 + 4 + 0 + 2 + 12 + 15 + 24 = 123.

Step 4: r = 123 / sqrt(138 x 164) = 123 / sqrt(22632) = 123 / 150.44 = 0.8176, given in the book as 0.82.

Answer: The coefficient of correlation between the ages of husbands and wives is r = 0.82, a strong positive correlation.

Q. 2 (b) The simple correlation coefficient r = s_xy / (s_x s_y) is given as r = sum x_i y_i - n x-bar y-bar / sqrt((sum x_i-squared - n x-bar-squared)(sum y_i-squared - n y-bar-squared)). The following table gives the demand and supply of a commodity. Supply x_i: 400, 200, 700, 100, 500, 300, 600. Demand y_i: 50, 60, 20, 70, 40, 30, 10. Calculate the coefficient of correlation by using the above formula. (Book answer: r = -0.857.)

Step 1: n = 7, sum x_i = 2800, sum y_i = 280, sum x_i y_i = 88000, sum x_i-squared = 1400000, sum y_i-squared = 14000.

Step 2: Numerator: sum x_i y_i - n x-bar y-bar = 88000 - 7 x 400 x 40 = 88000 - 112000 = -24000.

Step 3: Denominator: sqrt((1400000 - 7 x 400-squared) x (14000 - 7 x 40-squared))

= sqrt((1400000 - 1120000) x (14000 - 11200)) = sqrt(280000 x 2800) =

sqrt(784000000) = 28000.

Answer: r = -24000 / 28000 = -0.857, a strong negative correlation between supply and demand.

Q. 2 (c) The simple correlation coefficient r = s_xy / (s_x s_y) is given as r = sum x_i y_i - (sum x_i)(sum y_i) / n / sqrt((sum x_i-squared - (sum x_i)-squared / n)(sum y_i-squared - (sum y_i)-squared / n)). The following table gives the traffic density and accident rate. Traffic density x_i: 30, 35, 40, 45, 50, 60, 70, 80, 90. Accident rate y_i: 2, 4, 5, 5, 8, 15, 24, 30, 32. Calculate the coefficient of correlation by using the above formula. (Book answer: r = 0.983.)

Step 1: n = 9, sum x_i = 500, sum y_i = 125, sum x_i y_i = 8885, sum x_i-squared = 31250, sum y_i-squared = 2859.

Step 2: Numerator: sum x_i y_i - (sum x_i)(sum y_i) / n = 8885 - (500 x 125) / 9 = 8885 - 6944.44 = 1940.56.

Step 3: Denominator: sqrt((31250 - 500-squared / 9) x (2859 - 125-squared / 9)) = sqrt((31250 - 27777.78) x (2859 - 1736.11)) = sqrt(3472.22 x 1122.89) = sqrt(3898964) = 1974.58.

Answer: r = 1940.56 / 1974.58 = 0.9828, given in the book as 0.983, a very strong positive correlation between traffic density and accident rate.

Q. 2 (d) The simple correlation coefficient r = s_xy / (s_x s_y) is given as r = n sum

x_i y_i - (sum x_i)(sum y_i) / sqrt((n sum x_i-squared - (sum x_i)-squared)(n sum y_i-squared - (sum y_i)-squared)). The following table gives the number of persons employed and cloth manufactured in a textile mill. Persons employed x_i: 137, 209, 113, 189, 176, 200, 219. Cloth manufactured y_i: 23, 47, 22, 40, 39, 51, 49. Calculate the coefficient of correlation by using the above formula. (Book answer: r = 0.963.)

Step 1: n = 7, sum x_i = 1243, sum y_i = 271, sum x_i y_i = 50815, sum x_i-squared = 229877, sum y_i-squared = 11345.

Step 2: Numerator: n sum x_i y_i - (sum x_i)(sum y_i) = 7 x 50815 - 1243 x 271 = 355705 - 336853 = 18852.

Step 3: Denominator: sqrt((7 x 229877 - 1243-squared) x (7 x 11345 - 271-squared)) = sqrt((1609139 - 1545049) x (79415 - 73441)) = sqrt(64090 x 5974) = sqrt(382873660) = 19568.

Answer: r = 18852 / 19568 = 0.9635, given in the book as 0.963, a very strong positive correlation.

Exercise 14.2 (part 2) - Correlation, Q. 3-13 (book pp. 208-212)

Q. 3 (a) For a set of 22 pairs of observations, we have sum x_i = 983, sum y_i = 409, sum x_i-squared = 61339, sum y_i-squared = 8475, sum x_i y_i = 15811. Find the product moment correlation coefficient for the data. (Book answer: r = -0.6325.)

Step 1: n = 22. Numerator: n sum x_i y_i - (sum x_i)(sum y_i) = 22 x 15811 - 983 x 409 = 347842 - 402047 = -54205.

Step 2: Denominator: sqrt((22 x 61339 - 983-squared) x (22 x 8475 - 409-squared))

= sqrt((1349458 - 966289) x (186450 - 167281)) = sqrt(383169 x 19169) =

sqrt(7344975361) = 85696.

Answer: r = -54205 / 85696 = -0.6325, a fairly strong negative correlation.

Q. 3 (b) For a sample of 20 pairs of observations, we have x-bar = 2, y-bar = 8, sum x_i-squared = 180, sum y_i-squared = 3424, sum x_i y_i = 604. Calculate the coefficient of correlation. (Book answer: r = 0.6133.)

Step 1: n = 20. Sum (x_i - x-bar)-squared = sum x_i-squared - n x-bar-squared = 180 - 20 x 4 = 100. Sum (y_i - y-bar)-squared = 3424 - 20 x 64 = 3424 - 1280 = 2144.

Step 2: Sum (x_i - x-bar)(y_i - y-bar) = sum x_i y_i - n x-bar y-bar = 604 - 20 x 2 x 8 = 604 - 320 = 284.

Answer: r = 284 / sqrt(100 x 2144) = 284 / 463.04 = 0.6133, a moderately strong positive correlation.

Q. 3 (c) For a set of 8 pairs of observations, we have sum x_i = 448, sum y_i = 472, sum y_i-squared = 29958, sum x_i y_i = 26762, s_x = 16.6. Compute the product moment correlation coefficient. (Book answer: r = 0.15.)

Step 1: n = 8, y-bar = 472 / 8 = 59. Sum (y_i - y-bar)-squared = sum y_i-squared - n y-bar-squared = 29958 - 8 x 59-squared = 29958 - 27848 = 2110.

Step 2: Sum (x_i - x-bar)(y_i - y-bar) = sum x_i y_i - (sum x_i)(sum y_i) / n = 26762 - (448 x 472) / 8 = 26762 - 26432 = 330.

Step 3: Sum (x_i - x-bar)-squared = n s_x-squared = 8 x 16.6-squared = 8 x 275.56 = 2204.48.

Answer: r = 330 / sqrt(2204.48 x 2110) = 330 / 2156.42 = 0.153, given in the book as 0.15, a weak positive correlation.

Q. 4 (a) For a set of 50 pairs of observations, the standard deviations of x and y are 4.5 and 3.5 respectively. If the sum of products of deviations of x and y values from their respective means be 420, find the Karl Pearson's coefficient of correlation. (Book answer: r = 0.53.)

Step 1: r = sum (x_i - x-bar)(y_i - y-bar) / (n s_x s_y).

Step 2: n = 50, s_x = 4.5, s_y = 3.5, and the sum of the products of deviations is 420.

Answer: r = 420 / (50 x 4.5 x 3.5) = 420 / 787.5 = 0.5333, given in the book as 0.53.

Q. 4 (b) For a given set of data, we have s_x-squared = 9.102, s_y-squared = 2.204, s_xy = 1.694. Find the product moment correlation coefficient for the data. (Book answer: r = 0.378.)

Step 1: r = s_xy / (s_x s_y) = s_xy / sqrt(s_x-squared x s_y-squared).

Step 2: s_x = sqrt(9.102) = 3.017, s_y = sqrt(2.204) = 1.485.

Answer: r = 1.694 / (3.017 x 1.485) = 1.694 / 4.480 = 0.378.

Q. 5 (a) For a given set of data, we have r = 0.48, s_xy = 36, s_x-squared = 16. Find s_y. (Book answer: s_y = 18.75.)

Step 1: r = s_xy / (s_x s_y), so s_y = s_xy / (r s_x).

Step 2: s_x = sqrt(16) = 4.

Answer: s_y = 36 / (0.48 x 4) = 36 / 1.92 = 18.75.

Q. 5 (b) For a given set of data, we have r = 0.5, sum (x_i - x-bar)(y_i - y-bar) = 120, s_y = 8, sum (x_j - x-bar)-squared = 90. Find the number of pairs of values. (Book answer: n = 10.)

Step 1: s_xy = sum (x_i - x-bar)(y_i - y-bar) / n = 120 / n, and s_x = sqrt(sum (x_j - x-bar)-squared / n) = sqrt(90 / n).

Step 2: r = s_xy / (s_x s_y) gives 0.5 = (120 / n) / (sqrt(90 / n) x 8). So sqrt(90 / n) = (120 / n) / (0.5 x 8) = 30 / n.

Step 3: Squaring: 90 / n = 900 / n-squared, so 90 n = 900.

Answer: n = 10 pairs of values.

Q. 6 (a) A computer while calculating the correlation coefficient between two variables X and Y from 25 pairs of observations obtained the following results: sum x_i = 125, sum x_i-squared = 650, sum y_i = 100, sum y_i-squared = 460, sum x_i y_i = 508. It was, however, later discovered at the time of checking that he had copied down two pairs as (x_i, y_i) = (6, 14) and (8, 6), while the correct values were (8, 12) and (6, 8). Obtain the correct value of coefficient of correlation. (Book

answer: r = 0.67.)

Step 1: Correct the sums. Wrong pairs contribute sum x = 14, sum y = 20, sum x-squared = 36 + 64 = 100, sum y-squared = 196 + 36 = 232, sum xy = 84 + 48 = 132. Correct pairs contribute sum x = 14, sum y = 20, sum x-squared = 64 + 36 = 100, sum y-squared = 144 + 64 = 208, sum xy = 96 + 48 = 144.

Step 2: Corrected sums: sum x_i = 125 (unchanged), sum y_i = 100 (unchanged), sum x_i-squared = 650 (unchanged), sum y_i-squared = 460 - 232 + 208 = 436, sum x_i y_i = 508 - 132 + 144 = 520.

Step 3: Numerator: 25 x 520 - 125 x 100 = 13000 - 12500 = 500. Denominator: sqrt((25 x 650 - 125-squared) x (25 x 436 - 100-squared)) = sqrt((16250 - 15625) x (10900 - 10000)) = sqrt(625 x 900) = 750.

Answer: r = 500 / 750 = 0.6667, given in the book as 0.67.

Q. 6 (b) The following data show the marks in economics and marks in statistics obtained by ten students. Student i: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Economics x_i: 78, 36, 96, 25, 75, 82, 90, 62, 65, 39. Statistics y_i: 84, 51, 91, 60, 68, 62, 86, 58, 53, 47. (i) Compute the coefficient of correlation. (ii) Again compute the coefficient of correlation by taking the deviations of variable X from 50 and of variable Y from 60. (iii) Do the results in (i) and (ii) agree? (Book answers: (i) 0.775, (ii) 0.775, (iii) Yes.)

Step 1: n = 10, sum x_i = 648, sum y_i = 660, sum x_i y_i = 45422, sum x_i-squared = 47260, sum y_i-squared = 45784.

Step 2: (i) r = (10 x 45422 - 648 x 660) / sqrt((10 x 47260 - 648-squared) x (10 x 45784 - 660-squared)) = (454220 - 427680) / sqrt((472600 - 419904) x (457840 -

435600)) = 26540 / sqrt(52696 x 22240) = 26540 / 34231 = 0.7753, given in the book as 0.775.

Step 3: (ii) Taking deviations from 50 and 60 is a change of origin only; the sums of squared deviations and the sum of products of deviations are unchanged, so the computation gives the same value, r = 0.775.

Answer: (iii) Yes, the results agree, because the correlation coefficient is invariant under a change of origin.

Q. 6 (c) Compute the correlation coefficient between the variables X and Y represented in the following table: x_i: 2, 4, 5, 6, 8, 11. y_i: 18, 12, 10, 8, 7, 5. Multiply each x_i value by 2 and add 6. Multiply each y_i value by 3 and subtract 15. Find the correlation co-efficient between the two new sets of values, explaining why you do or do not obtain the same result as above. (Book answer: -0.92.)

Step 1: n = 6, sum x_i = 36, sum y_i = 60, sum x_i y_i = 293, sum x_i-squared = 266, sum y_i-squared = 706.

Step 2: r = (6 x 293 - 36 x 60) / sqrt((6 x 266 - 36-squared) x (6 x 706 - 60-squared)) = (1758 - 2160) / sqrt((1596 - 1296) x (4236 - 3600)) = -402 / sqrt(300 x 636) = -402 / 436.81 = -0.9203, given in the book as -0.92.

Step 3: The transformed variables are x-prime = 2x + 6 and y-prime = 3y - 15. Both multipliers (2 and 3) are positive, so the transformation is a change of origin and a positive change of scale for both variables.

Answer: The correlation coefficient between the transformed variables is again r =

-0.92. The same result is obtained because r is invariant under changes of origin and scale, provided the scale factors have the same sign; here both are positive, so the sign of r is preserved.

Q. 7 (a) Interpret the meaning when r = -1, r = 0, r = 1.

Step 1: r = 1 means perfect positive linear correlation: all points lie exactly on a straight line with positive slope, so an increase in X is accompanied by a proportional increase in Y.

Step 2: r = -1 means perfect negative linear correlation: all points lie exactly on a straight line with negative slope, so an increase in X is accompanied by a proportional decrease in Y.

Answer: r = 0 means no linear correlation: there is no linear relationship between X and Y (though a non-linear relationship may still exist).

Q. 7 (b) Sketch scatter diagrams which illustrate: (i) positive linear correlation, (ii) perfect positive linear correlation, (iii) negative linear correlation, (iv) perfect negative linear correlation, (v) no correlation, between two variables X and Y.

Step 1: (i) Positive linear correlation: the points form a band rising from left to right, clustering fairly closely about an upward-sloping straight line. (ii) Perfect positive linear correlation: all points lie exactly on an upward-sloping straight line.

Step 2: (iii) Negative linear correlation: the points form a band falling from left to right, clustering fairly closely about a downward-sloping straight line. (iv) Perfect negative linear correlation: all points lie exactly on a downward-sloping straight line.

Answer: (v) No correlation: the points are scattered over the plane with no discernible upward or downward trend, for example in a roughly circular cloud or a horizontal band.

Q. 8 (a) From the data given below, calculate the coefficient of correlation between the ages of husbands and ages of wives at the time of their marriage. Couple i: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Husband's age x_i: 28, 27, 28, 23, 29, 30, 36, 35, 33, 31. Wife's age y_i: 27, 20, 22, 18, 21, 29, 29, 28, 29, 27. Find the regression coefficients. Verify that r is the geometric mean of the two regression coefficients. (Book answers: r = 0.82, b_yx = 0.89, b_xy = 0.75.)

Step 1: n = 10, sum x_i = 300, sum y_i = 250, sum x_i y_i = 7623, sum x_i-squared = 9138, sum y_i-squared = 6414.

Step 2: r = (10 x 7623 - 300 x 250) / sqrt((10 x 9138 - 300-squared) x (10 x 6414 - 250-squared)) = (76230 - 75000) / sqrt((91380 - 90000) x (64140 - 62500)) = 1230 / sqrt(1380 x 1640) = 1230 / 1504.4 = 0.8176, given in the book as 0.82.

Step 3: Regression coefficients: sum (x_i - x-bar)(y_i - y-bar) = 7623 - 10 x 30 x 25 = 123; sum (x_i - x-bar)-squared = 9138 - 10 x 30-squared = 138; sum (y_i - y-bar)-squared = 6414 - 10 x 25-squared = 164. So b_yx = 123 / 138 = 0.8913, taken as 0.89; b_xy = 123 / 164 = 0.75.

Answer: The geometric mean of the two regression coefficients is sqrt(b_yx x b_xy) = sqrt(0.89 x 0.75) = sqrt(0.6675) = 0.817, which equals r = 0.82. So r is the geometric mean of the two regression coefficients.

Q. 8 (b) The two regression coefficients have following values, find r. (i) b_yx = 0.86,

b_xy = 0.95. (ii) b_yx = -0.52, b_xy = -1.02. (Book answers: (i) r = 0.90, (ii) r = -0.73.)

Step 1: r = sqrt(b_yx x b_xy), with the sign of r the same as the sign of the regression coefficients (both coefficients always have the same sign).

Step 2: (i) r = sqrt(0.86 x 0.95) = sqrt(0.817) = 0.9039, given in the book as 0.90.

Answer: (ii) r = -sqrt(0.52 x 1.02) = -sqrt(0.5304) = -0.7283, given in the book as -0.73. The negative sign is used because both regression coefficients are negative.

Q. 8 (c) Find the two regression coefficients in each of the following cases. (i) sum x_i = 17.6, sum y_i = 32.8, sum x_i y_i = 94.7, sum x_i-squared = 49.64, sum y_i-squared = 182, n = 8. (ii) n = 10, sum (x_i - x-bar)-squared = 170, sum (y_i - y-bar)-squared = 140, sum (x_i - x-bar)(y_i - y-bar) = 92. (iii) sum (x_i - x-bar)(y_i - y-bar) = 148, s_x = 7.933, s_y = 16.627, n = 15. (iv) n = 8, x-bar = 7, y-bar = 5, sum x_i y_i = 364, sum (x_i - x-bar)-squared = 132, sum (y_i - y-bar)-squared = 56. (v) r = 0.97, s_x = 17.08, s_y = 14.34. (Book answers: (i) b_yx = 2.06, b_xy = 0.47; (ii) b_yx = 0.54, b_xy = 0.66; (iii) b_yx = 0.16, b_xy = 0.04; (iv) b_yx = 0.64, b_xy = 1.5; (v) b_yx = 0.81, b_xy = 1.16.)

Step 1: (i) b_yx = (8 x 94.7 - 17.6 x 32.8) / (8 x 49.64 - 17.6-squared) = (757.6 - 577.28) / (397.12 - 309.76) = 180.32 / 87.36 = 2.0641, taken as 2.06. b_xy = 180.32 / (8 x 182 - 32.8-squared) = 180.32 / (1456 - 1075.84) = 180.32 / 380.16 = 0.4743, taken as 0.47.

Step 2: (ii) b_yx = 92 / 170 = 0.5412, taken as 0.54. b_xy = 92 / 140 = 0.6571, taken as 0.66.

Step 3: (iii) sum (x_i - x-bar)-squared = n s_x-squared = 15 x 7.933-squared = 943.9; sum (y_i - y-bar)-squared = 15 x 16.627-squared = 4147.4. b_yx = 148 / 943.9 = 0.1568, taken as 0.16. b_xy = 148 / 4147.4 = 0.0357, taken as 0.04.

Step 4: (iv) Sum (x_i - x-bar)(y_i - y-bar) = sum x_i y_i - n x-bar y-bar = 364 - 8 x 7 x 5 = 364 - 280 = 84. b_yx = 84 / 132 = 0.6364, taken as 0.64. b_xy = 84 / 56 = 1.5.

Answer: (v) b_yx = r s_y / s_x = 0.97 x 14.34 / 17.08 = 0.8144, taken as 0.81. b_xy = r s_x / s_y = 0.97 x 17.08 / 14.34 = 1.1553, taken as 1.16.

Q. 9 (a) Explain why the regression line of Y on X is not necessarily the same as the regression line of X on Y. How would you decide which is the appropriate regression in any particular situation? Answer the following: (i) When do the two lines coincide? (ii) When are they at right angles? (Book answers: (i) Exact linear relation. (ii) Uncorrelated X, Y (i.e., rho = 0).)

Step 1: The two lines are different because they minimise different quantities: the Y-on-X line minimises the sum of squared vertical deviations (errors in Y), while the X-on-Y line minimises the sum of squared horizontal deviations (errors in X). In general b_yx x b_xy = r-squared, which is less than 1, so the two slopes are not reciprocals.

Step 2: Which line is appropriate depends on the purpose: if Y is to be predicted from given values of X, use the regression of Y on X; if X is to be predicted from given values of Y, use the regression of X on Y. The variable whose value is known (controlled) is the independent variable.

Step 3: (i) The two lines coincide when there is an exact linear relation between X

and Y, that is, when r = +1 or r = -1, so that all points lie on one straight line.

Answer: (ii) The two lines are at right angles when X and Y are uncorrelated, that is, when rho = 0 (r = 0); then the Y-on-X line is horizontal and the X-on-Y line is vertical.

Q. 9 (b) Calculate the coefficient of correlation and obtain the lines of regression from the following data. Price x_i: 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Demand y_i: 25, 24, 20, 20, 19, 17, 16, 13, 10, 6. (Book answers: r = -0.98, y-hat = 31.45 - 1.93 x, x-hat = 16 - 0.5 y.)

Step 1: n = 10, sum x_i = 75, sum y_i = 170, sum x_i y_i = 1116, sum x_i-squared = 645, sum y_i-squared = 3212.

Step 2: r = (10 x 1116 - 75 x 170) / sqrt((10 x 645 - 75-squared) x (10 x 3212 - 170-squared)) = (11160 - 12750) / sqrt((6450 - 5625) x (32120 - 28900)) = -1590 / sqrt(825 x 3220) = -1590 / 1629.9 = -0.9755, given in the book as -0.98.

Step 3: b_yx = -1590 / 825 = -1.9273, taken as -1.93. a_yx = (170 - (-1.9273) x 75) / 10 = (170 + 144.55) / 10 = 31.4545, taken as 31.45. So y-hat = 31.45 - 1.93 x.

Step 4: b_xy = -1590 / 3220 = -0.4938, taken as -0.5. a_xy = (75 - (-0.4938) x 170) / 10 = (75 + 83.95) / 10 = 15.8944, taken as 16. So x-hat = 16 - 0.5 y.

Answer: The coefficient of correlation is r = -0.98, and the regression lines are y-hat = 31.45 - 1.93 x and x-hat = 16 - 0.5 y.

Q. 9 (c) Given the following data: n = 100, sum x_i = 5000, sum y_i = 6000, sum x_i y_i = 300300, sum x_i-squared = 250400, sum y_i-squared = 360900. Calculate (i)

s_x, s_y and r, (ii) regression lines, (iii) estimate the value of y for x = 55. (Book answers: (i) s_x = 2, s_y = 3, r = 0.5, (ii) y-hat = 22.5 + 0.75 x, x-hat = 30.2 + 0.33 y, (iii) 63.75.)

Step 1: x-bar = 5000 / 100 = 50, y-bar = 6000 / 100 = 60. s_x-squared = (250400 / 100) - 50-squared = 2504 - 2500 = 4, so s_x = 2. s_y-squared = (360900 / 100) - 60-squared = 3609 - 3600 = 9, so s_y = 3.

Step 2: s_xy = (300300 / 100) - 50 x 60 = 3003 - 3000 = 3. r = s_xy / (s_x s_y) = 3 / (2 x 3) = 0.5.

Step 3: b_yx = r s_y / s_x = 0.5 x 3 / 2 = 0.75. a_yx = y-bar - b_yx x-bar = 60 - 0.75 x 50 = 22.5. So y-hat = 22.5 + 0.75 x. b_xy = r s_x / s_y = 0.5 x 2 / 3 = 0.3333, taken as 0.33. a_xy = x-bar - b_xy y-bar = 50 - 0.3333 x 60 = 30.0, given in the book as 30.2 with the rounded slope. So x-hat = 30.2 + 0.33 y.

Answer: (iii) For x = 55, y-hat = 22.5 + 0.75 x 55 = 22.5 + 41.25 = 63.75.

Q. 10 (a) Given the following data: n = 10, sum x_i = 120, sum y_i = 250, sum x_i y_i = 3070.7, s_x = 3.5, s_y = 7.2. Calculate regression lines. (Book answers: y-hat = 18.04 + 0.58 x; x-hat = 8.50 + 0.14 y.)

Step 1: x-bar = 120 / 10 = 12, y-bar = 250 / 10 = 25. Sum (x_i - x-bar)-squared = n s_x-squared = 10 x 3.5-squared = 122.5. Sum (y_i - y-bar)-squared = 10 x 7.2-squared = 518.4.

Step 2: Sum (x_i - x-bar)(y_i - y-bar) = sum x_i y_i - n x-bar y-bar = 3070.7 - 10 x 12 x 25 = 3070.7 - 3000 = 70.7.

Step 3: b_yx = 70.7 / 122.5 = 0.5771, taken as 0.58. a_yx = 25 - 0.5771 x 12 = 25 - 6.9257 = 18.0743, taken as 18.04. So y-hat = 18.04 + 0.58 x.

Answer: b_xy = 70.7 / 518.4 = 0.1364, taken as 0.14. a_xy = 12 - 0.1364 x 25 = 12 - 3.4095 = 8.5905, taken as 8.50. So x-hat = 8.50 + 0.14 y.

Q. 10 (b) Given that means and variances of two series X and Y are: X-series mean 25, variance 25; Y-series mean 38, variance 36. The correlation coefficient between X and Y is 0.75. Estimate the most plausible value of Y for x = 40 and most plausible value of X for y = 58. (Book answers: y-hat = 15.5 + 0.9 x, 51.5; x-hat = 1.25 + 0.625 y, 37.5.)

Step 1: s_x = sqrt(25) = 5, s_y = sqrt(36) = 6, r = 0.75. b_yx = r s_y / s_x = 0.75 x 6 / 5 = 0.9. a_yx = y-bar - b_yx x-bar = 38 - 0.9 x 25 = 38 - 22.5 = 15.5. So y-hat = 15.5 + 0.9 x.

Step 2: For x = 40, y-hat = 15.5 + 0.9 x 40 = 15.5 + 36 = 51.5.

Step 3: b_xy = r s_x / s_y = 0.75 x 5 / 6 = 0.625. a_xy = x-bar - b_xy y-bar = 25 - 0.625 x 38 = 25 - 23.75 = 1.25. So x-hat = 1.25 + 0.625 y.

Answer: For y = 58, x-hat = 1.25 + 0.625 x 58 = 1.25 + 36.25 = 37.5.

Q. 10 (c) If the mean height of 500 fathers is 68.65 inches with standard deviation of 2.8 inches and the mean height of their youngest sons is 69.65 inches with standard deviation of 2.85 inches and the coefficient of correlation between them is 0.52 obtain the two equations of the lines of regression in the simplest form. (Book answers: y-hat = 33.27 + 0.53 x; x-hat = 33.13 + 0.51 y.)

Step 1: b_yx = r s_y / s_x = 0.52 x 2.85 / 2.8 = 0.5293, taken as 0.53. a_yx = y-bar - b_yx x-bar = 69.65 - 0.5293 x 68.65 = 69.65 - 33.34 = 33.31, given in the book as 33.27. So y-hat = 33.27 + 0.53 x.

Answer: b_xy = r s_x / s_y = 0.52 x 2.8 / 2.85 = 0.5109, taken as 0.51. a_xy = x-bar - b_xy y-bar = 68.65 - 0.5109 x 69.65 = 68.65 - 35.58 = 33.07, given in the book as 33.13. So x-hat = 33.13 + 0.51 y.

Q. 11 (a) Given the following data: x-bar = 54, y-bar = 28, b_yx = -1.5, b_xy = -0.2. Show that the two estimated lines of regression intersect at the point (x-bar, y-bar). Estimate the value of X when Y = 30 and the value of Y when X = 55. (Book answers: y-hat = 26.5; x-hat = 53.6.)

Step 1: The Y-on-X line is y-hat - y-bar = b_yx (x - x-bar), that is, y-hat = 28 - 1.5 (x - 54). The X-on-Y line is x-hat - x-bar = b_xy (y - y-bar), that is, x-hat = 54 - 0.2 (y - 28).

Step 2: At x = x-bar = 54, the Y-on-X line gives y-hat = 28 = y-bar. At y = y-bar = 28, the X-on-Y line gives x-hat = 54 = x-bar. So both lines pass through (x-bar, y-bar) = (54, 28), and therefore they intersect at that point.

Step 3: For Y = 30, x-hat = 54 - 0.2 (30 - 28) = 54 - 0.4 = 53.6.

Answer: For X = 55, y-hat = 28 - 1.5 (55 - 54) = 28 - 1.5 = 26.5.

Q. 11 (b) For a given set of data, the least squares regression lines are: Estimation regression line of Y on X: y-hat = 20.8 - 0.219 x. Estimation regression line of X on Y: x-hat = 16.2 - 0.785 y. Find the product moment correlation coefficient for the data.

(Book answer: r = -0.415.)

Step 1: b_yx = -0.219, b_xy = -0.785. r-squared = b_yx x b_xy = (-0.219) x (-0.785) = 0.1719.

Step 2: r = sqrt(0.1719) = 0.4146. Since both regression coefficients are negative, r is negative.

Answer: r = -0.415.

Q. 12 (a) For the following set of data, use u_i = x_i - 1000, and v_i = (y_i - 250) / 5 to find the product moment correlation coefficient and the least squares lines of regression of Y on X and of X on Y. x_i: 1000, 1012, 1009, 1007, 1010, 1015, 1010, 1011. y_i: 235, 240, 245, 250, 255, 260, 265, 270. (Book answers: 0.583; y-hat = -1386.1346 + 1.6235 x; x-hat = 956.326 + 0.2096 y.)

Step 1: Code the variables: u_i = 0, 12, 9, 7, 10, 15, 10, 11; v_i = -3, -2, -1, 0, 1, 2, 3, 4. n = 8, sum u_i = 74, sum v_i = 4, sum u_i v_i = 81, sum u_i-squared = 820, sum v_i-squared = 44.

Step 2: r is invariant under change of origin and scale, so r computed from (u, v) equals r for (x, y): r = (8 x 81 - 74 x 4) / sqrt((8 x 820 - 74-squared) x (8 x 44 - 4-squared)) = (648 - 296) / sqrt((6560 - 5476) x (352 - 16)) = 352 / sqrt(1084 x 336) = 352 / 603.5 = 0.5833, given in the book as 0.583.

Step 3: b_yx = 1.6236, taken as 1.6235. a_yx = -1386.1347, given in the book as -1386.1346. So y-hat = -1386.1346 + 1.6235 x.

Answer: b_xy = 0.2095, taken as 0.2096. a_xy = 956.3452, given in the book as

956.326. So x-hat = 956.326 + 0.2096 y.

Q. 12 (b) On each of 30 items, two measurements are made on the variables X and Y. The following summations are given: sum x_i = 15, sum y_i = -6, sum x_i-squared = 61, sum y_i-squared = 90, sum x_i y_i = 56. Calculate the product moment correlation coefficient and obtain the lines of regression of Y on X and X on Y. If the variable X is replaced by U where u_i = (x_i - 1) / 2, find the correlation coefficient between U and Y and the regression lines of Y on U and U on Y. (Book answers: 0.856; y-hat = -0.751 + 1.10 x; x-hat = 0.633 + 0.664 y; 0.856; y-hat = 0.351 + 2.21 u; u-hat = -0.184 + 0.332 y.)

Step 1: n = 30. Numerator: 30 x 56 - 15 x (-6) = 1680 + 90 = 1770. Denominator: sqrt((30 x 61 - 15-squared) x (30 x 90 - (-6)-squared)) = sqrt((1830 - 225) x (2700 - 36)) = sqrt(1605 x 2664) = sqrt(4275720) = 2067.78. r = 1770 / 2067.78 = 0.856.

Step 2: b_yx = 1770 / 1605 = 1.1028, taken as 1.10. a_yx = y-bar - b_yx x-bar = (-6/30) - 1.1028 x (15/30) = -0.2 - 0.5514 = -0.7514, taken as -0.751. So y-hat = -0.751 + 1.10 x. b_xy = 1770 / 2664 = 0.6644, taken as 0.664. a_xy = x-bar - b_xy y-bar = 0.5 - 0.6644 x (-0.2) = 0.5 + 0.1329 = 0.6329, taken as 0.633. So x-hat = 0.633 + 0.664 y.

Step 3: U = (X - 1) / 2 is a change of origin and a positive change of scale, so r_UY = r_XY = 0.856. X = 2U + 1, so y-hat = -0.751 + 1.10 (2u + 1) = 0.349 + 2.21 u, given in the book as 0.351 + 2.21 u. U-hat = (x-hat - 1) / 2 = (0.633 + 0.664 y - 1) / 2 = -0.1835 + 0.332 y, given in the book as -0.184 + 0.332 y.

Answer: The correlation coefficient between U and Y is 0.856, and the regression lines are y-hat = 0.351 + 2.21 u and u-hat = -0.184 + 0.332 y.

Q. 12 (c) The following table shows the marks in statistics and mathematics obtained by 10 students from a large group of students. Marks in Statistics x_i: 75, 80, 93, 65, 87, 71, 98, 68, 84, 77. Marks in Mathematics y_i: 82, 78, 86, 72, 91, 80, 95, 72, 89, 74. Estimate the linear regression function considering (i) X as independent variable, (ii) Y as independent variable. (Book answers: y-hat = 29.13 + 0.661 x; x-hat = -14.39 + 1.15 y.)

Step 1: n = 10, sum x_i = 798, sum y_i = 819, sum x_i y_i = 66045, sum x_i-squared = 64722, sum y_i-squared = 67675.

Step 2: (i) b_yx = (10 x 66045 - 798 x 819) / (10 x 64722 - 798-squared) = (660450 - 653562) / (647220 - 636804) = 6888 / 10416 = 0.6613, taken as 0.661. a_yx = (819 - 0.6613 x 798) / 10 = (819 - 527.72) / 10 = 29.13. So y-hat = 29.13 + 0.661 x.

Answer: (ii) b_xy = 6888 / (10 x 67675 - 819-squared) = 6888 / (676750 - 670761) = 6888 / 5989 = 1.1501, taken as 1.15. a_xy = (798 - 1.1501 x 819) / 10 = (798 - 941.93) / 10 = -14.39. So x-hat = -14.39 + 1.15 y.

Q. 13 (a) A random sample of 20 pairs of observations (x_i, y_i) gave the following: x-bar = 2, y-bar = 8, sum x_i-squared = 180, sum y_i-squared = 1424, sum x_i y_i = 404. Estimate the linear regression function taking (i) X as independent variable, (ii) Y as independent variable. (Book answers: y-hat = 6.32 + 0.84 x; x-hat = -2.67 + 0.583 y.)

Step 1: n = 20. Sum (x_i - x-bar)-squared = 180 - 20 x 4 = 100. Sum (y_i - y-bar)-squared = 1424 - 20 x 64 = 144. Sum (x_i - x-bar)(y_i - y-bar) = 404 - 20 x 2 x 8 = 84.

Step 2: (i) b_yx = 84 / 100 = 0.84. a_yx = 8 - 0.84 x 2 = 6.32. So y-hat = 6.32 +

0.84 x.

Answer: (ii) b_xy = 84 / 144 = 0.5833, taken as 0.583. a_xy = 2 - 0.5833 x 8 = 2 - 4.6667 = -2.6667, taken as -2.67. So x-hat = -2.67 + 0.583 y.

Q. 13 (b) Given the following data: n = 5, sum x_i = 15, sum y_i = 25, sum (x_i - x-bar)-squared = 10, sum (y_i - y-bar)-squared = 26, sum (x_i - x-bar)(y_i - y-bar) = 13. Determine the two regression lines. (Book answers: y-hat = 1.1 + 1.3 x; x-hat = 0.5 + 0.5 y.)

Step 1: x-bar = 15 / 5 = 3, y-bar = 25 / 5 = 5. b_yx = 13 / 10 = 1.3. a_yx = 5 - 1.3 x 3 = 1.1. So y-hat = 1.1 + 1.3 x.

Answer: b_xy = 13 / 26 = 0.5. a_xy = 3 - 0.5 x 5 = 0.5. So x-hat = 0.5 + 0.5 y.

Q. 13 (c) The correlation coefficient between the two variables X and Y is r = 0.60. If s_x = 1.50, s_y = 2.00, x-bar = 10 and y-bar = 20, find the equations of the two regression lines of Y on X and X on Y. (Book answers: y-hat = 12 + 0.8 x; x-hat = 1 + 0.45 y.)

Step 1: b_yx = r s_y / s_x = 0.60 x 2.00 / 1.50 = 0.8. a_yx = y-bar - b_yx x-bar = 20 - 0.8 x 10 = 12. So y-hat = 12 + 0.8 x.

Answer: b_xy = r s_x / s_y = 0.60 x 1.50 / 2.00 = 0.45. a_xy = x-bar - b_xy y-bar = 10 - 0.45 x 20 = 1. So x-hat = 1 + 0.45 y.

Exercise 14.2 — Objective Questions, Q. 1: Fill in the blanks (book pp. 212-213)

Q. 1 (i) The regression is a relationship that describes the dependence of the expected value of the dependent random variable for a given value of the independent non-random variable.

Q. 1 (ii) The variable, that forms the basis of estimation, is called regressor

Q. 1 (iii) The variable, whose resulting value depends upon the selected value of the independent variable, is called regressand

Q. 1 (iv) The scatter diagram is a set of points in a rectangular co-ordinate system, where each point represents an observed pair of values.

Q. 1 (v) The principle of least squares is used for finding the estimates a and b of the parameters alpha and beta.

Q. 1 (vi) The estimated regression line always passes through (x-bar, y-bar).

Exercise 14.2 — Objective Questions, Q. 2: True or False (book p. 213)

Q. 2 (i) The simple linear regression model contains two parameters alpha and beta.

False

Q. 2 (ii) The simple linear regression model contains four parameters alpha, beta, mu-y|x and sigma-squared.

True

Q. 2 (iii) The simple linear regression model is simple in that there is only one independent variable.

True

Q. 2 (iv) The parameter alpha is called the slope and the parameter beta is the intercept of the regression line.

False

Q. 2 (v) The regression coefficient is denoted by alpha.

False

Q. 2 (vi) The parameter alpha is called the y-intercept of the regression line.

True

Q. 2 (vii) In a regression analysis the independent variable is always prefixed while the dependent variable is random.

True

Q. 2 (viii) The principle of least squares says that the sum of squares of the residuals of observed values from their corresponding estimated values should be the least possible.

True

Q. 2 (ix) The principle of least squares is used for finding the estimates a and b of the parameters alpha and beta.

True

Q. 2 (x) The constant b estimates the parameter beta representing the slope of the regression line.

True

Q. 2 (xi) The regression coefficient b is independent of change in origin and scale.

False

Q. 2 (xii) The estimated regression equation of Y on x is used to estimate the mean value of Y for a given value of x.

True

Exercise 14.2 — Objective Questions, Q. 3: Fill in the blanks, remaining items (book pp. 213-214; 2 items already in the earlier block above)

Q. 3 (i) The correlation analysis is possible when both the variables X and Y are random

Q. 3 (ii) If the two variables move in the same direction, the correlation is positive.

Q. 3 (iii) If the two variables move in opposite directions, the correlation is negative.

Q. 3 (vi) r = 0 indicates that the two variables are linearly independent

Exercise 14.2 — Objective Questions, Q. 4: True or False, remaining items (book p. 214; 3 items already in the earlier block above. Book prints (xiii)-(xv) after (vii) — numbering as printed.)

Q. 4 (ii) The sample correlation coefficient R is a point estimator of the population correlation coefficient rho.

True

Q. 4 (iii) The correlation coefficient r is not symmetrical with respect to X and Y.

False

Q. 4 (iv) The correlation coefficient changes with a change in origin.

False

Q. 4 (vi) The correlation coefficient is not independent of the origin and the unit of measurement.

False

Q. 4 (xiii, as printed) The correlation coefficient r always lies between -1 and 1.

True

Q. 4 (xiv, as printed) r = 1 indicates perfect positive correlation between the two variables and slope is positive.

True

Q. 4 (xv, as printed) r = -1 indicates perfect negative correlation between the two variables and slope is negative.

True

Exercise 14.2 — Objective Questions, Q. 5: True or False (book p. 214)

Q. 5 (i) r = 0 indicates that one regression line is horizontal and the other regression line is vertical.

True

Q. 5 (ii) The correlation coefficient r is the geometric mean of the two regression

coefficients.

True

Q. 5 (iii) Each of the two estimated regression lines passes through the point (x-bar, y-bar).

True

Q. 5 (iv) We can always estimate the X and Y values from the regression equation of Y on X.

False

Q. 5 (v) The regression coefficient of X on Y is -1.2 and of Y on X is 0.3.

False

Q. 5 (vi) The regression coefficient of X on Y is -1.2 and of Y on X is -0.3.

True

Q. 5 (vii) The regression coefficient of X on Y, regression coefficient of Y on X and correlation coefficient have same sign.

True

Q. 5 (viii) If the regression coefficient of X on Y is -1.2 and of Y on X is -0.3, the correlation coefficient is 0.6.

False

Q. 5 (ix) The regression coefficient of X on Y is always equal to the regression coefficient of Y on X.

False

Chapter 15: Association

Exercise — Short Questions

What is association between two attributes?

Two attributes are associated when they are not independent of each other.

Association is studied from a contingency table of observed frequencies.

The coefficient of association, denoted by Q, measures association between two attributes.

What is the chi-square test of independence?

It measures the discrepancy between observed and expected frequencies.

The value of the chi-square statistic is always non-negative.

Rejection of H0 of independence indicates that the two criteria of classification are associated.

Exercise — True or False (book p. 240)

The two attributes A and B are associated, if (AB) = (A)(B) divided by n.

False

The coefficient of association, denoted by Q, is a measure of association between the two attributes.

True

The coefficient of association always lies between -1 and 1.

True

A measure of the discrepancy between the observed and expected frequencies is called a chi-square test of independence.

True

The value of chi-square statistic is always non-negative.

True

The rejection of H0 of independence indicates that two criteria of classification are associated.

True

Exercise 15.1 (book pp. 221-222)

Q. 1 (a) Distinguish between attribute and variable. Define positive classes, negative classes and ultimate classes.

Step 1: An attribute is a characteristic which varies only in quality from one individual to another, for example marital status, blindness or smoking; a variable is a characteristic which varies in quantity, for example height or weight.

Step 2: A class represented by positive attributes only is a positive class, for example A, B and AB; a class represented by negative attributes only is a negative class, for example alpha, beta and alpha beta; a class represented by both positive and negative attributes is a contrary class, for example A beta and alpha B.

Answer: The frequencies of the classes of the highest order are called ultimate class frequencies. For two attributes A and B the ultimate classes are AB, A beta, alpha B and alpha beta; their number is 2 to the power k for k attributes.

Q. 1 (b) Given the ultimate class frequencies (AB) = 95, (A beta) = 55, (alpha B) = 85, (alpha beta) = 45, find the frequencies of the positive and negative classes and the whole number of observations n.

Step 1: (A) = (AB) + (A beta) = 95 + 55 = 150.

Step 2: (B) = (AB) + (alpha B) = 95 + 85 = 180.

Step 3: (alpha) = (alpha B) + (alpha beta) = 85 + 45 = 130.

Step 4: (beta) = (A beta) + (alpha beta) = 55 + 45 = 100.

Answer: n = 95 + 55 + 85 + 45 = 280; (A) = 150, (B) = 180, (alpha) = 130, (beta) = 100, as printed in the book.

Q. 2 (a) Given n = 250, (A) = 80, (B) = 100, (AB) = 70, find the frequencies of the ultimate classes.

Step 1: (A beta) = (A) - (AB) = 80 - 70 = 10.

Step 2: (alpha B) = (B) - (AB) = 100 - 70 = 30.

Step 3: (alpha beta) = n - (A) - (B) + (AB) = 250 - 80 - 100 + 70 = 140.

Answer: (A beta) = 10, (alpha B) = 30, (alpha beta) = 140, with (AB) = 70, as printed in the book.

Q. 2 (b) Measurements are made on a thousand husbands and a thousand wives. If the measurements of husbands exceed the measurements of the wives in 800 cases for one measurement, in 700 cases for another, and in 660 cases for both measurements, in how many cases will both the measurements on the wife exceed the measurement on the husband?

Step 1: Let H1 be the set where the husband exceeds the wife in the first measurement (800 cases) and H2 the set for the second measurement (700 cases); both together occur in 660 cases.

Step 2: Cases where the husband exceeds the wife in at least one measurement = 800 + 700 - 660 = 840.

Step 3: The remaining cases, where the wife exceeds the husband in both

measurements, = 1000 - 840 = 160.

Answer: 160 cases, as printed in the book.

Q. 2 (c) Given that (A) = (alpha) = (B) = (beta) = n/2, show that (i) (AB) = (alpha beta) and (ii) (A beta) = (alpha B).

Step 1: (AB) + (A beta) = (A) = n/2 and (AB) + (alpha B) = (B) = n/2, so (A beta) = (alpha B). This proves (ii).

Step 2: (alpha beta) + (alpha B) = (alpha) = n/2 and (AB) + (alpha B) = (B) = n/2, so (alpha beta) = (AB). This proves (i).

Answer: Both equalities follow from the four given class frequencies.

Q. 3 (a) Define the consistence of the data.

Step 1: Class frequencies observed in one and the same population are said to be consistent if they conform with one another and do not conflict with each other.

Step 2: No class frequency can ever be negative, so the necessary and sufficient condition for consistence is that no ultimate class frequency should be negative.

Answer: To test consistence, calculate the ultimate class frequencies from the given data; if any one of them is negative the data are inconsistent, and if none is negative the data are consistent.

Q. 3 (b) Find whether the data given below in each case are consistent: (i) n = 120, (A) = 82, (AB) = 90; (ii) n = 50, (A) = 40, (B) = 32, (AB) = 15; (iii) n = 1000, (AB) =

200, (A beta) = 350, (alpha B) = 500.

Step 1 (i): (A beta) = (A) - (AB) = 82 - 90 = -8, which is negative.

Step 2 (ii): (alpha beta) = n - (A) - (B) + (AB) = 50 - 40 - 32 + 15 = -7, which is negative.

Step 3 (iii): (alpha beta) = n - (AB) - (A beta) - (alpha B) = 1000 - 200 - 350 - 500 = -50, which is negative.

Answer: (i), (ii) and (iii) are all NOT consistent, since an ultimate class frequency is negative in each case (-8, -7 and -50 respectively), as printed in the book.

Q. 3 (c) Comment on the following data contained in a report: 100 students appeared in a test of whom 80 passed in Statistics, 70 passed in Mathematics and 48 passed in both the subjects.

Step 1: Let A be passing Statistics and B passing Mathematics, so (A) = 80, (B) = 70, (AB) = 48 and n = 100.

Step 2: (alpha beta) = n - (A) - (B) + (AB) = 100 - 80 - 70 + 48 = -2, which is negative.

Answer: Not consistent, since (alpha beta) = -2, as printed in the book.

Q. 4 (a) What is meant by independence of attributes?

Step 1: Two attributes A and B are independent if the proportion of individuals possessing both attributes equals the product of the proportions possessing each attribute separately.

Step 2: In class frequencies this is (AB) / n = ((A) / n) x ((B) / n), that is (AB) = (A)(B) / n.

Answer: Under independence the 2 x 2 table is completely determined by (A), (B) and n; any departure from (AB) = (A)(B) / n means the attributes are associated.

Q. 4 (b) There are 240 A's and 270 B's in 600 observations. What would be the number of AB if A and B are independent?

Step 1: Under independence (AB) = (A)(B) / n = (240)(270) / 600.

Step 2: (240)(270) = 64800, and 64800 / 600 = 108.

Answer: (AB) = 108, as printed in the book.

Q. 4 (c) If A's are 60% and B's are 40% of the whole number of observations, what must be the percentage of AB's in order that we conclude that A and B are independent?

Step 1: Take n = 100, so (A) = 60 and (B) = 40.

Step 2: Under independence (AB) = (A)(B) / n = (60)(40) / 100 = 24.

Answer: There must be 24% AB's, as printed in the book.

Q. 5 (a) When are two attributes independent, positively associated, negatively associated?

Step 1: The attributes are independent if (AB) = (A)(B) / n; then Yule's coefficient of association Q = 0.

Step 2: They are positively associated (or simply associated) if (AB) > (A)(B) / n; then Q > 0.

Answer: They are negatively associated (or disassociated) if (AB) < (A)(B) / n; then Q < 0. Complete association gives Q = 1 and complete disassociation gives Q = -1.

Q. 5 (b) Given the following data, determine the nature of association between the attributes A and B: (i) (A) = 30, (B) = 60, (AB) = 12, n = 150; (ii) (AB) = 110, (alpha B) = 90, (A beta) = 290, (alpha beta) = 510; (iii) (A) = 415, (AB) = 147, (alpha) = 285, (alpha beta) = 170.

Step 1 (i): Expected (AB) under independence = (30)(60) / 150 = 12, which equals the observed 12.

Step 2 (ii): (A) = 110 + 290 = 400, (B) = 110 + 90 = 200, n = 1000, so expected (AB) = (400)(200) / 1000 = 80, which is less than the observed 110.

Step 3 (iii): (alpha B) = (alpha) - (alpha beta) = 285 - 170 = 115, so (B) = 147 + 115 = 262 and n = 415 + 285 = 700; expected (AB) = (415)(262) / 700 = 155.33, which is greater than the observed 147.

Answer: (i) Independent, (ii) Positively associated, (iii) Negatively associated, as printed in the book.

Q. 6 (a) What is meant by association of attributes?

Step 1: Two attributes A and B are said to be associated if they are not independent, that is if (AB) is not equal to (A)(B) / n.

Step 2: Association is studied from a contingency table of observed frequencies, and its strength is measured by Yule's coefficient Q, which lies between -1 and +1.

Answer: Association means that the presence of one attribute changes the chance of the other attribute being present; it is the attribute analogue of correlation between variables.

Q. 6 (b) Explain the difference between the following with examples: (i) Attribute and variable, (ii) Correlation and association, (iii) Positive association and negative association.

Step 1 (i): An attribute varies only in quality (for example, blind or not blind, married or unmarried), while a variable varies in quantity (for example, height in centimetres or income in rupees).

Step 2 (ii): Correlation measures the linear relationship between two variables and is measured by the correlation coefficient r; association measures the relationship between two attributes and is measured by Q or tested by chi-square.

Answer (iii): Positive association means (AB) > (A)(B) / n, so the attributes tend to occur together (for example, vaccination and exemption from attack would be positively associated if the vaccinated escaped attack); negative association means (AB) < (A)(B) / n, so one attribute tends to exclude the other.

Q. 7 (a) Find the association between injection against typhoid and exemption from attack from the following contingency table: Inoculated - Attacked 528, Not attacked 25; Not inoculated - Attacked 790, Not attacked 175.

Step 1: Let A be inoculated and B be attacked, so (AB) = 528, (A beta) = 25, (alpha

B) = 790, (alpha beta) = 175.

Step 2: Q = ((AB)(alpha beta) - (A beta)(alpha B)) / ((AB)(alpha beta) + (A beta)(alpha B)) = ((528)(175) - (25)(790)) / ((528)(175) + (25)(790)).

Step 3: (528)(175) = 92400 and (25)(790) = 19750, so Q = (92400 - 19750) / (92400 + 19750) = 72650 / 112150 = 0.65.

Answer: Q = 0.65, as printed in the book; with A = inoculated and B = attacked, the positive value means inoculation and attack are positively associated in this table.

Q. 7 (b) Calculate the coefficient of the association between the intelligence of fathers and sons in the following data: Intelligent fathers with intelligent sons = 265; Intelligent fathers with dull sons = 100; Dull fathers with intelligent sons = 95; Dull fathers with dull sons = 450.

Step 1: (AB) = 265, (A beta) = 100, (alpha B) = 95, (alpha beta) = 450.

Step 2: Q = ((265)(450) - (100)(95)) / ((265)(450) + (100)(95)) = (119250 - 9500) / (119250 + 9500).

Step 3: Q = 109750 / 128750 = 0.85.

Answer: Q = 0.85, as printed in the book; intelligence of fathers and sons is strongly positively associated.

Q. 7 (c) Find if there is any association between the tempers of brothers and sisters from the following data: Good natured brothers and good natured sisters = 1230;

Good natured brothers and sullen sisters = 850; Sullen brothers and good natured sisters = 530; Sullen brothers and sullen sisters = 980.

Step 1: (AB) = 1230, (A beta) = 850, (alpha B) = 530, (alpha beta) = 980.

Step 2: Q = ((1230)(980) - (850)(530)) / ((1230)(980) + (850)(530)) = (1205400 - 450500) / (1205400 + 450500).

Step 3: Q = 754900 / 1655900 = 0.46.

Answer: Q = 0.46, as printed in the book; the tempers of brothers and sisters are positively associated.

Q. 8 (a) 750 students appeared in an examination and 470 were successful. 465 had attended classes and 58 of them failed. Calculate the coefficient of association to discuss association between attending classes and success.

Step 1: Let A be attending classes and B success. Attended and successful = 465 - 58 = 407, so (AB) = 407 and (A beta) = 58.

Step 2: Not attended = 750 - 465 = 285; successful among them = 470 - 407 = 63, so (alpha B) = 63 and (alpha beta) = 285 - 63 = 222.

Step 3: Q = ((407)(222) - (58)(63)) / ((407)(222) + (58)(63)) = (90354 - 3654) / (90354 + 3654) = 86700 / 94008 = 0.92.

Answer: Q = 0.92, highly positive association, as printed in the book; attending classes is strongly associated with success.

Q. 8 (b) 100 students appeared in an examination, and 50 failed in Mathematics, 60 failed in Statistics and 40 failed in both. Find if there is any association between the failing in Mathematics and Statistics.

Step 1: Let A be failing Mathematics and B failing Statistics, so (AB) = 40, (A beta) = 50 - 40 = 10, (alpha B) = 60 - 40 = 20, (alpha beta) = 100 - 50 - 60 + 40 = 30.

Step 2: Q = ((40)(30) - (10)(20)) / ((40)(30) + (10)(20)) = (1200 - 200) / (1200 + 200) = 1000 / 1400 = 0.71.

Answer: Q = 0.71, as printed in the book; failing in Mathematics and failing in Statistics are positively associated.

Q. 8 (c) Can vaccination be regarded as preventive measure for small pox from the following data: Of 1482 persons in a locality exposed to small pox, 368 in all were attacked. Of 1482 persons, 343 persons had been vaccinated and of these 35 were attacked.

Step 1: Let A be vaccinated and B attacked, so (AB) = 35, (A beta) = 343 - 35 = 308, (alpha B) = 368 - 35 = 333, (alpha beta) = 1482 - 343 - 333 = 806.

Step 2: Q = ((35)(806) - (308)(333)) / ((35)(806) + (308)(333)) = (28210 - 102564) / (28210 + 102564) = -74354 / 130774 = -0.57.

Answer: Q = -0.57, as printed in the book. The negative association between vaccination and attack means the vaccinated were attacked much less often, so vaccination can be regarded as a preventive measure.

Exercise 15.2 (book pp. 232-235)

Q. 1 (a) Define contingency table and cell frequency. What is a 2 x 2 contingency table?

Step 1: A contingency table is a table of observed frequencies in which the observations are classified according to two criteria, with rows for the categories of one attribute and columns for the categories of the other.

Step 2: The frequency in any one cell of the table, that is the number of observations falling in one particular row and column combination, is the cell frequency; row totals and column totals are the marginal frequencies.

Answer: A 2 x 2 contingency table is the table for two attributes each with two categories, giving the four ultimate class frequencies (AB), (A beta), (alpha B) and (alpha beta), with total n.

Q. 1 (b) In an investigation into eye-colour and left or right handedness of a person, the following results were obtained: Blue - Left 15, Right 85; Brown - Left 20, Right 80. Do these results indicate, at the 5% level of significance, an association between eye colour and left or right handedness?

Step 1: H0: eye colour and handedness are independent; H1: they are associated. Row totals are 100 and 100, column totals are 35 (Left) and 165 (Right), n = 200.

Step 2: Expected frequencies are (100)(35) / 200 = 17.5 for each Left cell and (100)(165) / 200 = 82.5 for each Right cell.

Step 3: With Yates' correction for a 2 x 2 table, adjusted chi-square = sum of (|o - e| - 0.5)^2 / e = (2)^2 / 17.5 + (2)^2 / 82.5 + (2)^2 / 17.5 + (2)^2 / 82.5 = 0.55,

about 0.56 as printed.

Step 4: Degrees of freedom = (2 - 1)(2 - 1) = 1; critical value at 5% is 3.84. Since 0.56 < 3.84, we do not reject H0.

Answer: There is no association between eye colour and left or right handedness, as printed in the book.

Q. 1 (c) An investigation into colour-blindness and sex of a person gave the following results: Male - Colourblind 36, Not colourblind 964; Female - Colourblind 19, Not colourblind 981. Is there evidence, at the 5% level, of an association between the sex of a person and whether or not they are colourblind?

Step 1: H0: sex and colour-blindness are independent; H1: they are associated. Row totals are 1000 and 1000, column totals are 55 and 1945, n = 2000.

Step 2: Expected frequencies are (1000)(55) / 2000 = 27.5 for each Colourblind cell and (1000)(1945) / 2000 = 972.5 for each Not colourblind cell.

Step 3: With Yates' correction, adjusted chi-square = (8)^2 / 27.5 + (8)^2 / 972.5 + (8)^2 / 27.5 + (8)^2 / 972.5 = 4.79 as printed.

Step 4: Degrees of freedom = 1; critical value at 5% is 3.84. Since 4.79 > 3.84, we reject H0.

Answer: There is an association between sex of a person and colour-blindness, as printed in the book.

Q. 2 (a) A driving school examined the results of 100 candidates who were taking their driving test for the first time. They found that out of the 40 men, 28 passed and out of the 60 women, 34 passed. Do these results indicate, at the 5% level of significance, a relationship between the sex of a candidate and the ability to pass first time?

Step 1: Table: Men - Pass 28, Fail 12; Women - Pass 34, Fail 26. Column totals are 62 passed and 38 failed, n = 100.

Step 2: Expected frequencies: Men-Pass (40)(62) / 100 = 24.8, Men-Fail (40)(38) / 100 = 15.2, Women-Pass (60)(62) / 100 = 37.2, Women-Fail (60)(38) / 100 = 22.8.

Step 3: With Yates' correction each |o - e| - 0.5 = 2.7, so adjusted chi-square = (2.7)^2 / 24.8 + (2.7)^2 / 15.2 + (2.7)^2 / 37.2 + (2.7)^2 / 22.8 = 1.29 as printed.

Step 4: Degrees of freedom = 1; critical value at 5% is 3.84. Since 1.29 < 3.84, we do not reject H0.

Answer: There is no relationship between the sex of a candidate and the ability to pass first time, as printed in the book.

Q. 2 (b) Out of 1350 persons, 450 were literate and 600 had traveled beyond the limits of their district, 300 of the literates were among those who had traveled. Find out by calculating (i) coefficient of association, (ii) the value of chi-square, if there is any association between traveling and literacy.

Step 1: Table: Literate and traveled = 300, Literate and not traveled = 150, Illiterate and traveled = 300, Illiterate and not traveled = 600.

Step 2 (i): Q = ((300)(600) - (150)(300)) / ((300)(600) + (150)(300)) = (180000 - 45000) / (180000 + 45000) = 135000 / 225000 = 0.6 as printed.

Step 3 (ii): Expected frequencies: (450)(600) / 1350 = 200, (450)(750) / 1350 = 250, (900)(600) / 1350 = 400, (900)(750) / 1350 = 500.

Step 4: With Yates' correction each |o - e| - 0.5 = 99.5, so adjusted chi-square = (99.5)^2 / 200 + (99.5)^2 / 250 + (99.5)^2 / 400 + (99.5)^2 / 500 = 133.65 as printed.

Step 5: Degrees of freedom = 1; critical value at 5% is 3.84. Since 133.65 > 3.84, we reject H0.

Answer: Q = 0.6 and adjusted chi-square = 133.65; there is an association between traveling and literacy, as printed in the book.

Q. 3 (a) The following are the data on a random sample of 150 chickens, divided into two groups according to breed and into three groups according to yield of eggs: Rhode Red - High 46, Medium 29, Low 28; Leghorn White - High 27, Medium 14, Low 6. Are these data consistent with the hypothesis that yield is not affected by the type of breed?

Step 1: H0: yield is not affected by breed (independence); H1: yield is affected by breed. Row totals are 103 and 47, column totals are 73, 43 and 34, n = 150.

Step 2: Expected frequencies: Rhode Red - 50.13, 29.53, 23.35; Leghorn White - 22.87, 13.47, 10.65.

Step 3: Chi-square = sum of (o - e)^2 / e = 4.07 as printed.

Step 4: Degrees of freedom = (2 - 1)(3 - 1) = 2; critical value at 5% is 5.99. Since 4.07 < 5.99, we do not reject H0.

Answer: There is no association between chicken breed and yield of eggs, as printed in the book.

Q. 3 (b) The students of a college took three courses: arts, commerce and science. The students were classified according to sex: Male - Arts 200, Commerce 300, Science 100; Female - Arts 100, Commerce 200, Science 100. Use chi-square test whether there is any association between sex and choice of course of study.

Step 1: H0: sex and choice of course are independent; H1: they are associated. Row totals are 600 and 400, column totals are 300, 500 and 200, n = 1000.

Step 2: Expected frequencies: Male - 180, 300, 120; Female - 120, 200, 80.

Step 3: Chi-square = sum of (o - e)^2 / e = 13.89 as printed (13.888).

Step 4: Degrees of freedom = (2 - 1)(3 - 1) = 2; critical value at 5% is 5.99. Since 13.89 > 5.99, we reject H0.

Answer: There is an association between sex and course of study, as printed in the book.

Q. 4 (a) The following table shows liking of three colours: pink, white and blue in samples of males and females: Pink - Male 20, Female 40; White - Male 40, Female

20; Blue - Male 60, Female 20. Test whether there is any relation between sex and colour.

Step 1: H0: sex and liking of colours are independent; H1: they are associated. Row totals are 60, 60 and 80, column totals are 120 and 80, n = 200.

Step 2: Expected frequencies: Pink - 36, 24; White - 36, 24; Blue - 48, 32.

Step 3: Chi-square = sum of (o - e)^2 / e = 26.39 as printed (26.3889).

Step 4: Degrees of freedom = (3 - 1)(2 - 1) = 2; critical value at 5% is 5.99. Since 26.39 > 5.99, we reject H0.

Answer: There is a relation between sex and liking of colours, as printed in the book.

Q. 4 (b) The following table gives the condition at home and condition of the children: Clean - Clean 175, Not clean 143; Fairly clean - Clean 136, Not clean 116; Dirty - Clean 125, Not clean 145. Test for the association between the conditions at home and condition of children.

Step 1: H0: condition at home and condition of children are independent; H1: they are associated. Row totals are 318, 252 and 270, column totals are 436 and 404, n = 840.

Step 2: Expected frequencies: Clean - 165.06, 152.94; Fairly clean - 130.80, 121.20; Dirty - 140.14, 129.86.

Step 3: Chi-square = sum of (o - e)^2 / e = 5.08 by exact recomputation; the book prints 5.027, a small arithmetic difference that does not change the decision.

Step 4: Degrees of freedom = (3 - 1)(2 - 1) = 2; critical value at 5% is 5.99. Since 5.08 < 5.99, we do not reject H0.

Answer: There is no association between conditions at home and condition of children; the decision agrees with the book.

Q. 5 (a) The table given below shows the relation between the performance of students in economics and statistics. Test the hypothesis that the performance in economics is independent of the performance in statistics using 5% level of significance: Economics High - Statistics High 56, Medium 96, Low 28; Medium - High 48, Medium 168, Low 24; Low - High 16, Medium 86, Low 78.

Step 1: H0: performance in economics is independent of performance in statistics; H1: they are associated. Row totals are 180, 240 and 180, column totals are 120, 350 and 130, n = 600.

Step 2: Expected frequencies: High - 36, 105, 39; Medium - 48, 140, 52; Low - 36, 105, 39.

Step 3: Chi-square = sum of (o - e)^2 / e = 89.21 as printed (89.2112).

Step 4: Degrees of freedom = (3 - 1)(3 - 1) = 4; critical value at 5% is 9.49. Since 89.21 > 9.49, we reject H0.

Answer: There is an association between the performance of students in economics and statistics, as printed in the book.

Q. 5 (b) A thousand households are taken at random and divided into three groups

A, B and C, according to the total weekly income. The following table shows the numbers in each group having a colour television receiver, a black and white receiver, or no television at all: Colour television - A 56, B 51, C 93; Black and white - A 118, B 207, C 375; None - A 26, B 42, C 32. Calculate the expected frequencies if there is no association between total income and television ownership. Apply a test to find whether the observed frequencies suggest that there is such an association.

Step 1: Row totals are 200, 700 and 100; column totals are 200, 300 and 500; n = 1000.

Step 2: Expected frequencies: Colour television - 40, 60, 100; Black and white - 140, 210, 350; None - 20, 30, 50.

Step 3: Chi-square = sum of (o - e)^2 / e = 26.61 as printed (26.6).

Step 4: Degrees of freedom = (3 - 1)(3 - 1) = 4; critical value at 5% is 9.49. Since 26.61 > 9.49, we reject H0.

Answer: There is an association between television type and income group, as printed in the book.

Q. 6 (a) A random sample of 200 married men, all retired, were classified according to education and number of children as indicated below: Elementary - 0-1: 13, 2-3: 37, Over 3: 35; Secondary - 0-1: 19, 2-3: 42, Over 3: 14; College - 0-1: 12, 2-3: 17, Over 3: 11. Test the hypothesis, at 5% of significance, that the size of family is independent of the level of education attained by the father.

Step 1: H0: size of family is independent of the father's education; H1: they are associated. Row totals are 85, 75 and 40, column totals are 44, 96 and 60, n = 200.

Step 2: Expected frequencies: Elementary - 18.70, 40.80, 25.50; Secondary - 16.50, 36.00, 22.50; College - 8.80, 19.20, 12.00.

Step 3: Chi-square = sum of (o - e)^2 / e = 11.72 as printed (11.7194).

Step 4: Degrees of freedom = (3 - 1)(3 - 1) = 4; critical value at 5% is 9.49. Since 11.72 > 9.49, we reject H0.

Answer: There is an association between education and number of children, as printed in the book.

Q. 6 (b) A survey of 200 families known to be regular television viewers was undertaken. They were asked which of the three television channels they watched most during an average week: PTV 1 - North 29, East 16, South 42, West 23; PTV 2 - North 6, East 11, South 26, West 7; STN - North 15, East 3, South 12, West 10. Test the hypothesis that there is no association between the channel watched most and the region.

Step 1: H0: channel watched and region are independent; H1: they are associated. Row totals are 110, 50 and 40, column totals are 50, 30, 80 and 40, n = 200.

Step 2: Expected frequencies: PTV 1 - 27.50, 16.50, 44.00, 22.00; PTV 2 - 12.50, 7.50, 20.00, 10.00; STN - 10.00, 6.00, 16.00, 8.00.

Step 3: Chi-square = sum of (o - e)^2 / e = 13.45 as printed (13.446).

Step 4: Degrees of freedom = (3 - 1)(4 - 1) = 6; critical value at 5% is 12.59. Since 13.45 > 12.59, we reject H0.

Answer: There is an association between the channel and region, as printed in the book.

Q. 6 (c) From the following table showing the number of employees and condition of factory premises: A1 - Under 50: 84, 51-150: 133, 151-250: 49, Over 250: 62; A2 - Under 50: 87, 51-150: 82, 151-250: 20, Over 250: 25; A3 - Under 50: 26, 51-150: 9, 151-250: 9, Over 250: 5. Discuss the association between the condition of premises and the number of persons employed. Compute the coefficient of contingency.

Step 1: H0: condition of premises and number of persons employed are independent; H1: they are associated. Row totals are 328, 214 and 49, column totals are 197, 224, 78 and 92, n = 591.

Step 2: Expected frequencies: A1 - 109.33, 124.32, 43.29, 51.06; A2 - 71.33, 81.11, 28.24, 33.31; A3 - 16.33, 18.57, 6.47, 7.63.

Step 3: Chi-square = sum of (o - e)^2 / e = 30.06 as printed.

Step 4: Degrees of freedom = (3 - 1)(4 - 1) = 6; critical value at 5% is 12.59. Since 30.06 > 12.59, we reject H0.

Step 5: Pearson's coefficient of mean square contingency C = sqrt(chi-square / (n + chi-square)) = sqrt(30.06 / (591 + 30.06)) = sqrt(30.06 / 621.06) = 0.22 as printed.

Answer: There is an association between the condition of premises and the number of persons employed, with C = 0.22.

Exercise 15.3 (book pp. 237-238; the book prints no Q. 2 in this exercise)

Q. 1 (a) What is rank correlation?

Step 1: Rank correlation is the correlation between the ranks of individuals for two variables X and Y, instead of their measured values.

Step 2: When the observations are ranks, Spearman's coefficient of rank correlation is r_r = 1 - (6 sum of d_i squared) / (n (n squared - 1)), where d_i is the difference of the two ranks of the i-th individual.

Answer: It always lies between -1 and +1, and it is a nonparametric procedure that is useful when the basic data are not available in numerical form but only as ranks.

Q. 1 (b) The following table shows how 10 students, arranged in alphabetical order, were ranked according to their achievements in both laboratory and lecture portions of a statistics course. Find the coefficient of mark correlation: Laboratory ranks 8, 3, 9, 2, 7, 10, 4, 6, 1, 5; Lecture ranks 9, 5, 10, 1, 8, 7, 3, 4, 2, 6.

Step 1: Differences d = Laboratory rank - Lecture rank: -1, -2, -1, 1, -1, 3, 1, 2, -1, -1.

Step 2: Squares of d: 1, 4, 1, 1, 1, 9, 1, 4, 1, 1; sum of d squared = 24.

Step 3: r_r = 1 - (6)(24) / (10)(100 - 1) = 1 - 144 / 990 = 1 - 0.1455 = 0.8545.

Answer: r_r = 0.8545, as printed in the book.

Q. 1 (c) The ranks of the same 10 students in Mathematics and Economics were as follows: (1, 6); (2, 5); (3, 1); (4, 4); (5, 2); (6, 7); (7, 8); (8, 10); (9, 3); (10, 9); the two numbers within brackets denoting the ranks of the same students in

Mathematics and Economics respectively. Calculate the rank correlation coefficient for proficiencies of this group in two subjects.

Step 1: Differences d = Mathematics rank - Economics rank: -5, -3, 2, 0, 3, -1, -1, -2, 6, 1.

Step 2: Squares of d: 25, 9, 4, 0, 9, 1, 1, 4, 36, 1; sum of d squared = 90.

Step 3: $r_r = 1 - (6)(90) / (10)(100 - 1) = 1 - 540 / 990 = 1 - 0.5455 = 0.4545$.

Answer: $r_r = 0.45$, as printed in the book (0.4545 rounded).

Q. 3 (a) Five sacks of coal A, B, C, D and E have different weights, with A being heavier than B, B being heavier than C, and so on. A weight lifter ranks the sacks (heaviest first) in the order A, D, B, E, C. Calculate a coefficient of rank correlation.

Step 1: True ranks (heaviest first): A 1, B 2, C 3, D 4, E 5. The lifter's ranks: A 1, B 3, C 5, D 2, E 4.

Step 2: Differences d = true rank - lifter's rank: A 0, B -1, C -2, D 2, E 1; squares 0, 1, 4, 4, 1; sum of d squared = 10.

Step 3: $r_r = 1 - (6)(10) / (5)(25 - 1) = 1 - 60 / 120 = 1 - 0.5 = 0.5$.

Answer: $r_r = 0.5$, as printed in the book.

Q. 3 (b) Seven army recruits A, B, C, D, E, F and G were given two separate aptitude tests. Their orders of merit in each test were: First test - G, F, A, D, B, C, E; Second test - D, F, E, B, G, C, A. Find Spearman's coefficient of rank correlation between the

two orders and comment briefly on the correlation obtained.

Step 1: First-test ranks: A 3, B 5, C 6, D 4, E 7, F 2, G 1. Second-test ranks: A 7, B 4, C 6, D 1, E 3, F 2, G 5.

Step 2: Differences d = first - second: A -4, B 1, C 0, D 3, E 4, F 0, G -4; squares 16, 1, 0, 9, 16, 0, 16; sum of d squared = 58.

Step 3: $r_r = 1 - (6)(58) / (7)(49 - 1) = 1 - 348 / 336 = 1 - 1.0357 = -0.0357$.

Answer: $r_r = -0.036$, very little negative correlation, as printed in the book.

Q. 4 (a) Ten competitors in a beauty contest are ranked by three judges in the following order: Competitor A-J, Judge X: 1, 6, 5, 10, 3, 2, 4, 9, 7, 8; Judge Y: 3, 5, 8, 4, 7, 10, 2, 1, 6, 9; Judge Z: 6, 4, 9, 8, 1, 2, 3, 10, 5, 7. Use Spearman's rank correlation coefficient to discuss which pair of judges have the nearest approach to common tastes in beauty.

Step 1: For judges X and Y, differences d: -2, 1, -3, 6, -4, -8, 2, 8, 1, -1; squares 4, 1, 9, 36, 16, 64, 4, 64, 1, 1; sum = 200; $r_{xy} = 1 - (6)(200) / (10)(99) = 1 - 1200 / 990 = -0.21$.

Step 2: For judges Y and Z, differences d: -3, 1, -1, -4, 6, 8, -1, -9, 1, 2; squares 9, 1, 1, 16, 36, 64, 1, 81, 1, 4; sum = 214; $r_{yz} = 1 - (6)(214) / 990 = 1 - 1284 / 990 = -0.30$.

Step 3: For judges X and Z, differences d: -5, 2, -4, 2, 2, 0, 1, -1, 2, 1; squares 25, 4, 16, 4, 4, 0, 1, 1, 4, 1; sum = 60; $r_{xz} = 1 - (6)(60) / 990 = 1 - 360 / 990 = 0.64$.

Answer: r_xy = -0.21, r_yz = -0.30, r_xz = 0.64, as printed in the book. Judges X and Z have the nearest approach to common tastes in beauty, since their rank correlation is the largest positive value.

Q. 4 (b) The following table shows the grade point average awarded to six children in a competition by two different judges: Child A-F, Judge X: 6.8, 7.3, 8.1, 9.8, 7.1, 9.2; Judge Y: 7.8, 9.4, 7.9, 9.6, 8.9, 6.9. Calculate coefficient of rank correlation by Spearman's formula.

Step 1: Rank the averages from lowest to highest. Judge X ranks: A 1, B 3, C 4, D 6, E 2, F 5. Judge Y ranks: A 2, B 5, C 3, D 6, E 4, F 1.

Step 2: Differences d = X rank - Y rank: A -1, B -2, C 1, D 0, E -2, F 4; squares 1, 4, 1, 0, 4, 16; sum of d squared = 26.

Step 3: r_r = 1 - (6)(26) / (6)(36 - 1) = 1 - 156 / 210 = 1 - 0.7429 = 0.2571.

Answer: r_r = 0.26, as printed in the book.

Q. 4 (c) The following table shows the marks of six candidates in two subjects: Candidate A-F, Mathematics: 38, 62, 56, 42, 59, 48; Statistics: 64, 89, 84, 60, 73, 69. (i) Calculate the coefficient of rank correlation. (ii) Comment on the value of your result.

Step 1: Mathematics ranks (lowest to highest): A 1, B 6, C 4, D 2, E 5, F 3. Statistics ranks: A 2, B 6, C 5, D 1, E 4, F 3.

Step 2: Differences d = Mathematics rank - Statistics rank: A -1, B 0, C -1, D 1, E 1, F 0; squares 1, 0, 1, 1, 1, 0; sum of d squared = 4.

Step 3: r_r = 1 - (6)(4) / (6)(36 - 1) = 1 - 24 / 210 = 1 - 0.1143 = 0.8857.

Answer: (i) 0.886, as printed in the book. (ii) High positive correlation: candidates who rank high in Mathematics also rank high in Statistics.

Exercise 15.4 — Objective Questions, Q. 1: Fill in the blanks (book pp. 238-239; book skips labels (xviii), (xix), (xxi)-(xxiii) in print)

Q. 1 (i) A characteristic which varies in quantity from one individual to another is called a variable

Q. 1 (ii) A characteristic which varies in quality from one individual to another is called an attribute

Q. 1 (iii) The observations made on objects regarding an attribute are called qualitative data.

Q. 1 (iv) Dichotomy is a process of dividing the objects into two mutually exclusive classes of an attribute.

Q. 1 (v) The degree of linear relationship between the two variables is called correlation

Q. 1 (vi) The degree of relationship between the two attributes is called association

Q. 1 (vii) The two attributes A and B are independent , if (AB) = (A)(B) / n.

Q. 1 (viii) The two attributes A and B are associated , if (AB) is not equal to (A)(B) / n.

Q. 1 (ix) The two attributes A and B are positively associated, if (AB) > (A)(B) / n.

Q. 1 (x) The two attributes A and B are negatively associated, if (AB) < (A)(B) / n.

Q. 1 (xi) The coefficient of association, denoted by Q, is a measure of association between the two attributes

Q. 1 (xii) If the coefficient of association equals 0, the two attributes A and B are

independent

Q. 1 (xiii) If the coefficient of association is not equal to 0, the two attributes A and B are associated

Q. 1 (xiv) If the coefficient of association equals -1, the two attributes A and B are completely dissociated

Q. 1 (xv) If the coefficient of association equals 1, the two attributes A and B are completely associated

Q. 1 (xvi) A contingency table consisting of r rows and c columns is made up of the observed frequencies relative to two attributes and their categories.

Q. 1 (xvii) The two attributes are said to be independent , if for every cell of a contingency table the observed frequency o-i-j is equal to expected frequency e-i-j.

Q. 1 (xx, as printed) For an r x c contingency table, the chi-squared-statistic has degrees of freedom v = (r - 1)(c - 1)

Q. 1 (xxiv, as printed) The larger are the difference between the observed and expected frequencies, the larger will be the value of chi-squared which leads to the rejection of H0 of independence.

Q. 1 (xxv, as printed) The rejection of H0 of independence indicates that the two criteria of classification are associated

Q. 1 (xxvi, as printed) In a chi-square test for independence, no expected frequency should be less than 5.

Exercise 15.4 — Objective Questions, Q. 2: True or False, remaining items (book pp. 239-240; 6 items already in the earlier block above)

Q. 2 (i) A characteristic which varies in quantity from one individual to another is called an attribute.

False

Q. 2 (ii) The quantitative data relating to an attribute may be obtained simply by

noting its presence or absence in the objects.

True

Q. 2 (iii) The presence of attributes is denoted by capital Latin letters and their absence by Greek or small letters.

True

Q. 2 (iv) The class frequencies of the highest order are called ultimate class frequencies.

True

Q. 2 (vi) The two attributes A and B are positively associated, if (AB) < (A)(B) / n.

False

Q. 2 (vii) The coefficient of correlation, denoted by r, is a measure of the strength of linear relationship between two variables.

True

Q. 2 (x) A contingency table consisting of r rows and c columns is made up of the observed frequencies relative to two attributes and their categories.

True

Q. 2 (xi) The disassociation of two attributes means their independence.

False

Q. 2 (xiv) The larger are the differences between the observed and expected frequencies, the larger will be the value of chi-squared which leads to rejection of H0 of independence.

True

Chapter 16: Analysis of Time Series

Exercise — Short Questions

What is a time series? What is secular trend?

A time series is a set of observations recorded over time.

Secular trend is the long-term movement of the series.

It is measured by methods such as semi-averages, moving averages and least squares.

Exercise — Multiple Choice (book p. 268, Q. 3)

The graph of a time series is called

A. histogram

B. polygon

C. straight line

✓ D. historgram

The secular trend is measured by the method of semi-averages when

A. time series contains yearly values

✓ B. trend is linear

C. time series contains odd number of values

D. none of them

In the measurement of secular trend the moving averages

A. give the trend in a straight line

B. measure the seasonal variations

✓ C. smooth out a time series

D. none of them

For a least squares linear trend y-hat = a + b x, the b is the

A. variable

B. intercept

C. trend

✓ D. slope

For a least squares linear trend y-hat = a + b x, the sum of y-hat equals

A. sum of y less than sum of y-hat

B. 0

✓ C. sum of y

D. none of them

For a least squares linear trend, the sum of (y minus y-hat) squared = 0 when

✓ A. all the y-values lie on the line

B. all the y-values are positive

C. all the y-values lie above the line

D. none of them

Exercise 16.1 (book pp. 262-266)

Q. 1 (a) What is meant by a time series? What are different movements that may be present in a time series? Describe each of them carefully.

A time series is the sequence y_1, y_2, ..., y_n of observations of a variable Y recorded in accordance with their time of occurrence t_1, t_2, ..., t_n.

The four movements that may be present in a time series are: (i) Secular trend (T) - the long-term, steady, gradual movement in one direction, upward or downward, showing the general increase or decrease over a fairly long period. (ii) Seasonal variations (S) - short-term movements that recur regularly during a period of one year or less, with the same pattern and similar timing, for example increased sales of cold drinks in summer. (iii) Cyclical fluctuations (C) - long-term oscillations about the trend, recurring over periods longer than a year, with the four phases trough, expansion (recovery), peak (boom or prosperity) and recession (contraction); they are the business cycle. (iv) Irregular movements (I) - unpredictable, non-recurring, accidental changes caused by random events such as wars, floods, earthquakes, strikes or fires; they are also called erratic or random variations.

Q. 1 (b) Explain the difference between histogram and historgram.

A histogram is a graphic representation of a frequency distribution: the values of a variable are shown on the horizontal axis and their frequencies by bars, so it shows how observations are distributed at one time.

A historgram is a graphic representation of a time series: time is taken along the x-axis as the independent variable, the observed values of the variable are plotted as the dependent variable, and the plotted points are joined by line segments to show the movement of the series over time.

In short, a histogram displays a distribution, while a historgram displays the history (movement over time) of a series.

Q. 1 (c) Describe the following terms: (i) Secular trend, (ii) Seasonal variations, (iii) Cyclical fluctuations, (iv) Irregular movements.

(i) Secular trend: a line or curve that shows the general tendency of a time series to increase or decrease over a long period; it is the smooth long-term movement after short-term fluctuations are removed.

(ii) Seasonal variations: variations that repeat with a fixed periodicity, such as daily, weekly, monthly or yearly, and are caused by seasonal factors such as weather, festivals or customs.

(iii) Cyclical fluctuations: periodic up-and-down movements about the trend lasting more than a year, less predictable than seasonal variations and more dangerous to business because of their longer duration and varying amplitude.

(iv) Irregular movements: sudden, unpredictable changes that do not follow any pattern and are caused by chance events such as floods, earthquakes, wars or strikes.

Q. 2 (a) Describe various methods of measuring secular trend in a time series. Discuss the merits and demerits of the methods of smoothing the data.

The secular trend is measured by: (i) the method of free hand curve - plot the series

and draw a smooth curve through it by eye; (ii) the method of semi-averages - divide the series into two equal halves, average each half, plot the two average points and join them by a straight line; (iii) the method of moving averages - average successive overlapping groups of observations (for example 3-year or 4-year moving averages, centred when the period is even) to smooth out fluctuations; (iv) the method of least squares - fit the line y-hat = a + b x (or a parabola y-hat = a + b x + c x squared) that minimises the sum of squared deviations.

Merits of smoothing methods: the free hand method is simple, easy and quick and gives a close approximation; the semi-average method is simple, easy, quick and objective and smooths out seasonal variations; the moving average method is easy and simple, eliminates periodic fluctuations and can eliminate cyclical and seasonal fluctuations.

Demerits: the free hand method is rough and crude, is greatly affected by personal bias and gives no reliable estimates; the semi-average method uses the arithmetic mean, which is highly affected by extreme values, and can only be applied when the trend is linear; the moving average method gives no trend values at the beginning and end of the series, is highly affected by extreme values, provides no mathematical equation for the trend, makes forecasting subjective, and a wrongly chosen period may generate cycles.

Q. 2 (b) Plot the original time series to obtain a historgram. Year: 1982-1992; Value: 50.0, 36.5, 43.0, 44.5, 38.9, 38.1, 32.6, 38.7, 41.7, 41.1, 33.8. Draw a free-hand trend of the following data on the same graph paper.

Step 1: Take years along the x-axis and values along the y-axis on a suitable scale, and plot the eleven points (1982, 50.0), (1983, 36.5), (1984, 43.0), (1985, 44.5), (1986, 38.9), (1987, 38.1), (1988, 32.6), (1989, 38.7), (1990, 41.7), (1991, 41.1), (1992, 33.8).

Step 2: Join the plotted points by line segments; this is the historgram of the series, showing a fall from 50.0 to 36.5, a rise to 44.5, a fall to 32.6 and a partial recovery.

Step 3: Draw a smooth free-hand curve through the middle of the fluctuations, ignoring short ups and downs, so that it follows the general downward drift of the series.

Answer: The free-hand trend is a smooth curve sloping gently downward from about 50 in 1982 to about 34-38 in 1992; being drawn by eye, it has no single numerical equation, which is the known limitation of the free-hand method.

Q. 3 (a) What do you understand by the method of semi-averages utilized for smoothing of a time series? Give an example.

The method of semi-averages divides the observed values of a time series into two equal parts, averages each part, and places each average against the mid-point of its part; the straight line through the two plotted average points is the trend line. If the number of observations is odd, the middle value is either omitted or included in each half; if it is even, the two halves are simply the first half and the second half of the series.

Example (book Example 16.3, property damaged by road accidents in Punjab, 1973-1979): the first three values 201, 238, 392 average to 831 / 3 = 277, placed at 1974; the last three values 484, 649, 742 average to 1875 / 3 = 625, placed at 1978. The slope is (625 - 277) / (1978 - 1974) = 348 / 4 = 87, and the trend line is y-hat = 190 + 87 x with origin at 1973, since 277 - 87 x 1 = 190.

Q. 3 (b) The following table shows the property damaged by road accidents in Punjab for the years 1972 to 1982: 1972: 213, 1973: 203, 1974: 238, 1975: 392, 1976: 507, 1977: 441, 1978: 649, 1979: 473, 1980: 342, 1981: 365, 1982: 330.

Using the method of semi-averages, find the linear trend.

Step 1: Divide the eleven years into two halves of five, leaving the middle year 1977 aside for the averages: first half 1972-1976 values 213, 203, 238, 392, 507 sum to 1553, semi-average = 1553 / 5 = 310.6, placed at the middle year 1974.

Step 2: Second half 1978-1982 values 649, 473, 342, 365, 330 sum to 2159, semi-average = 2159 / 5 = 431.8, placed at the middle year 1980.

Step 3: Slope b = (431.8 - 310.6) / (1980 - 1974) = 121.2 / 6 = 20.2.

Step 4: With origin at 1972, the intercept a = 310.6 - 20.2 x 2 = 310.6 - 40.4 = 270.2.

Answer: y-hat = 270.2 + 20.2 x with origin at 1972, as printed in the book.

Q. 3 (c) The following table gives the number of books (in 000's) sold at a book stall for the year 1970 to 1981: 1970: 15, 1971: 18, 1972: 17, 1973: 42, 1974: 38, 1975: 40, 1976: 25, 1977: 20, 1978: 20, 1979: 16, 1980: 19, 1981: 17. Using semi-average method, find the trend line. Compute the trend values.

Step 1: First half 1970-1975 values 15, 18, 17, 42, 38, 40 sum to 170, semi-average = 170 / 6 = 28.33, placed at the middle of the first half, x = 2.5 years after the 1970 origin.

Step 2: Second half 1976-1981 values 25, 20, 20, 16, 19, 17 sum to 117, semi-average = 117 / 6 = 19.5, placed at x = 8.5.

Step 3: Slope b = (19.5 - 28.33) / (8.5 - 2.5) = -8.83 / 6 = -1.47.

Step 4: Intercept a = 28.33 + 1.47 x 2.5 = 32.01, so y-hat = 32.01 - 1.47 x with origin at 1970, as printed in the book.

Answer: Trend values for 1970-1981 (x = 0 to 11) are 32.01, 30.54, 29.07, 27.60, 26.13, 24.66, 23.19, 21.72, 20.25, 18.78, 17.31, 15.84.

Q. 4 (a) What are moving averages? How is a time series smoothed by moving average method? Give an example.

A moving average is the average of a fixed number of successive observations, recomputed by dropping the first observation of the group and including the next one, so that a sequence of averages is obtained.

A time series is smoothed by moving averages because each average removes short-term and periodic fluctuations: when the period of the moving average matches the periodicity of the series, the fluctuations are nearly eliminated and the averages show the trend.

When the period k is odd, each average is placed against the middle of its group; when k is even, the averages are first placed between periods and then centred by averaging successive pairs.

Example (book Example 16.5, production of silver utensils): the first 3-year moving total is 170.0 + 154.8 + 156.5 = 481.3, giving the moving average 481.3 / 3 = 160.43 placed at 1971; the next is 154.8 + 156.5 + 158.9 = 470.2, giving 156.73 placed at 1972, and so on.

Q. 4 (b) Draw a historgram of the following time series. Determine a trend line by a simple moving averages of 5-year from the following data: Year 1921-1930; Value: 102, 108, 130, 140, 158, 180, 196, 210, 220, 230.

Step 1: Plot the ten points on graph paper with years on the x-axis and values on the y-axis and join them; this is the historgram, showing a steadily rising series.

Step 2: 5-year moving totals: 102 + 108 + 130 + 140 + 158 = 638; 108 + 130 + 140 + 158 + 180 = 716; 130 + 140 + 158 + 180 + 196 = 804; 140 + 158 + 180 + 196 + 210 = 884; 158 + 180 + 196 + 210 + 220 = 964; 180 + 196 + 210 + 220 + 230 = 1036.

Step 3: Divide each total by 5 to get the 5-year moving averages, placed at the middle year of each group: 638 / 5 = 127.6 at 1923; 716 / 5 = 143.2 at 1924; 804 / 5 = 160.8 at 1925; 884 / 5 = 176.8 at 1926; 964 / 5 = 192.8 at 1927; 1036 / 5 = 207.2 at 1928.

Answer: The 5-year moving averages are (127.6, 143.2, 160.8, 176.8, 192.8, 207.2), as printed in the book; plotting them on the same graph gives the smoothed trend line.

Q. 5 (a) Calculate 7-day moving averages for the following record of attendances: Week 1 - Sun 24, Mon 50, Tues 30, Wed 48, Thurs 54, Fri 55, Sat 62; Week 2 - Sun 28, Mon 52, Tues 41, Wed 42, Thurs 50, Fri 41, Sat 42. Plot the given data and moving averages on the same graph.

Step 1: Write the fourteen daily attendances in order: 24, 50, 30, 48, 54, 55, 62, 28, 52, 41, 42, 50, 41, 42.

Step 2: 7-day moving totals: 24 + 50 + 30 + 48 + 54 + 55 + 62 = 323; then drop the first day and add the next, giving 327, 329, 340, 334, 330, 316, 296.

Step 3: Divide each total by 7: 323 / 7 = 46.14; 327 / 7 = 46.71; 329 / 7 = 47.00; 340 / 7 = 48.57; 334 / 7 = 47.71; 330 / 7 = 47.14; 316 / 7 = 45.14; 296 / 7 = 42.29.

Answer: The 7-day moving averages are (46.14, 46.71, 47.00, 48.57, 47.71, 47.14, 45.14, 42.29), as printed in the book; plotted on the same graph they smooth the daily ups and downs into the trend.

Q. 5 (b) The following table shows the United States average monthly production of bituminous coal in millions of short tons for the years 1981-91: 1981: 50.0, 1982: 36.5, 1983: 43.0, 1984: 44.5, 1985: 38.9, 1986: 38.1, 1987: 38.7, 1988: 32.6, 1989: 41.1, 1990: 41.7, 1991: 33.8. Construct (i) 4-year moving averages (ii) 4-year centred moving averages.

Step 1 (i): 4-year moving totals divided by 4: (50.0 + 36.5 + 43.0 + 44.5) / 4 = 174.0 / 4 = 43.50; (36.5 + 43.0 + 44.5 + 38.9) / 4 = 162.9 / 4 = 40.73; (43.0 + 44.5 + 38.9 + 38.1) / 4 = 164.5 / 4 = 41.12; (44.5 + 38.9 + 38.1 + 38.7) / 4 = 160.2 / 4 = 40.05; (38.9 + 38.1 + 38.7 + 32.6) / 4 = 148.3 / 4 = 37.08; (38.1 + 38.7 + 32.6 + 41.1) / 4 = 150.5 / 4 = 37.62; (38.7 + 32.6 + 41.1 + 41.7) / 4 = 154.1 / 4 = 38.53; (32.6 + 41.1 + 41.7 + 33.8) / 4 = 149.2 / 4 = 37.30.

Step 2 (ii): Centre each successive pair of 4-year averages: (43.50 + 40.73) / 2 = 42.11; (40.73 + 41.12) / 2 = 40.92; (41.12 + 40.05) / 2 = 40.59; (40.05 + 37.08) / 2 = 38.56; (37.08 + 37.62) / 2 = 37.35; (37.62 + 38.53) / 2 = 38.08; (38.53 + 37.30) / 2 = 37.91.

Answer: (i) 4-year moving averages: 43.50, 40.73, 41.12, 40.05, 37.08, 37.62, 38.53, 37.30. (ii) 4-year centred moving averages: 42.11, 40.92, 40.59, 38.56, 37.35, 38.08, 37.91. The book prints these rounded to one decimal place as (43.5, 40.7, 41.1, 40.0, 37.1, 37.6, 38.6, 37.3) and (42.1, 40.9, 40.6, 38.6, 37.4, 38.1, 38.0); its 38.6 and 38.0 follow from rounding the 4-year average 38.53 up to 38.6.

Q. 6 (a) Compute 4-month centred moving averages from the following: Month Jan-Oct; Value: 23, 26, 28, 30, 31, 35, 37, 32, 34, 38.

Step 1: 4-month moving totals divided by 4: (23 + 26 + 28 + 30) / 4 = 107 / 4 = 26.75; (26 + 28 + 30 + 31) / 4 = 115 / 4 = 28.75; (28 + 30 + 31 + 35) / 4 = 124 / 4 = 31.00; (30 + 31 + 35 + 37) / 4 = 133 / 4 = 33.25; (31 + 35 + 37 + 32) / 4 = 135 / 4 = 33.75; (35 + 37 + 32 + 34) / 4 = 138 / 4 = 34.50; (37 + 32 + 34 + 38) / 4 = 141 / 4 = 35.25.

Step 2: Centre successive pairs: (26.75 + 28.75) / 2 = 27.75; (28.75 + 31.00) / 2 = 29.88; (31.00 + 33.25) / 2 = 32.12; (33.25 + 33.75) / 2 = 33.50; (33.75 + 34.50) / 2 = 34.12; (34.50 + 35.25) / 2 = 34.88.

Answer: The 4-month centred moving averages are (27.75, 29.88, 32.12, 33.50, 34.12, 34.88), as printed in the book.

Q. 6 (b) Find 4-quarter centred moving averages for the following data: 1948 - Quarter I 71, II 72, III 78, IV 84; 1949 - I 72, II 69, III 75, IV 79; 1950 - I 73, II 80, III 85, IV 86. Plot the original data and the trend values on a graph.

Step 1: Write the twelve quarterly values in order: 71, 72, 78, 84, 72, 69, 75, 79, 73, 80, 85, 86.

Step 2: 4-quarter moving averages: 305 / 4 = 76.25; 306 / 4 = 76.50; 303 / 4 = 75.75; 300 / 4 = 75.00; 295 / 4 = 73.75; 296 / 4 = 74.00; 307 / 4 = 76.75; 317 / 4 = 79.25; 324 / 4 = 81.00.

Step 3: Centre successive pairs: (76.25 + 76.50) / 2 = 76.38; (76.50 + 75.75) / 2 = 76.12; (75.75 + 75.00) / 2 = 75.38; (75.00 + 73.75) / 2 = 74.38; (73.75 + 74.00) / 2

= 73.88; (74.00 + 76.75) / 2 = 75.38; (76.75 + 79.25) / 2 = 78.00; (79.25 + 81.00) / 2 = 80.12.

Answer: The 4-quarter centred moving averages are (76.38, 76.12, 75.38, 74.38, 73.88, 75.38, 78.00, 80.12), as printed in the book; plotted with the original quarterly data they show the trend.

Exercise 16.1 (book pp. 262-266)

Q. 7 (a) Explain the method of least squares utilized for finding a secular trend in a time series.

The method of least squares fits the trend line y-hat = a + b x (or the parabola y-hat = a + b x + c x squared) for which the sum of the squares of the deviations of the observed values y from the trend values y-hat is the least.

For a straight line the normal equations are: sum y = n a + b sum x, and sum x y = a sum x + b sum x squared. Solving them gives a and b.

If the coded years are chosen so that sum x = 0, the equations reduce to a = sum y / n = y-bar, and b = sum x y / sum x squared, which is the form used in most of the questions below.

Merits: it gives the most satisfactory measurement of the secular trend when the deviations are approximately normal, and its estimates are unbiased. Demerits: it gives too much weight to extremely large deviations, the line is best only for the period it covers, adding or removing a few periods may change its position, and the only real criterion for choosing a method is how well the trend line follows the general movement of the series.

Q. 7 (b) Given the following time series: Year 1968-1976; Value: 3, 4, 6, 8, 7, 7, 10, 13, 12. Determine the linear trend using least squares method by taking the origin

at the beginning period of the time period. Estimate the value for the year 1978.

Step 1: Take the origin at 1968, so coded years x = 0, 1, 2, 3, 4, 5, 6, 7, 8. Then n = 9, sum x = 36, sum x squared = 204, sum y = 70.

Step 2: sum x y = (0)(3) + (1)(4) + (2)(6) + (3)(8) + (4)(7) + (5)(7) + (6)(10) + (7)(13) + (8)(12) = 0 + 4 + 12 + 24 + 28 + 35 + 60 + 91 + 96 = 350.

Step 3: Slope b = (n sum x y - sum x sum y) / (n sum x squared - (sum x) squared) = (9 x 350 - 36 x 70) / (9 x 204 - 36 squared) = (3150 - 2520) / (1836 - 1296) = 630 / 540 = 1.17.

Step 4: Intercept a = (sum y - b sum x) / n = (70 - 1.1667 x 36) / 9 = (70 - 42) / 9 = 3.11.

Step 5: For 1978, x = 10, so the estimated value = 3.1111 + 1.1667 x 10 = 14.78 (the rounded equation 3.11 + 1.17 x gives 14.81).

Answer: y-hat = 3.11 + 1.17 x with origin at 1968; estimated value for 1978 = 14.78, as printed in the book.

Q. 8 (a) The following time series shows the number of road accidents in Punjab for the years 1977 to 1987: 1977: 2493, 1978: 2639, 1979: 2669, 1980: 3038, 1981: 3745, 1982: 4079, 1983: 4683, 1984: 4845, 1985: 4505, 1986: 4793, 1987: 4728. (i) Use the method of least squares to fit a straight line taking the origin at the middle of the time period. (ii) Find the trend values for this time series. (iii) Estimate the number of road accidents in 1989.

Step 1: The middle year is 1982, so coded years x = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5.

Then sum x = 0, sum x squared = 110, sum y = 42217.

Step 2: sum x y = (-5)(2493) + (-4)(2639) + (-3)(2669) + (-2)(3038) + (-1)(3745) + (0)(4079) + (1)(4683) + (2)(4845) + (3)(4505) + (4)(4793) + (5)(4728) = 29851.

Step 3: a = sum y / n = 42217 / 11 = 3837.91; b = sum x y / sum x squared = 29851 / 110 = 271.37.

Step 4: Trend values y-hat = 3837.91 + 271.37 x for x = -5 to 5 are: 2481.05, 2752.42, 3023.79, 3295.16, 3566.54, 3837.91, 4109.28, 4380.65, 4652.02, 4923.39, 5194.76, as printed in the book (the last value uses the rounded slope; the exact slope gives 5194.77).

Step 5: For 1989, x = 7, so the estimate = 3837.91 + 271.37 x 7 = 5737.50 (exact coefficients give 5737.52).

Answer: (i) y-hat = 3837.91 + 271.37 x with origin at 1982. (ii) Trend values as in Step 4. (iii) Estimated accidents in 1989 = 5737.50, as printed in the book.

Q. 8 (b) Fit a straight line y-hat = a + b x from the following results, for the years 1985-95 (both inclusive). Find out the trend values of y as well. sum x = 0, sum y = 438.9, sum x squared = 110, sum x y = -84.4.

Step 1: The years 1985-1995 are eleven years, so n = 11 and, with sum x = 0, a = sum y / n = 438.9 / 11 = 39.9.

Step 2: b = sum x y / sum x squared = -84.4 / 110 = -0.77 (exact value -0.7673).

Step 3: Trend values y-hat = 39.9 - 0.77 x for x = -5 to 5 are: 43.75, 42.98, 42.21, 41.44, 40.67, 39.90, 39.13, 38.36, 37.59, 36.82, 36.05.

Answer: y-hat = 39.9 - 0.77 x with origin at 1990; trend values as in Step 3, as printed in the book.

Q. 9 (a) Fit a straight line to the following data taking the origin at the middle of the time period and unit of measurement as 1/2 year and find the trend values: Year 1980-1985; Production (000): 10, 12, 8, 10, 14, 16.

Step 1: The middle of 1982 and 1983 is the origin, and with unit 1/2 year the coded years are x = -5, -3, -1, 1, 3, 5. Then sum x = 0, sum x squared = 70, sum y = 70.

Step 2: sum x y = (-5)(10) + (-3)(12) + (-1)(8) + (1)(10) + (3)(14) + (5)(16) = -50 - 36 - 8 + 10 + 42 + 80 = 38.

Step 3: a = 70 / 6 = 11.67 (book rounds to 11.66); b = 38 / 70 = 0.54 (exact 0.5429).

Step 4: Trend values from the book's rounded equation y-hat = 11.66 + 0.54 x are: 8.96, 10.04, 11.12, 12.20, 13.28, 14.36. With the exact coefficients they are 8.95, 10.04, 11.12, 12.21, 13.30, 14.38, differing by at most 0.02.

Answer: y-hat = 11.66 + 0.54 x with origin at the middle of 1982 and 1983 and unit of measurement as 1/2 year; trend values as in Step 4, as printed in the book.

Q. 9 (b) Fit a straight line to following data. Plot on the same graph paper the actual and trend values: Year 1970-1977; Value: 12, 15, 18, 25, 20, 22, 26, 30.

Step 1: Take the origin at January 1, 1974. An annual value is centred half a year from the origin's neighbouring years, so in half-year units the coded years are x = -7, -5, -3, -1, 1, 3, 5, 7. Then sum x = 0, sum x squared = 168, sum y = 168.

Step 2: sum x y = (-7)(12) + (-5)(15) + (-3)(18) + (-1)(25) + (1)(20) + (3)(22) + (5)(26) + (7)(30) = -84 - 75 - 54 - 25 + 20 + 66 + 130 + 210 = 188.

Step 3: a = 168 / 8 = 21; b = 188 / 168 = 1.12 (exact 1.1190).

Step 4: Trend values y-hat = 21 + 1.12 x are: 13.16, 15.40, 17.64, 19.88, 22.12, 24.36, 26.60, 28.84.

Answer: y-hat = 21 + 1.12 x with origin at January 1, 1974; trend values as in Step 4, as printed in the book. Plotting the actual values and these trend values on the same graph paper gives the straight trend line through the series.

Q. 9 (c) For the following time series, determine the trend by using the method of: (i) semi-average, (ii) 3-year moving average, (iii) least-squares for fitting a straight line. Year 1968-1976; Value: 2, 4, 6, 8, 7, 6, 8, 10, 12. Which of the trend do you prefer, and why?

Step 1 (i) Semi-average: first four values 2, 4, 6, 8 average to 20 / 4 = 5.0, placed at the middle of the first half, x = 1.5 after the 1968 origin; last four values 6, 8, 10, 12 average to 36 / 4 = 9.0, placed at x = 6.5. Slope = (9.0 - 5.0) / (6.5 - 1.5) = 4 / 5 = 0.8; intercept = 5.0 - 0.8 x 1.5 = 3.8. So y-hat = 3.8 + 0.8 x with origin at 1968, as printed in the book.

Step 2 (ii) 3-year moving averages: (2 + 4 + 6) / 3 = 4.0; (4 + 6 + 8) / 3 = 6.0; (6 +

8 + 7) / 3 = 7.0; (8 + 7 + 6) / 3 = 7.0; (7 + 6 + 8) / 3 = 7.0; (6 + 8 + 10) / 3 = 8.0; (8 + 10 + 12) / 3 = 10.0, as printed in the book.

Step 3 (iii) Least squares: with origin at the middle year 1972, coded years x = -4 to 4, sum x = 0, sum x squared = 60, sum y = 63, sum x y = 60. So a = 63 / 9 = 7 and b = 60 / 60 = 1, giving y-hat = 7 + x with origin at 1972, as printed in the book.

Answer: The least squares trend is preferred, because it uses all the observations, is free from personal bias, and gives a definite mathematical equation for the trend, whereas the semi-average method uses only two averages and the moving average method gives no equation and no trend values at the ends.

Q. 10 (a) Fit a second degree curve to the following time series and find the trend values: Year 1980-1990; Production: 23.2, 31.4, 39.8, 50.2, 62.9, 76.0, 92.0, 105.7, 122.8, 131.7, 151.1.

Step 1: Take the origin at 1985 with unit 1 year, so coded years x = -5 to 5. Then sum x = 0, sum x cubed = 0, sum x squared = 110, sum x to the fourth = 1958, sum y = 886.8, sum x y = 1429.8, sum x squared y = 9209.0.

Step 2: c = (n sum x squared y - sum x squared sum y) / (n sum x to the fourth - (sum x squared) squared) = (11 x 9209.0 - 110 x 886.8) / (11 x 1958 - 110 squared) = (101299 - 97548) / (21538 - 12100) = 3751 / 9438 = 0.3974.

Step 3: a = (sum y - c sum x squared) / n = (886.8 - 0.3974 x 110) / 11 = (886.8 - 43.71) / 11 = 76.64; b = sum x y / sum x squared = 1429.8 / 110 = 13.00.

Step 4: Trend values y-hat = 76.64 + 13.0 x + 0.3974 x squared for x = -5 to 5 are:

21.6, 31.0, 41.2, 52.2, 64.0, 76.6, 90.0, 104.2, 119.2, 135.0, 151.6.

Answer: y-hat = 76.64 + 13.0 x + 0.3974 x squared with origin at 1985 and unit of x as 1 year; trend values as in Step 4, as printed in the book.

Q. 10 (b) Fit a quadratic curve to the following time series: Year 1924, 1927, 1930, 1933, 1936, 1939, 1942; Index of coal price: 187, 142, 133, 129, 136, 169, 279. Use your results to estimate the values of the index for 1935.

Step 1: Take the origin at 1933 with unit 3 years, so coded years x = -3, -2, -1, 0, 1, 2, 3. Then sum x = 0, sum x squared = 28, sum x to the fourth = 196, sum y = 1175, sum x y = 333, sum x squared y = 5707.

Step 2: c = (7 x 5707 - 28 x 1175) / (7 x 196 - 28 squared) = (39949 - 32900) / (1372 - 784) = 7049 / 588 = 11.99 (exact 11.9881).

Step 3: a = (1175 - 11.99 x 28) / 7 = (1175 - 335.72) / 7 = 119.90; b = 333 / 28 = 11.89 (exact 11.8929).

Step 4: For 1935, which is 2 years after the 1933 origin with unit 3 years, x = 2 / 3. So y-hat = 119.9 + 11.89 x (2 / 3) + 11.99 x (4 / 9) = 119.9 + 7.93 + 5.33 = 133.16.

Answer: y-hat = 119.9 + 11.89 x + 11.99 x squared with origin at 1933 and unit of x as 3 years; estimated index for 1935 = 133.16, as printed in the book.

Q. 11 (a) Fit a second degree curve to the following time series: Year 1980-1987; Quantum Index: 100, 87, 96, 102, 139, 210, 289, 307.

Step 1: Take the origin at January 1, 1984 with unit 1/2 year, so coded years x = -7, -5, -3, -1, 1, 3, 5, 7. Then sum x = 0, sum x squared = 168, sum x to the fourth = 6216, sum y = 1330, sum x y = 2838, sum x squared y = 32338.

Step 2: c = (8 x 32338 - 168 x 1330) / (8 x 6216 - 168 squared) = (258704 - 223440) / (49728 - 28224) = 35264 / 21504 = 1.64 (exact 1.6399).

Step 3: a = (1330 - 1.64 x 168) / 8 = (1330 - 275.52) / 8 = 131.81 (book rounds to 131.8); b = 2838 / 168 = 16.89 (exact 16.8929).

Answer: y-hat = 131.8 + 16.89 x + 1.64 x squared with origin at January 1, 1984 and unit of x as 1/2 year, as printed in the book.

Q. 11 (b) Fit a quadratic curve (parabola) to the following data. Compute the trend values: Year 1931, 1933, 1935, 1937, 1939, 1941, 1943, 1945; Price index: 96, 87, 91, 102, 108, 139, 307, 289.

Step 1: Take the origin at 1938 with unit 1 year, so coded years x = -7, -5, -3, -1, 1, 3, 5, 7. Then sum x = 0, sum x squared = 168, sum x to the fourth = 6216, sum y = 1219, sum x y = 2601, sum x squared y = 30995.

Step 2: c = (8 x 30995 - 168 x 1219) / (8 x 6216 - 168 squared) = (247960 - 204792) / 21504 = 43168 / 21504 = 2.01 (exact 2.0074).

Step 3: a = (1219 - 2.01 x 168) / 8 = (1219 - 337.68) / 8 = 110.16 (exact coefficients give 110.22); b = 2601 / 168 = 15.48 (exact 15.4821).

Step 4: Trend values from the book's rounded equation y-hat = 110.16 + 15.48 x +

2.01 x squared are: 100.3, 83.0, 81.8, 96.7, 127.7, 174.7, 237.7, 317.0.

Answer: y-hat = 110.16 + 15.48 x + 2.01 x squared with origin at 1938 and unit of x as one year; trend values as in Step 4, as printed in the book.

Q. 12 (a) The following are the annual profits in thousands of rupees in certain business: Year 1977-1983; Profit: 88, 101, 105, 91, 113, 120, 132. (i) Fit a linear trend by the method of least-squares and make an estimate of the profits in 1985. (ii) Fit a parabolic trend. (iii) Determine which is the better fitting trend.

Step 1: Take the origin at 1980, so coded years x = -3 to 3. Then sum x = 0, sum x squared = 28, sum x to the fourth = 196, sum y = 750, sum x y = 178, sum x squared y = (9)(88) + (4)(101) + (1)(105) + (0)(91) + (1)(113) + (4)(120) + (9)(132) = 792 + 404 + 105 + 0 + 113 + 480 + 1188 = 3082.

Step 2 (i): a = 750 / 7 = 107.14; b = 178 / 28 = 6.36 (exact 6.3571). So the linear trend is y-hat = 107.14 + 6.36 x with origin at 1980 and unit of x as 1 year.

Step 3 (i, estimate): For 1985, x = 5, so the estimate is 107.14 + 6.36 x 5 = 138.94 (exact coefficients give 138.93). The book prints 139.04, which does not follow from its own equation; the computed value 138.94 is used here and the difference is flagged.

Step 4 (ii): c = (7 x 3082 - 28 x 750) / (7 x 196 - 28 squared) = (21574 - 21000) / (1372 - 784) = 574 / 588 = 0.976 (exact 0.9762); a = (750 - 0.976 x 28) / 7 = (750 - 27.33) / 7 = 103.24; b = 178 / 28 = 6.36. So the parabolic trend is y-hat = 103.24 + 6.36 x + 0.976 x squared with origin at 1980 and unit of x as 1 year, as printed in the book.

Step 5 (iii): The sum of squared residuals for the linear trend is 355.29, while for the parabolic trend it is 275.24. Since the parabolic trend leaves the smaller sum of squared residuals, it fits better.

Answer: (i) y-hat = 107.14 + 6.36 x; estimate for 1985 = 138.94 (book prints 139.04 - flagged above). (ii) y-hat = 103.24 + 6.36 x + 0.976 x squared. (iii) The parabolic trend is the better fitting trend.

Q. 12 (b) Fit a quadratic trend from the following results, for the years 1985-95 (both inclusive): sum x = sum x cubed = 0, sum x squared = 110, sum x to the fourth = 1958, sum y = 410, sum x y = 601, sum x squared y = 4587. Find out the trend values of y as well. Estimate the trend value for the year 1996.

Step 1: The years 1985-1995 are eleven years, so n = 11. c = (n sum x squared y - sum x squared sum y) / (n sum x to the fourth - (sum x squared) squared) = (11 x 4587 - 110 x 410) / (11 x 1958 - 110 squared) = (50457 - 45100) / (21538 - 12100) = 5357 / 9438 = 0.568 (exact 0.5676).

Step 2: a = (sum y - c sum x squared) / n = (410 - 0.568 x 110) / 11 = (410 - 62.48) / 11 = 31.60; b = sum x y / sum x squared = 601 / 110 = 5.46 (exact 5.4636).

Step 3: Trend values y-hat = 31.6 + 5.46 x + 0.568 x squared for x = -5 to 5 are: 18.50, 18.85, 20.33, 22.95, 26.71, 31.60, 37.63, 44.79, 53.09, 62.53, 73.10, as printed in the book.

Step 4: For 1996, x = 6, so the trend value = 31.6 + 5.46 x 6 + 0.568 x 36 = 31.6 + 32.76 + 20.45 = 84.81.

Answer: y-hat = 31.6 + 5.46 x + 0.568 x squared with origin at 1990 and unit of x as 1 year; trend values as in Step 3; estimated trend value for 1996 = 84.81, as printed in the book.

Q. 13 (a) Suppose that the linear trend equation is y-hat = 50 + 2 x, with origin at 1983 and unit of measurement for x is one year. Shift the origin at 1980.

Step 1: Let the new coded year be x-prime for origin 1980. A year t has old code x = t - 1983 and new code x-prime = t - 1980, so x-prime = x + 3, that is x = x-prime - 3.

Step 2: Substitute in the old equation: y-hat = 50 + 2 (x-prime - 3) = 50 + 2 x-prime - 6 = 44 + 2 x-prime.

Answer: y-hat = 44 + 2 x, with origin at the year 1980, as printed in the book.

Q. 13 (b) If the linear trend in the data for the years 1960 to 1965 both inclusive with origin at the middle of 1962 and 1963 is y-hat = 1306.667 + 73.428 x, the unit of x being one year, then determine the trend line with origin at 1960 and hence determine the trend values.

Step 1: The old origin is the middle of 1962 and 1963, that is 1962.5. A year t has old code x = t - 1962.5 and new code x-prime = t - 1960, so x-prime = x + 2.5, that is x = x-prime - 2.5.

Step 2: Substitute: y-hat = 1306.667 + 73.428 (x-prime - 2.5) = 1306.667 - 183.570 + 73.428 x-prime = 1123.097 + 73.428 x-prime.

Step 3: Trend values for 1960-1965 (x-prime = 0 to 5) are: 1123.097, 1196.525,

1269.953, 1343.381, 1416.809, 1490.237.

Answer: y-hat = 1123.097 + 73.428 x with origin at 1960; trend values as in Step 3, as printed in the book.

Q. 13 (c) The parabolic trend equation for the projects of a company (in thousand rupees) is y-hat = 10.4 + 0.6 x + 0.7 x squared, with origin at 1980 and unit of measurement for x is one year. Shift the origin to 1975.

Step 1: Let the new coded year be x-prime for origin 1975. A year t has old code x = t - 1980 and new code x-prime = t - 1975, so x-prime = x + 5, that is x = x-prime - 5.

Step 2: Substitute: y-hat = 10.4 + 0.6 (x-prime - 5) + 0.7 (x-prime - 5) squared = 10.4 + 0.6 x-prime - 3 + 0.7 (x-prime squared - 10 x-prime + 25).

Step 3: Collect terms: constant = 10.4 - 3 + 17.5 = 24.9; coefficient of x-prime = 0.6 - 7 = -6.4; coefficient of x-prime squared = 0.7.

Answer: y-hat = 24.9 - 6.4 x + 0.7 x squared, with origin at 1975, as printed in the book.

Exercise 16.2 — Objective Questions, Q. 1: Classify each movement (book pp. 266-267). T = secular trend, C = cyclical variation, S = seasonal variation, I = irregular variation

Q. 1 (i) Increased demand for foot-wears before Eid. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (ii) The decline in death rate due to advancement in science. — Which movement is it mainly associated with?

T — Secular trend.

Q. 1 (iii) A steel strike, delaying production for a week. — Which movement is it mainly associated with?

I — Irregular variation.

Q. 1 (iv) Rise in the prices of certain consumer goods due to tax increase in the annual budget. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (v) An era of prosperity in a business. — Which movement is it mainly associated with?

C — Cyclical variation.

Q. 1 (vi) The festival sale. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (vii) The production of sugar recorded for 1986, 1987, ..., 1992. — Which movement is it mainly associated with?

T — Secular trend.

Q. 1 (viii) The weekly statement of the sale of pens. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (ix) A fire in a factory delaying production for 3 weeks. — Which movement is it mainly associated with?

I — Irregular variation.

Q. 1 (x) An after Eid sale in a departmental store. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (xi) A need for increased wheat production due to a constant increase in population. — Which movement is it mainly associated with?

T — Secular trend.

Q. 1 (xii) The monthly rainfall in inches in a city over a 5-year period. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (xiii) A recession in a business. — Which movement is it mainly associated with?

C — Cyclical variation.

Q. 1 (xiv) An increase in employment during summer months. — Which movement is it mainly associated with?

S — Seasonal variation.

Q. 1 (xv) A continually increasing demand for smaller automobiles. — Which movement is it mainly associated with?

T — Secular trend.

Exercise 16.2 — Objective Questions, Q. 2: True or False (book p. 267; book prints the school-fee item with a repeated (iv) label — kept as printed)

Q. 2 (i) The graph of a time series is called histogram.

False

Q. 2 (ii) Secular trend is a short term variation.

False

Q. 2 (iii) Seasonal variations are regular in nature.

True

Q. 2 (iv) Secular trend has booms and depressions.

False

Q. 2 (v) Irregular variations are not regular in nature.

True

Q. 2 (iv, as printed a second time) The increase in the school fee in private schools is an irregular variation.

False

Q. 2 (vii) The increase in the number of patients in the hospitals is like secular trend in a time series.

True

Q. 2 (viii) The increase in the number of patients of heat stroke in summer is like secular trend in the time series.

False

Q. 2 (ix) The secular trend is measured by a straight line when a time series has an upward trend.

False

Q. 2 (x) The secular trend is measured by semi-averages method when trend is linear.

True

Q. 2 (xi) The straight line is fitted to a time series when the movements in the time series are linear.

True

Q. 2 (xii) In the measurement of secular trend by the method of least squares, the number of years must be odd.

False

Q. 2 (xiii) For a least squares linear trend y-hat = a + b x, the b is a variable and y-hat is the slope of the line.

False

Q. 2 (xiv) Seasonal variations can be measured only when the time series contains yearly values.

False

Chapter 17: Orientation of Computers

Exercise — Short Questions

What does this chapter teach about computers?

It gives an orientation to computers: their parts, storage and number system.

A hard disk is a secondary storage device.

All computers work on the binary number system.

Exercise — True or False (book p. 286)

A hard disk may also be referred to as a secondary storage device.

True

The central processing unit (CPU) contains a Control Unit that performs arithmetic and logic operations.

False

All computers work on a binary number system.

True

FORTRAN is a low-level language.

False

Exercise 17.1 (book p. 284)

Q. 1 (a) How are computers generally classified? What are the four major categories of computers?

Computers are generally classified by their size, speed and capacity, that is by the amount of data they can handle and the speed at which they process it. Depending

on their speed and memory size they are classified into different groups.

The four major categories are: (i) Micro Computers - used by one person at a time, commonly used in offices, homes and educational institutions, for example personal computers, laptops and notebooks. (ii) Mini Computers - larger than micro computers, used by large business organisations, banks, insurance companies and scientific forecasting bureaus, serving hundreds of users at a time. (iii) Mainframe Computers - very large computers used by large organisations for huge amounts of data, for example IBM S/390 mainframes. (iv) Super Computers - the most powerful, largest and fastest computers, performing more than 1 trillion calculations per second, used for nuclear research, weather forecasting and other highly complex problems, for example Cray T90.

Q. 1 (b) What is CPU? Why is it called the brain of the computer?

The Central Processing Unit (CPU) is the place where data is manipulated; in large computer systems it consists of several processing chips, while in a personal computer the entire CPU is a single chip called a micro processor.

It is called the brain of the computer because, like the brain, it controls and coordinates all the operations of the computer: it receives data and instructions, processes them, and directs the flow of data between the input devices, the memory, the ALU and the output devices.

The CPU has two parts: (i) the Control Unit, which manages all the resources of the computer, and (ii) the Arithmetic Logic Unit (ALU), in which arithmetic and logical operations are performed.

Q. 2 (a) Explain the working of Arithmetic Logical Unit (ALU).

The Arithmetic Logic Unit (ALU) is a part of the processor in which arithmetic and logical operations of the data are performed.

The arithmetic section of the ALU performs basic arithmetic operations such as addition, subtraction, multiplication, division and raising to a power.

The logical section performs logical operations: it compares data, for example whether the first number is greater than, less than, equal to, not equal to, greater than or equal to, or less than or equal to the second number.

The ALU includes a group of high speed memory locations called registers, which hold the data currently being processed: for example, the control unit might load two numbers from memory into registers in the ALU, tell the ALU to divide them, and then move the result back to memory or to an output device.

Q. 2 (a) [second (a) as printed] Explain the Control Unit.

The Control Unit is the part of the CPU that manages all the resources of the computer. Think of it as a traffic cop directing the flow of data through the CPU and to the other devices of the computer; it is the logical hub of the computer.

The CPU's instructions for carrying out commands are built into the control unit, and instructions set in machine code tell the control unit how to execute more complex operations.

Data must first be transferred from the input device or secondary storage to the main memory and then to the ALU for processing; the results then go back to the main memory and from there to the output device. The control unit takes care of all these activities, so it is like the nervous system of the body that supervises all the operations of the computer.

Q. 2 (c) What is secondary storage? How it differ from a primary storage?

Secondary storage is storage that holds data and programmes when they are not being used; it is also called auxiliary storage. Its purpose is to hold data permanently, even when the computer is switched off.

Primary storage (main memory, or RAM) temporarily stores data while it is being processed: the main memory holds the data and instructions that are currently being used, and when new data is stored in a location, the previous contents of that location are lost.

The difference is therefore: primary storage is temporary, fast and holds only the data in current use, while secondary storage is permanent and holds programmes and data for later use. The main types of secondary storage are magnetic storage (diskettes, hard disks, high-capacity floppy disks, disk cartridges, magnetic tape) and optical storage (CD-ROM, DVD-ROM, CD-R, CD-RW, Photo CD).

Q. 3 (a) Describe the various input and output devices with examples.

Input devices are hardware that translate data into a form that the computer can process: the most common input devices are the keyboard, mouse, scanners and digital cameras. The keyboard converts letters, numbers and characters into electrical signals; it has alphabet keys, numeric keys and special keys such as F1, F2, Alt, Ctrl, Shift and Tab. Pointing devices control the position of the cursor on the screen and include the mouse (point, click, drag, drop, right click), the trackball, the joystick, the touch pad and the light pen. Source data entry devices include scanners and bar-code readers.

Output devices are used to receive data from the CPU and convert it into a form that a user can understand or that can be stored. Output is of two kinds: softcopy output, which is temporary and is erased when the computer is switched off, for example the display on a monitor or sound from speakers; and hardcopy output, which is permanent, for example print on paper from a printer or a plotter.

The most commonly used softcopy devices are monitors (also called Visual Display Units), projectors and sound systems; the devices used to produce a hardcopy are the printer and the plotter.

Q. 4 (a) What is computers software?

Software is the set of instructions given to the computer to solve a problem or to control the operation of the computer; it is prepared in computer programming languages. In other words, software is the collection of programmes, procedures and documentation that tells the hardware what to do.

Hardware is the physical parts of the computer, such as the CPU, monitor, mouse and keyboard, while software is the non-physical part: without software, hardware alone cannot do any useful work.

Q. 4 (b) What are the functions of a system software?

System software consists of all programmes, including the operating system, that control the operations of the computer equipment. Its functions include: starting up the computer, loading, executing and storing application programs, storing and retrieving files, performing a variety of functions of the system, and translating programmes into machine language.

System software is classified into three major categories: operating systems, utilities and language translators. The operating system itself performs functions such as processor management, memory management, input/output management, file management, establishment and enforcement of job priority, automatic transition from job to job, interpretation of commands, coordination of compilers and assemblers, data security and integrity, production of dumps and error messages, maintenance of internal time clock and log of system usage, and providing an interface between the computer system and the operator.

Q. 5 (a) What do you know about DOS?

DOS stands for Disk Operating System, the most widely used operating system on personal computers. Several slightly different but compatible versions of DOS exist; the two most widely used, MS-DOS and PC-DOS, were originally developed by Microsoft Corporation in 1981.

MS-DOS is text driven and user unfriendly in the sense that the user types a line of text as a command and the computer then executes it; its commands are used to format a disk, copy, rename and delete files, organise and manage files on disk, and set up a tree structured hierarchical file management scheme. In MS-DOS 4.0 and above, additional enhanced commands and support for networks were added, and a

user interface called DOS shell provides a limited graphic interface and certain utility functions such as file maintenance.

Q. 5 (b) What does application software do and what are its generic types?

Application software consists of programmes that tell a computer how to produce information: when you think of the different ways people use computers in their careers or in their personal lives, you are thinking of examples of application software.

The most widely used personal computer application software are: (i) Word processing, (ii) Desktop publishing, (iii) Spreadsheet, (iv) Database, (v) Presentation graphics, (vi) Communications, (vii) Electronic mail, (viii) Personal information management, and (ix) Project management.

In short, application software does the particular job the user wants - writing, calculating, storing records, drawing or communicating - while system software manages the computer itself.

Q. 6 (a) What are computer languages and their types?

A programming language is a way of communication between the user and the computer: with the help of a programming language, the programmer writes programmes to solve problems with the computer. Each programming language has its own rules for writing a computer program, called the syntax of the language.

Programming languages are divided into two types: (i) Low Level Languages, which are close to machine language and include Machine Language, in which instructions are in the form of binary strings of 0's and 1's executed directly by the computer, and Assembly Language, in which symbolic codes (mnemonics) are used instead of binary codes. (ii) High Level Languages, which are close to human language and whose instructions are written in a form close to English, for example ALGOL (ALGOrithmic Language), BASIC (Beginners All-purpose Symbolic Instruction Code), COBOL (Common Business Oriented Language), PASCAL and FORTRAN (FORmula

TRANslation).

Q. 6 (b) What is an assembler?

An assembler is a language processor or translator that translates a programme written in an assembly language into machine code. Assembly language uses symbolic mnemonic codes instead of binary codes, so it is easier for a programmer to write than machine language, but the computer can only execute machine code; the assembler performs this translation, converting each assembly instruction into the corresponding machine instruction.

Q. 6 (c) What is a compiler?

A compiler is a translator that converts a programme written in a high-level language into machine code. Unlike an interpreter, which translates and executes one instruction at a time, a compiler translates the whole programme into machine code before it is executed.

A compiler therefore takes a high-level language programme as input and produces a complete machine language programme as output, which can then be run directly by the computer.

Q. 7 (a) What is Binary Number System? Why is it used in computer?

The Binary Number System is a number system whose base is 2 and which consists of only two digits, 0 and 1. In binary, any number greater than 1 is represented by a combination of binary digits: every digit in the number has a value that depends on its position and weight, for example 1011 = 1 x 2 cubed + 0 x 2 squared + 1 x 2 to the power 1 + 1.

It is used in computers for three reasons given in the chapter: (1) Electronic components work in binary mode by their very nature - a switch is either on (1) or off (0), a transistor is either conducting (1) or non-conducting (0), and magnetic material is either magnetised (1) or non-magnetised (0) - so the binary system, with only two digits, is the most suitable way to express these two possible states. (2)

Computer circuits only have to handle two binary digits rather than ten decimal digits, so the internal circuit design is simplified, making circuits less expensive and more reliable. (3) Everything that can be done with base 10 can also be done in binary.

The octal system (base 8) and the hexadecimal system (base 16) are used with computers because they represent binary values in a more compact form: an eight-digit binary number can be represented by a two-digit hexadecimal number.

Exercise 17.2 — Objective Questions, Q. 1: Fill in the blanks (book p. 285)

Q. 1 (i) keyboard is commonly used input device.

Q. 1 (ii) 1 MB equals 1,048,576 bytes.

Q. 1 (iii) Screen output is considered as a softcopy

Q. 1 (iv) CD-ROM is a type of Optical disk

Q. 1 (v) Software is a set of electronic instructions.

Q. 1 (vi) The most common type of computer memory is called RAM

Q. 1 (vii) A high speed memory that is built into the processor is called cache memory

Q. 1 (viii) RAM is called primary storage.

Q. 1 (ix, printed xi) Arithmetic operations are carried out by ALU unit.

Q. 1 (x) The monitor is the TV type screen that you view your programs on.

Q. 1 (xi) The keyboard allow you to type information into the computer.

Q. 1 (xii) Keyboard, mouse, scanner are the input devices.

Exercise 17.2 — Objective Questions, Q. 2: True or False, remaining items (book pp. 285-286; 4 items already in the earlier block above)

Q. 2 (i) 1 Kb = 1000 bytes.

True

Q. 2 (ii) Plotter is an input device to draw the graphs of the output.

False

Q. 2 (iii) A complete computer system has two parts: hardware and software.

True

Q. 2 (iv) The keyboard and monitor are examples of output devices.

False

Q. 2 (v) UNIX is a application software.

False

Q. 2 (vi) The purpose of a storage device is to hold data.

True

Q. 2 (vii) Base 2 is another name for the decimal number system.

False

Q. 2 (viii) A CD-ROM is an example of a magnetic storage device.

False